

It's hard to calculate M using a current in the little loop, so, exploiting the equality of the mutual inductances, I'll find the flux through the *little* loop when a current I flows in the *big* loop: $\Phi = MI$. The field of *one* long wire is $B = \frac{\mu_0 I}{2\pi s} \Rightarrow \Phi_1 = \frac{\mu_0 I}{2\pi} \int_a^{2a} \frac{1}{s} ds = \frac{\mu_0 I a}{2\pi} \ln 2$, so the *total* flux is

$$\Phi = 2\Phi_1 = \frac{\mu_0 I a \ln 2}{\pi} \Rightarrow M = \frac{\mu_0 a \ln 2}{\pi} \Rightarrow \boxed{\mathcal{E} = \frac{\mu_0 k a \ln 2}{\pi}}, \text{ in magnitude.}$$

Direction: The net flux (through the big loop), due to I in the little loop, is *into the page*. (Why? Field lines point *in*, for the inside of the little loop, and *out* everywhere outside the little loop. The big loop encloses *all* of the former, and only *part* of the latter, so *net* flux is *inward*.) This flux is *increasing*, so the induced current in the big loop is such that *its* field points *out* of the page: it flows counterclockwise.

Problem 7.24

$B = \mu_0 n I \Rightarrow \Phi_1 = \mu_0 n I \pi R^2$ (flux through a single turn). In a length l there are nl such turns, so the total flux is $\Phi = \mu_0 n^2 \pi R^2 I l$. The self-inductance is given by $\Phi = LI$, so the self-inductance per unit length is

$$\boxed{\mathcal{L} = \mu_0 n^2 \pi R^2}.$$

Problem 7.25

The field of one wire is $B_1 = \frac{\mu_0 I}{2\pi} \frac{1}{s}$, so $\Phi = 2 \cdot \frac{\mu_0 I}{2\pi} \cdot l \int_{\epsilon}^{d-\epsilon} \frac{ds}{s} = \frac{\mu_0 I l}{\pi} \ln\left(\frac{d-\epsilon}{\epsilon}\right)$. The ϵ in the numerator is negligible (compared to d), but in the denominator we *cannot* let $\epsilon \rightarrow 0$, else the flux is *infinite*.

$$\boxed{L = \frac{\mu_0 l}{\pi} \ln(d/\epsilon)}.$$
 Evidently the size of the wire itself is critical in determining L .

Problem 7.26

(a) In the quasistatic approximation $\mathbf{B} = \frac{\mu_0}{2\pi s} \hat{\phi}$. So $\Phi_1 = \frac{\mu_0 I}{2\pi} \int_a^b \frac{1}{s} h ds = \frac{\mu_0 I h}{2\pi} \ln(b/a)$.

This is the flux through *one* turn; the *total* flux is N times Φ_1 : $\Phi = \frac{\mu_0 N h}{2\pi} \ln(b/a) I_0 \cos(\omega t)$. So

$$\begin{aligned} \mathcal{E} &= -\frac{d\Phi}{dt} = \frac{\mu_0 N h}{2\pi} \ln(b/a) I_0 \omega \sin(\omega t) = \frac{(4\pi \times 10^{-7})(10^3)(10^{-2})}{2\pi} \ln(2)(0.5)(2\pi 60) \sin(\omega t) \\ &= \boxed{2.61 \times 10^{-4} \sin(\omega t)} \text{ (in volts), where } \omega = 2\pi 60 = 377/\text{s}. \quad I_r = \frac{\mathcal{E}}{R} = \frac{2.61 \times 10^{-4}}{500} \sin(\omega t) \\ &= \boxed{5.22 \times 10^{-7} \sin(\omega t)} \text{ (amperes).} \end{aligned}$$

(b) $\mathcal{E}_b = -L \frac{dI_r}{dt}$; where (Eq. 7.28) $L = \frac{\mu_0 N^2 h}{2\pi} \ln(b/a) = \frac{(4\pi \times 10^{-7})(10^6)(10^{-2})}{2\pi} \ln(2) = 1.39 \times 10^{-3}$ (henries).

Therefore $\mathcal{E}_b = -(1.39 \times 10^{-3})(5.22 \times 10^{-7} \omega) \cos(\omega t) = \boxed{-2.74 \times 10^{-7} \cos(\omega t)}$ (volts).

Ratio of amplitudes: $\frac{2.74 \times 10^{-7}}{2.61 \times 10^{-4}} = \boxed{1.05 \times 10^{-3}} = \frac{\mu_0 N^2 h \omega}{2\pi R} \ln(b/a)$.

Problem 7.27

With I positive clockwise, $\mathcal{E} = -L \frac{dI}{dt} = Q/C$, where Q is the charge on the capacitor; $I = \frac{dQ}{dt}$, so $\frac{d^2 Q}{dt^2} = -\frac{1}{LC} Q = -\omega^2 Q$, where $\omega = \frac{1}{\sqrt{LC}}$. The general solution is $Q(t) = A \cos \omega t + B \sin \omega t$. At $t = 0$,

$Q = CV$, so $A = CV$; $I(t) = \frac{dQ}{dt} = -A\omega \sin \omega t + B\omega \sin \omega t$. At $t = 0$, $I = 0$, so $B = 0$, and

$$I(t) = -CV\omega \sin \omega t = -V\sqrt{\frac{C}{L}} \sin\left(\frac{t}{\sqrt{LC}}\right).$$

If you put in a resistor, the oscillation is “damped”. This time $-L\frac{dI}{dt} = \frac{Q}{C} + IR$, so $L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{1}{C}Q = 0$. For an analysis of this case, see Purcell’s *Electricity and Magnetism* (Ch. 8) or any book on oscillations and waves.

Problem 7.28

(a) $W = \frac{1}{2}LI^2$. $L = \mu_0 n^2 \pi R^2 l$ (Prob. 7.24) $W = \frac{1}{2}\mu_0 n^2 \pi R^2 l I^2$.

(b) $W = \frac{1}{2} \oint (\mathbf{A} \cdot \mathbf{I}) dl$. $\mathbf{A} = (\mu_0 n I / 2) R \hat{\phi}$, at the surface (Eq. 5.72 or 5.73). So $W_1 = \frac{1}{2} \frac{\mu_0 n I}{2} R I \cdot 2\pi R$, for one turn. There are nl such turns in length l , so $W = \frac{1}{2} \mu_0 n^2 \pi R^2 l I^2$. ✓

(c) $W = \frac{1}{2\mu_0} \int B^2 d\tau$. $B = \mu_0 n I$, inside, and zero outside; $\int d\tau = \pi R^2 l$, so $W = \frac{1}{2\mu_0} \mu_0^2 n^2 I^2 \pi R^2 l = \frac{1}{2} \mu_0 n^2 \pi R^2 l I^2$. ✓

(d) $W = \frac{1}{2\mu_0} [\int B^2 d\tau - \oint (\mathbf{A} \times \mathbf{B}) \cdot d\mathbf{a}]$. This time $\int B^2 d\tau = \mu_0^2 n^2 I^2 \pi (R^2 - a^2) l$. Meanwhile,

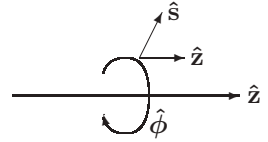
$\mathbf{A} \times \mathbf{B} = \mathbf{0}$ outside (at $s = b$). Inside, $\mathbf{A} = \frac{\mu_0 n I}{2} a \hat{\phi}$ (at $s = a$), while $\mathbf{B} = \mu_0 n I \hat{z}$.

$\mathbf{A} \times \mathbf{B} = \frac{1}{2} \mu_0^2 n^2 I^2 a (\underbrace{\hat{\phi} \times \hat{z}}_{\hat{s}})$

points inward (“out” of the volume)
↓

$\oint (\mathbf{A} \times \mathbf{B}) \cdot d\mathbf{a} = \int (\frac{1}{2} \mu_0^2 n^2 I^2 a \hat{s}) \cdot [a d\phi dz (-\hat{s})] = -\frac{1}{2} \mu_0^2 n^2 I^2 a^2 2\pi l$.

$W = \frac{1}{2\mu_0} [\mu_0^2 n^2 I^2 \pi (R^2 - a^2) l + \mu_0^2 n^2 I^2 \pi a^2 l] = \frac{1}{2} \mu_0 n^2 I^2 R^2 \pi l$. ✓



Problem 7.29

$$B = \frac{\mu_0 n I}{2\pi s}; \quad W = \frac{1}{2\mu_0} \int B^2 d\tau = \frac{1}{2\mu_0} \frac{\mu_0^2 n^2 I^2}{4\pi^2} \int \frac{1}{s^2} h s d\phi ds = \frac{\mu_0 n^2 I^2}{8\pi^2} h 2\pi \ln\left(\frac{b}{a}\right) = \boxed{\frac{1}{4\pi} \mu_0 n^2 I^2 h \ln(b/a)}.$$

$L = \frac{\mu_0}{2\pi} n^2 h \ln(b/a)$ (same as Eq. 7.28).

Problem 7.30

$$\oint \mathbf{B} \cdot d\mathbf{l} = B(2\pi s) = \mu_0 I_{\text{enc}} = \mu_0 I (s^2/R^2) \Rightarrow B = \frac{\mu_0 I s}{2\pi R^2}.$$

$$W = \frac{1}{2\mu_0} \int B^2 d\tau = \frac{1}{2\mu_0} \frac{\mu_0^2 I^2}{4\pi^2 R^4} \int_0^R s^2 (2\pi s) l ds = \frac{\mu_0 I^2 l}{4\pi R^4} \left(\frac{s^4}{4}\right) \Big|_0^R = \frac{\mu_0 l}{16\pi} I^2 = \frac{1}{2} L I^2.$$

So $L = \frac{\mu_0}{8\pi} l$, and $\mathcal{L} = L/l = \boxed{\mu_0/8\pi}$, independent of R !

Problem 7.31

(a) Initial current: $I_0 = \mathcal{E}_0/R$. So $-L\frac{dI}{dt} = IR \Rightarrow \frac{dI}{dt} = -\frac{R}{L}I \Rightarrow I = I_0 e^{-Rt/L}$, or $I(t) = \frac{\mathcal{E}_0}{R} e^{-Rt/L}$.

(b) $P = I^2 R = (\mathcal{E}_0/R)^2 e^{-2Rt/L} R = \frac{\mathcal{E}_0^2}{R} e^{-2Rt/L} = \frac{dW}{dt}$.

$$W = \frac{\mathcal{E}_0^2}{R} \int_0^\infty e^{-2Rt/L} dt = \frac{\mathcal{E}_0^2}{R} \left(-\frac{L}{2R} e^{-2Rt/L}\right) \Big|_0^\infty = \frac{\mathcal{E}_0^2}{R} (0 + L/2R) = \boxed{\frac{1}{2} L (\mathcal{E}_0/R)^2}.$$

(c) $W_0 = \frac{1}{2} L I_0^2 = \frac{1}{2} L (\mathcal{E}_0/R)^2$. ✓

Problem 7.32

(a) $\mathbf{B}_1 = \frac{\mu_0}{4\pi} \frac{1}{r^3} I_1 [3(\mathbf{a}_1 \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{a}_1]$, since $\mathbf{m}_1 = I_1 \mathbf{a}_1$. The flux through loop 2 is then

$\Phi_2 = \mathbf{B}_1 \cdot \mathbf{a}_2 = \frac{\mu_0}{4\pi} \frac{1}{r^3} I_1 [3(\mathbf{a}_1 \cdot \hat{\mathbf{r}})(\mathbf{a}_2 \cdot \hat{\mathbf{r}}) - \mathbf{a}_1 \cdot \mathbf{a}_2] = M I_1$. $M = \frac{\mu_0}{4\pi r^3} [3(\mathbf{a}_1 \cdot \hat{\mathbf{r}})(\mathbf{a}_2 \cdot \hat{\mathbf{r}}) - \mathbf{a}_1 \cdot \mathbf{a}_2]$.

(b) $\mathcal{E}_1 = -M \frac{dI_2}{dt}$, $\left. \frac{dW}{dt} \right|_1 = -\mathcal{E}_1 I_1 = M I_1 \frac{dI_2}{dt}$. (This is the work done per unit time *against* the mutual emf in loop 1—hence the minus sign.) So (since I_1 is constant) $W_1 = M I_1 I_2$, where I_2 is the final current in loop 2:

$$W = \frac{\mu_0}{4\pi\mathcal{L}} [3(\mathbf{m}_1 \cdot \hat{\mathbf{z}})(\mathbf{m}_2 \cdot \hat{\mathbf{z}}) - \mathbf{m}_1 \cdot \mathbf{m}_2].$$

Notice that this is *opposite in sign* to Eq. 6.35. In Prob. 6.21 we assumed that the magnitudes of the dipole moments were *fixed*, and we did not worry about the energy necessary to sustain the currents themselves—only the energy required to move them into position and rotate them into their final orientations. But in *this* problem we are including it *all*, and it is a curious fact that this merely changes the sign of the answer. For commentary on this subtle issue see R. H. Young, *Am. J. Phys.* **66**, 1043 (1998), and the references cited there.

Problem 7.33

(a) The (solenoid) magnetic field is

$$\mathbf{B} = \begin{cases} \mu_0 K \hat{\mathbf{z}} = \boxed{\mu_0 \sigma \omega R \hat{\mathbf{z}}} & (s < R), \\ \boxed{0} & (s > R). \end{cases}$$

From Example 7.7, the electric field is

$$\mathbf{E} = \begin{cases} -\frac{s}{2} \frac{dB}{dt} \hat{\phi} = \boxed{-\frac{sR}{2} \mu_0 \sigma \dot{\omega} \hat{\phi}} & (s < R), \\ -\frac{R^2}{2s} \frac{dB}{dt} \hat{\phi} = \boxed{-\frac{R^3}{2s} \mu_0 \sigma \dot{\omega} \hat{\phi}} & (s > R). \end{cases}$$

At the surface ($s = R$) $\mathbf{E} = -\frac{1}{2} \mu_0 R^2 \sigma \dot{\omega} \hat{\phi}$, so the torque on a length ℓ of the cylinder is

$$\mathbf{N} = -R(\sigma 2\pi R \ell) \left(\frac{1}{2} \mu_0 R^2 \sigma \dot{\omega} \right) \hat{\mathbf{z}} = -\pi \mu_0 \sigma^2 R^4 \dot{\omega} \ell \hat{\mathbf{z}},$$

and the work done per unit length is

$$\frac{W}{\ell} = -\pi \mu_0 \sigma^2 R^4 \int \frac{d\omega}{dt} d\phi.$$

But $d\phi = \omega dt$, and the integral becomes

$$\int_0^{\omega_f} \omega d\omega = \frac{1}{2} \omega_f^2; \quad \Rightarrow \quad \frac{W}{\ell} = -\boxed{\frac{\mu_0 \pi}{2} (\sigma \omega_f R^2)^2}.$$

(This is the work done *by the field*; the work *you* must do does not include the minus sign.)

(b) Because $\mathbf{B} = \mu_0 K \hat{\mathbf{z}} = \mu_0 \sigma \omega_f R \hat{\mathbf{z}}$ is *uniform* inside the solenoid (and zero outside), $W = \frac{1}{2\mu_0} B^2 \pi R^2 \ell$.

$$\frac{W}{\ell} = \frac{1}{2\mu_0} (\mu_0 \sigma \omega_f R)^2 \pi R^2 = \boxed{\frac{\mu_0 \pi}{2} (\sigma \omega_f R^2)^2}.$$

Problem 7.34

The displacement current density (Sect. 7.3.2) is $\mathbf{J}_d = \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \frac{I}{A} = \frac{I}{\pi a^2} \hat{\mathbf{z}}$. Drawing an “amperian loop” at radius s ,

$$\oint \mathbf{B} \cdot d\mathbf{l} = B \cdot 2\pi s = \mu_0 I_{d_{\text{enc}}} = \mu_0 \frac{I}{\pi a^2} \cdot \pi s^2 = \mu_0 I \frac{s^2}{a^2} \Rightarrow B = \frac{\mu_0 I s^2}{2\pi s a^2}; \quad \boxed{\mathbf{B} = \frac{\mu_0 I s}{2\pi a^2} \hat{\phi}.$$

Problem 7.35

$$(a) \mathbf{E} = \frac{\sigma(t)}{\epsilon_0} \hat{\mathbf{z}}; \quad \sigma(t) = \frac{Q(t)}{\pi a^2} = \frac{It}{\pi a^2}; \quad \boxed{\frac{It}{\pi \epsilon_0 a^2} \hat{\mathbf{z}}}.$$

$$(b) I_{d_{\text{enc}}} = J_d \pi s^2 = \epsilon_0 \frac{dE}{dt} \pi s^2 = \boxed{I \frac{s^2}{a^2}}. \quad \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{d_{\text{enc}}} \Rightarrow B 2\pi s = \mu_0 I \frac{s^2}{a^2} \Rightarrow \mathbf{B} = \boxed{\frac{\mu_0 I}{2\pi a^2} s \hat{\phi}}.$$

(c) A surface current flows radially outward over the left plate; let $I(s)$ be the total current crossing a circle of radius s . The charge density (at time t) is

$$\sigma(t) = \frac{[I - I(s)]t}{\pi s^2}.$$

Since we are told this is independent of s , it must be that $I - I(s) = \beta s^2$, for some constant β . But $I(a) = 0$, so $\beta a^2 = I$, or $\beta = I/a^2$. Therefore $I(s) = I(1 - s^2/a^2)$.

$$B 2\pi s = \mu_0 I_{\text{enc}} = \mu_0 [I - I(s)] = \mu_0 I \frac{s^2}{a^2} \Rightarrow \boxed{\mathbf{B} = \frac{\mu_0 I}{2\pi a^2} s \hat{\phi}}. \quad \checkmark$$

Problem 7.36

$$(a) \mathbf{J}_d = \epsilon_0 \frac{\mu_0 I_0 \omega^2}{2\pi} \cos(\omega t) \ln(a/s) \hat{\mathbf{z}}. \text{ But } I_0 \cos(\omega t) = I. \text{ So } \boxed{\mathbf{J}_d = \frac{\mu_0 \epsilon_0}{2\pi} \omega^2 I \ln(a/s) \hat{\mathbf{z}}}.$$

$$(b) I_d = \int \mathbf{J}_d \cdot d\mathbf{a} = \frac{\mu_0 \epsilon_0 \omega^2 I}{2\pi} \int_0^a \ln(a/s) (2\pi s ds) = \mu_0 \epsilon_0 \omega^2 I \int_0^a (s \ln a - s \ln s) ds$$

$$= \mu_0 \epsilon_0 \omega^2 I \left[(\ln a) \frac{s^2}{2} - \frac{s^2}{2} \ln s + \frac{s^2}{4} \right] \Big|_0^a = \mu_0 \epsilon_0 \omega^2 I \left[\cancel{\frac{a^2}{2} \ln a} - \cancel{\frac{a^2}{2} \ln a} + \frac{a^2}{4} \right] = \boxed{\frac{\mu_0 \epsilon_0 \omega^2 I a^2}{4}}.$$

$$(c) \boxed{\frac{I_d}{I} = \frac{\mu_0 \epsilon_0 \omega^2 a^2}{4}}. \quad \text{Since } \mu_0 \epsilon_0 = 1/c^2, I_d/I = (\omega a/2c)^2. \text{ If } a = 10^{-3} \text{ m, and } \frac{I_d}{I} = \frac{1}{100}, \text{ so that } \frac{\omega a}{2c} = \frac{1}{10},$$

$$\omega = \frac{2c}{10a} = \frac{3 \times 10^8 \text{ m/s}}{5 \times 10^{-3} \text{ m}}, \text{ or } \omega = 0.6 \times 10^{11} \text{ /s} = \boxed{6 \times 10^{10} \text{ /s}}; \quad \nu = \frac{\omega}{2\pi} \approx 10^{10} \text{ Hz, or } 10^4 \text{ megahertz. (This is the}$$

microwave region, *way* above radio frequencies.)

Problem 7.37

Physically, this is the field of a point charge q at the origin, out to an expanding spherical shell of radius vt ; outside this shell the field is zero. Evidently the shell carries the opposite charge, $-q$. *Mathematically*, using product rule #5 and Eq. 1.99:

$$\nabla \cdot \mathbf{E} = \theta(vt - r) \nabla \cdot \left(\frac{1}{4\pi \epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \right) + \frac{1}{4\pi \epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \cdot \nabla [\theta(vt - r)] = \frac{q}{\epsilon_0} \delta^3(\mathbf{r}) \theta(vt - r) + \frac{1}{4\pi \epsilon_0} \frac{q}{r^2} (\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}) \frac{\partial}{\partial r} \theta(vt - r).$$

But $\delta^3(\mathbf{r}) \theta(vt - r) = \delta^3(\mathbf{r}) \theta(t)$, and $\frac{\partial}{\partial r} \theta(vt - r) = -\delta(vt - r)$ (Prob. 1.46), so

$$\rho = \epsilon_0 \nabla \cdot \mathbf{E} = \boxed{q \delta^3(\mathbf{r}) \theta(t) - \frac{q}{4\pi r^2} \delta(vt - r)}.$$

(For $t < 0$ the field and the charge density are zero everywhere.)

Clearly $\nabla \cdot \mathbf{B} = 0$, and $\nabla \times \mathbf{E} = \mathbf{0}$ (since $\mathbf{E}$ has only an r component, and it is independent of θ and ϕ). There remains only the Ampère/Maxwell law, $\nabla \times \mathbf{B} = \mathbf{0} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial \mathbf{E} / \partial t$. Evidently

$$\mathbf{J} = -\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = -\epsilon_0 \left\{ \frac{q}{4\pi \epsilon_0 r^2} \frac{\partial}{\partial t} [\theta(vt - r)] \right\} \hat{\mathbf{r}} = \boxed{-\frac{q}{4\pi r^2} v \delta(vt - r) \hat{\mathbf{r}}}.$$

(The stationary charge at the origin does not contribute to $\mathbf{J}$, of course; for the expanding shell we have $\mathbf{J} = \rho \mathbf{v}$, as expected—Eq. 5.26.)

Problem 7.38

From $\nabla \cdot \mathbf{B} = \mu_0 \rho_m$ it follows that the field of a point monopole is $\mathbf{B} = \frac{\mu_0}{4\pi} \frac{q_m}{r^2} \hat{\mathbf{r}}$. The force law has the form $\mathbf{F} \propto q_m (\mathbf{B} - \frac{1}{c^2} \mathbf{v} \times \mathbf{E})$ (see Prob. 5.22—the c^2 is needed on dimensional grounds). The proportionality

constant must be 1 to reproduce “Coulomb’s law” for point charges at rest. So $\mathbf{F} = q_m \left(\mathbf{B} - \frac{1}{c^2} \mathbf{v} \times \mathbf{E} \right)$.

Problem 7.39

Integrate the “generalized Faraday law” (Eq. 7.44iii), $\nabla \times \mathbf{E} = -\mu_0 \mathbf{J}_m - \frac{\partial \mathbf{B}}{\partial t}$, over the surface of the loop:

$$\int (\nabla \times \mathbf{E}) \cdot d\mathbf{a} = \oint \mathbf{E} \cdot d\mathbf{l} = \mathcal{E} = -\mu_0 \int \mathbf{J}_m \cdot d\mathbf{a} - \frac{d}{dt} \int \mathbf{B} \cdot d\mathbf{a} = -\mu_0 I_{m_{\text{enc}}} - \frac{d\Phi}{dt}.$$

But $\mathcal{E} = -L \frac{dI}{dt}$, so $\frac{dI}{dt} = \frac{\mu_0}{L} I_{m_{\text{enc}}} + \frac{1}{L} \frac{d\Phi}{dt}$, or $I = \frac{\mu_0}{L} \Delta Q_m + \frac{1}{L} \Delta \Phi$, where ΔQ_m is the total magnetic charge passing through the surface, and $\Delta \Phi$ is the change in flux through the surface. If we use the *flat* surface, then $\Delta Q_m = q_m$ and $\Delta \Phi = 0$ (when the monopole is far away, $\Phi = 0$; the flux builds up to $\mu_0 q_m/2$ just before it passes through the loop; then it abruptly drops to $-\mu_0 q_m/2$, and rises back up to zero as the monopole disappears into the distance). If we use a huge balloon-shaped surface, so that q_m remains *inside* it on the far side, then $\Delta Q_m = 0$, but Φ rises monotonically from 0 to $\mu_0 q_m$. In either case,

$$I = \frac{\mu_0 q_m}{L}.$$

[The analysis is slightly different for a *superconducting* loop, but the conclusion is the same.]

Problem 7.40

$$E = \frac{V}{d} \Rightarrow J_c = \sigma E = \frac{1}{\rho} E = \frac{V}{\rho d}. \quad J_d = \frac{\partial D}{\partial t} = \frac{\partial}{\partial t} (\epsilon E) = \epsilon \frac{\partial}{\partial t} \left[\frac{V_0 \cos(2\pi \nu t)}{d} \right] = \frac{\epsilon V_0}{d} [-2\pi \nu \sin(2\pi \nu t)].$$

The ratio of the amplitudes is therefore:

$$\frac{J_c}{J_d} = \frac{V_0}{\rho d} \frac{d}{2\pi \nu \epsilon V_0} = \frac{1}{2\pi \nu \epsilon \rho} = [2\pi(4 \times 10^8)(81)(8.85 \times 10^{-12})(0.23)]^{-1} = \boxed{2.41}.$$

Problem 7.41

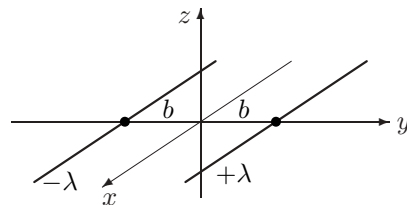
Begin with a different problem: two parallel wires carrying charges $+\lambda$ and $-\lambda$ as shown.

Field of one wire: $\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{\mathbf{s}}$; potential: $V = -\frac{\lambda}{2\pi\epsilon_0} \ln(s/a)$.

Potential of combination: $V = \frac{\lambda}{2\pi\epsilon_0} \ln(s_-/s_+)$,

or $V(y, z) = \frac{\lambda}{4\pi\epsilon_0} \ln \left\{ \frac{(y+b)^2 + z^2}{(y-b)^2 + z^2} \right\}$.

Find the locus of points of fixed V (i.e. equipotential surfaces):



$$e^{4\pi\epsilon_0 V/\lambda} \equiv \mu = \frac{(y+b)^2 + z^2}{(y-b)^2 + z^2} \implies \mu(y^2 - 2yb + b^2 + z^2) = y^2 + 2yb + b^2 + z^2;$$

$$y^2(\mu - 1) + b^2(\mu - 1) + z^2(\mu - 1) - 2yb(\mu + 1) = 0 \implies y^2 + z^2 + b^2 - 2yb\beta = 0 \quad \left(\beta \equiv \frac{\mu + 1}{\mu - 1} \right);$$

$$(y - b\beta)^2 + z^2 + b^2 - b^2\beta^2 = 0 \implies (y - b\beta^2) + z^2 = b^2(\beta^2 - 1).$$

This is a *circle*, with center at $y_0 = b\beta = b\left(\frac{\mu+1}{\mu-1}\right)$ and radius $= b\sqrt{\beta^2 - 1} = b\sqrt{\frac{(\mu^2+2\mu+1) - (\mu^2-2\mu+1)}{(\mu-1)^2}} = \frac{2b\sqrt{\mu}}{\mu-1}$.

This suggests an image solution to the problem at hand. We want $y_0 = d$, radius $= a$, and $V = V_0$. These determine the parameters b , μ , and λ of the image solution:

$$\frac{d}{a} = \frac{y_0}{\text{radius}} = \frac{b\left(\frac{\mu+1}{\mu-1}\right)}{\frac{2b\sqrt{\mu}}{\mu-1}} = \frac{\mu+1}{2\sqrt{\mu}}. \quad \text{Call } \frac{d}{a} \equiv \alpha.$$

$$4\alpha^2\mu = (\mu+1)^2 = \mu^2 + 2\mu + 1 \implies \mu^2 + (2-4\alpha^2)\mu + 1 = 0;$$

$$\mu = \frac{4\alpha^2 - 2 \pm \sqrt{4(1-2\alpha^2)^2 - 4}}{2} = 2\alpha^2 - 1 \pm \sqrt{1-4\alpha^2+4\alpha^4-1} = 2\alpha^2 - 1 \pm 2\alpha\sqrt{\alpha^2-1};$$

$$\frac{4\pi\epsilon_0 V_0}{\lambda} = \ln \mu \implies \lambda = \frac{4\pi\epsilon_0 V_0}{\ln(2\alpha^2 - 1 \pm 2\alpha\sqrt{\alpha^2-1})}. \quad \text{That's the line charge in the image problem.}$$

$$I = \int \mathbf{J} \cdot d\mathbf{a} = \sigma \int \mathbf{E} \cdot d\mathbf{a} = \sigma \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{\sigma}{\epsilon_0} \lambda l.$$

The current per unit length is $i = \frac{I}{l} = \frac{\sigma\lambda}{\epsilon_0} = \frac{4\pi\sigma V_0}{\ln(2\alpha^2 - 1 \pm 2\alpha\sqrt{\alpha^2-1})}$. Which sign do we want? Suppose the cylinders are far apart, $d \gg a$, so that $\alpha \gg 1$.

$$\begin{aligned} () &= 2\alpha^2 - 1 \pm 2\alpha^2\sqrt{1-1/\alpha^2} = 2\alpha^2 - 1 \pm 2\alpha^2 \left[1 - \frac{1}{2\alpha^2} - \frac{1}{8\alpha^4} + \dots \right] \\ &= 2\alpha^2(1 \pm 1) - (1 \pm 1) \mp \frac{1}{4\alpha^2} \pm \dots = \begin{cases} 4\alpha^2 - 2 - 1/2\alpha^2 + \dots \approx 4\alpha^2 & (+ \text{ sign}), \\ -1/4\alpha^2 & (- \text{ sign}). \end{cases} \end{aligned}$$

The current must surely *decrease* with increasing α , so evidently the $+$ sign is correct:

$$i = \frac{4\pi\sigma V_0}{\ln(2\alpha^2 - 1 + 2\alpha\sqrt{\alpha^2-1})}, \quad \text{where } \alpha = \frac{d}{a}.$$

Problem 7.42

$$\text{From Prob. 3.24, } \begin{cases} V_{\text{in}}(s, \phi) = \sum_{k=1}^{\infty} s^k b_k \sin(k\phi), & (s < a); \\ V_{\text{out}}(s, \phi) = \sum_{k=1}^{\infty} s^{-k} d_k \sin(k\phi), & (s > a). \end{cases}$$

(We don't need the cosine terms, because V is clearly an *odd* function of ϕ .) At $s = a$, $V_{\text{in}} = V_{\text{out}} = V_0\phi/2\pi$. Let's start with V_{in} , and use Fourier's trick to determine b_k :

$$\sum_{k=1}^{\infty} a^k b_k \sin(k\phi) = \frac{V_0\phi}{2\pi} \implies \sum_{k=1}^{\infty} a^k b_k \int_{-\pi}^{\pi} \sin(k\phi) \sin(k'\phi) d\phi = \frac{V_0}{2\pi} \int_{-\pi}^{\pi} \phi \sin(k'\phi) d\phi. \quad \text{But}$$

$$\int_{-\pi}^{\pi} \sin(k\phi) \sin(k'\phi) d\phi = \pi \delta_{kk'}, \quad \text{and}$$

$$\int_{-\pi}^{\pi} \phi \sin(k'\phi) d\phi = \left[\frac{1}{(k')^2} \sin(k'\phi) - \frac{\phi}{k'} \cos(k'\phi) \right] \Big|_{-\pi}^{\pi} = -\frac{2\pi}{k'} \cos(k'\phi) = -\frac{2\pi}{k'} (-1)^{k'}. \quad \text{So}$$

$$\pi a^k b_k = \frac{V_0}{2\pi} \left[-\frac{2\pi}{k} (-1)^k \right], \quad \text{or } b_k = -\frac{V_0}{\pi k} \left(-\frac{1}{a} \right)^k, \quad \text{and hence } V_{\text{in}}(s, \phi) = -\frac{V_0}{\pi} \sum_{k=1}^{\infty} \frac{1}{k} \left(-\frac{s}{a} \right)^k \sin(k\phi).$$

Similarly, $V_{\text{out}}(s, \phi) = -\frac{V_0}{\pi} \sum_{k=1}^{\infty} \frac{1}{k} \left(-\frac{a}{s}\right)^k \sin(k\phi)$. Both sums are of the form $S \equiv \sum_{k=1}^{\infty} \frac{1}{k} (-x)^k \sin(k\phi)$ (with $x = s/a$ for $r < a$ and $x = a/s$ for $r > a$). This series can be summed explicitly, using Euler's formula ($e^{i\theta} = \cos \theta + i \sin \theta$): $S = \text{Im} \sum_{k=1}^{\infty} \frac{1}{k} (-x)^k e^{ik\phi} = \text{Im} \sum_{k=1}^{\infty} \frac{1}{k} (-xe^{i\phi})^k$.

But $\ln(1+w) = w - \frac{1}{2}w^2 + \frac{1}{3}w^3 - \frac{1}{4}w^4 \dots = -\sum_{k=1}^{\infty} \frac{1}{k} (-w)^k$, so $S = -\text{Im} [\ln(1+xe^{i\phi})]$.

Now $\ln(Re^{i\theta}) = \ln R + i\theta$, so $S = -\theta$, where

$$\tan \theta = \frac{\text{Im}(1+xe^{i\phi})}{\text{Re}(1+xe^{i\phi})} = \frac{\frac{1}{2i} [(1+xe^{i\phi}) - (1+xe^{-i\phi})]}{\frac{1}{2} [(1+xe^{i\phi}) + (1+xe^{-i\phi})]} = \frac{x(e^{i\phi} - e^{-i\phi})}{i[2+x(e^{i\phi} + e^{-i\phi})]} = \frac{x \sin \phi}{1+x \cos \phi}.$$

Conclusion:
$$\begin{cases} V_{\text{in}}(s, \phi) = \frac{V_0}{\pi} \tan^{-1} \left(\frac{s \sin \phi}{a + s \cos \phi} \right), & (s < a); \\ V_{\text{out}}(s, \phi) = \frac{V_0}{\pi} \tan^{-1} \left(\frac{a \sin \phi}{s + a \cos \phi} \right), & (s > a). \end{cases}$$

(b) From Eq. 2.36, $\sigma(\phi) = -\epsilon_0 \left\{ \frac{\partial V_{\text{out}}}{\partial s} \Big|_{s=a} - \frac{\partial V_{\text{in}}}{\partial s} \Big|_{s=a} \right\}$.

$$\begin{aligned} \frac{\partial V_{\text{out}}}{\partial s} &= \frac{V_0}{\pi} \left\{ \frac{1}{\left[1 + \left(\frac{a \sin \phi}{s + a \cos \phi}\right)^2\right]} \frac{(-a \sin \phi)}{(s + a \cos \phi)^2} \right\} = -\frac{V_0}{\pi} \left[\frac{a \sin \phi}{(s + a \cos \phi)^2 + (a \sin \phi)^2} \right] \\ &= -\frac{V_0}{\pi} \left(\frac{a \sin \phi}{s^2 + 2as \cos \phi + a^2} \right); \\ \frac{\partial V_{\text{in}}}{\partial s} &= \frac{V_0}{\pi} \left\{ \frac{1}{\left[1 + \left(\frac{s \sin \phi}{a + s \cos \phi}\right)^2\right]} \frac{[(a + s \cos \phi) \sin \phi - s \sin \phi \cos \phi]}{(a + s \cos \phi)^2} \right\} = \frac{V_0}{\pi} \left[\frac{a \sin \phi}{(a + s \cos \phi)^2 + (s \sin \phi)^2} \right] \\ &= \frac{V_0}{\pi} \left(\frac{a \sin \phi}{s^2 + 2as \cos \phi + a^2} \right). \end{aligned}$$

$$\frac{\partial V_{\text{in}}}{\partial s} \Big|_{s=a} = -\frac{\partial V_{\text{out}}}{\partial s} \Big|_{s=a} = \frac{V_0}{2\pi a} \left(\frac{\sin \phi}{1 + \cos \phi} \right), \text{ so } \sigma(\phi) = \frac{\epsilon_0 V_0}{\pi a} \frac{\sin \phi}{(1 + \cos \phi)} = \boxed{\frac{\epsilon_0 V_0}{\pi a} \tan(\phi/2)}.$$

Problem 7.43

(a) $\nabla^2 V = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial(zf)}{\partial s} \right) + \frac{\partial^2(zf)}{\partial z^2} = \frac{z}{s} \frac{d}{ds} \left(s \frac{df}{ds} \right) = 0 \Rightarrow \frac{d}{ds} \left(s \frac{df}{ds} \right) = 0 \Rightarrow s \frac{df}{ds} = A$ (a constant) $\Rightarrow A \frac{ds}{s} = df \Rightarrow f = A \ln(s/s_0)$ (s_0 another constant). But (ii) $\Rightarrow f(b) = 0$, so $\ln(b/s_0) = 0$, so $s_0 = b$, and $V(s, z) = Az \ln(s/b)$. But (i) $\Rightarrow Az \ln(a/b) = -(I\rho z)/(\pi a^2)$, so $A = -\frac{I\rho}{\pi a^2} \frac{1}{\ln(a/b)}$; $V(s, z) = -\frac{I\rho z}{\pi a^2} \frac{\ln(s/b)}{\ln(a/b)}$.

$$(b) \mathbf{E} = -\nabla V = -\frac{\partial V}{\partial s} \hat{\mathbf{s}} - \frac{\partial V}{\partial z} \hat{\mathbf{z}} = \frac{I\rho z}{\pi a^2} \frac{1}{s \ln(a/b)} \hat{\mathbf{s}} + \frac{I\rho}{\pi a^2} \frac{\ln(s/b)}{\ln(a/b)} \hat{\mathbf{z}} = \boxed{\frac{I\rho}{\pi a^2 \ln(a/b)} \left(\frac{z}{s} \hat{\mathbf{s}} + \ln\left(\frac{s}{b}\right) \hat{\mathbf{z}} \right)}.$$

$$(c) \sigma(z) = \epsilon_0 [E_s(a^+) - E_s(a^-)] = \epsilon_0 \left[\frac{I\rho}{\pi a^2 \ln(a/b)} \left(\frac{z}{a} \right) - 0 \right] = \boxed{\frac{\epsilon_0 I\rho z}{\pi a^3 \ln(a/b)}}.$$

Problem 7.44

(a) Faraday's law says $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$, so $\mathbf{E} = \mathbf{0} \Rightarrow \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0} \Rightarrow \mathbf{B}(\mathbf{r})$ is independent of t .

(b) Faraday's law in integral form (Eq. 7.19) says $\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi/dt$. In the wire itself $\mathbf{E} = \mathbf{0}$, so Φ through the loop is constant.

(c) Ampère-Maxwell $\Rightarrow \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$, so $\mathbf{E} = \mathbf{0}$, $\mathbf{B} = \mathbf{0} \Rightarrow \mathbf{J} = \mathbf{0}$, and hence any current must be at the surface.

(d) From Eq. 5.70, a rotating shell produces a uniform magnetic field (inside): $\mathbf{B} = \frac{2}{3} \mu_0 \sigma \omega a \hat{\mathbf{z}}$. So to *cancel* such a field, we need $\sigma \omega a = -\frac{3B_0}{2\mu_0}$. Now $\mathbf{K} = \sigma \mathbf{v} = \sigma \omega a \sin \theta \hat{\phi}$, so $\boxed{\mathbf{K} = -\frac{3B_0}{2\mu_0} \sin \theta \hat{\phi}}.$

Problem 7.45

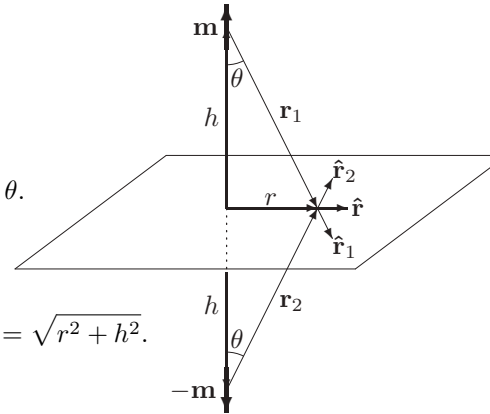
(a) To make the field *parallel* to the plane, we need image monopoles of the *same* sign (compare Figs. 2.13 and 2.14), so the image dipole points down $(-z)$.

(b) From Prob. 6.3 (with $r \rightarrow 2z$):

$$F = \boxed{\frac{3\mu_0}{2\pi} \frac{m^2}{(2z)^4}}. \quad \frac{3\mu_0}{2\pi} \frac{m^2}{(2h)^4} = Mg \Rightarrow h = \boxed{\frac{1}{2} \left(\frac{3\mu_0 m^2}{2\pi Mg} \right)^{1/4}}.$$

(c) Using Eq. 5.89, and referring to the figure:

$$\begin{aligned} \mathbf{B} &= \frac{\mu_0}{4\pi} \frac{1}{(r_1)^3} \{ [3(m \hat{\mathbf{z}} \cdot \hat{\mathbf{r}}_1) \hat{\mathbf{r}}_1 - m \hat{\mathbf{z}}] + [3(-m \hat{\mathbf{z}} \cdot \hat{\mathbf{r}}_2) \hat{\mathbf{r}}_2 + m \hat{\mathbf{z}}] \} \\ &= \frac{3\mu_0 m}{4\pi (r_1)^3} [(\hat{\mathbf{z}} \cdot \hat{\mathbf{r}}_1) \hat{\mathbf{r}}_1 - (\hat{\mathbf{z}} \cdot \hat{\mathbf{r}}_2) \hat{\mathbf{r}}_2]. \quad \text{But } \hat{\mathbf{z}} \cdot \hat{\mathbf{r}}_1 = -\hat{\mathbf{z}} \cdot \hat{\mathbf{r}}_2 = \cos \theta. \\ &= -\frac{3\mu_0 m}{4\pi (r_1)^3} \cos \theta (\hat{\mathbf{r}}_1 + \hat{\mathbf{r}}_2). \quad \text{But } \hat{\mathbf{r}}_1 + \hat{\mathbf{r}}_2 = 2 \sin \theta \hat{\mathbf{r}}. \\ &= -\frac{3\mu_0 m}{2\pi (r_1)^3} \sin \theta \cos \theta \hat{\mathbf{r}}. \quad \text{But } \sin \theta = \frac{r}{r_1}, \cos \theta = \frac{h}{r_1}, \text{ and } r_1 = \sqrt{r^2 + h^2}. \\ &= -\frac{3\mu_0 m h}{2\pi} \frac{r}{(r^2 + h^2)^{5/2}} \hat{\mathbf{r}}. \end{aligned}$$



Now $\mathbf{B} = \mu_0 (\mathbf{K} \times \hat{\mathbf{z}}) \Rightarrow \hat{\mathbf{z}} \times \mathbf{B} = \mu_0 \hat{\mathbf{z}} \times (\mathbf{K} \times \hat{\mathbf{z}}) = \mu_0 [\mathbf{K} - \hat{\mathbf{z}}(\mathbf{K} \cdot \hat{\mathbf{z}})] = \mu_0 \mathbf{K}$. (I used the BAC-CAB rule, and noted that $\mathbf{K} \cdot \hat{\mathbf{z}} = 0$, because the surface current is in the xy plane.)

$$\mathbf{K} = \frac{1}{\mu_0} (\hat{\mathbf{z}} \times \mathbf{B}) = -\frac{3mh}{2\pi} \frac{r}{(r^2 + h^2)^{5/2}} (\hat{\mathbf{z}} \times \hat{\mathbf{r}}) = -\frac{3mh}{2\pi} \frac{r}{(r^2 + h^2)^{5/2}} \hat{\phi}. \quad \text{qed}$$

Problem 7.46

Say the angle between the dipole ($\mathbf{m}_1$) and the z axis is θ (see diagram).

The field of the image dipole ($\mathbf{m}_2$) is

$$\mathbf{B}(z) = \frac{\mu_0}{4\pi} \frac{1}{(h+z)^3} [3(\mathbf{m}_2 \cdot \hat{\mathbf{z}}) \hat{\mathbf{z}} - \mathbf{m}_2]$$

for points on the z axis (Eq. 5.89). The torque on $\mathbf{m}_1$ is (Eq. 6.1)

$$\mathbf{N} = \mathbf{m}_1 \times \mathbf{B} = \frac{\mu_0}{4\pi(2h)^3} [3(\mathbf{m}_2 \cdot \hat{\mathbf{z}})(\mathbf{m}_1 \times \hat{\mathbf{z}}) - (\mathbf{m}_1 \times \mathbf{m}_2)].$$

But $\mathbf{m}_1 = m(\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{z}})$, $\mathbf{m}_2 = m(\sin \theta \hat{\mathbf{x}} - \cos \theta \hat{\mathbf{z}})$, so $\mathbf{m}_2 \cdot \hat{\mathbf{z}} = -m \cos \theta$, $\mathbf{m}_1 \times \hat{\mathbf{z}} = -m \sin \theta \hat{\mathbf{y}}$, and $\mathbf{m}_1 \times \mathbf{m}_2 = 2m^2 \sin \theta \cos \theta \hat{\mathbf{y}}$.

$$\mathbf{N} = \frac{\mu_0}{4\pi(2h)^3} [3m^2 \sin \theta \cos \theta \hat{\mathbf{y}} - 2m^2 \sin \theta \cos \theta \hat{\mathbf{y}}] = \frac{\mu_0 m^2}{4\pi(2h)^3} \sin \theta \cos \theta \hat{\mathbf{y}}.$$

Evidently the torque is zero for $\theta = 0$, $\pi/2$, or π . But 0 and π are clearly unstable, since the nearby ends of the dipoles (minus, in the figure) dominate, and they repel. The stable configuration is $\theta = \pi/2$: parallel to the surface (contrast Prob. 4.6).

In this orientation, $\mathbf{B}(z) = -\frac{\mu_0 m}{4\pi(h+z)^3} \hat{\mathbf{x}}$, and the force on $\mathbf{m}_1$ is (Eq. 6.3):

$$\mathbf{F} = \nabla \left[-\frac{\mu_0 m^2}{4\pi(h+z)^3} \right] \bigg|_{z=h} = \frac{3\mu_0 m^2}{4\pi(h+z)^4} \hat{\mathbf{z}} \bigg|_{z=h} = \frac{3\mu_0 m^2}{4\pi(2h)^4} \hat{\mathbf{z}}.$$

At equilibrium this force upward balances the weight Mg :

$$\frac{3\mu_0 m^2}{4\pi(2h)^4} = Mg \Rightarrow h = \frac{1}{2} \left(\frac{3\mu_0 m^2}{4\pi Mg} \right)^{1/4}.$$

Incidentally, this is $(1/2)^{1/4} = 0.84$ times the height it would adopt in the orientation *perpendicular* to the plane (Prob. 7.45b).

Problem 7.47

$\mathbf{f} = \mathbf{v} \times \mathbf{B}$; $\mathbf{v} = \omega a \sin \theta \hat{\phi}$; $\mathbf{f} = \omega a B_0 \sin \theta (\hat{\phi} \times \hat{\mathbf{z}})$. $\mathcal{E} = \int \mathbf{f} \cdot d\mathbf{l}$, and $d\mathbf{l} = a d\theta \hat{\theta}$.

So $\mathcal{E} = \omega a^2 B_0 \int_0^{\pi/2} \sin \theta (\hat{\phi} \times \hat{\mathbf{z}}) \cdot \hat{\theta} d\theta$. But $\hat{\theta} \cdot (\hat{\phi} \times \hat{\mathbf{z}}) = \hat{\mathbf{z}} \cdot (\hat{\theta} \times \hat{\phi}) = \hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = \cos \theta$.

$$\mathcal{E} = \omega a^2 B_0 \int_0^{\pi/2} \sin \theta \cos \theta d\theta = \omega a^2 B_0 \left[\frac{\sin^2 \theta}{2} \right]_0^{\pi/2} = \frac{1}{2} \omega a^2 B_0 \quad (\text{same as the rotating disk in Ex. 7.4}).$$

Problem 7.48

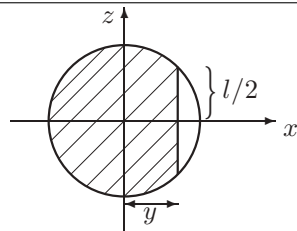
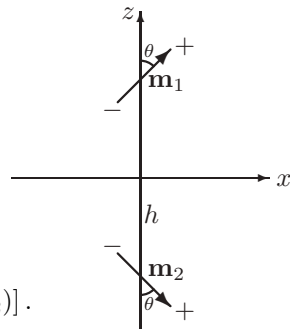
$F = IBl$; $\Phi = 2B \int_{-a}^y \sqrt{a^2 - x^2} dx$ (a = radius of circle).

$$\mathcal{E} = -\frac{d\Phi}{dt} = -2B \sqrt{a^2 - y^2} \frac{dy}{dt} = 2Bv \sqrt{a^2 - y^2} = IR.$$

$$I = \frac{2Bv}{R} \sqrt{a^2 - y^2}; l/2 = \sqrt{a^2 - y^2}. \text{ So } F = \frac{4B^2 v}{R} (a^2 - y^2) = mg.$$

$$v_{\text{circle}} = \frac{mgR}{4B^2(a^2 - y^2)};$$

$$t_{\text{circle}} = \int_{+a}^{-a} -\frac{dy}{v} = \frac{4B^2}{mgR} \int_{-a}^a (a^2 - y^2) dy = \frac{4B^2}{mgR} \left(a^2 y - \frac{1}{3} y^3 \right) \bigg|_{-a}^a = \frac{4B^2}{mgR} \left(\frac{4}{3} a^3 \right) = \frac{16}{3} \frac{B^2 a^3}{mgR}.$$



Problem 7.49

(a) From Prob. 5.52a,

$$\mathbf{A}(\mathbf{r}, t) = \frac{1}{4\pi} \int \frac{\mathbf{B}(\mathbf{r}', t) \times \hat{\mathbf{z}}}{r'^2} d\tau', \text{ so } \mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}.$$

[Check: $\nabla \times \mathbf{E} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}) = -\frac{\partial \mathbf{B}}{\partial t}$, and we recover Faraday's law.]

(b) The Coulomb field is zero inside and $\frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}} = \frac{1}{4\pi\epsilon_0} \frac{\sigma 4\pi R^2}{r^2} \hat{\mathbf{r}} = \frac{\sigma R^2}{\epsilon_0 r^2} \hat{\mathbf{r}}$ outside. The Faraday field is $-\frac{\partial \mathbf{A}}{\partial t}$, where $\mathbf{A}$ is given (in the quasistatic approximation) by Eq. 5.69, with ω a function of time. Letting $\dot{\omega} \equiv d\omega/dt$,

$$\mathbf{E}(r, \theta, \phi, t) = \begin{cases} -\frac{\mu_0 R \dot{\omega} \sigma}{3} r \sin \theta \hat{\phi} & (r < R), \\ \frac{\sigma R^2}{\epsilon_0 r^2} \hat{\mathbf{r}} - \frac{\mu_0 R^4 \dot{\omega} \sigma}{3} \frac{\sin \theta}{r^2} \hat{\phi} & (r > R). \end{cases}$$

Problem 7.50

$qBR = mv$ (Eq. 5.3). If R is to stay fixed, then $qR \frac{dB}{dt} = m \frac{dv}{dt} = ma = F = qE$, or $E = R \frac{dB}{dt}$. But $\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}$, so $E 2\pi R = -\frac{d\Phi}{dt}$, so $-\frac{1}{2\pi R} \frac{d\Phi}{dt} = R \frac{dB}{dt}$, or $B = -\frac{1}{2} \left(\frac{1}{\pi R^2} \Phi \right) + \text{constant}$. If at time $t = 0$ the field is off, then the constant is zero, and $B(R) = \frac{1}{2} \left(\frac{1}{\pi R^2} \Phi \right)$ (in magnitude). Evidently the field at R must be *half* the *average* field over the cross-section of the orbit. qed

Problem 7.51

In the quasistatic approximation the magnetic field of the wire is $\mathbf{B} = (\mu_0 I / 2\pi s) \hat{\phi}$, or, in Cartesian coordinates,

$$\mathbf{B} = \frac{\mu_0 I}{2\pi s} (-\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}) = \frac{\mu_0 I}{2\pi s} \left(-\frac{y}{s} \hat{\mathbf{x}} + \frac{x}{s} \hat{\mathbf{y}} \right) = \frac{\mu_0 I}{2\pi} \frac{(-y \hat{\mathbf{x}} + x \hat{\mathbf{y}})}{x^2 + y^2},$$

where x and y are measured *from the wire*. To convert to the stationary coordinates in the diagram, $y \rightarrow y - vt$:

$$\mathbf{B} = \frac{\mu_0 I}{2\pi} \frac{[-(y - vt) \hat{\mathbf{x}} + x \hat{\mathbf{y}}]}{x^2 + (y - vt)^2}.$$

Faraday's law says

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\frac{\mu_0 I}{2\pi} \left\{ \frac{v \hat{\mathbf{x}}}{x^2 + (y - vt)^2} - \frac{[-(y - vt) \hat{\mathbf{x}} + x \hat{\mathbf{y}}]}{[x^2 + (y - vt)^2]^2} (-2v)(y - vt) \right\}.$$

At $t = 0$, then,

$$\nabla \times \mathbf{E} = -\frac{\mu_0 I v}{2\pi} \left\{ \frac{\hat{\mathbf{x}}}{x^2 + y^2} + 2 \frac{[-y^2 \hat{\mathbf{x}} + xy \hat{\mathbf{y}}]}{[x^2 + y^2]^2} \right\} = -\frac{\mu_0 I v}{2\pi} \left\{ \frac{(x^2 - y^2) \hat{\mathbf{x}} + 2xy \hat{\mathbf{y}}}{[x^2 + y^2]^2} \right\} = -\frac{\mu_0 I v}{2\pi s^2} (\cos \phi \hat{\mathbf{s}} + \sin \phi \hat{\phi}).$$

Our problem is to find a vector function of s and ϕ (it obviously doesn't depend on z) whose divergence is zero, whose curl is as given above, that goes to zero at large s :

$$\mathbf{E}(s, \phi) = E_s(s, \phi) \hat{\mathbf{s}} + E_\phi(s, \phi) \hat{\phi} + E_z(s, \phi) \hat{\mathbf{z}},$$

with

$$\nabla \cdot \mathbf{E} = \frac{1}{s} \frac{\partial}{\partial s}(sE_s) + \frac{1}{s} \frac{\partial E_\phi}{\partial \phi} = 0,$$

$$(\nabla \times \mathbf{E})_s = \frac{1}{s} \frac{\partial E_z}{\partial \phi} = -\frac{\mu_0 I v}{2\pi s^2} \cos \phi$$

$$(\nabla \times \mathbf{E})_\phi = -\frac{\partial E_z}{\partial s} = -\frac{\mu_0 I v}{2\pi s^2} \sin \phi,$$

$$(\nabla \times \mathbf{E})_z = \frac{1}{s} \left[\frac{\partial}{\partial s}(sE_\phi) - \frac{\partial E_s}{\partial \phi} \right] = 0.$$

The first and last of these are satisfied if $E_s = E_\phi = 0$, the middle two are satisfied by $E_z = -\frac{\mu_0 I v}{2\pi s} \sin \phi$.

Evidently $\boxed{\mathbf{E} = -\frac{\mu_0 I v}{2\pi s} \sin \phi \hat{\mathbf{z}}}$. The electric and magnetic fields ride along with the wire.

Problem 7.52

Initially, $\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \Rightarrow T = \frac{1}{2}mv^2 = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{qQ}{r}$. After the magnetic field is on, the electron circles in a new orbit, of radius r_1 and velocity v_1 :

$$\frac{mv_1^2}{r_1} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r_1^2} + qv_1 B \Rightarrow T_1 = \frac{1}{2}mv_1^2 = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{qQ}{r_1} + \frac{1}{2}qv_1 r_1 B.$$

But $r_1 = r + dr$, so $(r_1)^{-1} = r^{-1} \left(1 + \frac{dr}{r}\right)^{-1} \cong r^{-1} \left(1 - \frac{dr}{r}\right)$, while $v_1 = v + dv$, $B = dB$. To first order, then,

$$T_1 = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{qQ}{r} \left(1 - \frac{dr}{r}\right) + \frac{1}{2}q(vr)dB, \text{ and hence } dT = T_1 - T = \frac{qvr}{2}dB - \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} dr.$$

Now, the induced electric field is $E = \frac{r}{2} \frac{dB}{dt}$ (Ex. 7.7), so $m \frac{dv}{dt} = qE = \frac{qr}{2} \frac{dB}{dt}$, or $m dv = \frac{qr}{2} dB$. The increase in kinetic energy is therefore $dT = d(\frac{1}{2}mv^2) = mv dv = \frac{qvr}{2} dB$. Comparing the two expressions, I conclude that $dr = 0$. qed

Problem 7.53

$\mathcal{E} = -\frac{d\Phi}{dt} = -\alpha$. So the current in R_1 and R_2 is $I = \frac{\alpha}{R_1 + R_2}$; by Lenz's law, it flows counterclockwise. Now

the voltage across R_1 (which voltmeter #1 measures) is $V_1 = IR_1 = \boxed{\frac{\alpha R_1}{R_1 + R_2}}$ (V_b is the *higher* potential),

and $V_2 = -IR_2 = \boxed{\frac{-\alpha R_2}{R_1 + R_2}}$ (V_b is *lower*).

Problem 7.54

(a) $\mathcal{E} = -\frac{d\Phi}{dt} = -\pi r^2 \frac{dB}{dt} = -\alpha \pi r^2 = IR \Rightarrow$ (in magnitude) $I = \boxed{\frac{\pi \alpha r^2}{R}}$. If $\mathbf{B}$ is *out* of the page, Lenz's law says the current is *clockwise*.

(b) Inside the shaded region, for a circle of radius s , apply Faraday's law:

$$\oint \mathbf{E} \cdot d\mathbf{l} = E 2\pi s = -\pi s^2 \alpha \Rightarrow \mathbf{E} = -\frac{\alpha s}{2} \hat{\boldsymbol{\phi}} = -\frac{\alpha s}{2} (-\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}) = \frac{\alpha}{2} (s \sin \phi \hat{\mathbf{x}} - s \cos \phi \hat{\mathbf{y}}) = \frac{\alpha}{2} (y \hat{\mathbf{x}} - x \hat{\mathbf{y}}).$$

Along the line from P to Q , $d\mathbf{l} = dx \hat{\mathbf{x}}$, and $y = r/\sqrt{2}$, so $V = -\int \mathbf{E} \cdot d\mathbf{l} = -\frac{\alpha}{2} \int y dx = -\frac{\alpha}{2} \frac{r}{\sqrt{2}} (r\sqrt{2})$. Thus

P is at the higher voltage, and the meter reads $\boxed{\frac{\alpha r^2}{2}}$.

That's the simplest way to do it. But you might instead regard the 3/4-circle+chord as a circuit, and use Kirchhoff's rule (the total emf around a closed loop is zero): $V + IR_1 - \alpha A_1 = 0$, where $R_1 = 3/4R$ is the resistance of the curved portion, and $A_1 = (3/4)\pi r^2 + r^2/2 = (r^2/4)(3\pi + 2)$ is the area (3/4 of the circle, plus the triangle). Then

$$V = \alpha \frac{r^2}{4}(3\pi + 2) - \frac{\pi r^2 \alpha}{R} \frac{3}{4}R = \frac{\alpha r^2}{4}(3\pi + 2 - 3\pi) = \frac{\alpha r^2}{2}. \quad \checkmark$$

Or we could do the same thing, using the small loop at the top: $-V + IR_2 - \alpha A_2 = 0$, where $R_2 = (1/4)R$ and $A_2 = (1/4)\pi r^2 - r^2/2 = (r^2/4)(\pi - 2)$.

$$V = \frac{\pi \alpha r^2}{R} \frac{R}{4} - \alpha \frac{r^2}{4}(\pi - 2) = \frac{\alpha r^2}{4}(\pi - \pi + 2) = \frac{\alpha r^2}{2}. \quad \checkmark$$

Problem 7.55

$$\mathcal{E} = vBh = -L \frac{dI}{dt}; \quad F = IhB = m \frac{dv}{dt}; \quad \frac{d^2v}{dt^2} = \frac{hB}{m} \frac{dI}{dt} = -\frac{hB}{m} \left(\frac{hB}{L} \right) v, \quad \boxed{\frac{d^2v}{dt^2} = -\omega^2 v}, \quad \text{with} \quad \boxed{\omega = \frac{hB}{\sqrt{mL}}}.$$

Problem 7.56

A point on the upper loop: $\mathbf{r}_2 = (a \cos \phi_2, a \sin \phi_2, z)$; a point on the lower loop: $\mathbf{r}_1 = (b \cos \phi_1, b \sin \phi_1, 0)$.

$$\begin{aligned} r^2 &= (\mathbf{r}_2 - \mathbf{r}_1)^2 = (a \cos \phi_2 - b \cos \phi_1)^2 + (a \sin \phi_2 - b \sin \phi_1)^2 + z^2 \\ &= a^2 \cos^2 \phi_2 - 2ab \cos \phi_2 \cos \phi_1 + b^2 \cos^2 \phi_1 + a^2 \sin^2 \phi_2 - 2ab \sin \phi_1 \sin \phi_2 + b^2 \sin^2 \phi_1 + z^2 \\ &= a^2 + b^2 + z^2 - 2ab(\cos \phi_2 \cos \phi_1 + \sin \phi_2 \sin \phi_1) = a^2 + b^2 + z^2 - 2ab \cos(\phi_2 - \phi_1) \\ &= (a^2 + b^2 + z^2)[1 - 2\beta \cos(\phi_2 - \phi_1)] = \frac{ab}{\beta}[1 - 2\beta \cos(\phi_2 - \phi_1)]. \end{aligned}$$

$dl_1 = b d\phi_1 \hat{\phi}_1 = b d\phi_1 [-\sin \phi_1 \hat{\mathbf{x}} + \cos \phi_1 \hat{\mathbf{y}}]$; $dl_2 = a d\phi_2 \hat{\phi}_2 = a d\phi_2 [-\sin \phi_2 \hat{\mathbf{x}} + \cos \phi_2 \hat{\mathbf{y}}]$, so
 $dl_1 \cdot dl_2 = ab d\phi_1 d\phi_2 [\sin \phi_1 \sin \phi_2 + \cos \phi_1 \cos \phi_2] = ab \cos(\phi_2 - \phi_1) d\phi_1 d\phi_2$.

$$M = \frac{\mu_0}{4\pi} \oint \oint \frac{dl_1 \cdot dl_2}{r} = \frac{\mu_0}{4\pi} \frac{ab}{\sqrt{ab/\beta}} \iint \frac{\cos(\phi_2 - \phi_1)}{\sqrt{1 - 2\beta \cos(\phi_2 - \phi_1)}} d\phi_2 d\phi_1.$$

Both integrals run from 0 to 2π . Do the ϕ_2 integral first, letting $u \equiv \phi_2 - \phi_1$:

$$\int_{-\phi_1}^{2\pi - \phi_1} \frac{\cos u}{\sqrt{1 - 2\beta \cos u}} du = \int_0^{2\pi} \frac{\cos u}{\sqrt{1 - 2\beta \cos u}} du$$

(since the integral runs over a complete cycle of $\cos u$, we may as well change the limits to $0 \rightarrow 2\pi$). Then the ϕ_1 integral is just 2π , and

$$M = \frac{\mu_0}{4\pi} \sqrt{ab\beta} 2\pi \int_0^{2\pi} \frac{\cos u}{\sqrt{1 - 2\beta \cos u}} du = \frac{\mu_0}{2} \sqrt{ab\beta} \int_0^{2\pi} \frac{\cos u}{\sqrt{1 - 2\beta \cos u}} du.$$

(a) If a is small, then $\beta \ll 1$, so (using the binomial theorem)

$$\frac{1}{\sqrt{1 - 2\beta \cos u}} \cong 1 + \beta \cos u, \quad \text{and} \quad \int_0^{2\pi} \frac{\cos u}{\sqrt{1 - 2\beta \cos u}} du \cong \int_0^{2\pi} \cos u du + \beta \int_0^{2\pi} \cos^2 u du = 0 + \beta\pi,$$

and hence $M = (\mu_0\pi/2)\sqrt{ab\beta^3}$. Moreover, $\beta \cong ab/(b^2 + z^2)$, so $M \cong \boxed{\frac{\mu_0\pi a^2 b^2}{2(b^2 + z^2)^{3/2}}}$ (same as in Prob. 7.22).

(b) More generally,

$$(1 + \epsilon)^{-1/2} = 1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \frac{5}{16}\epsilon^3 + \cdots \Rightarrow \frac{1}{\sqrt{1 - 2\beta \cos u}} = 1 + \beta \cos u + \frac{3}{2}\beta^2 \cos^2 u + \frac{5}{2}\beta^3 \cos^3 u + \cdots,$$

so

$$\begin{aligned} M &= \frac{\mu_0}{2} \sqrt{ab\beta} \left\{ \int_0^{2\pi} \cos u \, du + \beta \int_0^{2\pi} \cos^2 u \, du + \frac{3}{2}\beta^2 \int_0^{2\pi} \cos^3 u \, du + \frac{5}{2}\beta^3 \int_0^{2\pi} \cos^4 u \, du + \cdots \right\} \\ &= \frac{\mu_0}{2} \sqrt{ab\beta} \left[0 + \beta(\pi) + \frac{3}{2}\beta^2(0) + \frac{5}{2}\beta^3\left(\frac{3}{4}\pi\right) + \cdots \right] = \boxed{\frac{\mu_0\pi}{2} \sqrt{ab\beta^3} \left(1 + \frac{15}{8}\beta^2 + (\)\beta^4 + \cdots \right)}. \quad \text{qed} \end{aligned}$$

Problem 7.57

Let Φ be the flux of $\mathbf{B}$ through a *single* loop of either coil, so that $\Phi_1 = N_1\Phi$ and $\Phi_2 = N_2\Phi$. Then

$$\mathcal{E}_1 = -N_1 \frac{d\Phi}{dt}, \quad \mathcal{E}_2 = -N_2 \frac{d\Phi}{dt}, \quad \text{so } \frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{N_2}{N_1}. \quad \text{qed}$$

Problem 7.58

(a) Suppose current I_1 flows in coil 1, and I_2 in coil 2. Then (if Φ is the flux through *one* turn):

$$\Phi_1 = I_1 L_1 + M I_2 = N_1 \Phi; \quad \Phi_2 = I_2 L_2 + M I_1 = N_2 \Phi, \quad \text{or } \Phi = I_1 \frac{L_1}{N_1} + I_2 \frac{M}{N_1} = I_2 \frac{L_2}{N_2} + I_1 \frac{M}{N_2}.$$

In case $I_1 = 0$, we have $\frac{M}{N_1} = \frac{L_2}{N_2}$; if $I_2 = 0$, we have $\frac{L_1}{N_1} = \frac{M}{N_2}$. Dividing: $\frac{M}{N_1} = \frac{L_2}{M}$, or $L_1 L_2 = M^2$. qed

(b) $-\mathcal{E}_1 = \frac{d\Phi_1}{dt} = L_1 \frac{dI_1}{dt} + M \frac{dI_2}{dt} = V_1 \cos(\omega t)$; $-\mathcal{E}_2 = \frac{d\Phi_2}{dt} = L_2 \frac{dI_2}{dt} + M \frac{dI_1}{dt} = -I_2 R$. qed

(c) Multiply the first equation by L_2 : $L_1 L_2 \frac{dI_1}{dt} + L_2 \frac{dI_2}{dt} M = L_2 V_1 \cos \omega t$. Plug in $L_2 \frac{dI_2}{dt} = -I_2 R - M \frac{dI_1}{dt}$.

$$M^2 \frac{dI_1}{dt} - M R I_2 - M^2 \frac{dI_1}{dt} = L_2 V_1 \cos \omega t \Rightarrow \boxed{I_2(t) = -\frac{L_2 V_1}{M R} \cos \omega t.} \quad L_1 \frac{dI_1}{dt} + M \left(\frac{L_2 V_1}{M R} \omega \sin \omega t \right) = V_1 \cos \omega t.$$

$$\frac{dI_1}{dt} = \frac{V_1}{L_1} \left(\cos \omega t - \frac{L_2}{R} \omega \sin \omega t \right) \Rightarrow \boxed{I_1(t) = \frac{V_1}{L_1} \left(\frac{1}{\omega} \sin \omega t + \frac{L_2}{R} \cos \omega t \right).}$$

(d) $\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{I_2 R}{V_1 \cos \omega t} = \frac{-\frac{L_2 V_1}{M R} \cos \omega t R}{V_1 \cos \omega t} = -\frac{L_2}{M} = -\frac{N_2}{N_1}$. The ratio of the amplitudes is $\frac{N_2}{N_1}$. qed

(e) $P_{\text{in}} = V_{\text{in}} I_1 = (V_1 \cos \omega t) \left(\frac{V_1}{L_1} \right) \left(\frac{1}{\omega} \sin \omega t + \frac{L_2}{R} \cos \omega t \right) = \boxed{\frac{(V_1)^2}{L_1} \left(\frac{1}{\omega} \sin \omega t \cos \omega t + \frac{L_2}{R} \cos^2 \omega t \right)}.$

$P_{\text{out}} = V_{\text{out}} I_2 = (I_2)^2 R = \boxed{\frac{(L_2 V_1)^2}{M^2 R} \cos^2 \omega t.}$ Average of $\cos^2 \omega t$ is $1/2$; average of $\sin \omega t \cos \omega t$ is zero.

So $\langle P_{\text{in}} \rangle = \frac{1}{2} (V_1)^2 \left(\frac{L_2}{L_1 R} \right)$; $\langle P_{\text{out}} \rangle = \frac{1}{2} (V_1)^2 \left[\frac{(L_2)^2}{M^2 R} \right] = \frac{1}{2} (V_1)^2 \left[\frac{(L_2)^2}{L_1 L_2 R} \right]$; $\boxed{\langle P_{\text{in}} \rangle = \langle P_{\text{out}} \rangle = \frac{(V_1)^2 L_2}{2 L_1 R}}.$

Problem 7.59

(a) The charge flowing into dz in time dt is

$$dq = I(z) dt - I(z + dz) dt = -\frac{dI}{dz} dz dt = \lambda(t + dt) dz - \lambda(t) dz = \frac{d\lambda}{dt} dt dz \Rightarrow \frac{d\lambda}{dt} = -\frac{dI}{dz}. \quad \checkmark$$

Since the left side is a function only of t , and the right side is a function only of z , they must both be constant; call it k :

$$\frac{d\lambda}{dt} = k \Rightarrow \lambda(t) = kt + C_1; \quad \frac{dI}{dz} = -k \Rightarrow I(z) = -kz + C_2.$$

If $\lambda(0) = 0$ and $I(0) = 0$ the constants C_1 and C_2 must both be zero: $\lambda(t) = kt$, $I(z) = -kz$. $\checkmark$

(b) In the quasistatic approximation, $\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{\mathbf{s}} = \frac{kt}{2\pi\epsilon_0 s} \hat{\mathbf{s}}$; $\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi} = -\frac{\mu_0 kz}{2\pi s} \hat{\phi}$.

$$\nabla \cdot \mathbf{E} = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{kt}{2\pi\epsilon_0 s} \right) = 0 \quad \checkmark$$

$$\nabla \cdot \mathbf{B} = \frac{1}{s} \frac{\partial}{\partial \phi} \left(-\frac{\mu_0 kz}{2\pi s} \right) = 0 \quad \checkmark$$

$$\nabla \times \mathbf{E} = \frac{\partial E}{\partial z} \hat{\phi} - \frac{1}{s} \frac{\partial E}{\partial \phi} \hat{\mathbf{z}} = \mathbf{0} = -\frac{\partial \mathbf{B}}{\partial t} \quad \checkmark$$

$$\nabla \times \mathbf{B} = -\frac{\partial B}{\partial z} \hat{\phi} + \frac{1}{s} \frac{\partial (sB)}{\partial s} \hat{\mathbf{z}} = \frac{\mu_0 k}{2\pi s} \hat{\mathbf{z}} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad \checkmark$$

For a gaussian cylinder of radius s about the z axis, at height z and with width dz :

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{kt}{2\pi\epsilon_0 s} (2\pi s) dz = \frac{kt}{\epsilon_0} dz = \frac{\lambda dz}{\epsilon_0} = \frac{Q_{\text{enc}}}{\epsilon_0}. \quad \checkmark \quad \oint \mathbf{B} \cdot d\mathbf{a} = 0. \quad \checkmark$$

For a circular amperian loop of radius s about the z axis, at height z :

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0 = -\frac{d}{dt} \int \mathbf{B} \cdot d\mathbf{a}, \quad \checkmark \quad \oint \mathbf{B} \cdot d\mathbf{l} = -\frac{\mu_0 kz}{2\pi s} (2\pi s) = \mu_0 I = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d}{dt} \int \mathbf{E} \cdot d\mathbf{a}. \quad \checkmark$$

(Note that $\int \mathbf{B} \cdot d\mathbf{a}$ and $\int \mathbf{E} \cdot d\mathbf{a}$ are zero through this loop.)

Problem 7.60

(a) The continuity equation says $\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}$. Here the right side is independent of t , so we can integrate: $\rho(t) = (-\nabla \cdot \mathbf{J})t + \text{constant}$. The “constant” may be a function of $\mathbf{r}$ —it’s only constant with respect to t . So, putting in the $\mathbf{r}$ dependence explicitly, and noting that $\nabla \cdot \mathbf{J} = -\dot{\rho}(\mathbf{r}, 0)$, $\rho(\mathbf{r}, t) = \dot{\rho}(\mathbf{r}, 0)t + \rho(\mathbf{r}, 0)$. qed

(b) Suppose $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho \hat{\mathbf{z}}}{r^2} d\tau$ and $\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J} \times \hat{\mathbf{z}}}{r^2} d\tau$. We want to show that $\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$; $\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho$, and $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$, provided that $\mathbf{J}$ is independent of t .

We know from Ch. 2 that Coulomb’s law $\left(\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho \hat{\mathbf{z}}}{r^2} d\tau \right)$ satisfies $\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho$ and $\nabla \times \mathbf{E} = \mathbf{0}$. Since $\mathbf{B}$ is constant (in time), the $\nabla \cdot \mathbf{E}$ and $\nabla \times \mathbf{E}$ equations are satisfied. From Chapter 5 (specifically, Eqs. 5.47-5.50) we know that the Biot-Savart law satisfies $\nabla \cdot \mathbf{B} = 0$. It remains only to check $\nabla \times \mathbf{B}$. The argument in Sect. 5.3.2 carries through until the equation following Eq. 5.54, where I invoked $\nabla' \cdot \mathbf{J} = 0$. In its place we now put $\nabla' \cdot \mathbf{J} = -\dot{\rho}$:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} - \frac{\mu_0}{4\pi} \int \underbrace{(\mathbf{J} \cdot \nabla') \frac{\hat{\mathbf{z}}}{r^2}}_{(-\mathbf{J} \cdot \nabla') \frac{\hat{\mathbf{z}}}{r^2}} d\tau \quad (\text{Eqs. 5.51-5.53})$$

$$(-\mathbf{J} \cdot \nabla') \frac{\hat{\mathbf{z}}}{r^2} \quad (\text{Eq. 5.54})$$

Integration by parts yields two terms, one of which becomes a surface integral, and goes to zero. The other is

$$\frac{\hat{\mathbf{z}}}{z^3} \nabla' \cdot \mathbf{J} = \frac{\hat{\mathbf{z}}}{z^2} (-\dot{\rho}). \text{ So:}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} - \frac{\mu_0}{4\pi} \int \frac{\hat{\mathbf{z}}}{z^2} (-\dot{\rho}) d\tau = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left\{ \frac{1}{4\pi \epsilon_0} \int \frac{\rho \hat{\mathbf{z}}}{z^3} d\tau \right\} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad \text{qed}$$

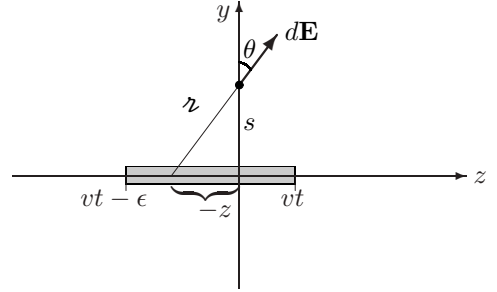
Problem 7.61

$$(a) dE_z = \frac{1}{4\pi\epsilon_0} \frac{(-\lambda)dz}{z^2} \sin \theta$$

$$\sin \theta = \frac{-z}{z}; \quad z = \sqrt{z^2 + s^2}$$

$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int \frac{z dz}{(z^2 + s^2)^{3/2}} = \frac{\lambda}{4\pi\epsilon_0} \left[\frac{-1}{\sqrt{z^2 + s^2}} \right] \Big|_{vt-\epsilon}^{vt}$$

$$E_z = \frac{\lambda}{4\pi\epsilon_0} \left\{ \frac{1}{\sqrt{(vt-\epsilon)^2 + s^2}} - \frac{1}{\sqrt{(vt)^2 + s^2}} \right\}.$$



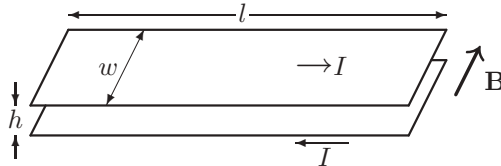
(b)

$$\begin{aligned} \Phi_E &= \frac{\lambda}{4\pi\epsilon_0} \int_0^a \left\{ \frac{1}{\sqrt{(vt-\epsilon)^2 + s^2}} - \frac{1}{\sqrt{(vt)^2 + s^2}} \right\} 2\pi s ds = \frac{\lambda}{2\epsilon_0} \left[\sqrt{(vt-\epsilon)^2 + s^2} - \sqrt{(vt)^2 + s^2} \right] \Big|_0^a \\ &= \frac{\lambda}{2\epsilon_0} \left[\sqrt{(vt-\epsilon)^2 + a^2} - \sqrt{(vt)^2 + a^2} - (\epsilon - vt) + (vt) \right]. \end{aligned}$$

$$(c) I_d = \epsilon_0 \frac{d\Phi_E}{dt} = \frac{\lambda}{2} \left\{ \frac{v(vt-\epsilon)}{\sqrt{(vt-\epsilon)^2 + a^2}} - \frac{v(vt)}{\sqrt{(vt)^2 + a^2}} + 2v \right\}.$$

As $\epsilon \rightarrow 0$, $vt < \epsilon$ also $\rightarrow 0$, so $I_d \rightarrow \frac{\lambda}{2}(2v) = \lambda v = I$. With an infinitesimal gap we attribute the magnetic field to *displacement current*, instead of real current, but we get the same answer. qed

Problem 7.62



$$(a) \text{ Parallel-plate capacitor: } E = \frac{1}{\epsilon_0} \sigma; \quad V = Eh = \frac{1}{\epsilon_0} \frac{Q}{wl} h \Rightarrow C = \frac{Q}{V} = \frac{\epsilon_0 w l}{h} \Rightarrow \boxed{\mathcal{C} = \frac{\epsilon_0 w}{h}}.$$

$$(b) B = \mu_0 K = \mu_0 \frac{I}{w}; \quad \Phi = Bh l = \frac{\mu_0 I}{w} h l = LI \Rightarrow L = \frac{\mu_0 h}{w} l \Rightarrow \boxed{\mathcal{L} = \frac{\mu_0 h}{w}}.$$

$$(c) \boxed{\mathcal{C}\mathcal{L} = \mu_0 \epsilon_0} = (4\pi \times 10^{-7})(8.85 \times 10^{-12}) = \boxed{1.112 \times 10^{-17} \text{ s}^2/\text{m}^2}.$$

$$(\text{Propagation speed } 1/\sqrt{\mathcal{C}\mathcal{L}} = 1/\sqrt{\mu_0 \epsilon_0} = 2.999 \times 10^8 \text{ m/s} = c.)$$

$$(d) \left. \begin{array}{l} D = \sigma, \quad E = D/\epsilon = \sigma/\epsilon, \text{ so just replace } \epsilon_0 \text{ by } \epsilon; \\ H = K, \quad B = \mu H = \mu K, \text{ so just replace } \mu_0 \text{ by } \mu. \end{array} \right\} \quad \boxed{\mathcal{L}\mathcal{C} = \epsilon\mu}; \quad \boxed{v = 1/\sqrt{\epsilon\mu}}.$$

Problem 7.63

(a) $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$; $\mathbf{J}$ finite, $\sigma = \infty \Rightarrow \mathbf{E} + (\mathbf{v} \times \mathbf{B}) = \mathbf{0}$. Take the curl: $\nabla \times \mathbf{E} + \nabla \times (\mathbf{v} \times \mathbf{B}) = \mathbf{0}$. But Faraday's law says $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$. So $\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B})$. qed

(b) $\nabla \cdot \mathbf{B} = 0 \Rightarrow \oint \mathbf{B} \cdot d\mathbf{a} = 0$ for any closed surface. Apply this at time $(t + dt)$ to the surface consisting of S , S' , and $\mathcal{R}$:

$$\int_{S'} \mathbf{B}(t + dt) \cdot d\mathbf{a} + \int_{\mathcal{R}} \mathbf{B}(t + dt) \cdot d\mathbf{a} - \int_S \mathbf{B}(t + dt) \cdot d\mathbf{a} = 0$$

(the sign change in the third term comes from switching *outward* $d\mathbf{a}$ to *inward* $d\mathbf{a}$).

$$d\Phi = \int_{S'} \mathbf{B}(t + dt) \cdot d\mathbf{a} - \int_S \mathbf{B}(t) \cdot d\mathbf{a} = \int_S \underbrace{[\mathbf{B}(t + dt) - \mathbf{B}(t)] \cdot d\mathbf{a}}_{\frac{\partial \mathbf{B}}{\partial t} dt \text{ (for infinitesimal } dt)} - \int_{\mathcal{R}} \mathbf{B}(t + dt) \cdot d\mathbf{a}$$

$$d\Phi = \left\{ \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a} \right\} dt - \int_{\mathcal{R}} \mathbf{B}(t + dt) \cdot [(d\mathbf{l} \times \mathbf{v}) dt] \quad (\text{Figure 7.13}).$$

Since the second term is already first order in dt , we can replace $\mathbf{B}(t + dt)$ by $\mathbf{B}(t)$ (the distinction would be second order):

$$d\Phi = dt \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a} - dt \oint_C \underbrace{\mathbf{B} \cdot (d\mathbf{l} \times \mathbf{v})}_{(\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}} = dt \left\{ \int_S \left(\frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{a} - \int_S \nabla \times (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{a} \right\}.$$

$$\frac{d\Phi}{dt} = \int_S \left[\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{v} \times \mathbf{B}) \right] \cdot d\mathbf{a} = 0. \quad \text{qed}$$

Problem 7.64

(a)

$$\begin{aligned} \nabla \cdot \mathbf{E}' &= (\nabla \cdot \mathbf{E}) \cos \alpha + c(\nabla \cdot \mathbf{B}) \sin \alpha = \frac{1}{\epsilon_0} \rho_e \cos \alpha + c\mu_0 \rho_m \sin \alpha \\ &= \frac{1}{\epsilon_0} (\rho_e \cos \alpha + c\mu_0 \epsilon_0 \rho_m \sin \alpha) = \frac{1}{\epsilon_0} (\rho_e \cos \alpha + \frac{1}{c} \rho_m \sin \alpha) = \frac{1}{\epsilon_0} \rho'_e. \quad \checkmark \end{aligned}$$

$$\begin{aligned} \nabla \cdot \mathbf{B}' &= (\nabla \cdot \mathbf{B}) \cos \alpha - \frac{1}{c} (\nabla \cdot \mathbf{E}) \sin \alpha = \mu_0 \rho_m \cos \alpha - \frac{1}{c\epsilon_0} \rho_e \sin \alpha \\ &= \mu_0 (\rho_m \cos \alpha - \frac{1}{c\mu_0 \epsilon_0} \rho_e \sin \alpha) = \mu_0 (\rho_m \cos \alpha - c\rho_e \sin \alpha) = \mu_0 \rho'_m. \quad \checkmark \end{aligned}$$

$$\begin{aligned} \nabla \times \mathbf{E}' &= (\nabla \times \mathbf{E}) \cos \alpha + c(\nabla \times \mathbf{B}) \sin \alpha = \left(-\mu_0 \mathbf{J}_m - \frac{\partial \mathbf{B}}{\partial t} \right) \cos \alpha + c \left(\mu_0 \mathbf{J}_e + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \sin \alpha \\ &= -\mu_0 (\mathbf{J}_m \cos \alpha - c\mathbf{J}_e \sin \alpha) - \frac{\partial}{\partial t} \left(\mathbf{B} \cos \alpha - \frac{1}{c} \mathbf{E} \sin \alpha \right) = -\mu_0 \mathbf{J}'_m - \frac{\partial \mathbf{B}'}{\partial t}. \quad \checkmark \end{aligned}$$

$$\begin{aligned} \nabla \times \mathbf{B}' &= (\nabla \times \mathbf{B}) \cos \alpha - \frac{1}{c} (\nabla \times \mathbf{E}) \sin \alpha = \left(\mu_0 \mathbf{J}_e + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \cos \alpha - \frac{1}{c} \left(-\mu_0 \mathbf{J}_m - \frac{\partial \mathbf{B}}{\partial t} \right) \sin \alpha \\ &= \mu_0 (\mathbf{J}_e \cos \alpha + \frac{1}{c} \mathbf{J}_m \sin \alpha) + \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \cos \alpha + c\mathbf{B} \sin \alpha) = \mu_0 \mathbf{J}'_e + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}'}{\partial t}. \quad \checkmark \end{aligned}$$

(b)

$$\begin{aligned}\mathbf{F}' &= q'_e(\mathbf{E}' + \mathbf{v} \times \mathbf{B}') + q'_m(\mathbf{B}' - \frac{1}{c^2}\mathbf{v} \times \mathbf{E}') \\&= \left(q_e \cos \alpha + \frac{1}{c} q_m \sin \alpha \right) \left[(\mathbf{E} \cos \alpha + c\mathbf{B} \sin \alpha) + \mathbf{v} \times \left(\mathbf{B} \cos \alpha - \frac{1}{c}\mathbf{E} \sin \alpha \right) \right] \\&\quad + (q_m \cos \alpha - c q_e \sin \alpha) \left[\left(\mathbf{B} \cos \alpha - \frac{1}{c}\mathbf{E} \sin \alpha \right) - \frac{1}{c^2}\mathbf{v} \times (\mathbf{E} \cos \alpha + c\mathbf{B} \sin \alpha) \right] \\&= q_e \left[(\mathbf{E} \cos^2 \alpha + c\mathbf{B} \sin \alpha \cos \alpha - c\mathbf{B} \sin \alpha \cos \alpha + \mathbf{E} \sin^2 \alpha) \right. \\&\quad \left. + \mathbf{v} \times \left(\mathbf{B} \cos^2 \alpha - \frac{1}{c}\mathbf{E} \sin \alpha \cos \alpha + \frac{1}{c}\mathbf{E} \sin \alpha \cos \alpha + \mathbf{B} \sin^2 \alpha \right) \right] \\&\quad + q_m \left[\left(\frac{1}{c}\mathbf{E} \sin \alpha \cos \alpha + \mathbf{B} \sin^2 \alpha + \mathbf{B} \cos^2 \alpha - \frac{1}{c}\mathbf{E} \sin \alpha \cos \alpha \right) \right. \\&\quad \left. + \mathbf{v} \times \left(\frac{1}{c}\mathbf{B} \sin \alpha \cos \alpha - \frac{1}{c^2}\mathbf{E} \sin^2 \alpha - \frac{1}{c^2}\mathbf{E} \cos^2 \alpha - \frac{1}{c}\mathbf{B} \sin \alpha \cos \alpha \right) \right] \\&= q_e (\mathbf{E} + \mathbf{v} \times \mathbf{B}) + q_m \left(\mathbf{B} - \frac{1}{c^2}\mathbf{v} \times \mathbf{E} \right) = \mathbf{F}. \quad \text{qed}\end{aligned}$$

Chapter 8

Conservation Laws

Problem 8.1

Example 7.13.

$$\left. \begin{aligned} \mathbf{E} &= \frac{\lambda}{2\pi\epsilon_0} \frac{1}{s} \hat{\mathbf{s}} \\ \mathbf{B} &= \frac{\mu_0 I}{2\pi} \frac{1}{s} \hat{\phi} \end{aligned} \right\} \mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{\lambda I}{4\pi^2 \epsilon_0} \frac{1}{s^2} \hat{\mathbf{z}};$$

$$P = \int \mathbf{S} \cdot d\mathbf{a} = \int_a^b S 2\pi s \, ds = \frac{\lambda I}{2\pi\epsilon_0} \int_a^b \frac{1}{s} \, ds = \frac{\lambda I}{2\pi\epsilon_0} \ln(b/a).$$

$$\text{But } V = \int_a^b \mathbf{E} \cdot d\mathbf{l} = \frac{\lambda}{2\pi\epsilon_0} \int_a^b \frac{1}{s} \, ds = \frac{\lambda}{2\pi\epsilon_0} \ln(b/a), \text{ so } \boxed{P = IV.}$$

Problem 7.62.

$$\left. \begin{aligned} \mathbf{E} &= \frac{\sigma}{\epsilon_0} \hat{\mathbf{z}} \\ \mathbf{B} &= \mu_0 K \hat{\mathbf{x}} = \frac{\mu_0 I}{w} \hat{\mathbf{x}} \end{aligned} \right\} \mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{\sigma I}{\epsilon_0 w} \hat{\mathbf{y}};$$

$$P = \int \mathbf{S} \cdot d\mathbf{a} = Swh = \frac{\sigma I h}{\epsilon_0}, \text{ but } V = \int \mathbf{E} \cdot d\mathbf{l} = \frac{\sigma}{\epsilon_0} h, \text{ so } \boxed{P = IV.}$$

Problem 8.2

$$\text{(a) } \mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{z}}; \sigma = \frac{Q}{\pi a^2}; Q(t) = It \Rightarrow \mathbf{E}(t) = \boxed{\frac{It}{\pi \epsilon_0 a^2} \hat{\mathbf{z}}}.$$

$$B 2\pi s = \mu_0 \epsilon_0 \frac{\partial E}{\partial t} \pi s^2 = \mu_0 \epsilon_0 \frac{I \pi s^2}{\pi \epsilon_0 a^2} \Rightarrow \mathbf{B}(s, t) = \boxed{\frac{\mu_0 I s}{2\pi a^2} \hat{\phi}}.$$

$$\text{(b) } u_{\text{em}} = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) = \frac{1}{2} \left[\epsilon_0 \left(\frac{It}{\pi \epsilon_0 a^2} \right)^2 + \frac{1}{\mu_0} \left(\frac{\mu_0 I s}{2\pi a^2} \right)^2 \right] = \boxed{\frac{\mu_0 I^2}{2\pi^2 a^4} [(ct)^2 + (s/2)^2]}.$$

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0} \left(\frac{It}{\pi \epsilon_0 a^2} \right) \left(\frac{\mu_0 I s}{2\pi a^2} \right) (-\hat{\mathbf{s}}) = \boxed{-\frac{I^2 t}{2\pi^2 \epsilon_0 a^4} s \hat{\mathbf{s}}}.$$

$$\frac{\partial u_{\text{em}}}{\partial t} = \frac{\mu_0 I^2}{2\pi^2 a^4} 2c^2 t = \frac{I^2 t}{\pi^2 \epsilon_0 a^4}; \quad -\nabla \cdot \mathbf{S} = \frac{I^2 t}{2\pi^2 \epsilon_0 a^4} \nabla \cdot (s \hat{\mathbf{s}}) = \frac{I^2 t}{\pi^2 \epsilon_0 a^4} = \frac{\partial u_{\text{em}}}{\partial t}. \quad \checkmark$$

$$\begin{aligned} \text{(c) } U_{\text{em}} &= \int u_{\text{em}} w 2\pi s ds = 2\pi w \frac{\mu_0 I^2}{2\pi^2 a^4} \int_0^b [(ct)^2 + (s/2)^2] s ds = \frac{\mu_0 w I^2}{\pi a^4} \left[(ct)^2 \frac{s^2}{2} + \frac{1}{4} \frac{s^4}{4} \right] \Big|_0^b \\ &= \boxed{\frac{\mu_0 w I^2 b^2}{2\pi a^4} \left[(ct)^2 + \frac{b^2}{8} \right]}. \end{aligned}$$

Over a surface at radius b : $P_{\text{in}} = - \int \mathbf{S} \cdot d\mathbf{a} = \frac{I^2 t}{2\pi^2 \epsilon_0 a^4} [b \hat{\mathbf{s}} \cdot (2\pi b w \hat{\mathbf{s}})] = \boxed{\frac{I^2 w t b^2}{\pi \epsilon_0 a^4}}.$

$$\frac{dU_{\text{em}}}{dt} = \frac{\mu_0 w I^2 b^2}{2\pi a^4} 2c^2 t = \frac{I^2 w t b^2}{\pi \epsilon_0 a^4} = P_{\text{in}}. \quad \checkmark \quad (\text{Set } b = a \text{ for } total.)$$

Problem 8.3

The force is clearly in the z direction, so we need

$$\begin{aligned} (\vec{T} \cdot d\mathbf{a})_z &= T_{zx} da_x + T_{zy} da_y + T_{zz} da_z = \frac{1}{\mu_0} \left(B_z B_x da_x + B_z B_y da_y + B_z B_z da_z - \frac{1}{2} B^2 da_z \right) \\ &= \frac{1}{\mu_0} \left[B_z (\mathbf{B} \cdot d\mathbf{a}) - \frac{1}{2} B^2 da_z \right]. \end{aligned}$$

Now $\mathbf{B} = \frac{2}{3} \mu_0 \sigma R \omega \hat{\mathbf{z}}$ (inside) and $\mathbf{B} = \frac{\mu_0 m}{4\pi r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}})$ (outside), where $m = \frac{4}{3} \pi R^3 (\sigma \omega R)$. (From Eq. 5.70, Prob. 5.37, and Eq. 5.88.) We want a surface that encloses the entire upper hemisphere—say a hemispherical cap just outside $r = R$ plus the equatorial circular disk.

Hemisphere:

$$\begin{aligned} B_z &= \frac{\mu_0 m}{4\pi R^3} [2 \cos \theta (\hat{\mathbf{r}})_z + \sin \theta (\hat{\boldsymbol{\theta}})_z] = \frac{\mu_0 m}{4\pi R^3} [2 \cos^2 \theta - \sin^2 \theta] = \frac{\mu_0 m}{4\pi R^3} (3 \cos^2 \theta - 1). \\ d\mathbf{a} &= R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}; \quad \mathbf{B} \cdot d\mathbf{a} = \frac{\mu_0 m}{4\pi R^3} (2 \cos \theta) R^2 \sin \theta d\theta d\phi; \quad da_z = R^2 \sin \theta d\theta d\phi \cos \theta; \\ B^2 &= \left(\frac{\mu_0 m}{4\pi R^3} \right)^2 (4 \cos^2 \theta + \sin^2 \theta) = \left(\frac{\mu_0 m}{4\pi R^3} \right)^2 (3 \cos^2 \theta + 1). \\ (\vec{T} \cdot d\mathbf{a})_z &= \frac{1}{\mu_0} \left(\frac{\mu_0 m}{4\pi R^3} \right)^2 \left[(3 \cos^2 \theta - 1) 2 \cos \theta R^2 \sin \theta d\theta d\phi - \frac{1}{2} (3 \cos^2 \theta + 1) R^2 \sin \theta \cos \theta d\theta d\phi \right] \\ &= \mu_0 \left(\frac{\sigma \omega R}{3} \right)^2 \left[\frac{1}{2} R^2 \sin \theta \cos \theta d\theta d\phi \right] (12 \cos^2 \theta - 4 - 3 \cos^2 \theta - 1) \\ &= \frac{\mu_0}{2} \left(\frac{\sigma \omega R^2}{3} \right)^2 (9 \cos^2 \theta - 5) \sin \theta \cos \theta d\theta d\phi. \\ (F_{\text{hemi}})_z &= \frac{\mu_0}{2} \left(\frac{\sigma \omega R^2}{3} \right)^2 2\pi \int_0^{\pi/2} (9 \cos^3 \theta - 5 \cos \theta) \sin \theta d\theta = \mu_0 \pi \left(\frac{\sigma \omega R^2}{3} \right)^2 \left[-\frac{9}{4} \cos^4 \theta + \frac{5}{2} \cos^2 \theta \right] \Big|_0^{\pi/2} \\ &= \mu_0 \pi \left(\frac{\sigma \omega R^2}{3} \right)^2 \left(0 + \frac{9}{4} - \frac{5}{2} \right) = -\frac{\mu_0 \pi}{4} \left(\frac{\sigma \omega R^2}{3} \right)^2. \end{aligned}$$

Disk:

$$B_z = \frac{2}{3}\mu_0\sigma R\omega; \quad d\mathbf{a} = r \, dr \, d\phi \, \hat{\phi} = -r \, dr \, d\phi \, \hat{\mathbf{z}};$$

$$\mathbf{B} \cdot d\mathbf{a} = -\frac{2}{3}\mu_0\sigma R\omega r \, dr \, d\phi; \quad B^2 = \left(\frac{2}{3}\mu_0\sigma R\omega\right)^2; \quad da_z = -r \, dr \, d\phi.$$

$$(\vec{T} \cdot d\mathbf{a})_z = \frac{1}{\mu_0} \left(\frac{2}{3}\mu_0\sigma R\omega\right)^2 \left[-r \, dr \, d\phi + \frac{1}{2}r \, dr \, d\phi\right] = -\frac{1}{2\mu_0} \left(\frac{2}{3}\mu_0\sigma R\omega\right)^2 r \, dr \, d\phi.$$

$$(F_{\text{disk}})_z = -2\mu_0 \left(\frac{\sigma\omega R}{3}\right)^2 2\pi \int_0^R r \, dr = -2\pi\mu_0 \left(\frac{\sigma\omega R^2}{3}\right)^2.$$

Total:

$$\mathbf{F} = -\pi\mu_0 \left(\frac{\sigma\omega R^2}{3}\right)^2 \left(2 + \frac{1}{4}\right) \hat{\mathbf{z}} = \boxed{-\pi\mu_0 \left(\frac{\sigma\omega R^2}{2}\right)^2 \hat{\mathbf{z}}} \quad (\text{agrees with Prob. 5.44}).$$

Alternatively, we could use a surface consisting of the entire equatorial plane, closing it with a hemispherical surface “at infinity,” where (since the field is zero out there) the contribution is zero. We have already done the integral over the disk; it remains to do rest of the integral over the plane, from R to ∞ . On the plane, $\theta = 0$, and (for $r > R$) $\mathbf{B} = \frac{\mu_0 m}{4\pi r^3} \hat{\boldsymbol{\theta}} = -\frac{\mu_0 m}{4\pi r^3} \hat{\mathbf{z}}$, so

$$(\vec{T} \cdot d\mathbf{a})_z = \frac{1}{\mu_0} \left[B_z(B_z \, da_z) - \frac{1}{2}B_z^2 \, da_z\right] = \frac{1}{2\mu_0} B_z^2 \, da_z = \frac{1}{2\mu_0} \left(\frac{\mu_0 m}{4\pi r^3}\right)^2 (-r \, dr \, d\phi).$$

The contribution from the rest of the plane is therefore

$$\begin{aligned} (F_{\text{rest}})_z &= -\frac{1}{2\mu_0} \left(\frac{\mu_0 m}{4\pi}\right)^2 2\pi \int_R^\infty \frac{1}{r^5} \, dr = -\frac{\mu_0}{16\pi} \left(\frac{4}{3}\pi R^4 \sigma\omega\right)^2 \left[-\frac{1}{4r^4}\right] \Big|_R^\infty = -\mu_0\pi \left(\frac{R^4 \sigma\omega}{3}\right)^2 \left[\frac{1}{4R^4}\right] \\ &= -\frac{\mu_0\pi}{4} \left(\frac{\sigma\omega R^2}{3}\right)^2. \end{aligned}$$

This is the same as $(F_{\text{hem}})_z$, so—when added to $(F_{\text{disk}})_z$ —it will yield the same total force as before.

Problem 8.4

(a)

$$(\vec{T} \cdot d\mathbf{a})_z = T_{zx} \, da_x + T_{zy} \, da_y + T_{zz} \, da_z.$$

But for the xy plane $da_x = da_y = 0$, and $da_z = -r \, dr \, d\phi$ (I'll calculate the force on the *upper* charge).

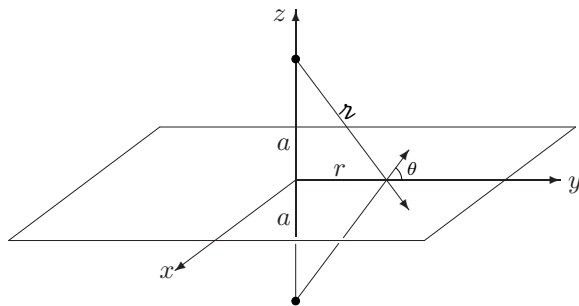
$$(\vec{T} \cdot d\mathbf{a})_z = \epsilon_0 \left(E_z E_z - \frac{1}{2}E^2\right) (-r \, dr \, d\phi).$$

Now $\mathbf{E} = \frac{1}{4\pi\epsilon_0} 2\frac{q}{r^2} \cos\theta \, \hat{\mathbf{r}}$, and $\cos\theta = \frac{r}{r}$, so $E_z =$

0, $E^2 = \left(\frac{q}{2\pi\epsilon_0}\right)^2 \frac{r^2}{(r^2 + a^2)^3}$. Therefore

$$F_z = \frac{1}{2}\epsilon_0 \left(\frac{q}{2\pi\epsilon_0}\right)^2 2\pi \int_0^\infty \frac{r^3 \, dr}{(r^2 + a^2)^3} = \frac{q^2}{4\pi\epsilon_0} \frac{1}{2} \int_0^\infty \frac{u \, du}{(u + a^2)^3} \quad (\text{letting } u \equiv r^2)$$

$$= \frac{q^2}{4\pi\epsilon_0} \frac{1}{2} \left[-\frac{1}{(u + a^2)} + \frac{a^2}{2(u + a^2)^3} \right] \Big|_0^\infty = \frac{q^2}{4\pi\epsilon_0} \frac{1}{2} \left[0 + \frac{1}{a^2} - \frac{a^2}{2a^4} \right] = \boxed{\frac{q^2}{4\pi\epsilon_0} \frac{1}{(2a)^2}}. \quad \checkmark$$



(b) In this case $\mathbf{E} = -\frac{1}{4\pi\epsilon_0} 2\frac{q}{r^2} \sin\theta \hat{\mathbf{z}}$, and $\sin\theta = \frac{a}{r}$, so

$$E^2 = E_z^2 = \left(\frac{qa}{2\pi\epsilon_0}\right)^2 \frac{1}{(r^2 + a^2)^3}, \text{ and hence } (\vec{T} \cdot d\mathbf{a})_z = -\frac{\epsilon_0}{2} \left(\frac{qa}{2\pi\epsilon_0}\right)^2 \frac{r dr d\phi}{(r^2 + a^2)^3}. \text{ Therefore}$$

$$F_z = -\frac{\epsilon_0}{2} \left(\frac{qa}{2\pi\epsilon_0}\right)^2 2\pi \int_0^\infty \frac{r dr}{(r^2 + a^2)^3} = -\frac{q^2 a^2}{4\pi\epsilon_0} \left[-\frac{1}{4} \frac{1}{(r^2 + a^2)^2} \right]_0^\infty = -\frac{q^2 a^2}{4\pi\epsilon_0} \left[0 + \frac{1}{4a^4} \right] = \boxed{-\frac{q^2}{4\pi\epsilon_0} \frac{1}{(2a)^2}}. \checkmark$$

Problem 8.5

(a) $\mathbf{E} = -\frac{\sigma}{\epsilon_0} \hat{\mathbf{z}}$, $\mathbf{B} = -\mu_0 \sigma v \hat{\mathbf{x}}$, $\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}) = \mu_0 \sigma^2 v \hat{\mathbf{y}}$, $\mathbf{p} = (dA)\mathbf{g} = \boxed{dA\mu_0 \sigma^2 v \hat{\mathbf{y}}}$.

(b) (i) There is a *magnetic* force, due to the (average) magnetic field at the upper plate:

$$\mathbf{F} = q(\mathbf{u} \times \mathbf{B}) = \sigma A [(-u \hat{\mathbf{z}}) \times (-\tfrac{1}{2}\mu_0 \sigma v \hat{\mathbf{x}})] = \tfrac{1}{2}\mu_0 \sigma^2 A v u \hat{\mathbf{y}},$$

$$\mathbf{I}_1 = \int \mathbf{F} dt = \tfrac{1}{2}\mu_0 \sigma^2 A v \hat{\mathbf{y}} \int u dt = \tfrac{1}{2}d\mu_0 \sigma^2 A v \hat{\mathbf{y}}.$$

[The velocity of the patch (of area A) is actually $\mathbf{v} + \mathbf{u} = v \hat{\mathbf{y}} - u \hat{\mathbf{z}}$, but the y component produces a magnetic force in the z direction (a repulsion of the plates) which reduces their (electrical) attraction but does not deliver (horizontal) momentum to the plates.]

(ii) Meanwhile, in the space immediately above the upper plate the magnetic field drops abruptly to zero (as the plate moves past), inducing an *electric* field by Faraday's law. The magnetic field in the vicinity of the top plate (at $d(t) = d_0 - ut$) can be written, using Problem 1.46(b),

$$\mathbf{B}(z, t) = -\mu_0 \sigma v \theta(d - z) \hat{\mathbf{x}}, \quad \Rightarrow \quad \frac{\partial \mathbf{B}}{\partial t} = \mu_0 \sigma v u \delta(d - z) \hat{\mathbf{x}}.$$

In the analogy at the beginning of Section 7.2.2, the Faraday-induced electric field is just like the magnetostatic field of a surface current $\mathbf{K} = -\sigma v u \hat{\mathbf{x}}$. Referring to Eq. 5.58, then,

$$\mathbf{E}_{\text{ind}} = \begin{cases} -\tfrac{1}{2}\mu_0 \sigma v u \hat{\mathbf{y}}, & \text{for } z < d, \\ +\tfrac{1}{2}\mu_0 \sigma v u \hat{\mathbf{y}}, & \text{for } z > d. \end{cases}$$

This induced electric field exerts a force on area A of the *bottom* plate, $\mathbf{F} = (-\sigma A)(-\tfrac{1}{2}\mu_0 \sigma v u \hat{\mathbf{y}})$, and delivers an impulse

$$\mathbf{I}_2 = \tfrac{1}{2}\mu_0 \sigma^2 A v \hat{\mathbf{y}} \int u dt = \tfrac{1}{2}d\mu_0 \sigma^2 A v d \hat{\mathbf{y}}.$$

(I dropped the subscript on d_0 , reverting to the original notation: d is the initial separation of the plates.)

The total impulse is thus $\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2 = \boxed{dA\mu_0 \sigma^2 v \hat{\mathbf{y}}}$, matching the momentum initially stored in the fields, from part (a). [I thank Michael Ligare for untangling this surprisingly subtle problem. Incidentally, there is also “hidden momentum” in the original configuration. It is not relevant here; it is (relativistic) mechanical momentum (see Example 12.13), and is delivered to the plates as they come together, so it does not affect the overall conservation of momentum.]

Problem 8.6

(a) $\mathbf{g}_{\text{em}} = \epsilon_0 (\mathbf{E} \times \mathbf{B}) = \epsilon_0 E B \hat{\mathbf{y}}$; $\mathbf{p}_{\text{em}} = \boxed{\epsilon_0 E B A d \hat{\mathbf{y}}}$.

(b) $\mathbf{I} = \int_0^\infty \mathbf{F} dt = \int_0^\infty I(1 \times \mathbf{B}) dt = \int_0^\infty I B d(\hat{\mathbf{z}} \times \hat{\mathbf{x}}) dt = (Bd \hat{\mathbf{y}}) \int_0^\infty \left(-\frac{dQ}{dt}\right) dt$

$= -(Bd\hat{\mathbf{y}})[Q(\infty) - Q(0)] = BQd\hat{\mathbf{y}}$. But the original field was $E = \sigma/\epsilon_0 = Q/\epsilon_0 A$, so $Q = \epsilon_0 EA$, and hence $\mathbf{I} = \boxed{\epsilon_0 EBAd\hat{\mathbf{y}}}$.

Problem 8.7

(a) $E_x = E_y = 0$, $E_z = -\sigma/\epsilon_0$. Therefore

$$T_{xy} = T_{xz} = T_{yz} = \cdots = 0; \quad T_{xx} = T_{yy} = -\frac{\epsilon_0}{2}E^2 = -\frac{\sigma^2}{2\epsilon_0}; \quad T_{zz} = \epsilon_0 \left(E_z^2 - \frac{1}{2}E^2 \right) = \frac{\epsilon_0}{2}E^2 = \frac{\sigma^2}{2\epsilon_0}.$$

$$\boxed{\vec{T} = \frac{\sigma^2}{2\epsilon_0} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & +1 \end{pmatrix}}.$$

(b) $\mathbf{F} = \oint \vec{T} \cdot d\mathbf{a}$ ($\mathbf{S} = \mathbf{0}$, since $\mathbf{B} = \mathbf{0}$); integrate over the xy plane: $d\mathbf{a} = -dx dy \hat{\mathbf{z}}$ (negative because *outward* with respect to a surface enclosing the upper plate). Therefore

$$F_z = \int T_{zz} da_z = -\frac{\sigma^2}{2\epsilon_0}A, \text{ and the force per unit area is } \mathbf{f} = \frac{\mathbf{F}}{A} = \boxed{-\frac{\sigma^2}{2\epsilon_0} \hat{\mathbf{z}}}.$$

(c) $-T_{zz} = \boxed{-\sigma^2/2\epsilon_0}$ is the momentum in the z direction crossing a surface perpendicular to z , per unit area, per unit time.

(d) The recoil force is the momentum delivered per unit time, so the force per unit area on the top plate is

$$\boxed{\mathbf{f} = -\frac{\sigma^2}{2\epsilon_0} \hat{\mathbf{z}}} \quad (\text{same as (b)}).$$

Problem 8.8

$\mathbf{B} = \mu_0 n I \hat{\mathbf{z}}$ (for $s < R$; outside the solenoid $B = 0$). The force on a segment ds of spoke is

$$d\mathbf{F} = I' d\mathbf{l} \times \mathbf{B} = I' \mu_0 n I ds (\hat{\mathbf{s}} \times \hat{\mathbf{z}}) = -I' \mu_0 n I ds \hat{\phi}.$$

The torque on the spoke is

$$\mathbf{N} = \int \mathbf{r} \times d\mathbf{F} = I' \mu_0 n I \int_a^R s ds (-\hat{\mathbf{s}} \times \hat{\phi}) = I' \mu_0 n I \frac{1}{2} (R^2 - a^2) (-\hat{\mathbf{z}}).$$

Therefore the angular momentum of the cylinders is $\mathbf{L} = \int \mathbf{N} dt = -\frac{1}{2} \mu_0 n I (R^2 - a^2) \hat{\mathbf{z}} \int I' dt$. But $\int I' dt = Q$, so

$$\boxed{\mathbf{L} = -\frac{1}{2} \mu_0 n I Q (R^2 - a^2) \hat{\mathbf{z}}} \quad (\text{in agreement with Eq. 8.34}).$$

Problem 8.9

(a) Between the shells, $\mathbf{E} = \frac{Q}{4\pi\epsilon_0} \frac{1}{r^2} \hat{\mathbf{r}}$, $\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}) = \frac{QB_0}{4\pi r^2} (\hat{\mathbf{r}} \times \hat{\mathbf{z}})$.

$$\mathbf{L} = \int (\mathbf{r} \times \mathbf{g}) d\tau = \frac{QB_0}{4\pi} \int \frac{1}{r^2} [\mathbf{r} \times (\hat{\mathbf{r}} \times \hat{\mathbf{z}})] r^2 \sin \theta dr d\theta d\phi = \frac{QB_0}{4\pi} \int r [\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \hat{\mathbf{z}})] \sin \theta dr d\theta d\phi.$$

Now $\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \hat{\mathbf{z}}) = \hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \hat{\mathbf{z}}) - \hat{\mathbf{z}}(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}) = \hat{\mathbf{r}} \cos \theta = \hat{\mathbf{z}}$, but $\mathbf{L}$ has to be along the z direction, so we pick off the z component of $\hat{\mathbf{r}}$: $[\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \hat{\mathbf{z}})]_z = \cos^2 \theta - 1 = -\sin^2 \theta$.

$$L_z = -\frac{QB_0}{4\pi} \int r \sin^3 \theta \, dr \, d\theta \, d\phi = -\frac{QB_0}{4\pi} 2\pi \int_0^\pi \sin^3 \theta \, d\theta \int_a^b r \, dr = -\frac{QB_0}{2} \left(\frac{4}{3}\right) \left(\frac{b^2 - a^2}{2}\right) = -\frac{1}{3} QB_0 (b^2 - a^2).$$

$$\mathbf{L} = \boxed{-\frac{1}{3} QB_0 (b^2 - a^2) \hat{\mathbf{z}}}.$$

(b) The changing magnetic field induces an electric field (Example 7.7). Assuming symmetry about the z axis, $\mathbf{E} = -\frac{s}{2} \frac{dB}{dt} \hat{\phi}$. The force on a patch of area da is $d\mathbf{F} = \mathbf{E}\sigma \, da$, and the torque on this patch is $d\mathbf{N} = \mathbf{s} \times d\mathbf{F}$. The net torque on the sphere at radius a is (using $\mathbf{s} = a \sin \theta \hat{\mathbf{s}}$ and $da = a^2 \sin \theta \, d\theta \, d\phi$):

$$\mathbf{N}_a = -\left(\frac{Q}{4\pi a^2}\right) \left(\frac{1}{2} \frac{dB}{dt} \hat{\mathbf{z}}\right) \int s^2 \, da = -\frac{Qa^2}{8\pi} \frac{dB}{dt} \hat{\mathbf{z}} 2\pi \int \sin^3 \theta \, d\theta = \frac{Qa^2}{3} \frac{dB}{dt} \hat{\mathbf{z}}.$$

Similarly, $\mathbf{N}_b = \frac{Qb^2}{3} \frac{dB}{dt} \hat{\mathbf{z}}$, so the total torque is $\mathbf{N} = \frac{Q}{3} \frac{dB}{dt} (b^2 - a^2) \hat{\mathbf{z}}$, and the angular momentum delivered to the spheres is

$$\mathbf{L} = \int \mathbf{N} \, dt = \frac{Q}{3} (b^2 - a^2) \hat{\mathbf{z}} \int_{B_0}^0 \frac{dB}{dt} \, dt = \boxed{-\frac{Q}{3} (b^2 - a^2) B_0 \hat{\mathbf{z}}} \quad \checkmark$$

Problem 8.10

(a)

$$\mathbf{E} = \begin{cases} \mathbf{0}, & (r < R) \\ \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}, & (r > R) \end{cases}; \quad \mathbf{B} = \begin{cases} \frac{2}{3} \mu_0 M \hat{\mathbf{z}}, & (r < R) \\ \frac{\mu_0}{4\pi} \frac{m}{r^3} [2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\theta}], & (r > R) \end{cases} \quad (\text{Ex. 6.1})$$

(where $m = \frac{4}{3} \pi R^3 M$); $\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}) = \frac{\mu_0}{(4\pi)^2} \frac{Qm}{r^5} (\hat{\mathbf{r}} \times \hat{\theta}) \sin \theta$, and $(\hat{\mathbf{r}} \times \hat{\theta}) = \hat{\phi}$, so

$$\boldsymbol{\ell} = \mathbf{r} \times \mathbf{g} = \frac{\mu_0}{(4\pi)^2} \frac{mQ}{r^4} \sin \theta (\hat{\mathbf{r}} \times \hat{\phi}).$$

But $(\hat{\mathbf{r}} \times \hat{\phi}) = -\hat{\theta}$, and only the z component will survive integration, so (since $(\hat{\theta})_z = -\sin \theta$):

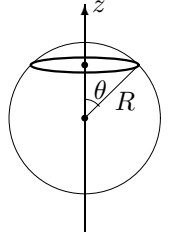
$$\mathbf{L} = \frac{\mu_0 m Q}{(4\pi)^2} \hat{\mathbf{z}} \int \frac{\sin^2 \theta}{r^4} (r^2 \sin \theta \, dr \, d\theta \, d\phi). \quad \int_0^{2\pi} d\phi = 2\pi; \quad \int_0^\pi \sin^3 \theta \, d\theta = \frac{4}{3}; \quad \int_R^\infty \frac{1}{r^2} \, dr = \left(-\frac{1}{r}\right) \Big|_R^\infty = \frac{1}{R}.$$

$$\mathbf{L} = \frac{\mu_0 m Q}{(4\pi)^2} \hat{\mathbf{z}} (2\pi) \left(\frac{4}{3}\right) \left(\frac{1}{R}\right) = \boxed{\frac{2}{9} \mu_0 M Q R^2 \hat{\mathbf{z}}}.$$

(b) Apply Faraday's law to the ring shown:

$$\oint \mathbf{E} \cdot d\mathbf{l} = E(2\pi r \sin \theta) = -\frac{d\Phi}{dt} = -\pi(r \sin \theta)^2 \left(\frac{2}{3}\mu_0 \frac{dM}{dt} \right)$$

$$\Rightarrow \boxed{\mathbf{E} = -\frac{\mu_0}{3} \frac{dM}{dt} (r \sin \theta) \hat{\phi}.}$$



The force on a patch of surface (da) is $d\mathbf{F} = \sigma \mathbf{E} da = -\frac{\mu_0 \sigma}{3} \frac{dM}{dt} (r \sin \theta) da \hat{\phi}$ ($\sigma = \frac{Q}{4\pi R^2}$).

The torque on the patch is $d\mathbf{N} = \mathbf{r} \times d\mathbf{F} = -\frac{\mu_0 \sigma}{3} \frac{dM}{dt} (r^2 \sin \theta) da (\hat{\mathbf{r}} \times \hat{\phi})$. But $(\hat{\mathbf{r}} \times \hat{\phi}) = -\hat{\theta}$, and we want only the z component ($\hat{\theta}_z = -\sin \theta$):

$$\mathbf{N} = -\frac{\mu_0 \sigma}{3} \frac{dM}{dt} \hat{\mathbf{z}} \int r^2 \sin^2 \theta (r^2 \sin \theta d\theta d\phi).$$

Here $r = R$; $\int_0^\pi \sin^3 \theta d\theta = \frac{4}{3}$; $\int_0^{2\pi} d\phi = 2\pi$, so $\mathbf{N} = -\frac{\mu_0 \sigma}{3} \frac{dM}{dt} \hat{\mathbf{z}} R^4 \left(\frac{4}{3} \right) (2\pi) = \boxed{-\frac{2\mu_0}{9} Q R^2 \frac{dM}{dt} \hat{\mathbf{z}}}$.

$$\mathbf{L} = \int \mathbf{N} dt = -\frac{2\mu_0}{9} Q R^2 \hat{\mathbf{z}} \int_M dM = \boxed{\frac{2\mu_0}{9} M Q R^2 \hat{\mathbf{z}}} \quad (\text{same as (a)}).$$

(c) Let the charge on the sphere at time t be $q(t)$; the charge *density* is $\sigma = \frac{q(t)}{4\pi R^2}$. The charge below (“south of”) the ring in the figure is

$$q_s = \sigma (2\pi R^2) \int_0^\pi \sin \theta' d\theta' = \frac{q}{2} (-\cos \theta') \Big|_0^\pi = \frac{q}{2} (1 + \cos \theta).$$

So the total current crossing the ring (flowing “north”) is $I(t) = -\frac{1}{2} \frac{dq}{dt} (1 + \cos \theta)$, and hence

$$\mathbf{K}(t) = \frac{I}{2\pi R \sin \theta} (-\hat{\theta}) = \frac{1}{4\pi R} \frac{dq}{dt} \frac{(1 + \cos \theta)}{\sin \theta} \hat{\theta}. \text{ The force on a patch of area } da \text{ is } d\mathbf{F} = (\mathbf{K} \times \mathbf{B}) da.$$

$$\mathbf{B}_{\text{ave}} = \left[\frac{2}{3} \mu_0 M \hat{\mathbf{z}} + \frac{\mu_0}{4\pi} \frac{\frac{4}{3} \pi R^3 M}{R^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\theta}) \right] \frac{1}{2} = \frac{\mu_0 M}{6} [2 \hat{\mathbf{z}} + 2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\theta}];$$

$$\mathbf{K} \times \mathbf{B} = \frac{1}{4\pi R} \frac{dq}{dt} \frac{\mu_0 M}{6} \frac{(1 + \cos \theta)}{\sin \theta} [2(\hat{\theta} \times \hat{\mathbf{z}}) + 2 \cos \theta \underbrace{(\hat{\theta} \times \hat{\mathbf{r}})}_{-\hat{\phi}}].$$

$$d\mathbf{N} = R \hat{\mathbf{r}} \times d\mathbf{F} = \frac{\mu_0 M}{24\pi} \left(\frac{dq}{dt} \right) \frac{(1 + \cos \theta)}{\sin \theta} 2 \left[\underbrace{\hat{\mathbf{r}} \times (\hat{\theta} \times \hat{\mathbf{z}})}_{\hat{\theta}(\hat{\mathbf{r}} \cdot \hat{\mathbf{z}}) - \hat{\mathbf{z}}(\hat{\mathbf{r}} \cdot \hat{\theta})} - \cos \theta \underbrace{(\hat{\mathbf{r}} \times \hat{\phi})}_{-\hat{\theta}} \right] R^2 \sin \theta d\theta d\phi$$

$$= \frac{\mu_0 M}{12\pi} \left(\frac{dq}{dt} \right) (1 + \cos \theta) R^2 [\cos \theta \hat{\theta} + \cos \theta \hat{\theta}] d\theta d\phi = \frac{\mu_0 M R^2}{6\pi} \left(\frac{dq}{dt} \right) (1 + \cos \theta) \cos \theta d\theta d\phi \hat{\theta}.$$

The x and y components integrate to zero; $(\hat{\boldsymbol{\theta}})_z = -\sin \theta$, so (using $\int_0^{2\pi} d\phi = 2\pi$):

$$\begin{aligned} N_z &= -\frac{\mu_0 M R^2}{6\pi} \left(\frac{dq}{dt} \right) (2\pi) \int_0^\pi (1 + \cos \theta) \cos \theta \sin \theta d\theta = -\frac{\mu_0 M R^2}{3} \left(\frac{dq}{dt} \right) \left(\frac{\sin^2 \theta}{2} - \frac{\cos^3 \theta}{3} \right) \Big|_0^\pi \\ &= -\frac{\mu_0 M R^2}{3} \left(\frac{dq}{dt} \right) \left(\frac{2}{3} \right) = -\frac{2\mu_0}{9} M R^2 \frac{dq}{dt}. \quad \boxed{\mathbf{N} = -\frac{2\mu_0}{9} M R^2 \frac{dq}{dt} \hat{\mathbf{z}}.} \end{aligned}$$

Therefore

$$\mathbf{L} = \int \mathbf{N} dt = -\frac{2\mu_0}{9} M R^2 \hat{\mathbf{z}} \int_Q^0 dq = \boxed{\frac{2\mu_0}{9} M R^2 Q \hat{\mathbf{z}}} \text{ (same as (a)).}$$

(I used the *average* field at the discontinuity—which is the correct thing to do—but in this case you’d get the same answer using either the inside field or the outside field.)

Problem 8.11 The magnetic field of the upper loop is given by Eq. 5.41: $\mathbf{B} = \frac{\mu_0 I_b}{2} \frac{b^2}{(b^2 + z^2)^{3/2}} \hat{\mathbf{z}}$. (I put the origin at the center of the upper loop, with the z axis pointing up, so $z = -h$ at the location of the lower loop.) Treat the lower loop as a magnetic dipole, with moment $\mathbf{m} = I_a \pi a^2 \hat{\mathbf{z}}$. The force on $\mathbf{m}$ is given by Eq. 6.3:

$$\begin{aligned} \mathbf{F} &= \nabla(\mathbf{m} \cdot \mathbf{B}) = \nabla \left(I_a \pi a^2 \frac{\mu_0 I_b}{2} \frac{b^2}{(b^2 + z^2)^{3/2}} \right) = \frac{\mu_0 \pi a^2 b^2 I_a I_b}{2} \frac{\partial}{\partial z} (b^2 + z^2)^{-3/2} \hat{\mathbf{z}} \\ &= \frac{\mu_0 \pi a^2 b^2 I_a I_b}{2} \left(-\frac{3}{2} \right) (b^2 + z^2)^{-5/2} (2z) \hat{\mathbf{z}} = \frac{3\pi}{2} \mu_0 I_a I_b \frac{a^2 b^2 h}{(b^2 + h^2)^{5/2}} \hat{\mathbf{z}}. \quad \checkmark \end{aligned}$$

Problem 8.12 Following the method of Section 7.2.4 (leading up to Eq. 7.30), the power delivered to the two loops is

$$\frac{dW}{dt} = -\mathcal{E}_a I_a - \mathcal{E}_b I_b$$

(where these are the *changing* values as the currents are turned on, not the final values). Now

$$\mathcal{E}_a = -L_a \frac{dI_a}{dt} - M \frac{dI_b}{dt}, \quad \mathcal{E}_b = -L_b \frac{dI_b}{dt} - M \frac{dI_a}{dt},$$

so

$$\frac{dW}{dt} = \left(L_a I_a \frac{dI_a}{dt} + M I_a \frac{dI_b}{dt} \right) + \left(L_b I_b \frac{dI_b}{dt} + M I_b \frac{dI_a}{dt} \right) = \frac{d}{dt} \left(\frac{1}{2} L_a I_a^2 + \frac{1}{2} L_b I_b^2 + M I_a I_b \right).$$

The total work done, then, to increase the currents from zero to their final values, is

$$W = \frac{1}{2} L_a I_a^2 + \frac{1}{2} L_b I_b^2 + M I_a I_b. \quad \text{qed}$$

Problem 8.13

(a) $\mathcal{E} = -\frac{d\Phi}{dt}$; $\Phi = \pi a^2 B$; $B = \mu_0 n I_s$; $\mathcal{E} = I_r R$. So $\boxed{I_r = -\frac{1}{R} (\mu_0 \pi a^2 n) \frac{dI_s}{dt}}.$

$$(b) \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt} \Rightarrow E(2\pi a) = -\mu_0 \pi a^2 n \frac{dI_s}{dt} \Rightarrow \mathbf{E} = -\frac{1}{2} \mu_0 a n \frac{dI_s}{dt} \hat{\phi}. \quad \mathbf{B} = \frac{\mu_0 I_r}{2} \frac{b^2}{(b^2 + z^2)^{3/2}} \hat{\mathbf{z}} \quad (\text{Eq. 5.41}).$$

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0} \left(-\frac{\mu_0 a n}{2} \frac{dI_s}{dt} \right) \left(\frac{\mu_0 I_r}{2} \frac{b^2}{(b^2 + z^2)^{3/2}} \right) (\hat{\phi} \times \hat{\mathbf{z}}) = \boxed{-\frac{1}{4} \mu_0 I_r \frac{dI_s}{dt} \frac{a b^2 n}{(b^2 + z^2)^{3/2}} \hat{\mathbf{s}}}.$$

Power:

$$P = \int \mathbf{S} \cdot d\mathbf{a} = \int_{-\infty}^{\infty} (S)(2\pi a) dz = -\frac{1}{2} \pi \mu_0 a^2 b^2 n I_r \frac{dI_s}{dt} \int_{-\infty}^{\infty} \frac{1}{(b^2 + z^2)^{3/2}} dz$$

$$\text{The integral is } \frac{z}{b^2 \sqrt{z^2 + b^2}} \Big|_{-\infty}^{\infty} = \frac{1}{b^2} - \left(-\frac{1}{b^2} \right) = \frac{2}{b^2}.$$

$$= - \left(\pi \mu_0 a^2 n \frac{dI_s}{dt} \right) I_r = (R I_r) I_r = I_r^2 R. \quad \text{qed}$$

Problem 8.14 The fields are zero for $s < a$ and $s > b$; between the cylinders,

$$\mathbf{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{s} \hat{\mathbf{s}}, \quad \mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi},$$

where $I = \lambda v$.

(a) From Eq. 8.5:

$$u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) = \frac{1}{2} \left[\epsilon_0 \left(\frac{1}{2\pi\epsilon_0} \right)^2 \frac{\lambda^2}{s^2} + \frac{1}{\mu_0} \left(\frac{\mu_0}{2\pi} \right)^2 \frac{\lambda^2 v^2}{s^2} \right] = \frac{\lambda^2}{8\pi^2 \epsilon_0} (1 + \epsilon_0 \mu_0 v^2) \frac{1}{s^2}.$$

Writing $\epsilon_0 \mu_0 = 1/c^2$, and integrating over the volume between the cylinders:

$$\frac{W}{\ell} = \frac{\lambda^2}{8\pi^2 \epsilon_0} \left(1 + \frac{v^2}{c^2} \right) \int_a^b \frac{1}{s^2} 2\pi s ds = \boxed{\frac{\lambda^2}{4\pi\epsilon_0} \left(1 + \frac{v^2}{c^2} \right) \ln(b/a)}.$$

(b) From Eq. 8.29:

$$\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}) = \epsilon_0 \left(\frac{\lambda}{2\pi\epsilon_0 s} \right) \left(\frac{\mu_0 \lambda v}{2\pi s} \right) \hat{\mathbf{z}} = \frac{\mu_0 \lambda^2 v}{4\pi^2 s^2} \hat{\mathbf{z}}.$$

$$\frac{\mathbf{p}}{\ell} = \frac{\mu_0 \lambda^2 v}{4\pi^2} \hat{\mathbf{z}} \int_a^b \frac{1}{s^2} 2\pi s ds = \boxed{\frac{\mu_0 \lambda^2 v}{2\pi} \ln(b/a) \hat{\mathbf{z}}}.$$

(c) From Eq. 8.10:

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\epsilon_0 \mu_0} \mathbf{g} = c^2 \mathbf{g}. \quad \frac{dW}{dt} = \int \mathbf{S} \cdot d\mathbf{a} = \boxed{\frac{\mu_0 \lambda^2 c^2 v}{2\pi} \ln(b/a) \hat{\mathbf{z}}}.$$

Problem 8.15

(a) The fields are $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$, $\mathbf{B} = \frac{\mu_0 NI}{2\pi s} \hat{\phi}$ (for points inside the toroid—see Eq. 5.60).

$$\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}) = \epsilon_0 \frac{q}{4\pi\epsilon_0 a^2} \frac{\mu_0 NI}{2\pi a} \hat{\mathbf{z}} = \frac{\mu_0}{8\pi^2} \frac{qNI}{a^3} \hat{\mathbf{z}}.$$

(Note that inside the toroid $\mathbf{r} \approx a \hat{\mathbf{s}}$ and $s \approx a$.)

$$\mathbf{p} = \left(\frac{\mu_0}{8\pi^2} \frac{qNI}{a^3} \hat{\mathbf{z}} \right) (2\pi a w h) = \boxed{\frac{\mu_0}{4\pi} \frac{qNIwh}{a^2} \hat{\mathbf{z}}}.$$

(b) The changing magnetic field induces an electric field, as given by Problem 7.19 (with $z = 0$):

$$\mathbf{E} = -\frac{\mu_0}{4\pi a^2} N h w \frac{dI}{dt} \hat{\mathbf{z}}. \text{ The impulse delivered to } q \text{ is therefore}$$

$$\mathbf{I} = \int \mathbf{F} dt = \int q \mathbf{E} dt = -q \frac{\mu_0}{4\pi a^2} N h w \hat{\mathbf{z}} \int \frac{dI}{dt} dt = \boxed{\frac{\mu_0 q}{4\pi a^2} N h w I \hat{\mathbf{z}}}. \quad \checkmark$$

Problem 8.16

According to Eqs. 3.104, 4.14, 5.89, and 6.16, the fields are

$$\mathbf{E} = \left\{ \begin{array}{ll} -\frac{1}{3\epsilon_0} \mathbf{P}, & (r < R), \\ \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{p}], & (r > R), \end{array} \right\} \quad \mathbf{B} = \left\{ \begin{array}{ll} \frac{2}{3} \mu_0 \mathbf{M}, & (r < R), \\ \frac{\mu_0}{4\pi} \frac{m}{r^3} [3(\mathbf{m} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{m}], & (r > R), \end{array} \right\}$$

where $\mathbf{p} = (4/3)\pi R^3 \mathbf{P}$, and $\mathbf{m} = (4/3)\pi R^3 \mathbf{M}$. Now $\mathbf{p} = \epsilon_0 \int (\mathbf{E} \times \mathbf{B}) d\tau$, and there are two contributions, one from inside the sphere and one from outside.

Inside:

$$\mathbf{p}_{\text{in}} = \epsilon_0 \int \left(-\frac{1}{3\epsilon_0} \mathbf{P} \right) \times \left(\frac{2}{3} \mu_0 \mathbf{M} \right) d\tau = -\frac{2}{9} \mu_0 (\mathbf{P} \times \mathbf{M}) \int d\tau = -\frac{2}{9} \mu_0 (\mathbf{P} \times \mathbf{M}) \frac{4}{3} \pi R^3 = \frac{8}{27} \mu_0 \pi R^3 (\mathbf{M} \times \mathbf{P}).$$

Outside:

$$\mathbf{p}_{\text{out}} = \epsilon_0 \frac{1}{4\pi\epsilon_0} \frac{\mu_0}{4\pi} \int \frac{1}{r^6} \{ [3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{p}] \times [3(\mathbf{m} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{m}] \} d\tau.$$

Now $\hat{\mathbf{r}} \times (\mathbf{p} \times \mathbf{m}) = \mathbf{p}(\hat{\mathbf{r}} \cdot \mathbf{m}) - \mathbf{m}(\hat{\mathbf{r}} \cdot \mathbf{p})$, so $\hat{\mathbf{r}} \times [\hat{\mathbf{r}} \times (\mathbf{p} \times \mathbf{m})] = (\hat{\mathbf{r}} \cdot \mathbf{m})(\hat{\mathbf{r}} \times \mathbf{p}) - (\hat{\mathbf{r}} \cdot \mathbf{p})(\hat{\mathbf{r}} \times \mathbf{m})$, whereas using the BAC-CAB rule directly gives $\hat{\mathbf{r}} \times [\hat{\mathbf{r}} \times (\mathbf{p} \times \mathbf{m})] = \hat{\mathbf{r}}[\hat{\mathbf{r}} \cdot (\mathbf{p} \times \mathbf{m})] - (\mathbf{p} \times \mathbf{m})(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}})$. So $\{ [3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{p}] \times [3(\mathbf{m} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{m}] \} = -3(\mathbf{p} \cdot \hat{\mathbf{r}})(\hat{\mathbf{r}} \times \mathbf{m}) + 3(\mathbf{m} \cdot \hat{\mathbf{r}})(\hat{\mathbf{r}} \times \mathbf{p}) + (\mathbf{p} \times \mathbf{m}) = 3[\hat{\mathbf{r}}[\hat{\mathbf{r}} \cdot (\mathbf{p} \times \mathbf{m})] - (\mathbf{p} \times \mathbf{m})] + (\mathbf{p} \times \mathbf{m}) = -2(\mathbf{p} \times \mathbf{m}) + 3\hat{\mathbf{r}}[\hat{\mathbf{r}} \cdot (\mathbf{p} \times \mathbf{m})]$.

$$\mathbf{p}_{\text{out}} = \frac{\mu_0}{16\pi^2} \int \frac{1}{r^6} \{ -2(\mathbf{p} \times \mathbf{m}) + 3\hat{\mathbf{r}}[\hat{\mathbf{r}} \cdot (\mathbf{p} \times \mathbf{m})] \} r^2 \sin \theta dr d\theta d\phi.$$

To evaluate the integral, set the z axis along $(\mathbf{p} \times \mathbf{m})$; then $\hat{\mathbf{r}} \cdot (\mathbf{p} \times \mathbf{m}) = |\mathbf{p} \times \mathbf{m}| \cos \theta$. Meanwhile, $\hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}$. But $\sin \phi$ and $\cos \phi$ integrate to zero, so the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ terms drop out, leaving

$$\begin{aligned} \mathbf{p}_{\text{out}} &= \frac{\mu_0}{16\pi^2} \left(\int_0^\infty \frac{1}{r^4} dr \right) \left\{ -2(\mathbf{p} \times \mathbf{m}) \int \sin \theta d\theta d\phi + 3|\mathbf{p} \times \mathbf{m}| \hat{\mathbf{z}} \int \cos^2 \theta \sin \theta d\theta d\phi \right\} \\ &= \frac{\mu_0}{16\pi^2} \left(-\frac{1}{3r^3} \right) \Big|_R^\infty \left[-2(\mathbf{p} \times \mathbf{m}) 4\pi + 3(\mathbf{p} \times \mathbf{m}) \frac{4\pi}{3} \right] = -\frac{\mu_0}{12\pi R^3} (\mathbf{p} \times \mathbf{m}) \\ &= -\frac{\mu_0}{12\pi R^3} \left(\frac{4}{3} \pi R^3 \mathbf{P} \right) \times \left(\frac{4}{3} \pi R^3 \mathbf{M} \right) = \frac{4\mu_0}{27} R^3 (\mathbf{M} \times \mathbf{P}). \end{aligned}$$

$$\mathbf{p}_{\text{tot}} = \left(\frac{8}{27} + \frac{4}{27} \right) \mu_0 R^3 (\mathbf{M} \times \mathbf{P}) = \boxed{\frac{4}{9} \mu_0 R^3 (\mathbf{M} \times \mathbf{P})}.$$

Problem 8.17

(a) From Eq. 5.70 and Prob. 5.37,

$$\begin{cases} r < R: \mathbf{E} = \mathbf{0}, \mathbf{B} = \frac{2}{3}\mu_0\sigma R\omega \hat{\mathbf{z}}, \text{ with } \sigma = \frac{e}{4\pi R^2}; \\ r > R: \mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{e}{r^2}\hat{\mathbf{r}}, \mathbf{B} = \frac{\mu_0}{4\pi}\frac{m}{r^3}(2\cos\theta\hat{\mathbf{r}} + \sin\theta\hat{\boldsymbol{\theta}}), \text{ with } m = \frac{4}{3}\pi\sigma\omega R^4. \end{cases}$$

The energy stored in the electric field is (Ex. 2.9):

$$W_E = \frac{1}{8\pi\epsilon_0} \frac{e^2}{R}.$$

The energy density of the internal magnetic field is:

$$u_B = \frac{1}{2\mu_0}B^2 = \frac{1}{2\mu_0} \left(\frac{2}{3}\mu_0 R\omega \frac{e}{4\pi R^2} \right)^2 = \frac{\mu_0\omega^2 e^2}{72\pi^2 R^2}, \text{ so } W_{B_{\text{in}}} = \frac{\mu_0\omega^2 e^2}{72\pi^2 R^2} \frac{4}{3}\pi R^3 = \frac{\mu_0 e^2 \omega^2 R}{54\pi}.$$

The energy density in the external magnetic field is:

$$u_B = \frac{1}{2\mu_0} \frac{\mu_0^2}{16\pi^2} \frac{m^2}{r^6} (4\cos^2\theta + \sin^2\theta) = \frac{e^2\omega^2 R^4 \mu_0}{18(16\pi^2)} \frac{1}{r^6} (3\cos^2\theta + 1), \text{ so}$$

$$W_{B_{\text{out}}} = \frac{\mu_0 e^2 \omega^2 R^4}{(18)(16)\pi^2} \int_R^\infty \frac{1}{r^6} r^2 dr \int_0^\pi (3\cos^2\theta + 1) \sin\theta d\theta \int_0^{2\pi} d\phi = \frac{\mu_0 e^2 \omega^2 R^4}{(18)(16)\pi^2} \left(\frac{1}{3R^3} \right) (4)(2\pi) = \frac{\mu_0 e^2 \omega^2 R}{108\pi}.$$

$$W_B = W_{B_{\text{in}}} + W_{B_{\text{out}}} = \frac{\mu_0 e^2 \omega^2 R}{108\pi} (2 + 1) = \frac{\mu_0 e^2 \omega^2 R}{36\pi}; \quad W = W_E + W_B = \boxed{\frac{1}{8\pi\epsilon_0} \frac{e^2}{R} + \frac{\mu_0 e^2 \omega^2 R}{36\pi}}.$$

(b) Same as Prob. 8.10(a), with $Q \rightarrow e$ and $m \rightarrow \frac{1}{3}e\omega R^2$: $\mathbf{L} = \frac{\mu_0 e^2 \omega R}{18\pi} \hat{\mathbf{z}}.$

$$(c) \frac{\mu_0 e^2}{18\pi} \omega R = \frac{\hbar}{2} \Rightarrow \omega R = \frac{9\pi\hbar}{\mu_0 e^2} = \frac{(9)(\pi)(1.05 \times 10^{-34})}{(4\pi \times 10^{-7})(1.60 \times 10^{-19})^2} = \boxed{9.23 \times 10^{10} \text{ m/s}}.$$

$$\frac{1}{8\pi\epsilon_0} \frac{e^2}{R} \left[1 + \frac{2}{9} \left(\frac{\omega R}{c} \right)^2 \right] = mc^2; \quad \left[1 + \frac{2}{9} \left(\frac{\omega R}{c} \right)^2 \right] = 1 + \frac{2}{9} \left(\frac{9.23 \times 10^{10}}{3 \times 10^8} \right)^2 = 2.10 \times 10^4;$$

$$R = \frac{(2.10 \times 10^4)(1.6 \times 10^{-19})^2}{8\pi(8.85 \times 10^{-12})(9.11 \times 10^{-31})(3 \times 10^8)^2} = \boxed{2.95 \times 10^{-11} \text{ m}}; \quad \omega = \frac{9.23 \times 10^{-10}}{2.95 \times 10^{-11}} = \boxed{3.13 \times 10^{21} \text{ rad/s}}.$$

Since ωR , the speed of a point on the equator, is 300 times the speed of light, this “classical” model is clearly unrealistic.

Problem 8.18

Maxwell's equations with magnetic charge (Eq. 7.44):

$$(i) \quad \nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho_e, \quad (iii) \quad \nabla \times \mathbf{E} = -\mu_0 \mathbf{J}_m - \frac{\partial \mathbf{B}}{\partial t},$$

$$(ii) \quad \nabla \cdot \mathbf{B} = \mu_0 \rho_m, \quad (iv) \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}_e + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

The Lorentz force law becomes (Eq. 8.44)

$$\mathbf{F} = q_e (\mathbf{E} + \mathbf{v} \times \mathbf{B}) + q_m \left(\mathbf{B} - \frac{1}{c^2} \mathbf{v} \times \mathbf{E} \right).$$

Following the argument in Section 8.1.2:

$$\mathbf{F} \cdot d\mathbf{l} = \left[q_e (\mathbf{E} + \mathbf{v} \times \mathbf{B}) + q_m \left(\mathbf{B} - \frac{1}{c^2} \mathbf{v} \times \mathbf{E} \right) \right] \cdot \mathbf{v} dt = (q_e \mathbf{E} + q_m \mathbf{B}) \cdot \mathbf{v} dt,$$

$$\frac{dW}{dt} = \int (\mathbf{E} \cdot \mathbf{J}_e + \mathbf{B} \cdot \mathbf{J}_m) d\tau$$

(which generalizes Eq. 8.6). Use (iii) and (iv) to eliminate $\mathbf{J}_e$ and $\mathbf{J}_m$:

$$(\mathbf{E} \cdot \mathbf{J}_e + \mathbf{B} \cdot \mathbf{J}_m) = \frac{1}{\mu_0} \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} - \frac{1}{\mu_0} \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \frac{1}{\mu_0} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t},$$

but $\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B})$, so

$$(\mathbf{E} \cdot \mathbf{J}_e + \mathbf{B} \cdot \mathbf{J}_m) = -\frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}) - \frac{1}{2} \frac{\partial}{\partial t} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right)$$

(generalizing Eq. 8.8). Thus

$$\frac{dW}{dt} = -\frac{d}{dt} \int \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) d\tau - \frac{1}{\mu_0} \oint (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a},$$

which is identical to Eq. 8.9. Evidently Poynting's theorem is unchanged(!), and

$$\boxed{u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right)}, \quad \boxed{\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B})},$$

the same as before.

To construct the stress tensor we begin with the generalization of Eq. 8.13:

$$\mathbf{F} = \int \left[(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \rho_e + \left(\mathbf{B} - \frac{1}{c^2} \mathbf{v} \times \mathbf{E} \right) \rho_m \right] d\tau.$$

The force per unit volume (Eq. 8.14) becomes

$$\begin{aligned} \mathbf{f} &= (\rho_e \mathbf{E} + \mathbf{J}_e \times \mathbf{B}) + \left(\rho_m \mathbf{B} - \frac{1}{c^2} \mathbf{J}_m \times \mathbf{E} \right) \\ &= \epsilon_0 (\nabla \cdot \mathbf{E}) \mathbf{E} + \left(\frac{1}{\mu_0} \nabla \times \mathbf{B} - \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \times \mathbf{B} + \frac{1}{\mu_0} (\nabla \cdot \mathbf{B}) \mathbf{B} - \mu_0 \epsilon_0 \left(-\frac{1}{\mu_0} \nabla \times \mathbf{E} - \frac{1}{\mu_0} \frac{\partial \mathbf{B}}{\partial t} \right) \times \mathbf{E} \\ &= \epsilon_0 [(\nabla \cdot \mathbf{E}) \mathbf{E} - \mathbf{E} \times (\nabla \times \mathbf{E})] + \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B}) \mathbf{B} - \mathbf{B} \times (\nabla \times \mathbf{B})] - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) \\ &= \epsilon_0 \left[(\nabla \cdot \mathbf{E}) \mathbf{E} - \frac{1}{2} \nabla (E^2) + (\mathbf{E} \cdot \nabla) \mathbf{E} \right] + \frac{1}{\mu_0} \left[(\nabla \cdot \mathbf{B}) \mathbf{B} - \frac{1}{2} \nabla (B^2) + (\mathbf{B} \cdot \nabla) \mathbf{B} \right] - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}). \end{aligned}$$

This is identical to Eq. 8.16, so the stress tensor is the same as before:

$$T_{ij} \equiv \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right).$$

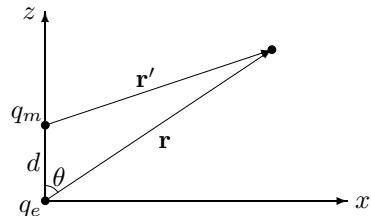
Likewise, Eq. 8.20 is still valid. In fact, this argument is *more straightforward* when you include magnetic charge, since you don't need artificially to insert the $(\nabla \cdot \mathbf{B})\mathbf{B}$ term (after Eq. 8.15).

The electromagnetic momentum density (Eq. 8.29) also stays the same, since the argument in Section 8.2.3 is formulated entirely in terms of the fields:

$$\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}).$$

Problem 8.19

$$\begin{aligned} \mathbf{E} &= \frac{q_e}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3}; \\ \mathbf{B} &= \frac{\mu_0 q_m}{4\pi} \frac{\mathbf{r}'}{r'^3} = \frac{\mu_0 q_m}{4\pi} \frac{(\mathbf{r} - d\hat{\mathbf{z}})}{(r^2 + d^2 - 2rd\cos\theta)^{3/2}}. \end{aligned}$$



Momentum density (Eq. 8.32):

$$\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}) = \frac{\mu_0 q_e q_m}{(4\pi)^2} \frac{(-d)(\mathbf{r} \times \hat{\mathbf{z}})}{r^3 (r^2 + d^2 - 2rd\cos\theta)^{3/2}}.$$

Angular momentum density (Eq. 8.33):

$$\ell = (\mathbf{r} \times \mathbf{g}) = -\frac{\mu_0 q_e q_m d}{(4\pi)^2} \frac{\mathbf{r} \times (\mathbf{r} \times \hat{\mathbf{z}})}{r^3 (r^2 + d^2 - 2rd\cos\theta)^{3/2}}. \quad \text{But } \mathbf{r} \times (\mathbf{r} \times \hat{\mathbf{z}}) = \mathbf{r}(\mathbf{r} \cdot \hat{\mathbf{z}}) - r^2 \hat{\mathbf{z}} = r^2 \cos\theta \hat{\mathbf{r}} - r^2 \hat{\mathbf{z}}.$$

The x and y components will integrate to zero; using $(\hat{\mathbf{r}})_z = \cos\theta$, we have:

$$\begin{aligned} \mathbf{L} &= -\frac{\mu_0 q_e q_m d}{(4\pi)^2} \hat{\mathbf{z}} \int \frac{r^2 (\cos^2\theta - 1)}{r^3 (r^2 + d^2 - 2rd\cos\theta)^{3/2}} r^2 \sin\theta dr d\theta d\phi. \quad \text{Let } u \equiv \cos\theta : \\ &= \frac{\mu_0 q_e q_m d}{(4\pi)^2} \hat{\mathbf{z}} (2\pi) \int_{-1}^1 \int_0^\infty \frac{r(1-u^2)}{(r^2 + d^2 - 2rdu)^{3/2}} du dr. \end{aligned}$$

Do the r integral first:

$$\int_0^\infty \frac{r dr}{(r^2 + d^2 - 2rdu)^{3/2}} = \frac{(ru - d)}{d(1-u^2)\sqrt{r^2 + d^2 - 2rdu}} \Big|_0^\infty = \frac{u}{d(1-u^2)} + \frac{d}{d(1-u^2)d} = \frac{u+1}{d(1-u^2)} = \frac{1}{d(1-u)}.$$

Then

$$\mathbf{L} = \frac{\mu_0 q_e q_m d}{8\pi} \hat{\mathbf{z}} \frac{1}{d} \int_{-1}^1 \frac{(1-u^2)}{(1-u)} du = \frac{\mu_0 q_e q_m}{8\pi} \hat{\mathbf{z}} \int_{-1}^1 (1+u) du = \frac{\mu_0 q_e q_m}{8\pi} \hat{\mathbf{z}} \left(u + \frac{u^2}{2} \right) \Big|_{-1}^1 = \boxed{\frac{\mu_0 q_e q_m}{4\pi} \hat{\mathbf{z}}}.$$

Problem 8.20

$$\begin{aligned}\mathbf{p} &= \epsilon_0 \int_{\mathcal{V}} (\mathbf{E} \times \mathbf{B}) d\tau = -\epsilon_0 \int_{\mathcal{V}} (\nabla V) \times \mathbf{B} d\tau = -\epsilon_0 \int_{\mathcal{V}} [\nabla \times (V\mathbf{B}) - V\nabla \times \mathbf{B}] d\tau \\ &= \epsilon_0 \oint_{\mathcal{S}} V\mathbf{B} \times d\mathbf{a} + \epsilon_0 \mu_0 \int_{\mathcal{V}} V\mathbf{J} d\tau = \frac{1}{c^2} \int_{\mathcal{V}} V\mathbf{J} d\tau.\end{aligned}$$

(I used Problem 1.61(b) in the penultimate step. Here $\mathcal{V}$ is all of space, and $\mathcal{S}$ is its surface at infinity, where $\mathbf{B} = \mathbf{0}$, so the surface integral vanishes.) Using $V(\mathbf{r}) \approx V(\mathbf{0}) + (\nabla V) \cdot \mathbf{r} = V(\mathbf{0}) - \mathbf{E}(\mathbf{0}) \cdot \mathbf{r}$,

$$\mathbf{p} = \frac{1}{c^2} V(\mathbf{0}) \int \mathbf{J} d\tau - \frac{1}{c^2} \int [\mathbf{E}(\mathbf{0}) \cdot \mathbf{r}] \mathbf{J} d\tau.$$

For a current loop, $\int \mathbf{J} d\tau \rightarrow \int \mathbf{I} dl = I \int d\mathbf{l} = \mathbf{0}$, and (Eq. 1.108):

$$\int [\mathbf{E}(\mathbf{0}) \cdot \mathbf{r}] \mathbf{J} d\tau \rightarrow \int [\mathbf{E}(\mathbf{0}) \cdot \mathbf{r}] \mathbf{I} dl = I \int [\mathbf{E}(\mathbf{0}) \cdot \mathbf{r}] d\mathbf{l} = I \mathbf{a} \times \mathbf{E}(\mathbf{0}) = \mathbf{m} \times \mathbf{E}.$$

So

$$\mathbf{p} = -\frac{1}{c^2} (\mathbf{m} \times \mathbf{E}). \quad \checkmark$$

Problem 8.21

(a) The rotating shell at radius b produces a solenoidal magnetic field:

$$\mathbf{B} = \mu_0 K \hat{\mathbf{z}}, \text{ where } K = \sigma_b \omega_b b, \text{ and } \sigma_b = -\frac{Q}{2\pi b l}. \text{ So } \mathbf{B} = -\frac{\mu_0 \omega_b Q}{2\pi l} \hat{\mathbf{z}} \text{ } (a < s < b).$$

(Note that if angular velocity is defined with respect to the z axis, then ω_b is a *negative* number.) The shell at a also produces a magnetic field $(\mu_0 \omega_a Q / 2\pi l) \hat{\mathbf{z}}$, in the region $s < a$, so the total field inside the inner shell is

$$\mathbf{B} = \frac{\mu_0 Q}{2\pi l} (\omega_a - \omega_b) \hat{\mathbf{z}}, \text{ } (s < a).$$

Meanwhile, the electric field is

$$\mathbf{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{s} \hat{\mathbf{s}} = \frac{Q}{2\pi\epsilon_0 l s} \hat{\mathbf{s}}, \text{ } (a < s < b).$$

$$\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}) = \epsilon_0 \left(\frac{Q}{2\pi\epsilon_0 l s} \right) \left(-\frac{\mu_0 \omega_b Q}{2\pi l} \right) (\hat{\mathbf{s}} \times \hat{\mathbf{z}}) = \frac{\mu_0 \omega_b Q^2}{4\pi^2 l^2 s} \hat{\phi}; \quad \boldsymbol{\ell} = \mathbf{r} \times \mathbf{g} = \frac{\mu_0 \omega_b Q^2}{4\pi^2 l^2 s} (\mathbf{r} \times \hat{\phi}).$$

Now $\mathbf{r} \times \hat{\phi} = (s \hat{\mathbf{s}} + z \hat{\mathbf{z}}) \times \hat{\phi} = s \hat{\mathbf{z}} - z \hat{\mathbf{s}}$, and the $\hat{\mathbf{s}}$ term integrates to zero, so

$$\mathbf{L} = \frac{\mu_0 \omega_b Q^2}{4\pi^2 l^2} \hat{\mathbf{z}} \int d\tau = \frac{\mu_0 \omega_b Q^2}{4\pi^2 l^2} \pi (b^2 - a^2) l \hat{\mathbf{z}} = \boxed{\frac{\mu_0 \omega_b Q^2 (b^2 - a^2)}{4\pi l} \hat{\mathbf{z}}}.$$

(b) The extra electric field induced by the changing magnetic field due to the rotating shells is given by

$$E 2\pi s = -\frac{d\Phi}{dt} \Rightarrow \mathbf{E} = -\frac{1}{2\pi s} \frac{d\Phi}{dt} \hat{\phi}, \text{ and in the region } a < s < b$$

$$\Phi = \frac{\mu_0 Q}{2\pi l} (\omega_a - \omega_b) \pi a^2 - \frac{\mu_0 Q \omega_b}{2\pi l} \pi (s^2 - a^2) = \frac{\mu_0 Q}{2l} (\omega_a a^2 - \omega_b s^2); \quad \mathbf{E}(s) = -\frac{1}{2\pi s} \frac{\mu_0 Q}{2l} \left(a^2 \frac{d\omega_a}{dt} - s^2 \frac{d\omega_b}{dt} \right) \hat{\phi}.$$

In particular,

$$\mathbf{E}(a) = -\frac{\mu_0 Q a}{4\pi l} \left(\frac{d\omega_a}{dt} - \frac{d\omega_b}{dt} \right) \hat{\phi}, \quad \text{and } \mathbf{E}(b) = -\frac{\mu_0 Q}{4\pi l b} \left(a^2 \frac{d\omega_a}{dt} - b^2 \frac{d\omega_b}{dt} \right) \hat{\phi}.$$

The torque on a shell is $\mathbf{N} = \mathbf{r} \times q\mathbf{E} = qsE \hat{\mathbf{z}}$, so

$$\begin{aligned} \mathbf{N}_a &= Qa \left(-\frac{\mu_0 Q a}{4\pi l} \right) \left(\frac{d\omega_a}{dt} - \frac{d\omega_b}{dt} \right) \hat{\mathbf{z}}; \quad \mathbf{L}_a = \int_0^\infty \mathbf{N}_a dt = -\frac{\mu_0 Q^2 a^2}{4\pi l} (\omega_a - \omega_b) \hat{\mathbf{z}}. \\ \mathbf{N}_b &= -Qb \left(-\frac{\mu_0 Q}{4\pi l b} \right) \left(a^2 \frac{d\omega_a}{dt} - b^2 \frac{d\omega_b}{dt} \right) \hat{\mathbf{z}}; \quad \mathbf{L}_b = \int_0^\infty \mathbf{N}_b dt = \frac{\mu_0 Q^2}{4\pi l} (a^2 \omega_a - b^2 \omega_b) \hat{\mathbf{z}}. \\ \mathbf{L}_{\text{tot}} &= \mathbf{L}_a + \mathbf{L}_b = \frac{\mu_0 Q^2}{4\pi l} (a^2 \omega_a - b^2 \omega_b - a^2 \omega_a + a^2 \omega_b) \hat{\mathbf{z}} = \boxed{-\frac{\mu_0 Q^2 \omega_b}{4\pi l} (b^2 - a^2) \hat{\mathbf{z}}}. \end{aligned}$$

Thus the reduction in the final mechanical angular momentum (b) is equal to the residual angular momentum in the fields (a). ✓

Problem 8.22

$$\mathbf{B} = \mu_0 n I \hat{\mathbf{z}}, \quad (s < R); \quad \mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{\boldsymbol{\mathbf{r}}}{r^3}, \quad \text{where } \boldsymbol{\mathbf{r}} = (x - a, y, z).$$

$$\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}) = \epsilon_0 (\mu_0 n I) \left(\frac{q}{4\pi\epsilon_0} \right) \frac{1}{r^3} (\boldsymbol{\mathbf{r}} \times \hat{\mathbf{z}}) = \frac{\mu_0 q n I}{4\pi r^3} [y \hat{\mathbf{x}} - (x - a) \hat{\mathbf{y}}].$$

Linear Momentum.

$$\begin{aligned} \mathbf{p} &= \int \mathbf{g} d\tau = \frac{\mu_0 q n I}{4\pi} \int \frac{y \hat{\mathbf{x}} - (x - a) \hat{\mathbf{y}}}{[(x - a)^2 + y^2 + z^2]^{3/2}} dx dy dz. \quad \text{The } \hat{\mathbf{x}} \text{ term is odd in } y; \text{ it integrates to zero.} \\ &= -\frac{\mu_0 q n I}{4\pi} \hat{\mathbf{y}} \int \frac{(x - a)}{[(x - a)^2 + y^2 + z^2]^{3/2}} dx dy dz. \quad \text{Do the } z \text{ integral first :} \\ &\quad \left. \frac{z}{[(x - a)^2 + y^2] \sqrt{(x - a)^2 + y^2 + z^2}} \right|_{-\infty}^{\infty} = \frac{2}{[(x - a)^2 + y^2]}. \\ &= -\frac{\mu_0 q n I}{2\pi} \hat{\mathbf{y}} \int \frac{(x - a)}{[(x - a)^2 + y^2]} dx dy. \quad \text{Switch to polar coordinates :} \\ &\quad x = s \cos \phi, \quad y = s \sin \phi, \quad dx dy \Rightarrow s ds d\phi; \quad [(x - a)^2 + y^2] = s^2 + a^2 - 2sa \cos \phi. \\ &= -\frac{\mu_0 q n I}{2\pi} \hat{\mathbf{y}} \int \frac{(s \cos \phi - a)}{(s^2 + a^2 - 2sa \cos \phi)} s ds d\phi \\ &\quad \text{Now } \int_0^{2\pi} \frac{\cos \phi d\phi}{(A + B \cos \phi)} = \frac{2\pi}{B} \left(1 - \frac{A}{\sqrt{A^2 - B^2}} \right); \quad \int_0^{2\pi} \frac{d\phi}{(A + B \cos \phi)} = \frac{2\pi}{\sqrt{A^2 - B^2}}. \\ &\quad \text{Here } A^2 - B^2 = (s^2 + a^2)^2 - 4s^2 a^2 = s^4 + 2s^2 a^2 + a^4 - 4s^2 a^2 = (s^2 - a^2)^2; \quad \sqrt{A^2 - B^2} = a^2 - s^2. \\ &= \frac{\mu_0 q n I}{2a} \hat{\mathbf{y}} \int \left[1 - \left(\frac{a^2 + s^2}{a^2 - s^2} \right) + \frac{2a^2}{(a^2 - s^2)} \right] s ds = \frac{\mu_0 q n I}{a} \hat{\mathbf{y}} \int_0^R s ds = \boxed{\frac{\mu_0 q n I R^2}{2a} \hat{\mathbf{y}}}. \end{aligned}$$

Angular Momentum.

$$\boldsymbol{\ell} = \mathbf{r} \times \mathbf{g} = \frac{\mu_0 q n I}{4\pi \mathcal{Z}^3} \mathbf{r} \times [y \hat{\mathbf{x}} - (x - a) \hat{\mathbf{y}}] = \frac{\mu_0 q n I}{4\pi \mathcal{Z}^3} \{z(x - a) \hat{\mathbf{x}} + zy \hat{\mathbf{y}} - [x(x - a) + y^2] \hat{\mathbf{z}}\}.$$

The $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ terms are odd in z , and integrate to zero, so

$$\begin{aligned} \mathbf{L} &= -\frac{\mu_0 q n I}{4\pi} \hat{\mathbf{z}} \int \frac{x^2 + y^2 - xa}{[(x - a)^2 + y^2 + z^2]^{3/2}} dx dy dz. \text{ The } z \text{ integral is the same as before.} \\ &= -\frac{\mu_0 q n I}{2\pi} \hat{\mathbf{z}} \int \frac{x^2 + y^2 - xa}{[(x - a)^2 + y^2]} dx dy = -\frac{\mu_0 q n I}{2\pi} \hat{\mathbf{z}} \int \frac{s - a \cos \phi}{(s^2 + a^2 - 2sa \cos \phi)} s^2 ds d\phi \\ &= -\mu_0 q n I \hat{\mathbf{z}} \int \left[\frac{s^2}{a^2 - s^2} + \left(1 - \frac{a^2 + s^2}{a^2 - s^2}\right) \right] s ds = -\mu_0 q n I \hat{\mathbf{z}} \int_0^R \frac{s^2 - s^2}{a^2 - s^2} s ds = \boxed{\text{zero.}} \end{aligned}$$

Problem 8.23

(a) If we're only interested in the work done on *free* charges and currents, Eq. 8.6 becomes $\frac{dW}{dt} = \int_V (\mathbf{E} \cdot \mathbf{J}_f) d\tau$. But $\mathbf{J}_f = \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t}$ (Eq. 7.56), so $\mathbf{E} \cdot \mathbf{J}_f = \mathbf{E} \cdot (\nabla \times \mathbf{H}) - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}$. From product rule #6, $\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H})$, while $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$, so $\mathbf{E} \cdot (\nabla \times \mathbf{H}) = -\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{H})$. Therefore $\mathbf{E} \cdot \mathbf{J}_f = -\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{H})$, and hence

$$\frac{dW}{dt} = - \int_V \left(\mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} + \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} \right) d\tau - \oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{a}.$$

This is Poynting's theorem for the fields in matter. Evidently the Poynting vector, representing the power per unit area transported by the fields, is $\mathbf{S} = \mathbf{E} \times \mathbf{H}$, and the rate of change of the electromagnetic energy density is $\frac{\partial u_{\text{em}}}{\partial t} = \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} + \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t}$.

For *linear* media, $\mathbf{D} = \epsilon \mathbf{E}$ and $\mathbf{H} = \frac{1}{\mu} \mathbf{B}$, with ϵ and μ constant (in time); then

$$\frac{\partial u_{\text{em}}}{\partial t} = \epsilon \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} + \frac{1}{\mu} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} = \frac{1}{2} \epsilon \frac{\partial}{\partial t} (\mathbf{E} \cdot \mathbf{E}) + \frac{1}{2\mu} \frac{\partial}{\partial t} (\mathbf{B} \cdot \mathbf{B}) = \frac{1}{2} \frac{\partial}{\partial t} (\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H}),$$

so $u_{\text{em}} = \frac{1}{2} (\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H})$. qed

(b) If we're only interested in the force on *free* charges and currents, Eq. 8.13 becomes $\mathbf{f} = \rho_f \mathbf{E} + \mathbf{J}_f \times \mathbf{B}$. But $\rho_f = \nabla \cdot \mathbf{D}$, and $\mathbf{J}_f = \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t}$, so $\mathbf{f} = \mathbf{E}(\nabla \cdot \mathbf{D}) + (\nabla \times \mathbf{H}) \times \mathbf{B} - \left(\frac{\partial \mathbf{D}}{\partial t} \right) \times \mathbf{B}$. Now $\frac{\partial}{\partial t} (\mathbf{D} \times \mathbf{B}) = \frac{\partial \mathbf{D}}{\partial t} \times \mathbf{B} + \mathbf{D} \times \left(\frac{\partial \mathbf{B}}{\partial t} \right)$, and $\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}$, so $\frac{\partial \mathbf{D}}{\partial t} \times \mathbf{B} = \frac{\partial}{\partial t} (\mathbf{D} \times \mathbf{B}) + \mathbf{D} \times (\nabla \times \mathbf{E})$, and hence $\mathbf{f} = \mathbf{E}(\nabla \cdot \mathbf{D}) - \mathbf{D} \times (\nabla \times \mathbf{E}) - \mathbf{B} \times (\nabla \times \mathbf{H}) - \frac{\partial}{\partial t} (\mathbf{D} \times \mathbf{B})$. As before, we can with impunity add the term $\mathbf{H}(\nabla \cdot \mathbf{B})$, so

$$\mathbf{f} = \{[\mathbf{E}(\nabla \cdot \mathbf{D}) - \mathbf{D} \times (\nabla \times \mathbf{E})] + [\mathbf{H}(\nabla \cdot \mathbf{B}) - \mathbf{B} \times (\nabla \times \mathbf{H})]\} - \frac{\partial}{\partial t} (\mathbf{D} \times \mathbf{B}).$$

The term in curly brackets can be written as the divergence of a stress tensor (as in Eq. 8.19), and the last term is (minus) the rate of change of the momentum density, $\mathbf{g} = \mathbf{D} \times \mathbf{B}$.

Problem 8.24

(a) Initially, the disk will rise like a helicopter. The force on one charge (velocity $\mathbf{v} = \omega R \hat{\phi} + v_z \hat{\mathbf{z}}$) is

$$\mathbf{F}_i = q(\mathbf{v} \times \mathbf{B}) = qk \begin{vmatrix} \hat{\mathbf{s}} & \hat{\phi} & \hat{\mathbf{z}} \\ 0 & \omega R & v_z \\ -R & 0 & 2z \end{vmatrix} = qk \left(2\omega R z \hat{\mathbf{s}} - R v_z \hat{\phi} + \omega R^2 \hat{\mathbf{z}} \right).$$

The net force on all the charges is

$$\mathbf{F} = \sum_{i=1}^n \mathbf{F}_i = nqkR^2\omega \hat{\mathbf{z}} = M \frac{d^2 z}{dt^2} \hat{\mathbf{z}}; \quad \frac{d^2 z}{dt^2} = \left(\frac{nqkR^2}{M} \right) \omega. \quad [1]$$

The net torque on the disk is

$$\mathbf{N} = \sum_{i=1}^n (\mathbf{r}_i \times \mathbf{F}_i) = n(R\hat{\mathbf{s}}) \times (-qkRv_z \hat{\phi}) = -nqkR^2v_z \hat{\mathbf{z}} = I \frac{d\omega}{dt} \hat{\mathbf{z}}$$

where I is the moment of inertia of the disk. So

$$\frac{d\omega}{dt} = - \left(\frac{nqkR^2}{I} \right) \frac{dz}{dt}. \quad [2]$$

Differentiate [2], and combine with [1]:

$$\frac{d^2 z}{dt^2} = - \left(\frac{I}{nqkR^2} \right) \frac{d^2 \omega}{dt^2} = \left(\frac{nqkR^2}{M} \right) \omega \quad \Rightarrow \quad \frac{d^2 \omega}{dt^2} = \alpha \omega, \quad \text{where} \quad \boxed{\alpha \equiv \frac{nqkR^2}{\sqrt{MI}}}.$$

The solution (with initial angular velocity ω_0 and initial angular acceleration 0) is

$$\boxed{\omega(t) = \omega_0 \cos \alpha t.}$$

Meanwhile,

$$\frac{dz}{dt} = - \left(\frac{I}{nqkR^2} \right) \frac{d\omega}{dt} = \left(\frac{I}{nqkR^2} \right) \omega_0 \alpha \sin \alpha t = \omega_0 \sqrt{\frac{I}{M}} \sin \alpha t.$$

$$z(t) = \omega_0 \sqrt{\frac{I}{M}} \int_0^t \sin \alpha t \, dt = \boxed{\frac{\omega_0}{\alpha} \sqrt{\frac{I}{M}} (1 - \cos \alpha t).}$$

[The problem is a little cleaner if you make the disk massless, and assign a mass m to each of the charges. Then $M \rightarrow nm$ and $I \rightarrow nmR^2$, so $\alpha = qkR/m$ and $z(t) \rightarrow (\omega_0 R/\alpha)(1 - \cos \alpha t)$.]

(b) The disk rises and falls harmonically, as its rotation slows down and speeds up. The total energy is

$$E = \frac{1}{2} M v_z^2 + \frac{1}{2} I \omega^2 = \frac{1}{2} M \omega_0^2 \frac{I}{M} \sin^2 \alpha t + \frac{1}{2} I \omega_0^2 \cos^2 \alpha t = \frac{1}{2} I \omega_0^2 (\sin^2 \alpha t + \cos^2 \alpha t) = \frac{1}{2} I \omega_0^2,$$

which is constant (and equal to the initial energy). [Of course, if you didn't notice that the rotation rate is changing, you might think the magnetic force is doing work, as the disk oscillates up and down.]

Chapter 9

Electromagnetic Waves

Problem 9.1

$$\begin{aligned}\frac{\partial f_1}{\partial z} &= -2Ab(z-vt)e^{-b(z-vt)^2}; \quad \frac{\partial^2 f_1}{\partial z^2} = -2Ab \left[e^{-b(z-vt)^2} - 2b(z-vt)^2 e^{-b(z-vt)^2} \right]; \\ \frac{\partial f_1}{\partial t} &= 2Abv(z-vt)e^{-b(z-vt)^2}; \quad \frac{\partial^2 f_1}{\partial t^2} = 2Abv \left[-ve^{-b(z-vt)^2} + 2bv(z-vt)^2 e^{-b(z-vt)^2} \right] = v^2 \frac{\partial^2 f_1}{\partial z^2}. \quad \checkmark \\ \frac{\partial f_2}{\partial z} &= Ab \cos[b(z-vt)]; \quad \frac{\partial^2 f_2}{\partial z^2} = -Ab^2 \sin[b(z-vt)]; \\ \frac{\partial f_2}{\partial t} &= -Abv \cos[b(z-vt)]; \quad \frac{\partial^2 f_2}{\partial t^2} = -Ab^2 v^2 \sin[b(z-vt)] = v^2 \frac{\partial^2 f_2}{\partial z^2}. \quad \checkmark \\ \frac{\partial f_3}{\partial z} &= \frac{-2Ab(z-vt)}{[b(z-vt)^2 + 1]^2}; \quad \frac{\partial^2 f_3}{\partial z^2} = \frac{-2Ab}{[b(z-vt)^2 + 1]^2} + \frac{8Ab^2(z-vt)^2}{[b(z-vt)^2 + 1]^3}; \\ \frac{\partial f_3}{\partial t} &= \frac{2Abv(z-vt)}{[b(z-vt)^2 + 1]^2}; \quad \frac{\partial^2 f_3}{\partial t^2} = \frac{-2Abv^2}{[b(z-vt)^2 + 1]^2} + \frac{8Ab^2 v^2 (z-vt)^2}{[b(z-vt)^2 + 1]^3} = v^2 \frac{\partial^2 f_3}{\partial z^2}. \quad \checkmark \\ \frac{\partial f_4}{\partial z} &= -2Ab^2 z e^{-b(bz^2+vt)}; \quad \frac{\partial^2 f_4}{\partial z^2} = -2Ab^2 \left[e^{-b(bz^2+vt)} - 2b^2 z^2 e^{-b(bz^2+vt)} \right]; \\ \frac{\partial f_4}{\partial t} &= -Abv e^{-b(bz^2+vt)}; \quad \frac{\partial^2 f_4}{\partial t^2} = Ab^2 v^2 e^{-b(bz^2+vt)} \neq v^2 \frac{\partial^2 f_4}{\partial z^2}. \\ \frac{\partial f_5}{\partial z} &= Ab \cos(bz) \cos(bvt)^3; \quad \frac{\partial^2 f_5}{\partial z^2} = -Ab^2 \sin(bz) \cos(bvt)^3; \quad \frac{\partial f_5}{\partial t} = -3Ab^3 v^3 t^2 \sin(bz) \sin(bvt)^3; \\ \frac{\partial^2 f_5}{\partial t^2} &= -6Ab^3 v^3 t \sin(bz) \sin(bvt)^3 - 9Ab^6 v^6 t^4 \sin(bz) \cos(bvt)^3 \neq v^2 \frac{\partial^2 f_5}{\partial z^2}.\end{aligned}$$

Problem 9.2

$$\begin{aligned}\frac{\partial f}{\partial z} &= Ak \cos(kz) \cos(kvt); \quad \frac{\partial^2 f}{\partial z^2} = -Ak^2 \sin(kz) \cos(kvt); \\ \frac{\partial f}{\partial t} &= -Akv \sin(kz) \sin(kvt); \quad \frac{\partial^2 f}{\partial t^2} = -Ak^2 v^2 \sin(kz) \cos(kvt) = v^2 \frac{\partial^2 f}{\partial z^2}. \quad \checkmark\end{aligned}$$

Use the trig identity $\sin \alpha \cos \beta = \frac{1}{2}[\sin(\alpha + \beta) + \sin(\alpha - \beta)]$ to write

$$f = \boxed{\frac{A}{2} \{ \sin[k(z + vt)] + \sin[k(z - vt)] \},}$$

which is of the form 9.6, with $g = (A/2) \sin[k(z - vt)]$ and $h = (A/2) \sin[k(z + vt)]$.

Problem 9.3

$$\begin{aligned} (A_3)^2 &= (A_3 e^{i\delta_3}) (A_3 e^{-i\delta_3}) = (A_1 e^{i\delta_1} + A_2 e^{i\delta_2}) (A_1 e^{-i\delta_1} + A_2 e^{-i\delta_2}) \\ &= (A_1)^2 + (A_2)^2 + A_1 A_2 (e^{i\delta_1} e^{-i\delta_2} + e^{-i\delta_1} e^{i\delta_2}) = (A_1)^2 + (A_2)^2 + A_1 A_2 2 \cos(\delta_1 - \delta_2); \end{aligned}$$

$$A_3 = \boxed{\sqrt{(A_1)^2 + (A_2)^2 + 2A_1 A_2 \cos(\delta_1 - \delta_2)}}.$$

$$\begin{aligned} A_3 e^{i\delta_3} &= A_3 (\cos \delta_3 + i \sin \delta_3) = A_1 (\cos \delta_1 + i \sin \delta_1) + A_2 (\cos \delta_2 + i \sin \delta_2) \\ &= (A_1 \cos \delta_1 + A_2 \cos \delta_2) + i(A_1 \sin \delta_1 + A_2 \sin \delta_2). \quad \tan \delta_3 = \frac{A_3 \sin \delta_3}{A_3 \cos \delta_3} = \frac{A_1 \sin \delta_1 + A_2 \sin \delta_2}{A_1 \cos \delta_1 + A_2 \cos \delta_2}; \end{aligned}$$

$$\delta_3 = \boxed{\tan^{-1} \left(\frac{A_1 \sin \delta_1 + A_2 \sin \delta_2}{A_1 \cos \delta_1 + A_2 \cos \delta_2} \right)}.$$

Problem 9.4

The wave equation (Eq. 9.2) says $\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}$. Look for solutions of the form $f(z, t) = Z(z)T(t)$. Plug this in: $T \frac{d^2 Z}{dz^2} = \frac{1}{v^2} Z \frac{d^2 T}{dt^2}$. Divide by ZT : $\frac{1}{Z} \frac{d^2 Z}{dz^2} = \frac{1}{v^2 T} \frac{d^2 T}{dt^2}$. The left side depends only on z , and the right side only on t , so both must be constant. Call the constant $-k^2$.

$$\left\{ \begin{array}{l} \frac{d^2 Z}{dz^2} = -k^2 Z \quad \Rightarrow Z(z) = A e^{ikz} + B e^{-ikz}, \\ \frac{d^2 T}{dt^2} = -(kv)^2 T \Rightarrow T(t) = C e^{ikvt} + D e^{-ikvt}. \end{array} \right\}$$

(Note that k must be *real*, else Z and T blow up; with no loss of generality we can assume k is *positive*.)

$$f(z, t) = (A e^{ikz} + B e^{-ikz}) (C e^{ikvt} + D e^{-ikvt}) = A_1 e^{i(kz+kv t)} + A_2 e^{i(kz-kv t)} + A_3 e^{i(-kz+kv t)} + A_4 e^{i(-kz-kv t)}.$$

The general linear combination of separable solutions is therefore

$$f(z, t) = \int_0^\infty \left[A_1(k) e^{i(kz+\omega t)} + A_2(k) e^{i(kz-\omega t)} + A_3(k) e^{i(-kz+\omega t)} + A_4(k) e^{i(-kz-\omega t)} \right] dk,$$

where $\omega \equiv kv$. But we can combine the third term with the first, by allowing k to run *negative* ($\omega = |k|v$ remains positive); likewise the second and the fourth:

$$f(z, t) = \int_{-\infty}^\infty \left[A_1(k) e^{i(kz+\omega t)} + A_2(k) e^{i(kz-\omega t)} \right] dk.$$

Because (in the end) we shall only want the *real part* of f , it suffices to keep only *one* of these terms (since k goes negative, both terms include waves traveling in both directions); the second is traditional (though either would do). Specifically,

$$\text{Re}(f) = \int_{-\infty}^\infty [\text{Re}(A_1) \cos(kz + \omega t) - \text{Im}(A_1) \sin(kz + \omega t) + \text{Re}(A_2) \cos(kz - \omega t) - \text{Im}(A_2) \sin(kz - \omega t)] dk.$$

The first term, $\cos(kz + \omega t) = \cos(-kz - \omega t)$, combines with the third, $\cos(kz - \omega t)$, since the negative k is picked up in the other half of the range of integration, and the second, $\sin(kz + \omega t) = -\sin(-kz - \omega t)$, combines with the fourth for the same reason. So the general solution, for our purposes, can be written in the form

$$\tilde{f}(z, t) = \int_{-\infty}^\infty \tilde{A}(k) e^{i(kz-\omega t)} dk \quad \text{qed} \quad (\text{the tilde's remind us that we want the real part}).$$

Problem 9.5

Equation 9.26 $\Rightarrow g_I(-v_1 t) + h_R(v_1 t) = g_T(-v_2 t)$. Now $\frac{\partial g_I}{\partial z} = -\frac{1}{v_1} \frac{\partial g_I}{\partial t}$; $\frac{\partial h_R}{\partial z} = \frac{1}{v_1} \frac{\partial h_R}{\partial t}$; $\frac{\partial g_T}{\partial z} = -\frac{1}{v_2} \frac{\partial g_T}{\partial t}$.
 Equation 9.27 $\Rightarrow -\frac{1}{v_1} \frac{\partial g_I(-v_1 t)}{\partial t} + \frac{1}{v_1} \frac{\partial h_R(v_1 t)}{\partial t} = -\frac{1}{v_2} \frac{\partial g_T(-v_2 t)}{\partial t} \Rightarrow g_I(-v_1 t) - h_R(v_1 t) = \frac{v_1}{v_2} g_T(-v_2 t) + \kappa$
 (where κ is a constant).

Adding these equations, we get $2g_I(-v_1 t) = \left(1 + \frac{v_1}{v_2}\right) g_T(-v_2 t) + \kappa$, or $g_T(-v_2 t) = \left(\frac{2v_2}{v_1 + v_2}\right) g_I(-v_1 t) + \kappa'$
 (where $\kappa' \equiv -\kappa \frac{v_2}{v_1 + v_2}$). Now $g_I(z, t)$, $g_T(z, t)$, and $h_R(z, t)$ are each functions of a single variable u (in the first case $u = z - v_1 t$, in the second $u = z - v_2 t$, and in the third $u = z + v_1 t$). Thus

$$g_T(u) = \left(\frac{2v_2}{v_1 + v_2}\right) g_I(v_1 u/v_2) + \kappa'.$$

Multiplying the first equation by v_1/v_2 and subtracting, $\left(1 - \frac{v_1}{v_2}\right) g_I(-v_1 t) - \left(1 + \frac{v_1}{v_2}\right) h_R(v_1 t) = \kappa \Rightarrow$
 $h_R(v_1 t) = \left(\frac{v_2 - v_1}{v_1 + v_2}\right) g_I(-v_1 t) - \kappa \left(\frac{v_2}{v_1 + v_2}\right)$, or $h_R(u) = \left(\frac{v_2 - v_1}{v_1 + v_2}\right) g_I(-u) + \kappa'$.

[The notation is tricky, so here's an example: for a sinusoidal wave,

$$\begin{cases} g_I = A_I \cos(k_1 z - \omega t) & = A_I \cos[k_1(z - v_1 t)] & \Rightarrow g_I(u) = A_I \cos(k_1 u). \\ g_T = A_T \cos(k_2 z - \omega t) & = A_T \cos[k_2(z - v_2 t)] & \Rightarrow g_T(u) = A_T \cos(k_2 u). \\ h_R = A_R \cos(-k_1 z - \omega t) & = A_R \cos[-k_1(z + v_1 t)] & \Rightarrow h_R(u) = A_R \cos(-k_1 u). \end{cases}$$

Here $\kappa' = 0$, and the boundary conditions say $\frac{A_T}{A_I} = \frac{2v_2}{v_1 + v_2}$, $\frac{A_R}{A_I} = \frac{v_2 - v_1}{v_1 + v_2}$ (same as Eq. 9.32), and $\frac{v_1}{v^2} k_1 = k_2$
 (consistent with Eq. 9.24).]

Problem 9.6

$$(a) T \sin \theta_+ - T \sin \theta_- = ma \Rightarrow T \left(\left. \frac{\partial f}{\partial z} \right|_{0+} - \left. \frac{\partial f}{\partial z} \right|_{0-} \right) = m \left. \frac{\partial^2 f}{\partial t^2} \right|_0.$$

$$(b) \tilde{A}_I + \tilde{A}_R = \tilde{A}_T; T[ik_2 \tilde{A}_T - ik_1(\tilde{A}_I - \tilde{A}_R)] = m(-\omega^2 \tilde{A}_T), \text{ or } k_1(\tilde{A}_I - \tilde{A}_R) = \left(k_2 - \frac{i m \omega^2}{T}\right) \tilde{A}_T.$$

Multiply first equation by k_1 and add: $2k_1 \tilde{A}_I = \left(k_1 + k_2 - i \frac{m \omega^2}{T}\right) \tilde{A}_T$, or $\tilde{A}_T = \left(\frac{2k_1}{k_1 + k_2 - i m \omega^2/T}\right) \tilde{A}_I$.

$$\tilde{A}_R = \tilde{A}_T - \tilde{A}_I = \frac{2k_1 - (k_1 + k_2 - i m \omega^2/T)}{k_1 + k_2 - i m \omega^2/T} \tilde{A}_I = \left(\frac{k_1 - k_2 + i m \omega^2/T}{k_1 + k_2 - i m \omega^2/T}\right) \tilde{A}_I.$$

If the second string is massless, so $v_2 = \sqrt{T/\mu_2} = \infty$, then $k_2/k_1 = 0$, and we have $\tilde{A}_T = \left(\frac{2}{1 - i\beta}\right) \tilde{A}_I$,

$$\tilde{A}_R = \left(\frac{1 + i\beta}{1 - i\beta}\right) \tilde{A}_I, \text{ where } \beta \equiv \frac{m \omega^2}{k_1 T} = \frac{m(k_1 v_1)^2}{k_1 T} = \frac{m k_1}{T} \frac{T}{\mu_1}, \text{ or } \boxed{\beta = m \frac{k_1}{\mu_1}}. \text{ Now } \left(\frac{1 + i\beta}{1 - i\beta}\right) = A e^{i\phi}, \text{ with}$$

$$A^2 = \left(\frac{1 + i\beta}{1 - i\beta}\right) \left(\frac{1 - i\beta}{1 + i\beta}\right) = 1 \Rightarrow A = 1, \text{ and } e^{i\phi} = \frac{(1 + i\beta)^2}{(1 - i\beta)(1 + i\beta)} = \frac{1 + 2i\beta - \beta^2}{1 + \beta^2} \Rightarrow$$

$$\tan \phi = \frac{2\beta}{1 - \beta^2}. \text{ Thus } A_R e^{i\delta_R} = e^{i\phi} A_I e^{i\delta_I} \Rightarrow \boxed{A_R = A_I}, \boxed{\delta_R = \delta_I + \tan^{-1} \left(\frac{2\beta}{1 - \beta^2}\right)}.$$

$$\text{Similarly, } \left(\frac{2}{1 - i\beta}\right) = A e^{i\phi} \Rightarrow A^2 = \left(\frac{2}{1 - i\beta}\right) \left(\frac{2}{1 + i\beta}\right) = \frac{4}{1 + \beta^2} \Rightarrow A = \frac{2}{\sqrt{1 + \beta^2}}.$$

$$Ae^{i\phi} = \frac{2(1+i\beta)}{(1-i\beta)(1+i\beta)} = \frac{2(1+i\beta)}{(1+\beta^2)} \Rightarrow \tan \phi = \beta. \text{ So } A_T e^{i\delta_T} = \frac{2}{\sqrt{1+\beta^2}} e^{i\phi} A_I e^{i\delta_I};$$

$$\boxed{A_T = \frac{2}{\sqrt{1+\beta^2}} A_I; \quad \delta_T = \delta_I + \tan^{-1} \beta.}$$

Problem 9.7

(a) $F = T \frac{\partial^2 f}{\partial z^2} \Delta z - \gamma \frac{\partial f}{\partial t} \Delta z = \mu \Delta z \frac{\partial^2 f}{\partial t^2}$, or $\boxed{T \frac{\partial^2 f}{\partial z^2} = \mu \frac{\partial^2 f}{\partial t^2} + \gamma \frac{\partial f}{\partial t}}.$

(b) Let $\tilde{f}(z, t) = \tilde{F}(z) e^{-i\omega t}$; then $T e^{-i\omega t} \frac{d^2 \tilde{F}}{dz^2} = \mu(-\omega^2) \tilde{F} e^{-i\omega t} + \gamma(-i\omega) \tilde{F} e^{-i\omega t} \Rightarrow$
 $T \frac{d^2 \tilde{F}}{dz^2} = -\omega(\mu\omega + i\gamma) \tilde{F}$, $\frac{d^2 \tilde{F}}{dz^2} = -\tilde{k}^2 \tilde{F}$, where $\tilde{k}^2 \equiv \frac{\omega}{T}(\mu\omega + i\gamma)$. Solution: $\tilde{F}(z) = \tilde{A} e^{i\tilde{k}z} + \tilde{B} e^{-i\tilde{k}z}$.

Resolve $\tilde{k}$ into its real and imaginary parts: $\tilde{k} = k + i\kappa \Rightarrow \tilde{k}^2 = k^2 - \kappa^2 + 2ik\kappa = \frac{\omega}{T}(\mu\omega + i\gamma)$.

$2k\kappa = \frac{\omega\gamma}{T} \Rightarrow \kappa = \frac{\omega\gamma}{2kT}$; $k^2 - \kappa^2 = k^2 - \left(\frac{\omega\gamma}{2T}\right)^2 \frac{1}{k^2} = \frac{\mu\omega^2}{T}$; or $k^4 - k^2(\mu\omega^2/T) - (\omega\gamma/2T)^2 = 0 \Rightarrow$

$k^2 = \frac{1}{2} \left[(\mu\omega^2/T) \pm \sqrt{(\mu\omega^2/T)^2 + 4(\omega\gamma/2T)^2} \right] = \frac{\mu\omega^2}{2T} \left[1 \pm \sqrt{1 + (\gamma/\mu\omega)^2} \right]$. But k is real, so k^2 is positive, so

we need the plus sign: $k = \omega \sqrt{\frac{\mu}{2T}} \sqrt{1 + \sqrt{1 + (\gamma/\mu\omega)^2}}$. $\kappa = \frac{\omega\gamma}{2kT} = \frac{\gamma}{\sqrt{2T\mu}} \left[1 + \sqrt{1 + (\gamma/\mu\omega)^2} \right]^{-1/2}$.

Plugging this in, $\tilde{F} = A e^{i(k+i\kappa)z} + B e^{-i(k+i\kappa)z} = A e^{-\kappa z} e^{ikz} + B e^{\kappa z} e^{-ikz}$. But the B term gives an exponentially *increasing* function, which we don't want (I assume the waves are propagating in the $+z$ direction), so $B = 0$, and the solution is $\boxed{\tilde{f}(z, t) = \tilde{A} e^{-\kappa z} e^{i(kz - \omega t)}}$. (The actual displacement of the string is the real part of this, of course.)

(c) The wave is attenuated by the factor $e^{-\kappa z}$, which becomes $1/e$ when

$z = \frac{1}{\kappa} = \boxed{\frac{\sqrt{2T\mu}}{\gamma} \sqrt{1 + \sqrt{1 + (\gamma/\mu\omega)^2}}}$; this is the characteristic penetration depth.

(d) This is the same as before, except that $k_2 \rightarrow k + i\kappa$. From Eq. 9.29, $\tilde{A}_R = \left(\frac{k_1 - k - i\kappa}{k_1 + k + i\kappa} \right) \tilde{A}_I$;

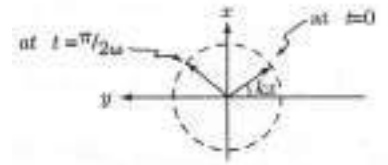
$\left(\frac{A_R}{A_I} \right)^2 = \left(\frac{k_1 - k - i\kappa}{k_1 + k + i\kappa} \right) \left(\frac{k_1 - k + i\kappa}{k_1 + k - i\kappa} \right) = \frac{(k_1 - k)^2 + \kappa^2}{(k_1 + k)^2 + \kappa^2}$. $\boxed{A_R = \sqrt{\frac{(k_1 - k)^2 + \kappa^2}{(k_1 + k)^2 + \kappa^2}} A_I}$

(where $k_1 = \omega/v_1 = \omega\sqrt{\mu_1/T}$, while k and κ are defined in part b). Meanwhile

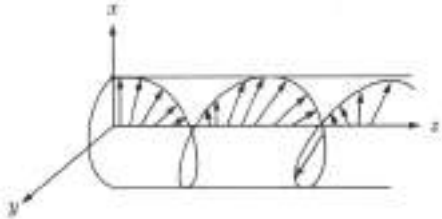
$\left(\frac{k_1 - k - i\kappa}{k_1 + k + i\kappa} \right) = \frac{(k_1 - k - i\kappa)(k_1 + k + i\kappa)}{(k_1 + k)^2 + \kappa^2} = \frac{(k_1)^2 - k^2 - \kappa^2 - 2i\kappa k_1}{(k_1 + k)^2 + \kappa^2} \Rightarrow \delta_R = \tan^{-1} \left(\frac{-2k_1\kappa}{(k_1)^2 - k^2 - \kappa^2} \right).$

Problem 9.8

(a) $\mathbf{f}_v(z, t) = A \cos(kz - \omega t) \hat{\mathbf{x}}$; $\mathbf{f}_h(z, t) = A \cos(kz - \omega t + 90^\circ) \hat{\mathbf{y}} = -A \sin(kz - \omega t) \hat{\mathbf{y}}$. Since $f_v^2 + f_h^2 = A^2$, the vector sum $\mathbf{f} = \mathbf{f}_v + \mathbf{f}_h$ lies on a circle of radius A . At time $t = 0$, $\mathbf{f} = A \cos(kz) \hat{\mathbf{x}} - A \sin(kz) \hat{\mathbf{y}}$. At time $t = \pi/2\omega$, $\mathbf{f} = A \cos(kz - 90^\circ) \hat{\mathbf{x}} - A \sin(kz - 90^\circ) \hat{\mathbf{y}} = A \sin(kz) \hat{\mathbf{x}} + A \cos(kz) \hat{\mathbf{y}}$. Evidently it circles counterclockwise. To make a wave circling the other way, use $\delta_h = -90^\circ$.



(b)

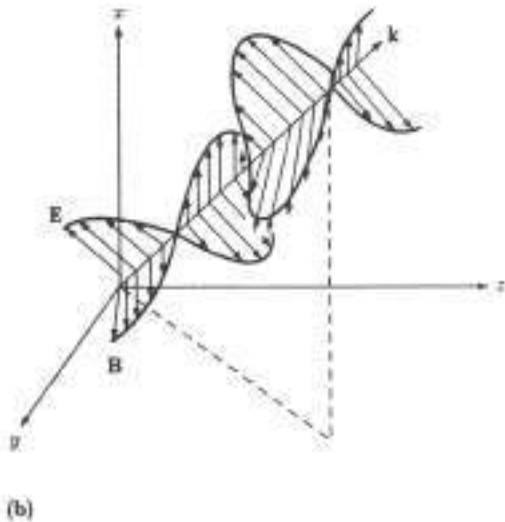
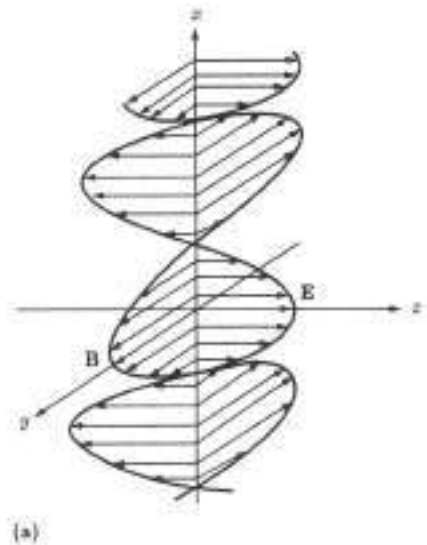


(c) Shake it around in a circle, instead of up and down.

Problem 9.9

(a) $\mathbf{k} = -\frac{\omega}{c} \hat{\mathbf{x}}; \quad \hat{\mathbf{n}} = \hat{\mathbf{z}}. \quad \mathbf{k} \cdot \mathbf{r} = \left(-\frac{\omega}{c} \hat{\mathbf{x}}\right) \cdot (x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}) = -\frac{\omega}{c} x; \quad \mathbf{k} \times \hat{\mathbf{n}} = -\hat{\mathbf{x}} \times \hat{\mathbf{z}} = \hat{\mathbf{y}}.$

$$\mathbf{E}(x, t) = E_0 \cos\left(\frac{\omega}{c}x + \omega t\right) \hat{\mathbf{z}}; \quad \mathbf{B}(x, t) = \frac{E_0}{c} \cos\left(\frac{\omega}{c}x + \omega t\right) \hat{\mathbf{y}}.$$



(b) $\mathbf{k} = \frac{\omega}{c} \left(\frac{\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}}{\sqrt{3}} \right); \quad \hat{\mathbf{n}} = \frac{\hat{\mathbf{x}} - \hat{\mathbf{z}}}{\sqrt{2}}.$ (Since $\hat{\mathbf{n}}$ is parallel to the xz plane, it must have the form $\alpha \hat{\mathbf{x}} + \beta \hat{\mathbf{z}}$;

since $\hat{\mathbf{n}} \cdot \mathbf{k} = 0, \beta = -\alpha$; and since it is a unit vector, $\alpha = 1/\sqrt{2}$.)

$$\mathbf{k} \cdot \mathbf{r} = \frac{\omega}{\sqrt{3}c} (\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}) \cdot (x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}) = \frac{\omega}{\sqrt{3}c} (x + y + z); \quad \hat{\mathbf{k}} \times \hat{\mathbf{n}} = \frac{1}{\sqrt{6}} \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 1 & 1 & 1 \\ 1 & 0 & -1 \end{vmatrix} = \frac{1}{\sqrt{6}} (-\hat{\mathbf{x}} + 2\hat{\mathbf{y}} - \hat{\mathbf{z}}).$$

$$\begin{aligned} \mathbf{E}(x, y, z, t) &= E_0 \cos \left[\frac{\omega}{\sqrt{3}c} (x + y + z) - \omega t \right] \left(\frac{\hat{\mathbf{x}} - \hat{\mathbf{z}}}{\sqrt{2}} \right); \\ \mathbf{B}(x, y, z, t) &= \frac{E_0}{c} \cos \left[\frac{\omega}{\sqrt{3}c} (x + y + z) - \omega t \right] \left(\frac{-\hat{\mathbf{x}} + 2\hat{\mathbf{y}} - \hat{\mathbf{z}}}{\sqrt{6}} \right). \end{aligned}$$

Problem 9.10

$$P = \frac{I}{c} = \frac{1.3 \times 10^3}{3.0 \times 10^8} = \boxed{4.3 \times 10^{-6} \text{ N/m}^2}.$$

For a perfect reflector the pressure is twice as great:

$$\boxed{8.6 \times 10^{-6} \text{ N/m}^2}.$$

Atmospheric pressure is $1.03 \times 10^5 \text{ N/m}^2$, so the pressure of light on a reflector is

$$(8.6 \times 10^{-6}) / (1.03 \times 10^5) = \boxed{8.3 \times 10^{-11} \text{ atmospheres}}.$$

Problem 9.11 The fields are $\mathbf{E}(z, t) = E_0 \cos(kz - \omega t) \hat{\mathbf{x}}$, $\mathbf{B}(z, t) = \frac{1}{c} E_0 \cos(kz - \omega t) \hat{\mathbf{y}}$, with $\omega = ck$.

(a) The electric force is $\mathbf{F}_e = q\mathbf{E} = qE_0 \cos(kz - \omega t) \hat{\mathbf{x}} = m\mathbf{a} = m \frac{d\mathbf{v}}{dt}$, so

$$\mathbf{v} = \frac{qE_0}{m} \hat{\mathbf{x}} \int \cos(kz - \omega t) dt = -\frac{qE_0}{m\omega} \sin(kz - \omega t) \hat{\mathbf{x}} + \mathbf{C}.$$

But $\mathbf{v}_{\text{ave}} = \mathbf{C} = \mathbf{0}$, so $\mathbf{v} = \boxed{-\frac{qE_0}{m\omega} \sin(kz - \omega t) \hat{\mathbf{x}}}.$

(b) The magnetic force is

$$\mathbf{F}_m = q(\mathbf{v} \times \mathbf{B}) = q \left(-\frac{qE_0}{m\omega} \right) \left(\frac{E_0}{c} \right) \sin(kz - \omega t) \cos(kz - \omega t) (\hat{\mathbf{x}} \times \hat{\mathbf{y}}) = \boxed{-\frac{q^2 E_0^2}{m\omega c} \sin(kz - \omega t) \cos(kz - \omega t) \hat{\mathbf{z}}}.$$

(c) The (time) average force is $(\mathbf{F}_m)_{\text{ave}} = -\frac{q^2 E_0^2}{m\omega c} \hat{\mathbf{z}} \int_0^T \sin(kz - \omega t) \cos(kz - \omega t) dt$, where $T = 2\pi/\omega$ is the period. The integral is $-\frac{1}{2\omega} \sin^2(kz - \omega t) \Big|_0^T = -\frac{1}{2\omega} [\sin^2(kz - 2\pi) - \sin^2(kz)] = 0$, so $\boxed{(\mathbf{F}_m)_{\text{ave}} = \mathbf{0}}.$

(d) Adding in the damping term,

$$\mathbf{F} = q\mathbf{E} - \gamma m\mathbf{v} = qE_0 \cos(kz - \omega t) \hat{\mathbf{x}} - \gamma m\mathbf{v} = m \frac{d\mathbf{v}}{dt} \Rightarrow \frac{d\mathbf{v}}{dt} + \gamma \mathbf{v} = \frac{qE_0}{m} \cos(kz - \omega t) \hat{\mathbf{x}}.$$

The steady state solution has the form $\mathbf{v} = A \cos(kz - \omega t + \theta) \hat{\mathbf{x}}$, $\frac{d\mathbf{v}}{dt} = A\omega \sin(kz - \omega t + \theta) \hat{\mathbf{x}}$. Putting this in, and using the trig identity $\cos u = \cos \theta \cos(u + \theta) + \sin \theta \sin(u + \theta)$,

$$A\omega \sin(kz - \omega t + \theta) + \gamma A \cos(kz - \omega t + \theta) = \frac{qE_0}{m} [\cos \theta \cos(kz - \omega t + \theta) + \sin \theta \sin(kz - \omega t + \theta)].$$

Equating like terms:

$$A\omega = \frac{qE_0}{m} \sin \theta, \quad A\gamma = \frac{qE_0}{m} \cos \theta \Rightarrow \tan \theta = \frac{\omega}{\gamma}, \quad A^2(\omega^2 + \gamma^2) = \left(\frac{qE_0}{m} \right)^2 \Rightarrow A = \frac{qE_0}{m\sqrt{\omega^2 + \gamma^2}}.$$

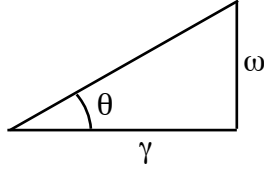
So

$$\mathbf{v} = \boxed{\frac{qE_0}{m\sqrt{\omega^2 + \gamma^2}} \cos(kz - \omega t + \theta) \hat{\mathbf{x}}}, \quad \theta \equiv \tan^{-1}(\omega/\gamma); \quad \mathbf{F}_m = \boxed{\frac{q^2 E_0^2}{mc\sqrt{\omega^2 + \gamma^2}} \cos(kz - \omega t + \theta) \cos(kz - \omega t) \hat{\mathbf{z}}}.$$

To calculate the time average, write $\cos(kz - \omega t + \theta) = \cos \theta \cos(kz - \omega t) - \sin \theta \sin(kz - \omega t)$. We already know that the average of $\cos(kz - \omega t) \sin(kz - \omega t)$ is zero, so

$$(\mathbf{F}_m)_{\text{ave}} = \frac{q^2 E_0^2}{mc\sqrt{\omega^2 + \gamma^2}} \hat{\mathbf{z}} \cos \theta \int_0^T \cos^2(kz - \omega t) dt.$$

The integral is $T/2 = \pi/\omega$, and $\cos \theta = \gamma/\sqrt{\omega^2 + \gamma^2}$ (see figure), so $(\mathbf{F}_m)_{\text{ave}} = \boxed{\frac{\pi\gamma q^2 E_0^2}{m\omega c(\omega^2 + \gamma^2)} \hat{\mathbf{z}}}$.



Problem 9.12

$$\begin{aligned} \langle fg \rangle &= \frac{1}{T} \int_0^T a \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta_a) b \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta_b) dt \\ &= \frac{ab}{2T} \int_0^T [\cos(2\mathbf{k} \cdot \mathbf{r} - 2\omega t + \delta_a + \delta_b) + \cos(\delta_a - \delta_b)] dt = \frac{ab}{2T} \cos(\delta_a - \delta_b) T = \frac{1}{2} ab \cos(\delta_a - \delta_b). \end{aligned}$$

Meanwhile, in the complex notation: $\tilde{f} = \tilde{a}e^{i\mathbf{k} \cdot \mathbf{r} - \omega t}$, $\tilde{g} = \tilde{b}e^{i\mathbf{k} \cdot \mathbf{r} - \omega t}$, where $\tilde{a} = ae^{i\delta_a}$, $\tilde{b} = be^{i\delta_b}$. So $\frac{1}{2}\tilde{f}\tilde{g}^* = \frac{1}{2}\tilde{a}e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}\tilde{b}^*e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)} = \frac{1}{2}\tilde{a}\tilde{b}^* = \frac{1}{2}abe^{i(\delta_a - \delta_b)}$, $\text{Re}\left(\frac{1}{2}\tilde{f}\tilde{g}^*\right) = \frac{1}{2}ab \cos(\delta_a - \delta_b) = \langle fg \rangle$. qed

Problem 9.13

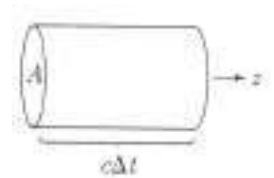
$$T_{ij} = \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right).$$

With the fields in Eq. 9.48, $\mathbf{E}$ has only an x component, and $\mathbf{B}$ only a y component. So all the “off-diagonal” ($i \neq j$) terms are zero. As for the “diagonal” elements:

$$\begin{aligned} T_{xx} &= \epsilon_0 \left(E_x E_x - \frac{1}{2} E^2 \right) + \frac{1}{\mu_0} \left(-\frac{1}{2} B^2 \right) = \frac{1}{2} \left(\epsilon_0 E^2 - \frac{1}{\mu_0} B^2 \right) = 0. \\ T_{yy} &= \epsilon_0 \left(-\frac{1}{2} E^2 \right) + \frac{1}{\mu_0} \left(B_y B_y - \frac{1}{2} B^2 \right) = \frac{1}{2} \left(-\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) = 0. \\ T_{zz} &= \epsilon_0 \left(-\frac{1}{2} E^2 \right) + \frac{1}{\mu_0} \left(-\frac{1}{2} B^2 \right) = -u. \end{aligned}$$

So $\boxed{T_{zz} = -\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)}$ (all other elements zero).

The momentum of these fields is in the z direction, and it is being *transported* in the z direction, so *yes*, it does make sense that T_{zz} should be the only nonzero element in T_{ij} . According to Sect. 8.2.3, $-\vec{T} \cdot d\mathbf{a}$ is the rate at which momentum crosses an area $d\mathbf{a}$. Here we have *no* momentum crossing areas oriented in the x or y direction; the momentum per unit time per unit area flowing across a surface oriented in the z direction is $-T_{zz} = u = \mathbf{g}c$ (Eq. 9.59), so $\Delta p = \mathbf{g}cA\Delta t$, and hence $\Delta p/\Delta t = \mathbf{g}cA =$ momentum per unit time crossing area A . Evidently momentum flux density = energy density. ✓



Problem 9.14

$$R = \left(\frac{E_{0R}}{E_{0I}} \right)^2 \quad (\text{Eq. 9.86}) \Rightarrow \boxed{R = \left(\frac{1 - \beta}{1 + \beta} \right)^2} \quad (\text{Eq. 9.82), where } \beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2}. \quad T = \frac{\epsilon_2 v_2}{\epsilon_1 v_1} \left(\frac{E_{0T}}{E_{0I}} \right)^2 \quad (\text{Eq. 9.87})$$

$$\Rightarrow \boxed{T = \beta \left(\frac{2}{1 + \beta} \right)^2} \quad (\text{Eq. 9.82}). \quad [\text{Note that } \frac{\epsilon_2 v_2}{\epsilon_1 v_1} = \frac{\mu_1}{\mu_2} \frac{\epsilon_2 \mu_2}{\epsilon_1 \mu_1} \frac{v_2}{v_1} = \frac{\mu_1}{\mu_2} \left(\frac{v_1}{v_2} \right)^2 \frac{v_2}{v_1} = \frac{\mu_1 v_1}{\mu_2 v_2} = \beta.]$$

$$T + R = \frac{1}{(1 + \beta)^2} [4\beta + (1 - \beta)^2] = \frac{1}{(1 + \beta)^2} (4\beta + 1 - 2\beta + \beta^2) = \frac{1}{(1 + \beta)^2} (1 + 2\beta + \beta^2) = 1. \quad \checkmark$$

Problem 9.15

Equation 9.78 is replaced by $\tilde{E}_{0_I} \hat{\mathbf{x}} + \tilde{E}_{0_R} \hat{\mathbf{n}}_R = \tilde{E}_{0_T} \hat{\mathbf{n}}_T$, and Eq. 9.80 becomes $\tilde{E}_{0_I} \hat{\mathbf{y}} - \tilde{E}_{0_R} (\hat{\mathbf{z}} \times \hat{\mathbf{n}}_R) = \beta \tilde{E}_{0_T} (\hat{\mathbf{z}} \times \hat{\mathbf{n}}_T)$. The y component of the first equation is $\tilde{E}_{0_R} \sin \theta_R = \tilde{E}_{0_T} \sin \theta_T$; the x component of the second is $\tilde{E}_{0_R} \sin \theta_R = -\beta \tilde{E}_{0_T} \sin \theta_T$. Comparing these two, we conclude that $\sin \theta_R = \sin \theta_T = 0$, and hence $\theta_R = \theta_T = 0$. $\quad \text{qed}$

Problem 9.16

$Ae^{iax} + Be^{ibx} = Ce^{icx}$ for all x , so (using $x = 0$), $A + B = C$.

Differentiate: $iaAe^{iax} + ibBe^{ibx} = icCe^{icx}$, so (using $x = 0$), $aA + bB = cC$.

Differentiate again: $-a^2 Ae^{iax} - b^2 Be^{ibx} = -c^2 Ce^{icx}$, so (using $x = 0$), $a^2 A + b^2 B = c^2 C$.

$a^2 A + b^2 B = c(cC) = c(aA + bB)$; $(A + B)(a^2 A + b^2 B) = (A + B)c(aA + bB) = cC(aA + bB)$;

$a^2 A^2 + b^2 AB + a^2 AB + b^2 B^2 = (aA + bB)^2 = a^2 A^2 + 2abAB + b^2 B^2$, or $(a^2 + b^2 - 2ab)AB = 0$, or

$(a - b)^2 AB = 0$. But A and B are nonzero, so $a = b$. Therefore $(A + B)e^{iax} = Ce^{icx}$.

$a(A + B) = cC$, or $aC = cC$, so (since $C \neq 0$) $a = c$. *Conclusion:* $a = b = c$. $\quad \text{qed}$

Problem 9.17

$$\left\{ \begin{array}{l} \tilde{\mathbf{E}}_I = \tilde{E}_{0_I} e^{i(\mathbf{k}_I \cdot \mathbf{r} - \omega t)} \hat{\mathbf{y}}, \\ \tilde{\mathbf{B}}_I = \frac{1}{v_1} \tilde{E}_{0_I} e^{i(\mathbf{k}_I \cdot \mathbf{r} - \omega t)} (-\cos \theta_1 \hat{\mathbf{x}} + \sin \theta_1 \hat{\mathbf{z}}); \end{array} \right\}$$

$$\left\{ \begin{array}{l} \tilde{\mathbf{E}}_R = \tilde{E}_{0_R} e^{i(\mathbf{k}_R \cdot \mathbf{r} - \omega t)} \hat{\mathbf{y}}, \\ \tilde{\mathbf{B}}_R = \frac{1}{v_1} \tilde{E}_{0_R} e^{i(\mathbf{k}_R \cdot \mathbf{r} - \omega t)} (\cos \theta_1 \hat{\mathbf{x}} + \sin \theta_1 \hat{\mathbf{z}}); \end{array} \right\}$$

$$\left\{ \begin{array}{l} \tilde{\mathbf{E}}_T = \tilde{E}_{0_T} e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)} \hat{\mathbf{y}}, \\ \tilde{\mathbf{B}}_T = \frac{1}{v_2} \tilde{E}_{0_T} e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)} (-\cos \theta_2 \hat{\mathbf{x}} + \sin \theta_2 \hat{\mathbf{z}}); \end{array} \right\}$$

$$\text{Boundary conditions: } \left\{ \begin{array}{ll} \text{(i)} \quad \epsilon_1 E_1^\perp = \epsilon_2 E_2^\perp, & \text{(iii)} \quad \mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel, \\ \text{(ii)} \quad B_1^\perp = B_2^\perp, & \text{(iv)} \quad \frac{1}{\mu_1} \mathbf{B}_1^\parallel = \frac{1}{\mu_2} \mathbf{B}_2^\parallel. \end{array} \right.$$

Law of refraction: $\frac{\sin \theta_2}{\sin \theta_1} = \frac{v_2}{v_1}$. [Note: $\mathbf{k}_I \cdot \mathbf{r} - \omega t = \mathbf{k}_R \cdot \mathbf{r} - \omega t = \mathbf{k}_T \cdot \mathbf{r} - \omega t$, at $z = 0$, so we can drop all exponential factors in applying the boundary conditions.]

Boundary condition (i): $0 = 0$ (trivial). Boundary condition (iii): $\boxed{\tilde{E}_{0_I} + \tilde{E}_{0_R} = \tilde{E}_{0_T}}$.

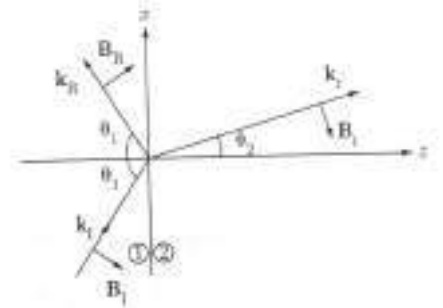
Boundary condition (ii): $\frac{1}{v_1} \tilde{E}_{0_I} \sin \theta_1 + \frac{1}{v_1} \tilde{E}_{0_R} \sin \theta_1 = \frac{1}{v_2} \tilde{E}_{0_T} \sin \theta_2 \Rightarrow \tilde{E}_{0_I} + \tilde{E}_{0_R} = \left(\frac{v_1 \sin \theta_2}{v_2 \sin \theta_1} \right) \tilde{E}_{0_T}$.

But the term in parentheses is 1, by the law of refraction, so this is the same as (iii).

Boundary condition (iv): $\frac{1}{\mu_1} \left[\frac{1}{v_1} \tilde{E}_{0_I} (-\cos \theta_1) + \frac{1}{v_1} \tilde{E}_{0_R} \cos \theta_1 \right] = \frac{1}{\mu_2 v_2} \tilde{E}_{0_T} (-\cos \theta_2) \Rightarrow$

$$\tilde{E}_{0_I} - \tilde{E}_{0_R} = \left(\frac{\mu_1 v_1 \cos \theta_2}{\mu_2 v_2 \cos \theta_1} \right) \tilde{E}_{0_T}. \quad \text{Let } \alpha \equiv \frac{\cos \theta_2}{\cos \theta_1}; \beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2}. \quad \text{Then } \boxed{\tilde{E}_{0_I} - \tilde{E}_{0_R} = \alpha \beta \tilde{E}_{0_T}}.$$

Solving for $\tilde{E}_{0_R}$ and $\tilde{E}_{0_T}$: $2\tilde{E}_{0_I} = (1 + \alpha\beta)\tilde{E}_{0_T} \Rightarrow \tilde{E}_{0_T} = \left(\frac{2}{1 + \alpha\beta} \right) \tilde{E}_{0_I}$;



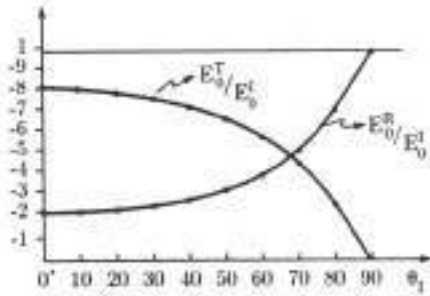
$$\tilde{E}_{0R} = \tilde{E}_{0T} - \tilde{E}_{0I} = \left(\frac{2}{1 + \alpha\beta} - \frac{1 + \alpha\beta}{1 + \alpha\beta} \right) \tilde{E}_{0I} \Rightarrow \tilde{E}_{0R} = \left(\frac{1 - \alpha\beta}{1 + \alpha\beta} \right) \tilde{E}_{0I}.$$

Since α and β are positive, it follows that $2/(1 + \alpha\beta)$ is positive, and hence the *transmitted* wave is *in phase* with the incident wave, and the (real) amplitudes are related by $E_{0T} = \left(\frac{2}{1 + \alpha\beta} \right) E_{0I}$. The *reflected* wave is

in phase if $\alpha\beta < 1$ and 180° out of phase if $\alpha\beta > 1$; the (real) amplitudes are related by $E_{0R} = \left| \frac{1 - \alpha\beta}{1 + \alpha\beta} \right| E_{0I}$.

These are the **Fresnel equations** for polarization perpendicular to the plane of incidence.

To construct the graphs, note that $\alpha\beta = \beta \frac{\sqrt{1 - \sin^2 \theta / \beta^2}}{\cos \theta} = \frac{\sqrt{\beta^2 - \sin^2 \theta}}{\cos \theta}$, where θ is the angle of incidence, so, for $\beta = 1.5$, $\alpha\beta = \frac{\sqrt{2.25 - \sin^2 \theta}}{\cos \theta}$. [In the figure, the minus signs on the vertical axis should be decimal points.]



Is there a Brewster's angle? Well, $E_{0R} = 0$ would mean that $\alpha\beta = 1$, and hence that

$$\alpha = \frac{\sqrt{1 - (v_2/v_1)^2 \sin^2 \theta}}{\cos \theta} = \frac{1}{\beta} = \frac{\mu_2 v_2}{\mu_1 v_1}, \text{ or } 1 - \left(\frac{v_2}{v_1} \right)^2 \sin^2 \theta = \left(\frac{\mu_2 v_2}{\mu_1 v_1} \right)^2 \cos^2 \theta, \text{ so}$$

$1 = \left(\frac{v_2}{v_1} \right)^2 [\sin^2 \theta + (\mu_2/\mu_1)^2 \cos^2 \theta]$. Since $\mu_1 \approx \mu_2$, this means $1 \approx (v_2/v_1)^2$, which is only true for optically indistinguishable media, in which case there is of *course* no reflection—but that would be true at *any* angle, not just at a special “Brewster's angle”. [If μ_2 were substantially different from μ_1 , and the relative velocities were just right, it *would* be possible to get a Brewster's angle for this case, at

$$\left(\frac{v_1}{v_2} \right)^2 = 1 - \cos^2 \theta + \left(\frac{\mu_2}{\mu_1} \right)^2 \cos^2 \theta \Rightarrow \cos^2 \theta = \frac{(v_1/v_2)^2 - 1}{(\mu_2/\mu_1)^2 - 1} = \frac{(\mu_2 \epsilon_2 / \mu_1 \epsilon_1) - 1}{(\mu_2/\mu_1)^2 - 1} = \frac{(\epsilon_2/\epsilon_1) - (\mu_1/\mu_2)}{(\mu_2/\mu_1) - (\mu_1/\mu_2)}.$$

But the media would be very peculiar.]

By the same token, δ_R is either always 0, or always π , for a given interface—it does not switch over as you change θ , the way it does for polarization in the plane of incidence. In particular, if $\beta = 3/2$, then $\alpha\beta > 1$, for

$$\alpha\beta = \frac{\sqrt{2.25 - \sin^2 \theta}}{\cos \theta} > 1 \text{ if } 2.25 - \sin^2 \theta > \cos^2 \theta, \text{ or } 2.25 > \sin^2 \theta + \cos^2 \theta = 1. \checkmark$$

In general, for $\beta > 1$, $\alpha\beta > 1$, and hence $\delta_R = \pi$. For $\beta < 1$, $\alpha\beta < 1$, and $\delta_R = 0$.

At normal incidence, $\alpha = 1$, so Fresnel's equations reduce to $E_{0T} = \left(\frac{2}{1 + \beta} \right) E_{0I}$; $E_{0R} = \left| \frac{1 - \beta}{1 + \beta} \right| E_{0I}$, consistent with Eq. 9.82.

Reflection and Transmission coefficients: $R = \left(\frac{E_{0R}}{E_{0I}} \right)^2 = \left(\frac{1 - \alpha\beta}{1 + \alpha\beta} \right)^2$. Referring to Eq. 9.116,

$$T = \frac{\epsilon_2 v_2}{\epsilon_1 v_1} \alpha \left(\frac{E_{0T}}{E_{0I}} \right)^2 = \alpha \beta \left(\frac{2}{1 + \alpha\beta} \right)^2.$$

$$R + T = \frac{(1 - \alpha\beta)^2 + 4\alpha\beta}{(1 + \alpha\beta)^2} = \frac{1 - 2\alpha\beta + \alpha^2\beta^2 + 4\alpha\beta}{(1 + \alpha\beta)^2} = \frac{(1 + \alpha\beta)^2}{(1 + \alpha\beta)^2} = 1. \checkmark$$

Problem 9.18

Equation 9.106 $\Rightarrow \beta = 2.42$; Eq. 9.110 $\Rightarrow$

$$\alpha = \frac{\sqrt{1 - (\sin \theta / 2.42)^2}}{\cos \theta}.$$

(a) $\theta = 0 \Rightarrow \alpha = 1$. Eq. 9.109 $\Rightarrow \left(\frac{E_{0R}}{E_{0I}} \right) = \frac{\alpha - \beta}{\alpha + \beta}$

$$= \frac{1 - 2.42}{1 + 2.42} = -\frac{1.42}{3.42} = \boxed{-0.415};$$

$$\left(\frac{E_{0T}}{E_{0I}} \right) = \frac{2}{\alpha + \beta} = \frac{2}{3.42} = \boxed{0.585}.$$

(b) Equation 9.112 $\Rightarrow \theta_B = \tan^{-1}(2.42) = \boxed{67.5^\circ}$.

(c) $E_{0R} = E_{0T} \Rightarrow \alpha - \beta = 2 \Rightarrow \alpha = \beta + 2 = 4.42$;

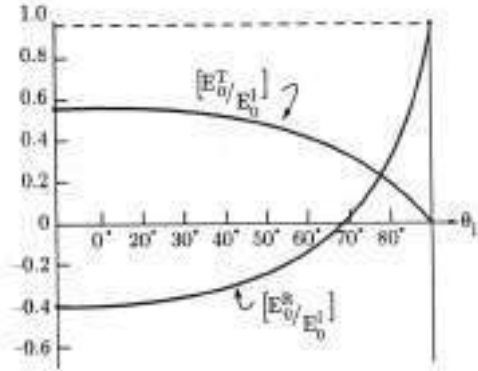
$$(4.42)^2 \cos^2 \theta = 1 - \sin^2 \theta / (2.42)^2;$$

$$(4.42)^2 (1 - \sin^2 \theta) = (4.42)^2 - (4.42)^2 \sin^2 \theta$$

$$= 1 - 0.171 \sin^2 \theta; 19.5 - 1 = (19.5 - 0.17) \sin^2 \theta;$$

$$18.5 = 19.3 \sin^2 \theta; \sin^2 \theta = 18.5 / 19.3 = 0.959;$$

$$\sin \theta = 0.979; \boxed{\theta = 78.3^\circ}.$$



Problem 9.19

(a) Equation 9.120 $\Rightarrow \tau = \epsilon / \sigma$. Now $\epsilon = \epsilon_0 \epsilon_r$ (Eq. 4.34), $\epsilon_r \cong n^2$ (Eq. 9.70), and for glass the index of refraction is typically around 1.5, so $\epsilon \approx (1.5)^2 \times 8.85 \times 10^{-12} = 2 \times 10^{-11} \text{ C}^2/\text{N m}^2$, while $\sigma = 1/\rho \approx 10^{-12} (\Omega \text{ m})^{-1}$ (Table 7.1). Then $\tau = (2 \times 10^{-11}) / 10^{-12} = \boxed{20 \text{ s}}$. (But the resistivity of glass varies enormously from one type to another, so this answer could be off by a factor of 100 in either direction.)

(b) For silver, $\rho = 1.59 \times 10^{-8}$ (Table 7.1), and $\epsilon \approx \epsilon_0$, so $\omega\epsilon = 2\pi \times 10^{10} \times 8.85 \times 10^{-12} = 0.56$. Since $\sigma = 1/\rho = 6.25 \times 10^7 \gg \omega\epsilon$, the skin depth (Eq. 9.128) is

$$d = \frac{1}{\kappa} \cong \sqrt{\frac{2}{\omega\sigma\mu}} = \sqrt{\frac{2}{2\pi \times 10^{10} \times 6.25 \times 10^7 \times 4\pi \times 10^{-7}}} = 6.4 \times 10^{-7} \text{ m} = 6.4 \times 10^{-4} \text{ mm}.$$

I'd plate silver to a depth of about $\boxed{0.001 \text{ mm}}$; there's no point in making it any thicker, since the fields don't penetrate much beyond this anyway.

(c) For copper, Table 7.1 gives $\sigma = 1/(1.68 \times 10^{-8}) = 6 \times 10^7$, $\omega\epsilon_0 = (2\pi \times 10^6) \times (8.85 \times 10^{-12}) = 6 \times 10^{-5}$. Since $\sigma \gg \omega\epsilon$, Eq. 9.126 $\Rightarrow k \approx \sqrt{\frac{\omega\sigma\mu}{2}}$, so (Eq. 9.129)

$$\lambda = 2\pi \sqrt{\frac{2}{\omega\sigma\mu_0}} = 2\pi \sqrt{\frac{2}{2\pi \times 10^6 \times 6 \times 10^7 \times 4\pi \times 10^{-7}}} = 4 \times 10^{-4} \text{ m} = \boxed{0.4 \text{ mm}}.$$

From Eq. 9.129, the propagation speed is $v = \frac{\omega}{k} = \frac{\omega}{2\pi} \lambda = \lambda \nu = (4 \times 10^{-4}) \times 10^6 = \boxed{400 \text{ m/s.}}$ In vacuum, $\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{10^6} = \boxed{300 \text{ m;}}$ $v = c = \boxed{3 \times 10^8 \text{ m/s.}}$ (But really, in a good conductor the skin depth is so small, compared to the wavelength, that the notions of “wavelength” and “propagation speed” lose their meaning.)

Problem 9.20

(a) Use the binomial expansion for the square root in Eq. 9.126:

$$\kappa \cong \omega \sqrt{\frac{\epsilon\mu}{2}} \left[1 + \frac{1}{2} \left(\frac{\sigma}{\epsilon\omega} \right)^2 - 1 \right]^{1/2} = \omega \sqrt{\frac{\epsilon\mu}{2}} \frac{1}{\sqrt{2}} \frac{\sigma}{\epsilon\omega} = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}.$$

So (Eq. 9.128) $d = \frac{1}{\kappa} \cong \frac{2}{\sigma} \sqrt{\frac{\epsilon}{\mu}}$. qed

For pure water, $\begin{cases} \epsilon = \epsilon_r \epsilon_0 = 80.1 \epsilon_0 & (\text{Table 4.2}), \\ \mu = \mu_0(1 + \chi_m) = \mu_0(1 - 9.0 \times 10^{-6}) \cong \mu_0 & (\text{Table 6.1}), \\ \sigma = 1/(2.5 \times 10^5) & (\text{Table 7.1}). \end{cases}$

So $d = (2)(2.5 \times 10^5) \sqrt{\frac{(80.1)(8.85 \times 10^{-12})}{4\pi \times 10^{-7}}} = \boxed{1.19 \times 10^4 \text{ m.}}$

(b) In this case $(\sigma/\epsilon\omega)^2$ dominates, so (Eq. 9.126) $k \cong \kappa$, and hence (Eqs. 9.128 and 9.129)

$\lambda = \frac{2\pi}{k} \cong \frac{2\pi}{\kappa} = 2\pi d$, or $d = \frac{\lambda}{2\pi}$. qed

Meanwhile $\kappa \cong \omega \sqrt{\frac{\epsilon\mu}{2}} \sqrt{\frac{\sigma}{\epsilon\omega}} = \sqrt{\frac{\omega\mu\sigma}{2}} = \sqrt{\frac{(10^{15})(4\pi \times 10^{-7})(10^7)}{2}} = 8 \times 10^7$; $d = \frac{1}{\kappa} = \frac{1}{8 \times 10^7} = 1.3 \times 10^{-8} = \boxed{13 \text{ nm.}}$ So the fields do not penetrate far into a metal—which is what accounts for their opacity.

(c) Since $k \cong \kappa$, as we found in (b), Eq. 9.134 says $\phi = \tan^{-1}(1) = 45^\circ$. qed

Meanwhile, Eq. 9.137 says $\frac{B_0}{E_0} \cong \sqrt{\epsilon\mu \frac{\sigma}{\epsilon\omega}} = \sqrt{\frac{\sigma\mu}{\omega}}$. For a typical metal, then, $\frac{B_0}{E_0} = \sqrt{\frac{(10^7)(4\pi \times 10^{-7})}{10^{15}}} = \boxed{10^{-7} \text{ s/m.}}$ (In vacuum, the ratio is $1/c = 1/(3 \times 10^8) = 3 \times 10^{-9} \text{ s/m}$, so the magnetic field is comparatively about 100 times larger in a metal.)

Problem 9.21

(a) $u = \frac{1}{2} \left(\epsilon E^2 + \frac{1}{\mu} B^2 \right) = \frac{1}{2} e^{-2\kappa z} \left[\epsilon E_0^2 \cos^2(kz - \omega t + \delta_E) + \frac{1}{\mu} B_0^2 \cos^2(kz - \omega t + \delta_E + \phi) \right]$. Averaging over a full cycle, using $\langle \cos^2 \rangle = \frac{1}{2}$ and Eq. 9.137:

$$\langle u \rangle = \frac{1}{2} e^{-2\kappa z} \left[\frac{\epsilon}{2} E_0^2 + \frac{1}{2\mu} B_0^2 \right] = \frac{1}{4} e^{-2\kappa z} \left[\epsilon E_0^2 + \frac{1}{\mu} E_0^2 \epsilon \mu \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega} \right)^2} \right] = \frac{1}{4} e^{-2\kappa z} \epsilon E_0^2 \left[1 + \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega} \right)^2} \right].$$

But Eq. 9.126 $\Rightarrow 1 + \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega} \right)^2} = \frac{2}{\epsilon\mu} \frac{k^2}{\omega^2}$, so $\langle u \rangle = \frac{1}{4} e^{-2\kappa z} \epsilon E_0^2 \frac{2}{\epsilon\mu} \frac{k^2}{\omega^2} = \boxed{\frac{k^2}{2\mu\omega^2} E_0^2 e^{-2\kappa z}}$. So the ratio of the magnetic contribution to the electric contribution is

$$\frac{\langle u_{\text{mag}} \rangle}{\langle u_{\text{elec}} \rangle} = \frac{B_0^2/\mu}{E_0^2\epsilon} = \frac{1}{\mu\epsilon} \mu\epsilon \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega} \right)^2} = \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega} \right)^2} > 1. \quad \text{qed}$$

(b) $\mathbf{S} = \frac{1}{\mu}(\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu} E_0 B_0 e^{-2\kappa z} \cos(kz - \omega t + \delta_E) \cos(kz - \omega t + \delta_E + \phi) \hat{\mathbf{z}}$; $\langle \mathbf{S} \rangle = \frac{1}{2\mu} E_0 B_0 e^{-2\kappa z} \cos \phi \hat{\mathbf{z}}$. [The average of the product of the cosines is $(1/2\pi) \int_0^{2\pi} \cos \theta \cos(\theta + \phi) d\theta = (1/2) \cos \phi$.] So $I = \frac{1}{2\mu} E_0 B_0 e^{-2\kappa z} \cos \phi = \frac{1}{2\mu} E_0^2 e^{-2\kappa z} \left(\frac{K}{\omega} \cos \phi \right)$, while, from Eqs. 9.133 and 9.134, $K \cos \phi = k$, so $I = \frac{k}{2\mu\omega} E_0^2 e^{-2\kappa z}$. qed

Problem 9.22

According to Eq. 9.147, $R = \left| \frac{\tilde{E}_{0R}}{\tilde{E}_{0I}} \right|^2 = \left| \frac{1 - \tilde{\beta}}{1 + \tilde{\beta}} \right|^2 = \left(\frac{1 - \tilde{\beta}}{1 + \tilde{\beta}} \right) \left(\frac{1 - \tilde{\beta}^*}{1 + \tilde{\beta}^*} \right)$, where $\tilde{\beta} = \frac{\mu_1 v_1}{\mu_2 \omega} \tilde{k}_2$
 $= \frac{\mu_1 v_1}{\mu_2 \omega} (k_2 + i\kappa_2)$ (Eqs. 9.125 and 9.146). Since silver is a good conductor ($\sigma \gg \epsilon\omega$), Eq. 9.126 reduces to
 $\kappa_2 \cong k_2 \cong \omega \sqrt{\frac{\epsilon_2 \mu_2}{2}} \sqrt{\frac{\sigma}{\epsilon_2 \omega}} = \sqrt{\frac{\sigma \omega \mu_2}{2}}$, so $\tilde{\beta} = \frac{\mu_1 v_1}{\mu_2 \omega} \sqrt{\frac{\sigma \omega \mu_2}{2}} (1 + i) = \mu_1 v_1 \sqrt{\frac{\sigma}{2\mu_2 \omega}} (1 + i)$.
Let $\gamma \equiv \mu_1 v_1 \sqrt{\frac{\sigma}{2\mu_2 \omega}} = \mu_0 c \sqrt{\frac{\sigma}{2\mu_0 \omega}} = c \sqrt{\frac{\sigma \mu_0}{2\omega}} = (3 \times 10^8) \sqrt{\frac{(6 \times 10^7)(4\pi \times 10^{-7})}{(2)(4 \times 10^{15})}} = 29$. Then
 $R = \left(\frac{1 - \gamma - i\gamma}{1 + \gamma + i\gamma} \right) \left(\frac{1 - \gamma + i\gamma}{1 + \gamma - i\gamma} \right) = \frac{(1 - \gamma)^2 + \gamma^2}{(1 + \gamma)^2 + \gamma^2} = \boxed{0.93}$. Evidently 93% of the light is reflected.

Problem 9.23

(a) We are told that $v = \alpha\sqrt{\lambda}$, where α is a constant. But $\lambda = 2\pi/k$ and $v = \omega/k$, so
 $\omega = \alpha k \sqrt{2\pi/k} = \alpha \sqrt{2\pi k}$. From Eq. 9.150, $v_g = \frac{d\omega}{dk} = \alpha \sqrt{2\pi} \frac{1}{2\sqrt{k}} = \frac{1}{2} \alpha \sqrt{\frac{2\pi}{k}} = \frac{1}{2} \alpha \sqrt{\lambda} = \frac{1}{2} v$, or $\boxed{v = 2v_g}$.
(b) $\frac{i(px - Et)}{\hbar} = i(kx - \omega t) \Rightarrow k = \frac{p}{\hbar}$, $\omega = \frac{E}{\hbar} = \frac{p^2}{2m\hbar} = \frac{\hbar k^2}{2m}$. Therefore $\boxed{v = \frac{\omega}{k} = \frac{E}{p} = \frac{p}{2m} = \frac{\hbar k}{2m}}$;
 $v_g = \frac{d\omega}{dk} = \frac{2\hbar k}{2m} = \frac{\hbar k}{m} = \boxed{\frac{p}{m}}$. So $\boxed{v = \frac{1}{2} v_g}$. Since $p = mv_c$ (where v_c is the classical speed of the particle), it follows that $\boxed{v_g \text{ (not } v) \text{ corresponds to the classical velocity.}}$

Problem 9.24

$E = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3} \Rightarrow F = -qE = -\left(\frac{1}{4\pi\epsilon_0} \frac{q^2}{a^3} \right) x = -k_{\text{spring}} x = -m\omega_0^2 x$ (Eq. 9.151). So $\boxed{\omega_0 = \sqrt{\frac{q^2}{4\pi\epsilon_0 m a^3}}}$.
 $\nu_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{(1.6 \times 10^{-19})^2}{4\pi(8.85 \times 10^{-12})(9.11 \times 10^{-31})(0.5 \times 10^{-10})^3}} = \boxed{7.16 \times 10^{15} \text{ Hz}}$. This is $\boxed{\text{ultraviolet}}$.

From Eqs. 9.173 and 9.174,

$$A = \frac{nq^2}{2m\epsilon_0 \omega_0^2} \cdot \begin{cases} N = \# \text{ of molecules per unit volume} = \frac{\text{Avogadro's \#}}{22.4 \text{ liters}} = \frac{6.02 \times 10^{23}}{22.4 \times 10^{-3}} = 2.69 \times 10^{25}, \\ f = \# \text{ of electrons per molecule} = 2 \text{ (for H}_2\text{)}. \end{cases}$$

$$= \frac{(2.69 \times 10^{25})(1.6 \times 10^{-19})^2}{(9.11 \times 10^{-31})(8.85 \times 10^{-12})(4.5 \times 10^{16})^2} = \boxed{4.2 \times 10^{-5}} \text{ (which is about 1/3 the actual value);}$$

$$B = \left(\frac{2\pi c}{\omega_0} \right)^2 = \left(\frac{2\pi \times 3 \times 10^8}{4.5 \times 10^{16}} \right)^2 = \boxed{1.8 \times 10^{-15} \text{ m}^2} \text{ (which is about 1/4 the actual value).}$$

So even this extremely crude model is in the right ball park.

Problem 9.25

Equation 9.170 $\Rightarrow n = 1 + \frac{Nq^2}{2m\epsilon_0} \frac{(\omega_0^2 - \omega^2)}{[(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2]}$. Let the denominator $\equiv D$. Then

$$\frac{dn}{d\omega} = \frac{Nq^2}{2m\epsilon_0} \left\{ \frac{-2\omega}{D} - \frac{(\omega_0^2 - \omega^2)}{D^2} [2(\omega_0^2 - \omega^2)(-2\omega) + \gamma^2 2\omega] \right\} = 0 \Rightarrow 2\omega D = (\omega_0^2 - \omega^2) [2(\omega_0^2 - \omega^2) - \gamma^2] 2\omega;$$

$$(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2 = 2(\omega_0^2 - \omega^2)^2 - \gamma^2(\omega_0^2 - \omega^2), \text{ or } (\omega_0^2 - \omega^2)^2 = \gamma^2(\omega^2 + \omega_0^2 - \omega^2) = \gamma^2\omega_0^2 \Rightarrow (\omega_0^2 - \omega^2) = \pm\omega_0\gamma;$$

$$\omega^2 = \omega_0^2 \mp \omega_0\gamma, \quad \omega = \omega_0\sqrt{1 \mp \gamma/\omega_0} \cong \omega_0(1 \mp \gamma/2\omega_0) = \omega_0 \mp \gamma/2. \text{ So } \omega_2 = \omega_0 + \gamma/2, \quad \omega_1 = \omega_0 - \gamma/2, \text{ and the width of the anomalous region is } \boxed{\Delta\omega = \omega_2 - \omega_1 = \gamma.}$$

From Eq. 9.171, $\alpha = \frac{Nq^2\omega^2}{m\epsilon_0 c} \frac{\gamma}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}$, so at the maximum ($\omega = \omega_0$), $\alpha_{\max} = \frac{Nq^2}{m\epsilon_0 c\gamma}$.

At ω_1 and ω_2 , $\omega^2 = \omega_0^2 \mp \omega_0\gamma$, so $\alpha = \frac{Nq^2\omega^2}{m\epsilon_0 c} \frac{\gamma}{\gamma^2\omega_0^2 + \gamma^2\omega^2} = \alpha_{\max} \left(\frac{\omega^2}{\omega^2 + \omega_0^2} \right)$. But

$$\frac{\omega^2}{\omega^2 + \omega_0^2} = \frac{\omega_0^2 \mp \omega_0\gamma}{2\omega_0^2 \mp \omega_0\gamma} = \frac{1}{2} \frac{(1 \mp \gamma/\omega_0)}{(1 \mp \gamma/2\omega_0)} \cong \frac{1}{2} \left(1 \mp \frac{\gamma}{\omega_0} \right) \left(1 \pm \frac{\gamma}{2\omega_0} \right) \cong \frac{1}{2} \left(1 \mp \frac{\gamma}{2\omega_0} \right) \cong \frac{1}{2}.$$

So $\alpha \cong \frac{1}{2}\alpha_{\max}$ at ω_1 and ω_2 . qed

Problem 9.26

Equation 9.170 for a single resonance:

$$\frac{ck}{\omega} = 1 + \left(\frac{Nq^2}{2m\epsilon_0} \right) \frac{(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} \Rightarrow ck = \omega \left[1 + \frac{a\omega_0^2(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} \right] \text{ where } a \equiv \frac{Nq^2}{2m\epsilon_0\omega_0^2}.$$

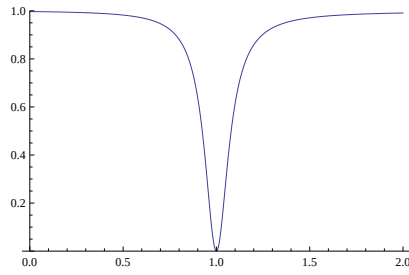
From Eq. 9.150, the group velocity is $v_g = d\omega/dk$, so

$$\begin{aligned} \frac{c}{v_g} &= c \frac{dk}{d\omega} \\ &= \left[1 + \frac{a\omega_0^2(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} \right] + a\omega\omega_0^2 \frac{[(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2](-2\omega) - (\omega_0^2 - \omega^2)[2(\omega_0^2 - \omega^2)(-2\omega) + \gamma^2(2\omega)]}{[(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2]^2} \\ &= 1 + a\omega_0^2 \frac{(\omega_0^2 - \omega^2)[(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2] - 2\omega^2[(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2] + 4\omega^2(\omega_0^2 - \omega^2)^2 - 2\gamma^2\omega^2(\omega_0^2 - \omega^2)}{[(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2]^2} \\ &= 1 + a\omega_0^2 \frac{(\omega_0^2 - \omega^2)^3 + 2\omega^2(\omega_0^2 - \omega^2)^2 - \gamma^2\omega^2(\omega_0^2 - \omega^2) - 2\gamma^2\omega^4}{[(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2]^2} \\ &= 1 + a\omega_0^2(\omega_0^2 + \omega^2) \frac{(\omega_0^2 - \omega^2)^2 - \gamma^2\omega^2}{[(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2]^2}. \end{aligned}$$

$$v_g = \left[c \left\{ 1 + a\omega_0^2(\omega_0^2 + \omega^2) \frac{(\omega_0^2 - \omega^2)^2 - \gamma^2\omega^2}{[(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2]^2} \right\}^{-1} \right].$$

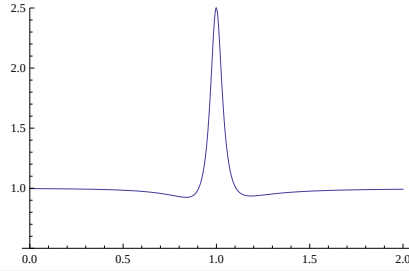
(a) For $\gamma = 0$ and $a = 0.003$:

$$\frac{v_g}{c} = \left[1 + a\omega_0^2 \frac{(\omega_0^2 + \omega^2)}{(\omega_0^2 - \omega^2)^2} \right]^{-1} \Rightarrow y = \left[1 + 0.003 \frac{1+x}{(1-x)^2} \right]^{-1}.$$



(b) For $\gamma = (0.1)\omega_0$ and $a = 0.003$:

$$y = \left\{ 1 + 0.003(1+x) \frac{(1-x)^2 - 0.01x}{[(1-x)^2 + 0.01x]^2} \right\}^{-1}.$$



Problem 9.27

(a) From Eqs. 9.176 and 9.177, $\nabla \times \tilde{\mathbf{E}} = -\frac{\partial \tilde{\mathbf{B}}}{\partial t} = i\omega \tilde{\mathbf{B}}_0 e^{i(kz-\omega t)}$; $\nabla \times \tilde{\mathbf{B}} = \frac{1}{c^2} \frac{\partial \tilde{\mathbf{E}}}{\partial t} = -\frac{i\omega}{c^2} \tilde{\mathbf{E}}_0 e^{i(kz-\omega t)}$.

In the terminology of Eq. 9.178:

$$(\nabla \times \tilde{\mathbf{E}})_x = \frac{\partial \tilde{E}_z}{\partial y} - \frac{\partial \tilde{E}_y}{\partial z} = \left(\frac{\partial \tilde{E}_{0z}}{\partial y} - ik\tilde{E}_{0y} \right) e^{i(kz-\omega t)}. \quad \text{So (ii) } \frac{\partial E_z}{\partial y} - ikE_y = i\omega B_x.$$

$$(\nabla \times \tilde{\mathbf{E}})_y = \frac{\partial \tilde{E}_x}{\partial z} - \frac{\partial \tilde{E}_z}{\partial x} = \left(ik\tilde{E}_{0x} - \frac{\partial \tilde{E}_{0z}}{\partial x} \right) e^{i(kz-\omega t)}. \quad \text{So (iii) } ikE_x - \frac{\partial E_z}{\partial x} = i\omega B_y.$$

$$(\nabla \times \tilde{\mathbf{E}})_z = \frac{\partial \tilde{E}_y}{\partial x} - \frac{\partial \tilde{E}_x}{\partial y} = \left(\frac{\partial \tilde{E}_{0y}}{\partial x} - \frac{\partial \tilde{E}_{0x}}{\partial y} \right) e^{i(kz-\omega t)}. \quad \text{So (i) } \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = i\omega B_z.$$

$$(\nabla \times \tilde{\mathbf{B}})_x = \frac{\partial \tilde{B}_z}{\partial y} - \frac{\partial \tilde{B}_y}{\partial z} = \left(\frac{\partial \tilde{B}_{0z}}{\partial y} - ik\tilde{B}_{0y} \right) e^{i(kz-\omega t)}. \quad \text{So (v) } \frac{\partial B_z}{\partial y} - ikB_y = -\frac{i\omega}{c^2} E_x.$$

$$(\nabla \times \tilde{\mathbf{B}})_y = \frac{\partial \tilde{B}_x}{\partial z} - \frac{\partial \tilde{B}_z}{\partial x} = \left(ik\tilde{B}_{0x} - \frac{\partial \tilde{B}_{0z}}{\partial x} \right) e^{i(kz-\omega t)}. \quad \text{So (vi) } ikB_x - \frac{\partial B_z}{\partial x} = -\frac{i\omega}{c^2} E_y.$$

$$(\nabla \times \tilde{\mathbf{B}})_z = \frac{\partial \tilde{B}_y}{\partial x} - \frac{\partial \tilde{B}_x}{\partial y} = \left(\frac{\partial \tilde{B}_{0y}}{\partial x} - \frac{\partial \tilde{B}_{0x}}{\partial y} \right) e^{i(kz-\omega t)}. \quad \text{So (iv) } \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = -\frac{i\omega}{c^2} E_z.$$

This confirms Eq. 9.179. Now multiply (iii) by k , (v) by ω , and subtract: $ik^2 E_x - k \frac{\partial E_z}{\partial x} - \omega \frac{\partial B_z}{\partial y} + i\omega k B_y = ik\omega B_y + \frac{i\omega^2}{c^2} E_x \Rightarrow i \left(k^2 - \frac{\omega^2}{c^2} \right) E_x = k \frac{\partial E_z}{\partial x} + \omega \frac{\partial B_z}{\partial y}$, or (i) $E_x = \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial E_z}{\partial x} + \omega \frac{\partial B_z}{\partial y} \right)$.

Multiply (ii) by k , (vi) by ω , and add: $k \frac{\partial E_z}{\partial y} - ik^2 E_y + i\omega k B_x - \omega \frac{\partial B_z}{\partial x} = i\omega k B_x - \frac{i\omega^2}{c^2} E_y \Rightarrow i \left(\frac{\omega^2}{c^2} - k^2 \right) E_y =$

$$-k \frac{\partial E_z}{\partial y} + \omega \frac{\partial B_z}{\partial x}, \text{ or (ii) } E_y = \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial E_z}{\partial y} - \omega \frac{\partial B_z}{\partial x} \right).$$

$$\text{Multiply (ii) by } \omega/c^2, \text{ (vi) by } k, \text{ and add: } \frac{\omega}{c^2} \frac{\partial E_z}{\partial y} - i \frac{\omega k}{c^2} E_y + ik^2 B_x - k \frac{\partial B_z}{\partial x} = i \frac{\omega^2}{c^2} B_x - i \frac{\omega k}{c^2} E_y \Rightarrow$$

$$i \left(k^2 - \frac{\omega^2}{c^2} \right) B_x = k \frac{\partial B_z}{\partial x} - \frac{\omega}{c^2} \frac{\partial E_z}{\partial y}, \text{ or (iii) } B_x = \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial B_z}{\partial x} - \frac{\omega}{c^2} \frac{\partial E_z}{\partial y} \right).$$

$$\text{Multiply (iii) by } \omega/c^2, \text{ (v) by } k, \text{ and subtract: } i \frac{\omega k}{c^2} E_x - \frac{\omega}{c^2} \frac{\partial E_z}{\partial x} - k \frac{\partial B_z}{\partial y} + ik^2 B_y = i \frac{\omega^2}{c^2} B_y + \frac{i \omega k}{c^2} E_x \Rightarrow$$

$$i \left(k^2 - \frac{\omega^2}{c^2} \right) B_y = \frac{\omega}{c^2} \frac{\partial E_z}{\partial x} + k \frac{\partial B_z}{\partial y}, \text{ or (iv) } B_y = \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial B_z}{\partial y} + \frac{\omega}{c^2} \frac{\partial E_z}{\partial x} \right).$$

This completes the confirmation of Eq. 9.180.

$$(b) \nabla \cdot \tilde{\mathbf{E}} = \frac{\partial \tilde{E}_x}{\partial x} + \frac{\partial \tilde{E}_y}{\partial y} + \frac{\partial \tilde{E}_z}{\partial z} = \left(\frac{\partial \tilde{E}_{0x}}{\partial x} + \frac{\partial \tilde{E}_{0y}}{\partial y} + ik \tilde{E}_{0z} \right) e^{i(kz - \omega t)} = 0 \Rightarrow \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + ik E_z = 0.$$

$$\text{Using Eq. 9.180, } \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial^2 E_z}{\partial x^2} + \omega \frac{\partial^2 B_z}{\partial x \partial y} \right) + \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial^2 E_z}{\partial^2 y} - \omega \frac{\partial^2 B_z}{\partial x \partial y} \right) + ik E_z = 0,$$

$$\text{or } \frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial^2 y} + [(\omega/c)^2 - k^2] E_z = 0.$$

$$\text{Likewise, } \nabla \cdot \tilde{\mathbf{B}} = 0 \Rightarrow \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + ik B_z = 0 \Rightarrow$$

$$\frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial^2 B_z}{\partial x^2} - \frac{\omega}{c^2} \frac{\partial^2 E_z}{\partial x \partial y} \right) + \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial^2 B_z}{\partial y^2} + \frac{\omega}{c^2} \frac{\partial^2 E_z}{\partial x \partial y} \right) + ik B_z = 0 \Rightarrow$$

$$\frac{\partial^2 B_z}{\partial x^2} + \frac{\partial^2 B_z}{\partial^2 y} + [(\omega/c)^2 - k^2] B_z = 0.$$

This confirms Eqs. 9.181. [You can also do it by putting Eq. 9.180 into Eq. 9.179 (i) and (iv).]

Problem 9.28

Here $E_z = 0$ (TE) and $\omega/c = k$ ($n = m = 0$), so Eq. 9.179(ii) $\Rightarrow E_y = -c B_x$, Eq. 9.179(iii) $\Rightarrow E_x = c B_y$, Eq. 9.179(v) $\Rightarrow \frac{\partial B_z}{\partial y} = i \left(k B_y - \frac{\omega}{c^2} E_x \right) = i \left(k B_y - \frac{\omega}{c} B_y \right) = 0$, Eq. 9.179(vi) $\Rightarrow \frac{\partial B_z}{\partial x} = i \left(k B_x + \frac{\omega}{c^2} E_y \right) = i \left(k B_x - \frac{\omega}{c} B_x \right) = 0$. So $\frac{\partial B_z}{\partial x} = \frac{\partial B_z}{\partial y} = 0$, and since B_z is a function only of x and y , this says B_z is in fact

a *constant* (as Eq. 9.186 also suggests). Now Faraday's law (in integral form) says $\oint \mathbf{E} \cdot d\mathbf{l} = - \int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a}$,

and Eq. 9.176 $\Rightarrow \frac{\partial \mathbf{B}}{\partial t} = -i\omega \mathbf{B}$, so $\oint \mathbf{E} \cdot d\mathbf{l} = i\omega \int \mathbf{B} \cdot d\mathbf{a}$. Applied to a cross-section of the waveguide this gives

$$\oint \mathbf{E} \cdot d\mathbf{l} = i\omega e^{i(kz - \omega t)} \int B_z da = i\omega B_z e^{i(kz - \omega t)} (ab) \text{ (since } B_z \text{ is constant, it comes outside the integral). But}$$

if the boundary is just inside the metal, where $\mathbf{E} = 0$, it follows that $B_z = 0$. So this would be a TEM mode, which we already know cannot exist for this guide.

Problem 9.29

Here $a = 2.28 \text{ cm}$ and $b = 1.01 \text{ cm}$, so $\nu_{10} = \frac{1}{2\pi} \omega_{10} = \frac{c}{2a} = 0.66 \times 10^{10} \text{ Hz}$; $\nu_{20} = 2 \frac{c}{2a} = 1.32 \times 10^{10} \text{ Hz}$;
 $\nu_{30} = 3 \frac{c}{2a} = 1.97 \times 10^{10} \text{ Hz}$; $\nu_{01} = \frac{c}{2b} = 1.49 \times 10^{10} \text{ Hz}$; $\nu_{02} = 2 \frac{c}{2b} = 2.97 \times 10^{10} \text{ Hz}$; $\nu_{11} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} =$
 $1.62 \times 10^{10} \text{ Hz}$. Evidently just four modes occur: 10, 20, 01, and 11.

To get only *one* mode you must drive the waveguide at a frequency between ν_{10} and ν_{20} :

$$\boxed{0.66 \times 10^{10} < \nu < 1.32 \times 10^{10} \text{ Hz.}} \quad \lambda = \frac{c}{\nu}, \text{ so } \lambda_{10} = 2a; \lambda_{20} = a. \quad \boxed{2.28 \text{ cm} < \lambda < 4.56 \text{ cm.}}$$

Problem 9.30

From Prob. 9.11, $\langle \mathbf{S} \rangle = \frac{1}{2\mu_0} \text{Re}(\tilde{\mathbf{E}} \times \tilde{\mathbf{B}}^*)$. Here (Eq. 9.176) $\tilde{\mathbf{E}} = \tilde{\mathbf{E}}_0 e^{i(kz - \omega t)}$, $\tilde{\mathbf{B}}^* = \tilde{\mathbf{B}}_0^* e^{-i(kz - \omega t)}$, and, for the TE_{mn} mode (Eqs. 9.180 and 9.186)

$$B_x^* = \frac{-ik}{(\omega/c)^2 - k^2} \left(\frac{-m\pi}{a} \right) B_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right);$$

$$B_y^* = \frac{-ik}{(\omega/c)^2 - k^2} \left(\frac{-n\pi}{b} \right) B_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right);$$

$$B_z^* = B_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right);$$

$$E_x = \frac{i\omega}{(\omega/c)^2 - k^2} \left(\frac{-n\pi}{b} \right) B_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right);$$

$$E_y = \frac{-i\omega}{(\omega/c)^2 - k^2} \left(\frac{-m\pi}{a} \right) B_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right);$$

$$E_z = 0.$$

So

$$\langle \mathbf{S} \rangle = \frac{\omega k \pi^2 B_0^2}{[(\omega/c)^2 - k^2]^2} \left[\left(\frac{n}{b} \right)^2 \cos^2\left(\frac{m\pi x}{a}\right) \sin^2\left(\frac{n\pi y}{b}\right) + \left(\frac{m}{a} \right)^2 \sin^2\left(\frac{m\pi x}{a}\right) \cos^2\left(\frac{n\pi y}{b}\right) \right] \hat{\mathbf{z}} \}.$$

$$\int \langle \mathbf{S} \rangle \cdot d\mathbf{a} = \boxed{\frac{1}{8\mu_0} \frac{\omega k \pi^2 B_0^2}{[(\omega/c)^2 - k^2]^2} ab \left[\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2 \right]}. \quad \text{[In the last step I used}$$

$$\int_0^a \sin^2(m\pi x/a) dx = \int_0^a \cos^2(m\pi x/a) dx = a/2; \quad \int_0^b \sin^2(n\pi y/b) dy = \int_0^b \cos^2(n\pi y/b) dy = b/2.]$$

Similarly,

$$\begin{aligned} \langle u \rangle &= \frac{1}{4} \left(\epsilon_0 \tilde{\mathbf{E}} \cdot \tilde{\mathbf{E}}^* + \frac{1}{\mu_0} \tilde{\mathbf{B}} \cdot \tilde{\mathbf{B}}^* \right) \\ &= \frac{\epsilon_0}{4} \frac{\omega^2 \pi^2 B_0^2}{[(\omega/c)^2 - k^2]^2} \left[\left(\frac{n}{b} \right)^2 \cos^2\left(\frac{m\pi x}{a}\right) \sin^2\left(\frac{n\pi y}{b}\right) + \left(\frac{m}{a} \right)^2 \sin^2\left(\frac{m\pi x}{a}\right) \cos^2\left(\frac{n\pi y}{b}\right) \right] \\ &\quad + \frac{1}{4\mu_0} \left\{ B_0^2 \cos^2\left(\frac{m\pi x}{a}\right) \cos^2\left(\frac{n\pi y}{b}\right) \right. \\ &\quad \left. + \frac{k^2 \pi^2 B_0^2}{[(\omega/c)^2 - k^2]^2} \left[\left(\frac{n}{b} \right)^2 \cos^2\left(\frac{m\pi x}{a}\right) \sin^2\left(\frac{n\pi y}{b}\right) + \left(\frac{m}{a} \right)^2 \sin^2\left(\frac{m\pi x}{a}\right) \cos^2\left(\frac{n\pi y}{b}\right) \right] \right\}. \end{aligned}$$

$$\int \langle u \rangle da = \boxed{\frac{ab}{4} \left\{ \frac{\epsilon_0}{4} \frac{\omega^2 \pi^2 B_0^2}{[(\omega/c)^2 - k^2]^2} \left[\left(\frac{n}{b} \right)^2 + \left(\frac{m}{a} \right)^2 \right] + \frac{B_0^2}{4\mu_0} + \frac{1}{4\mu_0} \frac{k^2 \pi^2 B_0^2}{[(\omega/c)^2 - k^2]^2} \left[\left(\frac{n}{b} \right)^2 + \left(\frac{m}{a} \right)^2 \right] \right\}}.$$

These results can be simplified, using Eq. 9.190 to write $[(\omega/c)^2 - k^2] = (\omega_{mn}/c)^2$, $\epsilon_0 \mu_0 = 1/c^2$ to eliminate ϵ_0 , and Eq. 9.188 to write $[(m/a)^2 + (n/b)^2] = (\omega_{mn}/\pi c)^2$:

$$\int \langle \mathbf{S} \rangle \cdot d\mathbf{a} = \frac{\omega k abc^2}{8\mu_0 \omega_{mn}^2} B_0^2; \quad \int \langle u \rangle da = \frac{\omega^2 ab}{8\mu_0 \omega_{mn}^2} B_0^2.$$

Evidently

$$\frac{\text{energy per unit time}}{\text{energy per unit length}} = \frac{\int \langle \mathbf{S} \rangle \cdot d\mathbf{a}}{\int \langle u \rangle da} = \frac{kc^2}{\omega} = \frac{c}{\omega} \sqrt{\omega^2 - \omega_{mn}^2} = v_g \text{ (Eq. 9.192).} \quad \text{qed}$$

Problem 9.31

Following Sect. 9.5.2, the problem is to solve Eq. 9.181 with $E_z \neq 0, B_z = 0$, subject to the boundary conditions 9.175. Let $E_z(x, y) = X(x)Y(y)$; as before, we obtain $X(x) = A \sin(k_x x) + B \cos(k_x x)$. But the boundary condition requires $E_z = 0$ (and hence $X = 0$) when $x = 0$ and $x = a$, so $B = 0$ and $k_x = m\pi/a$. But this time $m = 1, 2, 3, \dots$, but *not* zero, since $m = 0$ would kill X entirely. The same goes for $Y(y)$. Thus

$$E_z = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \quad \text{with } n, m = 1, 2, 3, \dots$$

The rest is the same as for TE waves: $\omega_{mn} = c\pi\sqrt{(m/a)^2 + (n/b)^2}$ is the cutoff frequency, the wave velocity is $v = c/\sqrt{1 - (\omega_{mn}/\omega)^2}$, and the group velocity is $v_g = c\sqrt{1 - (\omega_{mn}/\omega)^2}$. The lowest TM mode is 11, with cutoff frequency $\omega_{11} = c\pi\sqrt{(1/a)^2 + (1/b)^2}$. So the ratio of the lowest TM frequency to the lowest TE frequency is $\frac{c\pi\sqrt{(1/a)^2 + (1/b)^2}}{(c\pi/a)} = \sqrt{1 + (a/b)^2}$.

Problem 9.32

$$\begin{aligned} \text{(a)} \quad \nabla \cdot \mathbf{E} &= \frac{1}{s} \frac{\partial}{\partial s}(sE_s) = 0 \checkmark; \nabla \cdot \mathbf{B} = \frac{1}{s} \frac{\partial}{\partial \phi}(B_\phi) = 0 \checkmark; \nabla \times \mathbf{E} = \frac{\partial E_s}{\partial z} \hat{\phi} - \frac{1}{s} \frac{\partial E_s}{\partial \phi} \hat{\mathbf{z}} = -\frac{E_0 k \sin(kz - \omega t)}{s} \hat{\phi} \stackrel{?}{=} \\ -\frac{\partial \mathbf{B}}{\partial t} &= -\frac{E_0 \omega}{c} \frac{\sin(kz - \omega t)}{s} \hat{\phi} \checkmark \text{ (since } k = \omega/c); \nabla \times \mathbf{B} = -\frac{\partial B_\phi}{\partial z} \hat{\mathbf{s}} + \frac{1}{s} \frac{\partial}{\partial s}(sB_\phi) \hat{\mathbf{z}} = \frac{E_0 k}{c} \frac{\sin(kz - \omega t)}{s} \hat{\mathbf{s}} \stackrel{?}{=} \\ \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} &= \frac{E_0 \omega}{c^2} \frac{\sin(kz - \omega t)}{s} \hat{\mathbf{s}} \checkmark. \text{ Boundary conditions: } E^\parallel = E_z = 0 \checkmark; B^\perp = B_s = 0 \checkmark. \end{aligned}$$

(b) To determine λ , use Gauss's law for a cylinder of radius s and length dz :

$$\oint \mathbf{E} \cdot d\mathbf{a} = E_0 \frac{\cos(kz - \omega t)}{s} (2\pi s) dz = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \lambda dz \Rightarrow \boxed{\lambda = 2\pi\epsilon_0 E_0 \cos(kz - \omega t)}.$$

To determine I , use Ampère's law for a circle of radius s (note that the displacement current through this loop is zero, since $\mathbf{E}$ is in the $\hat{\mathbf{s}}$ direction): $\oint \mathbf{B} \cdot d\mathbf{l} = \frac{E_0 \cos(kz - \omega t)}{c} (2\pi s) = \mu_0 I_{\text{enc}} \Rightarrow \boxed{I = \frac{2\pi E_0}{\mu_0 c} \cos(kz - \omega t)}.$

The charge and current on the outer conductor are precisely the opposite of these, since $\mathbf{E} = \mathbf{B} = 0$ *inside* the metal, and hence the *total* enclosed charge and current must be zero.

Problem 9.33

$$\begin{aligned} \tilde{f}(z, 0) &= \int_{-\infty}^{\infty} \tilde{A}(k) e^{ikz} dk \Rightarrow \tilde{f}(z, 0)^* = \int_{-\infty}^{\infty} \tilde{A}(k)^* e^{-ikz} dk. \text{ Let } l \equiv -k; \text{ then } \tilde{f}(z, 0)^* = \\ \int_{-\infty}^{\infty} \tilde{A}(-l)^* e^{ilz} (-dl) &= \int_{-\infty}^{\infty} \tilde{A}(-l)^* e^{ilz} dl = \int_{-\infty}^{\infty} \tilde{A}(-k)^* e^{ikz} dk \text{ (renaming the dummy variable } l \rightarrow k). \\ f(z, 0) &= \text{Re} [\tilde{f}(z, 0)] = \frac{1}{2} [\tilde{f}(z, 0) + \tilde{f}(z, 0)^*] = \int_{-\infty}^{\infty} \frac{1}{2} [\tilde{A}(k) + \tilde{A}(-k)^*] e^{ikz} dk. \text{ Therefore} \\ \frac{1}{2} [\tilde{A}(k) + \tilde{A}(-k)^*] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(z, 0) e^{-ikz} dz. \end{aligned}$$

$$\text{Meanwhile, } \dot{\tilde{f}}(z, t) = \int_{-\infty}^{\infty} \tilde{A}(k) (-i\omega) e^{i(kz - \omega t)} dk \Rightarrow \dot{\tilde{f}}(z, 0) = \int_{-\infty}^{\infty} [-i\omega \tilde{A}(k)] e^{ikz} dk.$$

(Note that $\omega = |k|v$, here, so it does *not* come outside the integral.)

$$\begin{aligned}\dot{f}(z, 0)^* &= \int_{-\infty}^{\infty} [i\omega \tilde{A}(k)^*] e^{-ikz} dk = \int_{-\infty}^{\infty} [i|k|v \tilde{A}(k)^*] e^{-ikz} dk = \int_{-\infty}^{\infty} [i|l|v \tilde{A}(-l)^*] e^{ilz} (-dl) \\ &= \int_{-\infty}^{\infty} [i|k|v \tilde{A}(-k)^*] e^{ikz} dk = \int_{-\infty}^{\infty} [i\omega \tilde{A}(-k)^*] e^{ikz} dk.\end{aligned}$$

$$\dot{f}(z, 0) = \text{Re} [\dot{f}(z, 0)] = \frac{1}{2} [\dot{f}(z, 0) + \dot{f}(z, 0)^*] = \int_{-\infty}^{\infty} \frac{1}{2} [-i\omega \tilde{A}(k) + i\omega \tilde{A}(-k)^*] e^{ikz} dk.$$

$$\frac{-i\omega}{2} [\tilde{A}(k) - \tilde{A}(-k)^*] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \dot{f}(z, 0) e^{-ikz} dz, \text{ or } \frac{1}{2} [\tilde{A}(k) - \tilde{A}(-k)^*] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{i}{\omega} \dot{f}(z, 0) \right] e^{-ikz} dz.$$

Adding these two results, we get $\tilde{A}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[f(z, 0) + \frac{i}{\omega} \dot{f}(z, 0) \right] e^{-ikz} dz.$ qed

Problem 9.34

(a) Since $(\mathbf{E} \times \mathbf{B})$ points in the direction of propagation, $\mathbf{B} = \frac{E_0}{c} [\cos(kz - \omega t) + \cos(kz + \omega t)] \hat{\mathbf{y}}.$

(b) From Eq. 7.63, $\mathbf{K} \times (-\hat{\mathbf{z}}) = \frac{1}{\mu_0} \mathbf{B} = \frac{E_0}{\mu_0 c} [2 \cos(\omega t)] \hat{\mathbf{y}}, \mathbf{K} = \frac{2E_0}{\mu_0 c} \cos(\omega t) \hat{\mathbf{x}}.$

(c) The force per unit area is $\mathbf{f} = \mathbf{K} \times \mathbf{B}_{\text{ave}} = \frac{2E_0^2}{\mu_0 c^2} [\cos(\omega t) \hat{\mathbf{x}}] \times [\cos(\omega t) \hat{\mathbf{y}}] = \frac{2\epsilon_0 E_0^2 \cos^2(\omega t) \hat{\mathbf{z}}}{\mu_0 c^2}.$ The time average of $\cos^2(\omega t)$ is $1/2$, so

$$\mathbf{f}_{\text{ave}} = \epsilon_0 E_0^2 \hat{\mathbf{z}}.$$

This is twice the pressure in Eq. 9.64, but that was for a perfect absorber, whereas this is a perfect reflector.

Problem 9.35

(a) (i) Gauss's law: $\nabla \cdot \mathbf{E} = \frac{1}{r \sin \theta} \frac{\partial E_\phi}{\partial \phi} = 0. \checkmark$

(ii) Faraday's law:

$$\begin{aligned}-\frac{\partial \mathbf{B}}{\partial t} &= \nabla \times \mathbf{E} = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta E_\phi) \hat{\mathbf{r}} - \frac{1}{r} \frac{\partial}{\partial r} (r E_\phi) \hat{\boldsymbol{\theta}} \\ &= \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left[E_0 \frac{\sin^2 \theta}{r} \left(\cos u - \frac{1}{kr} \sin u \right) \right] \hat{\mathbf{r}} - \frac{1}{r} \frac{\partial}{\partial r} \left[E_0 \sin \theta \left(\cos u - \frac{1}{kr} \sin u \right) \right] \hat{\boldsymbol{\theta}}.\end{aligned}$$

$$\text{But } \frac{\partial}{\partial r} \cos u = -k \sin u; \frac{\partial}{\partial r} \sin u = k \cos u.$$

$$= \frac{1}{r \sin \theta} \frac{E_0}{r} 2 \sin \theta \cos \theta \left(\cos u - \frac{1}{kr} \sin u \right) \hat{\mathbf{r}} - \frac{1}{r} E_0 \sin \theta \left(-k \sin u + \frac{1}{kr^2} \sin u - \frac{1}{r} \cos u \right) \hat{\boldsymbol{\theta}}.$$

Integrating with respect to t , and noting that $\int \cos u dt = -\frac{1}{\omega} \sin u$ and $\int \sin u dt = \frac{1}{\omega} \cos u$, we obtain

$$\mathbf{B} = \frac{2E_0 \cos \theta}{\omega r^2} \left(\sin u + \frac{1}{kr} \cos u \right) \hat{\mathbf{r}} + \frac{E_0 \sin \theta}{\omega r} \left(-k \cos u + \frac{1}{kr^2} \cos u + \frac{1}{r} \sin u \right) \hat{\boldsymbol{\theta}}.$$

(iii) *Divergence of $\mathbf{B}$:*

$$\begin{aligned}
\nabla \cdot \mathbf{B} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 B_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta B_\theta) \\
&= \frac{1}{r^2} \frac{\partial}{\partial r} \left[\frac{2E_0 \cos \theta}{\omega} \left(\sin u + \frac{1}{kr} \cos u \right) \right] + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left[\frac{E_0 \sin^2 \theta}{\omega r} \left(-k \cos u + \frac{1}{kr^2} \cos u + \frac{1}{r} \sin u \right) \right] \\
&= \frac{1}{r^2} \frac{2E_0 \cos \theta}{\omega} \left(k \cos u - \frac{1}{kr^2} \cos u - \frac{1}{r} \sin u \right) \\
&\quad + \frac{1}{r \sin \theta} \frac{2E_0 \sin \theta \cos \theta}{\omega r} \left(-k \cos u + \frac{1}{kr^2} \cos u + \frac{1}{r} \sin u \right) \\
&= \frac{2E_0 \cos \theta}{\omega r^2} \left(k \cos u - \frac{1}{kr^2} \cos u - \frac{1}{r} \sin u - k \cos u + \frac{1}{kr^2} \cos u + \frac{1}{r} \sin u \right) = 0. \checkmark
\end{aligned}$$

(iv) *Ampère/Maxwell:*

$$\begin{aligned}
\nabla \times \mathbf{B} &= \frac{1}{r} \left[\frac{\partial}{\partial r} (r B_\theta) - \frac{\partial B_r}{\partial \theta} \right] \hat{\phi} \\
&= \frac{1}{r} \left\{ \frac{\partial}{\partial r} \left[\frac{E_0 \sin \theta}{\omega} \left(-k \cos u + \frac{1}{kr^2} \cos u + \frac{1}{r} \sin u \right) \right] - \frac{\partial}{\partial \theta} \left[\frac{2E_0 \cos \theta}{\omega r^2} \left(\sin u + \frac{1}{kr} \cos u \right) \right] \right\} \hat{\phi} \\
&= \frac{E_0 \sin \theta}{\omega r} \left(k^2 \sin u - \frac{2}{kr^3} \cos u - \frac{1}{r^2} \sin u - \frac{1}{r^2} \sin u + \frac{k}{r} \cos u + \frac{2}{r^2} \sin u + \frac{2}{kr^3} \cos u \right) \hat{\phi} \\
&= \frac{k E_0 \sin \theta}{\omega r} \left(k \sin u + \frac{1}{r} \cos u \right) \hat{\phi} = \frac{1}{c} \frac{E_0 \sin \theta}{r} \left(k \sin u + \frac{1}{r} \cos u \right) \hat{\phi}. \\
\frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} &= \frac{1}{c^2} \frac{E_0 \sin \theta}{r} \left(\omega \sin u + \frac{\omega}{kr} \cos u \right) \hat{\phi} = \frac{1}{c^2} \frac{\omega E_0 \sin \theta}{k r} \left(k \sin u + \frac{1}{r} \cos u \right) \hat{\phi} \\
&= \frac{1}{c} \frac{E_0 \sin \theta}{r} \left(k \sin u + \frac{1}{r} \cos u \right) \hat{\phi} = \nabla \times \mathbf{B}. \checkmark
\end{aligned}$$

(b) *Poynting Vector:*

$$\begin{aligned}
\mathbf{S} &= \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{E_0 \sin \theta}{\mu_0 r} \left(\cos u - \frac{1}{kr} \sin u \right) \left[\frac{2E_0 \cos \theta}{\omega r^2} \left(\sin u + \frac{1}{kr} \cos u \right) \hat{\theta} \right. \\
&\quad \left. + \frac{E_0 \sin \theta}{\omega r} \left(-k \cos u + \frac{1}{kr^2} \cos u + \frac{1}{r} \sin u \right) (-\hat{\mathbf{r}}) \right] \\
&= \frac{E_0^2 \sin \theta}{\mu_0 \omega r^2} \left\{ \frac{2 \cos \theta}{r} \left[\sin u \cos u + \frac{1}{kr} (\cos^2 u - \sin^2 u) - \frac{1}{k^2 r^2} \sin u \cos u \right] \hat{\theta} \right. \\
&\quad \left. - \sin \theta \left(-k \cos^2 u + \frac{1}{kr^2} \cos^2 u + \frac{1}{r} \sin u \cos u + \frac{1}{r} \sin u \cos u - \frac{1}{k^2 r^3} \sin u \cos u - \frac{1}{kr^2} \sin^2 u \right) \hat{\mathbf{r}} \right\} \\
&= \boxed{ \frac{E_0^2 \sin \theta}{\mu_0 \omega r^2} \left\{ \frac{2 \cos \theta}{r} \left[\left(1 - \frac{1}{k^2 r^2} \right) \sin u \cos u + \frac{1}{kr} (\cos^2 u - \sin^2 u) \right] \hat{\theta} \right. } \\
&\quad \left. + \sin \theta \left[\left(-\frac{2}{r} + \frac{1}{k^2 r^3} \right) \sin u \cos u + k \cos^2 u + \frac{1}{kr^2} (\sin^2 u - \cos^2 u) \right] \hat{\mathbf{r}} \right\} }.
\end{aligned}$$

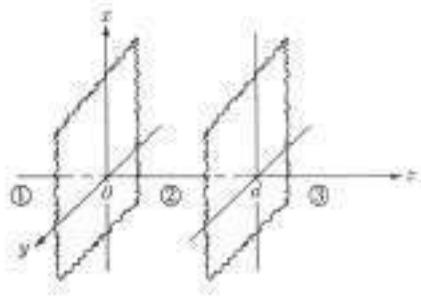
Averaging over a full cycle, using $\langle \sin u \cos u \rangle = 0$, $\langle \sin^2 u \rangle = \langle \cos^2 u \rangle = \frac{1}{2}$, we get the intensity:

$$\mathbf{I} = \langle \mathbf{S} \rangle = \frac{E_0^2 \sin \theta}{\mu_0 \omega r^2} \left(\frac{k}{2} \sin \theta \right) \hat{\mathbf{r}} = \boxed{\frac{E_0^2 \sin^2 \theta}{2\mu_0 c r^2} \hat{\mathbf{r}}}.$$

It points in the $\hat{\mathbf{r}}$ direction, and falls off as $1/r^2$, as we would expect for a spherical wave.

$$(c) P = \int \mathbf{I} \cdot d\mathbf{a} = \frac{E_0^2}{2\mu_0 c} \int \frac{\sin^2 \theta}{r^2} r^2 \sin \theta d\theta d\phi = \frac{E_0^2}{2\mu_0 c} 2\pi \int_0^\pi \sin^3 \theta d\theta = \boxed{\frac{4\pi}{3} \frac{E_0^2}{\mu_0 c}}.$$

Problem 9.36



$$z < 0 : \quad \begin{cases} \tilde{\mathbf{E}}_I(z, t) = \tilde{E}_I e^{i(k_1 z - \omega t)} \hat{\mathbf{x}}, & \tilde{\mathbf{B}}_I(z, t) = \frac{1}{v_1} \tilde{E}_I e^{i(k_1 z - \omega t)} \hat{\mathbf{y}} \\ \tilde{\mathbf{E}}_R(z, t) = \tilde{E}_R e^{i(-k_1 z - \omega t)} \hat{\mathbf{x}}, & \tilde{\mathbf{B}}_R(z, t) = -\frac{1}{v_1} \tilde{E}_R e^{i(-k_1 z - \omega t)} \hat{\mathbf{y}}. \end{cases}$$

$$0 < z < d : \quad \begin{cases} \tilde{\mathbf{E}}_r(z, t) = \tilde{E}_r e^{i(k_2 z - \omega t)} \hat{\mathbf{x}}, & \tilde{\mathbf{B}}_r(z, t) = \frac{1}{v_2} \tilde{E}_r e^{i(k_2 z - \omega t)} \hat{\mathbf{y}} \\ \tilde{\mathbf{E}}_l(z, t) = \tilde{E}_l e^{i(-k_2 z - \omega t)} \hat{\mathbf{x}}, & \tilde{\mathbf{B}}_l(z, t) = -\frac{1}{v_2} \tilde{E}_l e^{i(-k_2 z - \omega t)} \hat{\mathbf{y}}. \end{cases}$$

$$z > d : \quad \begin{cases} \tilde{\mathbf{E}}_T(z, t) = \tilde{E}_T e^{i(k_3 z - \omega t)} \hat{\mathbf{x}}, & \tilde{\mathbf{B}}_T(z, t) = \frac{1}{v_3} \tilde{E}_T e^{i(k_3 z - \omega t)} \hat{\mathbf{y}}. \end{cases}$$

Boundary conditions: $\mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel$, $\mathbf{B}_1^\parallel = \mathbf{B}_2^\parallel$, at each boundary (assuming $\mu_1 = \mu_2 = \mu_3 = \mu_0$):

$$z = 0 : \quad \begin{cases} \tilde{E}_I + \tilde{E}_R = \tilde{E}_r + \tilde{E}_l; \\ \frac{1}{v_1} \tilde{E}_I - \frac{1}{v_1} \tilde{E}_R = \frac{1}{v_2} \tilde{E}_r - \frac{1}{v_2} \tilde{E}_l \Rightarrow \tilde{E}_I - \tilde{E}_R = \beta(\tilde{E}_r - \tilde{E}_l), \text{ where } \beta \equiv v_1/v_2. \end{cases}$$

$$z = d : \quad \begin{cases} \tilde{E}_r e^{ik_2 d} + \tilde{E}_l e^{-ik_2 d} = \tilde{E}_T e^{ik_3 d}, \\ \frac{1}{v_2} \tilde{E}_r e^{ik_2 d} - \frac{1}{v_2} \tilde{E}_l e^{-ik_2 d} = \frac{1}{v_3} \tilde{E}_T e^{ik_3 d} \Rightarrow \tilde{E}_r e^{ik_2 d} - \tilde{E}_l e^{-ik_2 d} = \alpha \tilde{E}_T e^{ik_3 d}, \text{ where } \alpha \equiv v_2/v_3. \end{cases}$$

We have here four equations; the problem is to eliminate $\tilde{E}_R$, $\tilde{E}_r$, and $\tilde{E}_l$, to obtain a single equation for $\tilde{E}_T$ in terms of $\tilde{E}_I$.

$$\text{Add the first two to eliminate } \tilde{E}_R : \quad 2\tilde{E}_I = (1 + \beta)\tilde{E}_r + (1 - \beta)\tilde{E}_l;$$

$$\text{Add the last two to eliminate } \tilde{E}_l : \quad 2\tilde{E}_r e^{ik_2 d} = (1 + \alpha)\tilde{E}_T e^{ik_3 d};$$

$$\text{Subtract the last two to eliminate } \tilde{E}_r : \quad 2\tilde{E}_l e^{-ik_2 d} = (1 - \alpha)\tilde{E}_T e^{ik_3 d}.$$

Plug the last two of these into the first:

$$\begin{aligned}
2\tilde{E}_I &= (1 + \beta)\frac{1}{2}e^{-ik_2d}(1 + \alpha)\tilde{E}_Te^{ik_3d} + (1 - \beta)\frac{1}{2}e^{ik_2d}(1 - \alpha)\tilde{E}_Te^{ik_3d} \\
4\tilde{E}_I &= [(1 + \alpha)(1 + \beta)e^{-ik_2d} + (1 - \alpha)(1 - \beta)e^{ik_2d}] \tilde{E}_Te^{ik_3d} \\
&= [(1 + \alpha\beta)(e^{-ik_2d} + e^{ik_2d}) + (\alpha + \beta)(e^{-ik_2d} - e^{ik_2d})] \tilde{E}_Te^{ik_3d} \\
&= 2[(1 + \alpha\beta)\cos(k_2d) - i(\alpha + \beta)\sin(k_2d)] \tilde{E}_Te^{ik_3d}.
\end{aligned}$$

Now the transmission coefficient is $T = \frac{v_3\epsilon_3 E_{T_0}^2}{v_1\epsilon_1 E_{I_0}^2} = \frac{v_3}{v_1} \left(\frac{\mu_0\epsilon_3}{\mu_0\epsilon_1} \right) \frac{|\tilde{E}_T|^2}{|\tilde{E}_I|^2} = \frac{v_1}{v_3} \frac{|\tilde{E}_T|^2}{|\tilde{E}_I|^2} = \alpha\beta \frac{|\tilde{E}_T|^2}{|\tilde{E}_I|^2}$, so

$$\begin{aligned}
T^{-1} &= \frac{1}{\alpha\beta} \frac{|\tilde{E}_I|^2}{|\tilde{E}_T|^2} = \frac{1}{\alpha\beta} \left| \frac{1}{2} [(1 + \alpha\beta)\cos(k_2d) - i(\alpha + \beta)\sin(k_2d)] e^{ik_3d} \right|^2 \\
&= \frac{1}{4\alpha\beta} [(1 + \alpha\beta)^2 \cos^2(k_2d) + (\alpha + \beta)^2 \sin^2(k_2d)]. \quad \text{But } \cos^2(k_2d) = 1 - \sin^2(k_2d). \\
&= \frac{1}{4\alpha\beta} [(1 + \alpha\beta)^2 + (\alpha^2 + 2\alpha\beta + \beta^2 - 1 - 2\alpha\beta - \alpha^2\beta^2) \sin^2(k_2d)] \\
&= \frac{1}{4\alpha\beta} [(1 + \alpha\beta)^2 - (1 - \alpha^2)(1 - \beta^2) \sin^2(k_2d)]. \\
\text{But } n_1 &= \frac{c}{v_1}, \quad n_2 = \frac{c}{v_2}, \quad n_3 = \frac{c}{v_3}, \quad \text{so } \alpha = \frac{n_3}{n_2}, \quad \beta = \frac{n_2}{n_1}. \\
&= \boxed{\frac{1}{4n_1n_3} \left[(n_1 + n_3)^2 + \frac{(n_1^2 - n_2^2)(n_3^2 - n_2^2)}{n_2^2} \sin^2(k_2d) \right]}.
\end{aligned}$$

Problem 9.37

$T = 1 \Rightarrow \sin kd = 0 \Rightarrow kd = 0, \pi, 2\pi, \dots$ The *minimum* (nonzero) thickness is $d = \pi/k$. But $k = \omega/v = 2\pi\nu/v = 2\pi\nu n/c$, and $n = \sqrt{\epsilon\mu/\epsilon_0\mu_0}$ (Eq. 9.69), where (presumably) $\mu \approx \mu_0$. So $n = \sqrt{\epsilon/\epsilon_0} = \sqrt{\epsilon_r}$, and hence $d = \frac{\pi c}{2\pi\nu\sqrt{\epsilon_r}} = \frac{c}{2\nu\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{2(10 \times 10^9)\sqrt{2.5}} = 9.49 \times 10^{-3} \text{ m}$, or 9.5 mm.

Problem 9.38

From Eq. 9.199,

$$\begin{aligned}
T^{-1} &= \frac{1}{4(4/3)(1)} \left\{ [(4/3) + 1]^2 + \frac{[(16/9) - (9/4)][1 - (9/4)]}{(9/4)} \sin^2(3\omega d/2c) \right\} \\
&= \frac{3}{16} \left[\frac{49}{9} + \frac{(-17/36)(-5/4)}{(9/4)} \sin^2(3\omega d/2c) \right] = \frac{49}{48} + \frac{85}{(48)(36)} \sin^2(3\omega d/2c). \\
T &= \frac{48}{49 + (85/36) \sin^2(3\omega d/2c)}.
\end{aligned}$$

Since $\sin^2(3\omega d/2c)$ ranges from 0 to 1, $T_{\min} = \frac{48}{49 + (85/36)} = \boxed{0.935}$; $T_{\max} = \frac{48}{49} = \boxed{0.980}$. Not much variation, and the transmission is good (over 90%) for *all* frequencies. Since Eq. 9.199 is unchanged when you switch 1 and 3, the transmission is the same either direction, and the fish sees you just as well as you see it.

Problem 9.39

(a) Equation 9.91 $\Rightarrow \tilde{\mathbf{E}}_T(\mathbf{r}, t) = \tilde{\mathbf{E}}_{0T} e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)}$; $\mathbf{k}_T \cdot \mathbf{r} = k_T(\sin \theta_T \hat{\mathbf{x}} + \cos \theta_T \hat{\mathbf{z}}) \cdot (x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}) = k_T(x \sin \theta_T + z \cos \theta_T) = x k_T \sin \theta_T + i z k_T \sqrt{\sin^2 \theta_T - 1} = kx + i\kappa z$, where

$$k \equiv k_T \sin \theta_T = \left(\frac{\omega n_2}{c} \right) \frac{n_1}{n_2} \sin \theta_I = \frac{\omega n_1}{c} \sin \theta_I,$$

$$\kappa \equiv k_T \sqrt{\sin^2 \theta_T - 1} = \frac{\omega n_2}{c} \sqrt{(n_1/n_2)^2 \sin^2 \theta_I - 1} = \frac{\omega}{c} \sqrt{n_1^2 \sin^2 \theta_I - n_2^2}. \quad \text{So}$$

$$\tilde{\mathbf{E}}_T(\mathbf{r}, t) = \tilde{\mathbf{E}}_{0T} e^{-\kappa z} e^{i(kx - \omega t)}. \quad \text{qed}$$

(b) $R = \left| \frac{\tilde{E}_{0R}}{\tilde{E}_{0I}} \right|^2 = \left| \frac{\alpha - \beta}{\alpha + \beta} \right|^2$. Here β is real (Eq. 9.106) and α is purely imaginary (Eq. 9.108); write $\alpha = ia$,

with a real: $R = \left(\frac{ia - \beta}{ia + \beta} \right) \left(\frac{-ia - \beta}{-ia + \beta} \right) = \frac{a^2 + \beta^2}{a^2 + \beta^2} = \boxed{1.}$

(c) From Prob. 9.17, $E_{0R} = \left| \frac{1 - \alpha\beta}{1 + \alpha\beta} \right| E_{0I}$, so $R = \left| \frac{1 - \alpha\beta}{1 + \alpha\beta} \right|^2 = \left| \frac{1 - ia\beta}{1 + ia\beta} \right|^2 = \frac{(1 - ia\beta)(1 + ia\beta)}{(1 + ia\beta)(1 - ia\beta)} = \boxed{1.}$

(d) From the solution to Prob. 9.17, the transmitted wave is

$$\tilde{\mathbf{E}}(\mathbf{r}, t) = \tilde{E}_{0T} e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)} \hat{\mathbf{y}}, \quad \tilde{\mathbf{B}}(\mathbf{r}, t) = \frac{1}{v_2} \tilde{E}_{0T} e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)} (-\cos \theta_T \hat{\mathbf{x}} + \sin \theta_T \hat{\mathbf{z}}).$$

Using the results in (a): $\mathbf{k}_T \cdot \mathbf{r} = kx + i\kappa z$, $\sin \theta_T = \frac{ck}{\omega n_2}$, $\cos \theta_T = i \frac{c\kappa}{\omega n_2}$:

$$\tilde{\mathbf{E}}(\mathbf{r}, t) = \tilde{E}_{0T} e^{-\kappa z} e^{i(kx - \omega t)} \hat{\mathbf{y}}, \quad \tilde{\mathbf{B}}(\mathbf{r}, t) = \frac{1}{v_2} \tilde{E}_{0T} e^{-\kappa z} e^{i(kx - \omega t)} \left(-i \frac{c\kappa}{\omega n_2} \hat{\mathbf{x}} + \frac{ck}{\omega n_2} \hat{\mathbf{z}} \right).$$

We may as well choose the phase constant so that $\tilde{E}_{0T}$ is *real*. Then

$$\mathbf{E}(\mathbf{r}, t) = E_0 e^{-\kappa z} \cos(kx - \omega t) \hat{\mathbf{y}};$$

$$\begin{aligned} \mathbf{B}(\mathbf{r}, t) &= \frac{1}{v_2} E_0 e^{-\kappa z} \frac{c}{\omega n_2} \text{Re} \{ [\cos(kx - \omega t) + i \sin(kx - \omega t)] [-i\kappa \hat{\mathbf{x}} + k \hat{\mathbf{z}}] \} \\ &= \frac{1}{\omega} E_0 e^{-\kappa z} [\kappa \sin(kx - \omega t) \hat{\mathbf{x}} + k \cos(kx - \omega t) \hat{\mathbf{z}}]. \quad \text{qed} \end{aligned}$$

(I used $v_2 = c/n_2$ to simplify $\mathbf{B}$.)

$$(e) \quad (i) \quad \nabla \cdot \mathbf{E} = \frac{\partial}{\partial y} [E_0 e^{-\kappa z} \cos(kx - \omega t)] = 0. \quad \checkmark$$

$$(ii) \quad \nabla \cdot \mathbf{B} = \frac{\partial}{\partial x} \left[\frac{E_0}{\omega} e^{-\kappa z} \kappa \sin(kx - \omega t) \right] + \frac{\partial}{\partial z} \left[\frac{E_0}{\omega} e^{-\kappa z} k \cos(kx - \omega t) \right]$$

$$= \frac{E_0}{\omega} [e^{-\kappa z} \kappa k \cos(kx - \omega t) - \kappa e^{-\kappa z} k \cos(kx - \omega t)] = 0. \quad \checkmark$$

$$(iii) \quad \nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & E_y & 0 \end{vmatrix} = -\frac{\partial E_y}{\partial z} \hat{\mathbf{x}} + \frac{\partial E_y}{\partial x} \hat{\mathbf{z}}$$

$$= \kappa E_0 e^{-\kappa z} \cos(kx - \omega t) \hat{\mathbf{x}} - E_0 e^{-\kappa z} k \sin(kx - \omega t) \hat{\mathbf{z}}.$$

$$-\frac{\partial \mathbf{B}}{\partial t} = -\frac{E_0}{\omega} e^{-\kappa z} [-\kappa \omega \cos(kx - \omega t) \hat{\mathbf{x}} + k \omega \sin(kx - \omega t) \hat{\mathbf{z}}]$$

$$= \kappa E_0 e^{-\kappa z} \cos(kx - \omega t) \hat{\mathbf{x}} - k E_0 e^{-\kappa z} \sin(kx - \omega t) \hat{\mathbf{z}} = \nabla \times \mathbf{E}. \quad \checkmark$$

$$(iv) \quad \nabla \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ B_x & 0 & B_z \end{vmatrix} = \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \hat{\mathbf{y}}$$

$$= \left[-\frac{E_0}{\omega} \kappa^2 e^{-\kappa z} \sin(kx - \omega t) + \frac{E_0}{\omega} e^{-\kappa z} k^2 \sin(kx - \omega t) \right] \hat{\mathbf{y}} = (k^2 - \kappa^2) \frac{E_0}{\omega} e^{-\kappa z} \sin(kx - \omega t) \hat{\mathbf{y}}.$$

$$\text{Eq. 9.202} \Rightarrow k^2 - \kappa^2 = \left(\frac{\omega}{c} \right)^2 [n_1^2 \sin^2 \theta_I - (n_1 \sin \theta_I)^2 + (n_2)^2] = \left(\frac{n_2 \omega}{c} \right)^2 = \omega^2 \epsilon_2 \mu_2.$$

$$= \epsilon_2 \mu_2 \omega E_0 e^{-\kappa z} \sin(kx - \omega t) \hat{\mathbf{y}}.$$

$$\mu_2 \epsilon_2 \frac{\partial \mathbf{E}}{\partial t} = \mu_2 \epsilon_2 E_0 e^{-\kappa z} \omega \sin(kx - \omega t) \hat{\mathbf{y}} = \nabla \times \mathbf{B} \quad \checkmark.$$

(f)

$$\mathbf{S} = \frac{1}{\mu_2} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_2} \frac{E_0^2}{\omega} e^{-2\kappa z} \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & \cos(kx - \omega t) & 0 \\ \kappa \sin(kx - \omega t) & 0 & k \cos(kx - \omega t) \end{vmatrix}$$

$$= \boxed{\frac{E_0^2}{\mu_2 \omega} e^{-2\kappa z} [k \cos^2(kx - \omega t) \hat{\mathbf{x}} - \kappa \sin(kx - \omega t) \cos(kx - \omega t) \hat{\mathbf{z}}]}.$$

Averaging over a complete cycle, using $\langle \cos^2 \rangle = 1/2$ and $\langle \sin \cos \rangle = 0$, $\langle \mathbf{S} \rangle = \frac{E_0^2 k}{2\mu_2 \omega} e^{-2\kappa z} \hat{\mathbf{x}}$. On average, then, no energy is transmitted in the z direction, only in the x direction (parallel to the interface). qed

Problem 9.40

Look for solutions of the form $\mathbf{E} = \mathbf{E}_0(x, y, z) e^{-i\omega t}$, $\mathbf{B} = \mathbf{B}_0(x, y, z) e^{-i\omega t}$, subject to the boundary conditions $\mathbf{E}^\parallel = \mathbf{0}$, $B^\perp = 0$ at all surfaces. Maxwell's equations, in the form of Eq. 9.177, give

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = 0 \Rightarrow \nabla \cdot \mathbf{E}_0 = 0; \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \Rightarrow \nabla \times \mathbf{E}_0 = i\omega \mathbf{B}_0; \\ \nabla \cdot \mathbf{B} = 0 \Rightarrow \nabla \cdot \mathbf{B}_0 = 0; \quad \nabla \times \mathbf{B} = \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} \Rightarrow \nabla \times \mathbf{B}_0 = -\frac{i\omega}{c^2} \mathbf{E}_0. \end{array} \right\}$$

From now on I'll leave off the subscript (0). The problem is to solve the (time independent) equations

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = 0; \quad \nabla \times \mathbf{E} = i\omega \mathbf{B}; \\ \nabla \cdot \mathbf{B} = 0; \quad \nabla \times \mathbf{B} = -\frac{i\omega}{c^2} \mathbf{E}. \end{array} \right\}$$

From $\nabla \times \mathbf{E} = i\omega \mathbf{B}$ it follows that I can get $\mathbf{B}$ once I know $\mathbf{E}$, so I'll concentrate on the latter for the moment.

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E} = \nabla \times (i\omega \mathbf{B}) = i\omega \left(-\frac{i\omega}{c^2} \mathbf{E} \right) = \frac{\omega^2}{c^2} \mathbf{E}. \text{ So}$$

$\nabla^2 E_x = -\left(\frac{\omega}{c}\right)^2 E_x$; $\nabla^2 E_y = -\left(\frac{\omega}{c}\right)^2 E_y$; $\nabla^2 E_z = -\left(\frac{\omega}{c}\right)^2 E_z$. Solve each of these by separation of variables:

$$E_x(x, y, z) = X(x)Y(y)Z(z) \Rightarrow YZ \frac{d^2 X}{dx^2} + ZX \frac{d^2 Y}{dy^2} + XY \frac{d^2 Z}{dz^2} = -\left(\frac{\omega}{c}\right)^2 XYZ, \text{ or } \frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = -(\omega/c)^2. \text{ Each term must be a constant, so } \frac{d^2 X}{dx^2} = -k_x^2 X, \frac{d^2 Y}{dy^2} = -k_y^2 Y, \frac{d^2 Z}{dz^2} = -k_z^2 Z, \text{ with}$$

$$k_x^2 + k_y^2 + k_z^2 = (\omega/c)^2. \text{ The solution is}$$

$$E_x(x, y, z) = [A \sin(k_x x) + B \cos(k_x x)][C \sin(k_y y) + D \cos(k_y y)][E \sin(k_z z) + F \cos(k_z z)].$$

But $\mathbf{E}^\parallel = 0$ at the boundaries $\Rightarrow E_x = 0$ at $y = 0$ and $z = 0$, so $D = F = 0$, and $E_x = 0$ at $y = b$ and $z = d$, so $k_y = n\pi/b$ and $k_z = l\pi/d$, where n and l are integers. A similar argument applies to E_y and E_z . *Conclusion:*

$$E_x(x, y, z) = [A \sin(k_x x) + B \cos(k_x x)] \sin(k_y y) \sin(k_z z),$$

$$E_y(x, y, z) = \sin(k_x x) [C \sin(k_y y) + D \cos(k_y y)] \sin(k_z z),$$

$$E_z(x, y, z) = \sin(k_x x) \sin(k_y y) [E \sin(k_z z) + F \cos(k_z z)],$$

where $k_x = m\pi/a$. (Actually, there is no reason at this stage to assume that k_x , k_y , and k_z are the *same* for all three components, and I should really affix a second subscript (x for E_x , y for E_y , and z for E_z), but in a moment we shall see that *in fact* they *do* have to be the same, so to avoid cumbersome notation I'll *assume* they are from the start.)

Now $\nabla \cdot \mathbf{E} = 0 \Rightarrow k_x [A \cos(k_x x) - B \sin(k_x x)] \sin(k_y y) \sin(k_z z) + k_y \sin(k_x x) [C \cos(k_y y) - D \sin(k_y y)] \sin(k_z z) + k_z \sin(k_x x) \sin(k_y y) [E \cos(k_z z) - F \sin(k_z z)] = 0$. In particular, putting in $x = 0$, $k_x A \sin(k_y y) \sin(k_z z) = 0$, and hence $A = 0$. Likewise $y = 0 \Rightarrow C = 0$ and $z = 0 \Rightarrow E = 0$. (Moreover, if the k 's were *not* equal for different components, then by Fourier analysis this equation could not be satisfied (for all x , y , and z) unless the other three constants were *also* zero, and we'd be left with no field at all.) It follows that $-(Bk_x + Dk_y + Fk_z) = 0$ (in order that $\nabla \cdot \mathbf{E} = 0$), and we are left with

$$\mathbf{E} = B \cos(k_x x) \sin(k_y y) \sin(k_z z) \hat{\mathbf{x}} + D \sin(k_x x) \cos(k_y y) \sin(k_z z) \hat{\mathbf{y}} + F \sin(k_x x) \sin(k_y y) \cos(k_z z) \hat{\mathbf{z}},$$

with $k_x = (m\pi/a)$, $k_y = (n\pi/b)$, $k_z = (l\pi/d)$ (l, m, n all integers), and $Bk_x + Dk_y + Fk_z = 0$.

The corresponding magnetic field is given by $\mathbf{B} = -(i/\omega) \nabla \times \mathbf{E}$:

$$B_x = -\frac{i}{\omega} \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) = -\frac{i}{\omega} [Fk_y \sin(k_x x) \cos(k_y y) \cos(k_z z) - Dk_z \sin(k_x x) \cos(k_y y) \cos(k_z z)],$$

$$B_y = -\frac{i}{\omega} \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) = -\frac{i}{\omega} [Bk_z \cos(k_x x) \sin(k_y y) \cos(k_z z) - Fk_x \cos(k_x x) \sin(k_y y) \cos(k_z z)],$$

$$B_z = -\frac{i}{\omega} \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) = -\frac{i}{\omega} [Dk_x \cos(k_x x) \cos(k_y y) \sin(k_z z) - Bk_y \cos(k_x x) \cos(k_y y) \sin(k_z z)].$$

Or:

$$\mathbf{B} = -\frac{i}{\omega} (Fk_y - Dk_z) \sin(k_x x) \cos(k_y y) \cos(k_z z) \hat{\mathbf{x}} - \frac{i}{\omega} (Bk_z - Fk_x) \cos(k_x x) \sin(k_y y) \cos(k_z z) \hat{\mathbf{y}}$$

$$- \frac{i}{\omega} (Dk_x - Bk_y) \cos(k_x x) \cos(k_y y) \sin(k_z z) \hat{\mathbf{z}}.$$

These *automatically* satisfy the boundary condition $B^\perp = 0$ ($B_x = 0$ at $x = 0$ and $x = a$, $B_y = 0$ at $y = 0$ and $y = b$, and $B_z = 0$ at $z = 0$ and $z = d$).

As a check, let's see if $\nabla \cdot \mathbf{B} = 0$:

$$\begin{aligned}\nabla \cdot \mathbf{B} &= -\frac{i}{\omega}(Fk_y - Dk_z)k_x \cos(k_x x) \cos(k_y y) \cos(k_z z) - \frac{i}{\omega}(Bk_z - Fk_x)k_y \cos(k_x x) \cos(k_y y) \cos(k_z z) \\ &\quad - \frac{i}{\omega}(Dk_x - Bk_y)k_z \cos(k_x x) \cos(k_y y) \cos(k_z z) \\ &= -\frac{i}{\omega}(Fk_x k_y - Dk_x k_z + Bk_z k_y - Fk_x k_y + Dk_x k_z - Bk_y k_z) \cos(k_x x) \cos(k_y y) \cos(k_z z) = 0. \quad \checkmark\end{aligned}$$

The boxed equations satisfy all of Maxwell's equations, and they meet the boundary conditions. For TE modes, we pick $E_z = 0$, so $F = 0$ (and hence $Bk_x + Dk_y = 0$, leaving only the overall amplitude undetermined, for given l , m , and n); for TM modes we want $B_z = 0$ (so $Dk_x - Bk_y = 0$, again leaving only one amplitude undetermined, since $Bk_x + Dk_y + Fk_z = 0$). In either case (TE $_{lmn}$ or TM $_{lmn}$), the frequency is given by

$$\omega^2 = c^2(k_x^2 + k_y^2 + k_z^2) = c^2[(m\pi/a)^2 + (n\pi/b)^2 + (l\pi/d)^2], \text{ or } \boxed{\omega = c\pi\sqrt{(m/a)^2 + (n/b)^2 + (l/d)^2}}.$$

Chapter 10

Potentials and Fields

Problem 10.1

$$\begin{aligned}\square^2 V + \frac{\partial L}{\partial t} &= \nabla^2 V - \mu_0 \epsilon_0 \frac{\partial^2 V}{\partial t^2} + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) + \mu_0 \epsilon_0 \frac{\partial^2 V}{\partial t^2} = \nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = -\frac{1}{\epsilon_0} \rho. \quad \checkmark \\ \square^2 \mathbf{A} - \nabla L &= \nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} - \nabla \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) = -\mu_0 \mathbf{J}. \quad \checkmark\end{aligned}$$

Problem 10.2

(a) $W = \frac{1}{2} \int \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) d\tau$. At $t_1 = d/c$, $x \geq d = ct_1$, so $\mathbf{E} = 0$, $\mathbf{B} = 0$, and hence $\boxed{W(t_1) = 0}$.

At $T_2 = (d + h)/c$, $ct_2 = d + h$:

$$\mathbf{E} = -\frac{\mu_0 k}{2}(d + h - x) \hat{\mathbf{z}}, \quad \mathbf{B} = \frac{1}{c} \frac{\mu_0 k}{2}(d + h - x) \hat{\mathbf{y}},$$

so $B^2 = \frac{1}{c^2} E^2$, and

$$\left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) = \epsilon_0 \left(E^2 + \frac{1}{\mu_0 \epsilon_0} \frac{1}{c^2} E^2 \right) = 2\epsilon_0 E^2.$$

Therefore

$$W(t_2) = \frac{1}{2} (2\epsilon_0) \frac{\mu_0^2 k^2}{4} \int_d^{(d+h)} (d + h - x)^2 dx (lw) = \frac{\epsilon_0 \mu_0^2 k^2 lw}{4} \left[-\frac{(d + h - x)^3}{3} \right]_d^{d+h} = \boxed{\frac{\epsilon_0 \mu_0^2 k^2 lw h^3}{12}}.$$

(b) $\mathbf{S}(x) = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0 c} E^2 [-\hat{\mathbf{z}} \times (\pm \hat{\mathbf{y}})] = \pm \frac{1}{\mu_0 c} E^2 \hat{\mathbf{x}} = \boxed{\pm \frac{\mu_0 k^2}{4c} (ct - |x|)^2 \hat{\mathbf{x}}}$

(plus sign for $x > 0$, as here). For $|x| > ct$, $\mathbf{S} = 0$.

So the energy per unit time entering the box in this time interval is

$$\frac{dW}{dt} = P = \int \mathbf{S}(d) \cdot d\mathbf{a} = \boxed{\frac{\mu_0 k^2 lw}{4c} (ct - d)^2}.$$

Note that no energy flows out the top, since $\mathbf{S}(d + h) = 0$.

$$(c) \quad W = \int_{t_1}^{t_2} P dt = \frac{\mu_0 k^2 l w}{4c} \int_{d/c}^{(d+h)/c} (ct-d)^2 dt = \frac{\mu_0 k^2 l w}{4c} \left[\frac{(ct-d)^3}{3c} \right]_{d/c}^{(d+h)/c} = \boxed{\frac{\mu_0 k^2 l w h^3}{12c^2}}.$$

Since $1/c^2 = \mu_0 \epsilon_0$, this agrees with the answer to (a).

Problem 10.3

(a)

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} = \boxed{\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}}, \quad \mathbf{B} = \nabla \times \mathbf{A} = \boxed{\mathbf{0}}.$$

This is a funny set of potentials for a stationary point charge q at the origin. ($V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$, $\mathbf{A} = \mathbf{0}$ would, of course, be the customary choice.) Evidently $\rho = q\delta^3(\mathbf{r})$; $\mathbf{J} = \mathbf{0}$.

(b)

$$V' = V - \frac{\partial \lambda}{\partial t} = 0 - \left(-\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right) = \boxed{\frac{1}{4\pi\epsilon_0} \frac{q}{r}}; \quad \mathbf{A}' = \mathbf{A} + \nabla \lambda = -\frac{1}{4\pi\epsilon_0} \frac{qt}{r^2} \hat{\mathbf{r}} + \left(-\frac{1}{4\pi\epsilon_0} qt \right) \left(-\frac{1}{r^2} \hat{\mathbf{r}} \right) = \boxed{\mathbf{0}}.$$

This gauge function transforms the “funny” potentials into the “ordinary” potentials of a stationary point charge.

Problem 10.4

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} = -A_0 \cos(kx - \omega t) \hat{\mathbf{y}}(-\omega) = \boxed{A_0 \omega \cos(kx - \omega t) \hat{\mathbf{y}}},$$

$$\mathbf{B} = \nabla \times \mathbf{A} = \hat{\mathbf{z}} \frac{\partial}{\partial x} [A_0 \sin(kx - \omega t)] = \boxed{A_0 k \cos(kx - \omega t) \hat{\mathbf{z}}}.$$

Hence $\nabla \cdot \mathbf{E} = 0$ ✓, $\nabla \cdot \mathbf{B} = 0$ ✓.

$$\nabla \times \mathbf{E} = \hat{\mathbf{z}} \frac{\partial}{\partial x} [A_0 \omega \cos(kx - \omega t)] = -A_0 \omega k \sin(kx - \omega t) \hat{\mathbf{z}}, \quad -\frac{\partial \mathbf{B}}{\partial t} = -A_0 \omega k \sin(kx - \omega t) \hat{\mathbf{z}},$$

so $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ ✓.

$$\nabla \times \mathbf{B} = -\hat{\mathbf{y}} \frac{\partial}{\partial x} [A_0 k \cos(kx - \omega t)] = A_0 k^2 \sin(kx - \omega t) \hat{\mathbf{y}}, \quad \frac{\partial \mathbf{E}}{\partial t} = A_0 \omega^2 \sin(kx - \omega t) \hat{\mathbf{y}}.$$

So $\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$ *provided* $k^2 = \mu_0 \epsilon_0 \omega^2$, or, since $c^2 = 1/\mu_0 \epsilon_0$, $\omega = ck$.

Problem 10.5

Ex. 10.1: $\nabla \cdot \mathbf{A} = 0$; $\frac{\partial V}{\partial t} = 0$. Both Coulomb and Lorentz.

Prob. 10.3: $\nabla \cdot \mathbf{A} = -\frac{qt}{4\pi\epsilon_0} \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = -\frac{qt}{\epsilon_0} \delta^3(\mathbf{r})$; $\frac{\partial V}{\partial t} = 0$. Neither.

Prob. 10.4: $\nabla \cdot \mathbf{A} = 0$; $\frac{\partial V}{\partial t} = 0$. Both.

Problem 10.6

Suppose $\nabla \cdot \mathbf{A} \neq -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$. (Let $\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} = \Phi$ —some known function.) We want to pick λ such that $\mathbf{A}'$ and V' (Eq. 10.7) *do* obey $\nabla \cdot \mathbf{A}' = -\mu_0 \epsilon_0 \frac{\partial V'}{\partial t}$.

$$\nabla \cdot \mathbf{A}' + \mu_0 \epsilon_0 \frac{\partial V'}{\partial t} = \nabla \cdot \mathbf{A} + \nabla^2 \lambda + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} - \mu_0 \epsilon_0 \frac{\partial^2 \lambda}{\partial t^2} = \Phi + \square^2 \lambda.$$

This will be zero provided we pick for λ the solution to $\square^2 \lambda = -\Phi$, which by hypothesis (and in fact) we know how to solve.

We *could* always find a gauge in which $V' = 0$, simply by picking $\lambda = \int_0^t V dt'$. We *cannot* in general pick $\mathbf{A} = \mathbf{0}$ —this would make $\mathbf{B} = \mathbf{0}$. [Finding such a gauge function would amount to expressing $\mathbf{A}$ as $-\nabla \lambda$, and we know that vector functions *cannot* in general be written as gradients—only if they happen to have curl zero, which $\mathbf{A}$ (ordinarily) does *not*.]

Problem 10.7

(a) Using Eq. 1.99,

$$\nabla \cdot \mathbf{J} = -\frac{\dot{q}}{4\pi} \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = -\frac{\dot{q}}{4\pi} 4\pi \delta^3(\mathbf{r}) = -\dot{q} \delta^3(\mathbf{r}) = -\frac{\partial \rho}{\partial t}. \quad \checkmark$$

(b) From Eq. 10.10,

$$V(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \int \frac{\delta^3(\mathbf{r}')}{r} d\tau' = \boxed{\frac{1}{4\pi\epsilon_0} \frac{q(t)}{r}}.$$

By symmetry, $\mathbf{B} = \mathbf{0}$ (what direction could it point?), so $\nabla \times \mathbf{A} = \mathbf{0}$, $\nabla \cdot \mathbf{A} = 0$, and $\mathbf{A} \rightarrow \mathbf{0}$ at infinity. Evidently $\boxed{\mathbf{A} = \mathbf{0}}$.

(c) $\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} = \boxed{\frac{1}{4\pi\epsilon_0} \frac{q(t)}{r^2} \hat{\mathbf{r}}}$. $\boxed{\mathbf{B} = \mathbf{0}}$. Checking Maxwell's equations:

$$\nabla \cdot \mathbf{E} = \frac{q}{4\pi\epsilon_0} \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = \frac{q}{4\pi\epsilon_0} 4\pi \delta^3(\mathbf{r}) = \frac{q \delta^3(\mathbf{r})}{\epsilon_0} = \frac{\rho}{\epsilon_0}. \quad \checkmark$$

$$\nabla \cdot \mathbf{B} = 0. \quad \checkmark$$

$$\nabla \times \mathbf{E} = \mathbf{0} = -\frac{\partial \mathbf{B}}{\partial t}. \quad \checkmark$$

$$\nabla \times \mathbf{B} = \mathbf{0}; \quad \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \left(-\frac{1}{4\pi} \frac{\dot{q}}{r^2} \hat{\mathbf{r}} \right) + \mu_0 \epsilon_0 \left(\frac{\dot{q}}{4\pi\epsilon_0} \frac{1}{r^2} \hat{\mathbf{r}} \right) = \mathbf{0}. \quad \checkmark$$

[Note that the displacement current exactly cancels the conduction current. Physically, this configuration is a point charge at the origin that is changing with time as current flows in symmetrically (from infinity).]

Problem 10.8

Noting the $\mathbf{A}$ is independent of t and $\mathbf{B}$ is independent of $\mathbf{r}$, use Eq. 10.19:

$$\begin{aligned} \frac{d\mathbf{A}}{dt} &= \frac{\partial \mathbf{A}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{A} = (\mathbf{v} \cdot \nabla) \left[-\frac{1}{2} (\mathbf{r} \times \mathbf{B}) \right] \\ &= -\frac{1}{2} \left[v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z} \right] [(yB_z - zB_y) \hat{\mathbf{x}} + (zB_x - xB_z) \hat{\mathbf{y}} + (xB_y - yB_x) \hat{\mathbf{z}}] \\ &= -\frac{1}{2} [v_x(-B_z \hat{\mathbf{y}} + B_y \hat{\mathbf{z}}) + v_y(B_z \hat{\mathbf{x}} - B_x \hat{\mathbf{z}}) + v_z(-B_y \hat{\mathbf{x}} + B_x \hat{\mathbf{y}})] \\ &= -\frac{1}{2} [(v_y B_z - v_z B_y) \hat{\mathbf{x}} + (v_z B_x - v_x B_z) \hat{\mathbf{y}} + (v_x B_y - v_y B_x) \hat{\mathbf{z}}] = -\frac{1}{2} (\mathbf{v} \times \mathbf{B}). \quad \checkmark \end{aligned}$$

Equation 10.20 says

$$\frac{d}{dt}(\mathbf{p} + q\mathbf{A}) = \frac{d\mathbf{p}}{dt} - \frac{q}{2}(\mathbf{v} \times \mathbf{B}) = q\nabla(\mathbf{v} \cdot \mathbf{A}) = -\frac{q}{2}\nabla[\mathbf{v} \cdot (\mathbf{r} \times \mathbf{B})],$$

or

$$\frac{d\mathbf{p}}{dt} = \frac{q}{2}(\mathbf{v} \times \mathbf{B}) - \frac{q}{2}\nabla[\mathbf{r} \cdot (\mathbf{B} \times \mathbf{v})].$$

Now, for a vector $\mathbf{c}$ that is independent of position,

$$\nabla(\mathbf{r} \cdot \mathbf{c}) = \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) (xc_x + yc_y + zc_z) = c_x \hat{\mathbf{x}} + c_y \hat{\mathbf{y}} + c_z \hat{\mathbf{z}} = \mathbf{c}.$$

In this case $\mathbf{c} = (\mathbf{B} \times \mathbf{v}) = -(\mathbf{v} \times \mathbf{B})$, so

$$\frac{d\mathbf{p}}{dt} = \frac{q}{2}(\mathbf{v} \times \mathbf{B}) + \frac{q}{2}(\mathbf{v} \times \mathbf{B}) = q(\mathbf{v} \times \mathbf{B}). \quad \checkmark$$

Problem 10.9

$$\frac{d}{dt}(T + qV) = \frac{d}{dt} \left(\frac{1}{2}mv^2 \right) + q \frac{dV}{dt} = m\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} + q \left[\frac{\partial V}{\partial t} + (\mathbf{v} \cdot \nabla)V \right] = \mathbf{v} \cdot \mathbf{F} + q \left[\frac{\partial V}{\partial t} + (\mathbf{v} \cdot \nabla)V \right].$$

But Eq. 10.17 says

$$\mathbf{v} \cdot \mathbf{F} = q\mathbf{v} \cdot \left[-\nabla V - \frac{\partial \mathbf{A}}{\partial t} + \mathbf{v} \times (\nabla \times \mathbf{A}) \right] = -q \left[\mathbf{v} \cdot \nabla V + \mathbf{v} \cdot \frac{\partial \mathbf{A}}{\partial t} \right],$$

so

$$\frac{d}{dt}(T + qV) = -q \left[(\mathbf{v} \cdot \nabla)V + \mathbf{v} \cdot \frac{\partial \mathbf{A}}{\partial t} \right] + q \left[\frac{\partial V}{\partial t} + (\mathbf{v} \cdot \nabla)V \right] = q \left(\frac{\partial V}{\partial t} - \mathbf{v} \cdot \frac{\partial \mathbf{A}}{\partial t} \right) = \frac{\partial}{\partial t} [q(V - \mathbf{v} \cdot \mathbf{A})].$$

Problem 10.10

From the product rule:

$$\nabla \cdot \left(\frac{\mathbf{J}}{r} \right) = \frac{1}{r}(\nabla \cdot \mathbf{J}) + \mathbf{J} \cdot \left(\nabla \frac{1}{r} \right), \quad \nabla' \cdot \left(\frac{\mathbf{J}}{r} \right) = \frac{1}{r}(\nabla' \cdot \mathbf{J}) + \mathbf{J} \cdot \left(\nabla' \frac{1}{r} \right).$$

But $\nabla \frac{1}{r} = -\nabla' \frac{1}{r}$, since $\mathbf{r} = \mathbf{r} - \mathbf{r}'$. So

$$\nabla \cdot \left(\frac{\mathbf{J}}{r} \right) = \frac{1}{r}(\nabla \cdot \mathbf{J}) - \mathbf{J} \cdot \left(\nabla' \frac{1}{r} \right) = \frac{1}{r}(\nabla \cdot \mathbf{J}) + \frac{1}{r}(\nabla' \cdot \mathbf{J}) - \nabla' \cdot \left(\frac{\mathbf{J}}{r} \right).$$

But

$$\nabla \cdot \mathbf{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} = \frac{\partial J_x}{\partial t_r} \frac{\partial t_r}{\partial x} + \frac{\partial J_y}{\partial t_r} \frac{\partial t_r}{\partial y} + \frac{\partial J_z}{\partial t_r} \frac{\partial t_r}{\partial z},$$

and

$$\frac{\partial t_r}{\partial x} = -\frac{1}{c} \frac{\partial r}{\partial x}, \quad \frac{\partial t_r}{\partial y} = -\frac{1}{c} \frac{\partial r}{\partial y}, \quad \frac{\partial t_r}{\partial z} = -\frac{1}{c} \frac{\partial r}{\partial z},$$

so

$$\nabla \cdot \mathbf{J} = -\frac{1}{c} \left[\frac{\partial J_x}{\partial t_r} \frac{\partial z}{\partial x} + \frac{\partial J_y}{\partial t_r} \frac{\partial z}{\partial y} + \frac{\partial J_z}{\partial t_r} \frac{\partial z}{\partial z} \right] = -\frac{1}{c} \frac{\partial \mathbf{J}}{\partial t_r} \cdot (\nabla z).$$

Similarly,

$$\nabla' \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} - \frac{1}{c} \frac{\partial \mathbf{J}}{\partial t_r} \cdot (\nabla' z).$$

[The first term arises when we differentiate with respect to the *explicit* $\mathbf{r}'$, and use the continuity equation.] thus

$$\nabla \cdot \left(\frac{\mathbf{J}}{z} \right) = \frac{1}{z} \left[-\frac{1}{c} \frac{\partial \mathbf{J}}{\partial t_r} \cdot (\nabla' z) \right] + \frac{1}{z} \left[-\frac{\partial \rho}{\partial t} - \frac{1}{c} \frac{\partial \mathbf{J}}{\partial t_r} \cdot (\nabla' z) \right] - \nabla \cdot \left(\frac{\mathbf{J}}{z} \right) = -\frac{1}{z} \frac{\partial \rho}{\partial t} - \nabla' \cdot \left(\frac{\mathbf{J}}{z} \right)$$

(the other two terms cancel, since $\nabla z = -\nabla' z$). Therefore:

$$\nabla \cdot \mathbf{A} = \frac{\mu_0}{4\pi} \left[-\frac{\partial}{\partial t} \int \frac{\rho}{z} d\tau - \int \nabla' \cdot \left(\frac{\mathbf{J}}{z} \right) d\tau \right] = -\mu_0 \epsilon_0 \frac{\partial}{\partial t} \left[\frac{1}{4\pi \epsilon_0} \int \frac{\rho}{z} d\tau \right] - \frac{\mu_0}{4\pi} \oint \frac{\mathbf{J}}{z} \cdot d\mathbf{a}.$$

The last term is over the surface at “infinity”, where $\mathbf{J} = 0$, so it's zero. Therefore $\nabla \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$. ✓

Problem 10.11

(a) As in Ex. 10.2, for $t < s/c$, $\mathbf{A} = \mathbf{0}$; for $t > s/c$,

$$\begin{aligned} \mathbf{A}(s, t) &= \left(\frac{\mu_0}{4\pi} \hat{\mathbf{z}} \right) 2 \int_0^{\sqrt{(ct)^2 - s^2}} \frac{k(t - \sqrt{s^2 + z^2}/c)}{\sqrt{s^2 + z^2}} dz = \frac{\mu_0 k}{2\pi} \hat{\mathbf{z}} \left\{ t \int_0^{\sqrt{(ct)^2 - s^2}} \frac{dz}{\sqrt{s^2 + z^2}} - \frac{1}{c} \int_0^{\sqrt{(ct)^2 - s^2}} dz \right\} \\ &= \left(\frac{\mu_0 k}{2\pi} \hat{\mathbf{z}} \right) \left[t \ln \left(\frac{ct + \sqrt{(ct)^2 - s^2}}{s} \right) - \frac{1}{c} \sqrt{(ct)^2 - s^2} \right]. \quad \text{Accordingly,} \end{aligned}$$

$$\begin{aligned} \mathbf{E}(s, t) &= -\frac{\partial \mathbf{A}}{\partial t} = -\frac{\mu_0 k}{2\pi} \hat{\mathbf{z}} \left\{ \ln \left(\frac{ct + \sqrt{(ct)^2 - s^2}}{s} \right) + \right. \\ &\quad \left. t \left(\frac{s}{ct + \sqrt{(ct)^2 - s^2}} \right) \left(\frac{1}{s} \right) \left(c + \frac{1}{2} \frac{2c^2 t}{\sqrt{(ct)^2 - s^2}} \right) - \frac{1}{2c} \frac{2c^2 t}{\sqrt{(ct)^2 - s^2}} \right\} \\ &= -\frac{\mu_0 k}{2\pi} \hat{\mathbf{z}} \left\{ \ln \left(\frac{ct + \sqrt{(ct)^2 - s^2}}{s} \right) + \frac{ct}{\sqrt{(ct)^2 - s^2}} - \frac{ct}{\sqrt{(ct)^2 - s^2}} \right\} \\ &= \boxed{-\frac{\mu_0 k}{2\pi} \ln \left(\frac{ct + \sqrt{(ct)^2 - s^2}}{s} \right) \hat{\mathbf{z}}} \quad (\text{or zero, for } t < s/c). \end{aligned}$$

$$\begin{aligned} \mathbf{B}(s, t) &= -\frac{\partial A_z}{\partial s} \hat{\phi} \\ &= -\frac{\mu_0 k}{2\pi} \left\{ t \left(\frac{s}{ct + \sqrt{(ct)^2 - s^2}} \right) \left[\frac{s \frac{1}{2} \frac{(-2s)}{\sqrt{(ct)^2 - s^2}} - ct - \sqrt{(ct)^2 - s^2}}{s^2} \right] - \frac{1}{2c} \frac{(-2s)}{\sqrt{(ct)^2 - s^2}} \right\} \hat{\phi} \\ &= -\frac{\mu_0 k}{2\pi} \left\{ \frac{-ct^2}{s\sqrt{(ct)^2 - s^2}} + \frac{s}{c\sqrt{(ct)^2 - s^2}} \right\} \hat{\phi} = -\frac{\mu_0 k}{2\pi} \frac{(-c^2 t^2 + s^2)}{sc\sqrt{(ct)^2 - s^2}} \hat{\phi} = \boxed{\frac{\mu_0 k}{2\pi sc} \sqrt{(ct)^2 - s^2} \hat{\phi}}. \end{aligned}$$

(b) $\mathbf{A}(s, t) = \frac{\mu_0}{4\pi} \hat{\mathbf{z}} \int_{-\infty}^{\infty} \frac{q_0 \delta(t - \mathcal{Z}/c)}{\mathcal{Z}} dz$. But $\mathcal{Z} = \sqrt{s^2 + z^2}$, so the integrand is even in z :

$$\mathbf{A}(s, t) = \left(\frac{\mu_0 q_0}{4\pi} \hat{\mathbf{z}} \right) 2 \int_0^{\infty} \frac{\delta(t - \mathcal{Z}/c)}{\mathcal{Z}} dz.$$

Now $z = \sqrt{\mathcal{Z}^2 - s^2} \Rightarrow dz = \frac{1}{2} \frac{2\mathcal{Z} d\mathcal{Z}}{\sqrt{\mathcal{Z}^2 - s^2}} = \frac{\mathcal{Z} d\mathcal{Z}}{\sqrt{\mathcal{Z}^2 - s^2}}$, and $z = 0 \Rightarrow \mathcal{Z} = s$, $z = \infty \Rightarrow \mathcal{Z} = \infty$. So:

$$\mathbf{A}(s, t) = \frac{\mu_0 q_0}{2\pi} \hat{\mathbf{z}} \int_s^{\infty} \frac{1}{\mathcal{Z}} \delta\left(t - \frac{\mathcal{Z}}{c}\right) \frac{\mathcal{Z} d\mathcal{Z}}{\sqrt{\mathcal{Z}^2 - s^2}}.$$

Now $\delta(t - \mathcal{Z}/c) = c\delta(\mathcal{Z} - ct)$ (Ex. 1.15); therefore $\mathbf{A} = \frac{\mu_0 q_0}{2\pi} \hat{\mathbf{z}} c \int_s^{\infty} \frac{\delta(\mathcal{Z} - ct)}{\sqrt{\mathcal{Z}^2 - s^2}} d\mathcal{Z}$, so

$$\mathbf{A}(s, t) = \frac{\mu_0 q_0 c}{2\pi} \frac{1}{\sqrt{(ct)^2 - s^2}} \hat{\mathbf{z}} \quad (\text{or zero, if } ct < s);$$

$$\mathbf{E}(s, t) = -\frac{\partial \mathbf{A}}{\partial t} = -\frac{\mu_0 q_0 c}{2\pi} \left(-\frac{1}{2}\right) \frac{2c^2 t}{[(ct)^2 - s^2]^{3/2}} \hat{\mathbf{z}} = \boxed{\frac{\mu_0 q_0 c^3 t}{2\pi [(ct)^2 - s^2]^{3/2}} \hat{\mathbf{z}}} \quad (\text{or zero, for } t < s/c);$$

$$\mathbf{B}(s, t) = -\frac{\partial \mathbf{A}_z}{\partial t} \hat{\phi} = -\frac{\mu_0 q_0 c}{2\pi} \left(-\frac{1}{2}\right) \frac{-2s}{[(ct)^2 - s^2]^{3/2}} \hat{\phi} = \boxed{\frac{-\mu_0 q_0 c s}{2\pi [(ct)^2 - s^2]^{3/2}} \hat{\phi}} \quad (\text{or zero, for } t < s/c).$$

Problem 10.12

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I}(t_r)}{\mathcal{Z}} d\mathbf{l} = \frac{\mu_0 k}{4\pi} \int \frac{(t - \mathcal{Z}/c)}{\mathcal{Z}} d\mathbf{l} = \frac{\mu_0 k}{4\pi} \left\{ t \int \frac{d\mathbf{l}}{\mathcal{Z}} - \frac{1}{c} \int d\mathbf{l} \right\}.$$

But for the complete loop, $\int d\mathbf{l} = 0$, so $\mathbf{A} = \frac{\mu_0 k t}{4\pi} \left\{ \frac{1}{a} \int_1 d\mathbf{l} + \frac{1}{b} \int_2 d\mathbf{l} + 2\hat{\mathbf{x}} \int_a^b \frac{dx}{x} \right\}$. Here $\int_1 d\mathbf{l} = 2a\hat{\mathbf{x}}$ (inner circle), $\int_2 d\mathbf{l} = -2b\hat{\mathbf{x}}$ (outer circle), so

$$\mathbf{A} = \frac{\mu_0 k t}{4\pi} \left[\frac{1}{a}(2a) + \frac{1}{b}(-2b) + 2\ln(b/a) \right] \hat{\mathbf{x}} \Rightarrow \boxed{\mathbf{A} = \frac{\mu_0 k t}{2\pi} \ln(b/a) \hat{\mathbf{x}}}, \quad \mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = \boxed{-\frac{\mu_0 k}{2\pi} \ln(b/a) \hat{\mathbf{x}}}.$$

The changing magnetic field induces the electric field. Since we only know $\mathbf{A}$ at *one point* (the center), we can't compute $\nabla \times \mathbf{A}$ to get $\mathbf{B}$.

Problem 10.13

In this case $\dot{\rho}(\mathbf{r}, t) = \dot{\rho}(\mathbf{r}, 0)$ and $\dot{\mathbf{J}}(\mathbf{r}, t) = 0$, so Eq. 10.36 $\Rightarrow$

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= \frac{1}{4\pi\epsilon_0} \int \left[\frac{\rho(\mathbf{r}', 0) + \dot{\rho}(\mathbf{r}', 0)t_r}{\mathcal{Z}^2} + \frac{\dot{\rho}(\mathbf{r}', 0)}{c\mathcal{Z}} \right] \hat{\mathbf{z}} d\tau', \text{ but } t_r = t - \frac{\mathcal{Z}}{c} \text{ (Eq. 10.18), so} \\ &= \frac{1}{4\pi\epsilon_0} \int \left[\frac{\rho(\mathbf{r}', 0) + \dot{\rho}(\mathbf{r}', 0)t}{\mathcal{Z}^2} - \frac{\dot{\rho}(\mathbf{r}', 0)(\mathcal{Z}/c)}{\mathcal{Z}^2} + \frac{\dot{\rho}(\mathbf{r}', 0)}{c\mathcal{Z}} \right] \hat{\mathbf{z}} d\tau' = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t)}{\mathcal{Z}^2} \hat{\mathbf{z}} d\tau'. \quad \text{qed} \end{aligned}$$

Problem 10.14

In this approximation we're dropping the higher derivatives of $\mathbf{J}$, so $\dot{\mathbf{J}}(t_r) = \dot{\mathbf{J}}(t)$, and Eq. 10.38 $\Rightarrow$

$$\begin{aligned}\mathbf{B}(\mathbf{r}, t) &= \frac{\mu_0}{4\pi} \int \frac{1}{r^2} \left[\mathbf{J}(\mathbf{r}', t) + (t_r - t)\dot{\mathbf{J}}(\mathbf{r}', t) + \frac{r}{c}\dot{\mathbf{J}}(\mathbf{r}', t) \right] \times \hat{\mathbf{r}} \, d\tau', \text{ but } t_r - t = -\frac{r}{c} \text{ (Eq. 10.25), so} \\ &= \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t) \times \hat{\mathbf{r}}}{r^2} \, d\tau'. \quad \text{qed}\end{aligned}$$

Problem 10.15

At time t the charge is at $\mathbf{r}(t) = a[\cos(\omega t)\hat{\mathbf{x}} + \sin(\omega t)\hat{\mathbf{y}}]$, so $\mathbf{v}(t) = \omega a[-\sin(\omega t)\hat{\mathbf{x}} + \cos(\omega t)\hat{\mathbf{y}}]$. Therefore $\mathbf{r} = z\hat{\mathbf{z}} - a[\cos(\omega t_r)\hat{\mathbf{x}} + \sin(\omega t_r)\hat{\mathbf{y}}]$, and hence $r^2 = z^2 + a^2$ (of course), and $r = \sqrt{z^2 + a^2}$.

$$\hat{\mathbf{r}} \cdot \mathbf{v} = \frac{1}{r}(\mathbf{r} \cdot \mathbf{v}) = \frac{1}{r} \{ -\omega a^2 [-\sin(\omega t_r)\cos(\omega t_r) + \sin(\omega t_r)\cos(\omega t_r)] \} = 0, \text{ so } \left(1 - \frac{\hat{\mathbf{r}} \cdot \mathbf{v}}{c} \right) = 1.$$

Therefore

$$V(z, t) = \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{z^2 + a^2}}; \quad \mathbf{A}(z, t) = \frac{q\omega a}{4\pi\epsilon_0 c^2 \sqrt{z^2 + a^2}} [-\sin(\omega t_r)\hat{\mathbf{x}} + \cos(\omega t_r)\hat{\mathbf{y}}], \text{ where } t_r = t - \frac{\sqrt{z^2 + a^2}}{c}.$$

Problem 10.16

Term under square root in (Eq. 10.49) is:

$$\begin{aligned}I &= c^4 t^2 - 2c^2 t(\mathbf{r} \cdot \mathbf{v}) + (\mathbf{r} \cdot \mathbf{v})^2 + c^2 r^2 - c^4 t^2 - v^2 r^2 + v^2 c^2 t^2 \\ &= (\mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)r^2 + c^2(vt)^2 - 2c^2(\mathbf{r} \cdot \mathbf{v}t). \quad \text{put in } \mathbf{v}t = \mathbf{r} - \mathbf{R}. \\ &= (\mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)r^2 + c^2(r^2 + R^2 - 2\mathbf{r} \cdot \mathbf{R}) - 2c^2(r^2 - \mathbf{r} \cdot \mathbf{R}) = (\mathbf{r} \cdot \mathbf{v})^2 - r^2 v^2 + c^2 R^2.\end{aligned}$$

but

$$\begin{aligned}(\mathbf{r} \cdot \mathbf{v})^2 - r^2 v^2 &= ((\mathbf{R} + \mathbf{v}t) \cdot \mathbf{v})^2 - (\mathbf{R} + \mathbf{v}t)^2 v^2 \\ &= (\mathbf{R} \cdot \mathbf{v})^2 + v^4 t^2 + 2(\mathbf{R} \cdot \mathbf{v})v^2 t - R^2 v^2 - 2(\mathbf{R} \cdot \mathbf{v})tv^2 - v^2 t^2 v^2 \\ &= (\mathbf{R} \cdot \mathbf{v})^2 - R^2 v^2 = R^2 v^2 \cos^2 \theta - R^2 v^2 = -R^2 v^2 (1 - \cos^2 \theta) \\ &= -R^2 v^2 \sin^2 \theta.\end{aligned}$$

Therefore

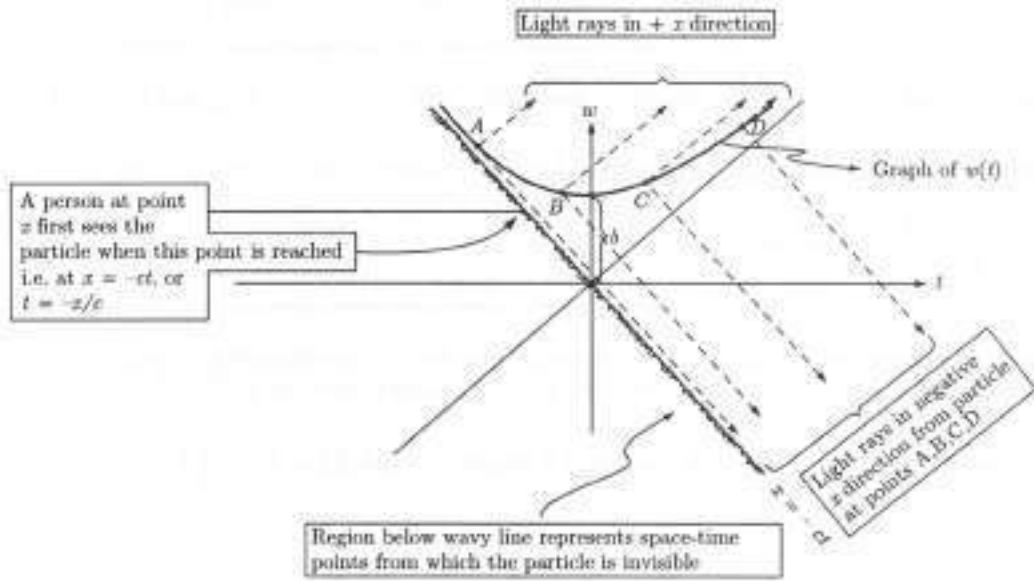
$$I = -R^2 v^2 \sin^2 \theta + c^2 R^2 = c^2 R^2 \left(1 - \frac{v^2}{c^2} \sin^2 \theta \right).$$

Hence

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{q}{R\sqrt{1 - \frac{v^2}{c^2} \sin^2 \theta}}. \quad \text{qed}$$

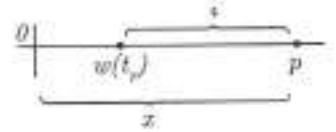
Problem 10.17

Once seen, from a given point x , the particle will forever remain in view—to disappear it would have to travel faster than light.



Problem 10.18

First calculate t_r : $t_r = t - |\mathbf{r} - \mathbf{w}(t_r)|/c \Rightarrow$
 $-c(t_r - t) = x - \sqrt{b^2 + c^2 t_r^2} \Rightarrow c(t_r - t) + x = \sqrt{b^2 + c^2 t_r^2};$
 $c^2 t_r^2 - 2c^2 t_r t + c^2 t^2 + 2xct_r - 2xct + x^2 = b^2 + c^2 t_r^2;$
 $2ct_r(x - ct) + (x^2 - 2xct + c^2 t^2) = b^2;$
 $2ct_r(x - ct) = b^2 - (x - ct)^2, \text{ or } t_r = \frac{b^2 - (x - ct)^2}{2c(x - ct)}.$



Now $V(x, t) = \frac{1}{4\pi\epsilon_0} \frac{qc}{(c - \mathbf{r} \cdot \mathbf{v})}$, and $c - \mathbf{r} \cdot \mathbf{v} = c(c - v)$; $c = c(t - t_r)$.

$$v = \frac{1}{2} \frac{1}{\sqrt{b^2 + c^2 t_r^2}} 2c^2 t_r = \frac{c^2 t_r}{c(t_r - t) + x} = \frac{c^2 t_r}{ct_r + (x - ct)}; \quad (c - v) = \frac{c^2 t_r + c(x - ct) - c^2 t_r}{ct_r + (x - ct)} = \frac{c(x - ct)}{ct_r + (x - ct)};$$

$$c - \mathbf{r} \cdot \mathbf{v} = \frac{c(t - t_r)c(x - ct)}{ct_r + (x - ct)} = \frac{c^2(t - t_r)(x - ct)}{ct_r + (x - ct)}; \quad ct_r + (x - ct) = \frac{b^2 - (x - ct)^2}{2(x - ct)} + (x - ct) = \frac{b^2 + (x - ct)^2}{2(x - ct)};$$

$$t - t_r = \frac{2ct(x - ct) - b^2 + (x - ct)^2}{2c(x - ct)} = \frac{(x - ct)(x + ct) - b^2}{2c(x - ct)} = \frac{(x^2 - c^2 t^2 - b^2)}{2c(x - ct)}.$$
 Therefore
$$\frac{1}{c - \mathbf{r} \cdot \mathbf{v}} = \left[\frac{b^2 + (x - ct)^2}{2(x - ct)} \right] \frac{1}{c^2(x - ct)} \frac{2c(x - ct)}{b^2 + (x - ct)^2} = \frac{b^2 + (x - ct)^2}{c(x - ct) [2ct(x - ct) - b^2 + (x - ct)^2]}.$$

The term in square brackets simplifies to $(2ct + x - ct)(x - ct) - b^2 = (x + ct)(x - ct) - b^2 = x^2 - c^2 t^2 - b^2$.

So $V(x, t) = \frac{q}{4\pi\epsilon_0} \frac{b^2 + (x - ct)^2}{(x - ct)(x^2 - c^2 t^2 - b^2)}.$

Meanwhile

$$\mathbf{A} = \frac{V}{c^2} \mathbf{v} = \frac{c^2 t_r}{ct_r + (x - ct)} \frac{V}{c^2} \hat{\mathbf{x}} = \left[\frac{b^2 - (x - ct)^2}{2c(x - ct)} \right] \frac{2(x - ct)}{b^2 + (x - ct)^2} \frac{q}{4\pi\epsilon_0} \frac{b^2 + (x - ct)^2}{(x - ct)(x^2 - c^2 t^2 - b^2)} \hat{\mathbf{x}}$$

$$= \frac{q}{4\pi\epsilon_0 c} \frac{b^2 - (x - ct)^2}{(x - ct)(x^2 - c^2 t^2 - b^2)} \hat{\mathbf{x}}.$$

Problem 10.19

From Eq. 10.44, $c(t - t_r) = \mathbf{r} \cdot \hat{\mathbf{r}} \Rightarrow c^2(t - t_r)^2 = \mathbf{r} \cdot \mathbf{r} = r^2$. Differentiate with respect to t :

$2c^2(t - t_r) \left(1 - \frac{\partial t_r}{\partial t}\right) = 2\mathbf{r} \cdot \frac{\partial \hat{\mathbf{r}}}{\partial t}$, or $c\mathbf{r} \left(1 - \frac{\partial t_r}{\partial t}\right) = \mathbf{r} \cdot \frac{\partial \mathbf{r}}{\partial t}$. Now $\mathbf{r} = \mathbf{r} - \mathbf{w}(t_r)$, so

$$\frac{\partial \mathbf{r}}{\partial t} = -\frac{\partial \mathbf{w}}{\partial t} = -\frac{\partial \mathbf{w}}{\partial t_r} \frac{\partial t_r}{\partial t} = -\mathbf{v} \frac{\partial t_r}{\partial t}; \quad c\mathbf{r} \left(1 - \frac{\partial t_r}{\partial t}\right) = -\mathbf{r} \cdot \mathbf{v} \frac{\partial t_r}{\partial t}; \quad c\mathbf{r} = \frac{\partial t_r}{\partial t}(c\mathbf{r} - \mathbf{r} \cdot \mathbf{v}) = \frac{\partial t_r}{\partial t}(\mathbf{r} \cdot \mathbf{u}) \text{ (Eq. 10.71), and hence } \frac{\partial t_r}{\partial t} = \frac{c\mathbf{r}}{\mathbf{r} \cdot \mathbf{u}}. \quad \text{qed}$$

Now Eq. 10.47 says $\mathbf{A}(\mathbf{r}, t) = \frac{\mathbf{v}}{c^2} V(\mathbf{r}, t)$, so

$$\begin{aligned} \frac{\partial \mathbf{A}}{\partial t} &= \frac{1}{c^2} \left(\frac{\partial \mathbf{v}}{\partial t} V + \mathbf{v} \frac{\partial V}{\partial t} \right) = \frac{1}{c^2} \left(\frac{\partial \mathbf{v}}{\partial t_r} \frac{\partial t_r}{\partial t} V + \mathbf{v} \frac{\partial V}{\partial t} \right) \\ &= \frac{1}{c^2} \left[\mathbf{a} \frac{\partial t_r}{\partial t} \frac{1}{4\pi\epsilon_0} \frac{qc}{\mathbf{r} \cdot \mathbf{u}} + \mathbf{v} \frac{1}{4\pi\epsilon_0} \frac{-qc}{(\mathbf{r} \cdot \mathbf{u})^2} \frac{\partial}{\partial t} (\mathbf{r} \cdot \mathbf{u} - \mathbf{r} \cdot \mathbf{v}) \right] \\ &= \frac{1}{c^2} \frac{qc}{4\pi\epsilon_0} \left[\frac{\mathbf{a}}{\mathbf{r} \cdot \mathbf{u}} \frac{\partial t_r}{\partial t} - \frac{\mathbf{v}}{(\mathbf{r} \cdot \mathbf{u})^2} \left(c \frac{\partial \mathbf{r}}{\partial t} - \frac{\partial \mathbf{r}}{\partial t} \cdot \mathbf{v} - \mathbf{r} \cdot \frac{\partial \mathbf{v}}{\partial t} \right) \right]. \end{aligned}$$

But $\mathbf{r} = c(t - t_r) \Rightarrow \frac{\partial \mathbf{r}}{\partial t} = c \left(1 - \frac{\partial t_r}{\partial t}\right)$, $\mathbf{r} = \mathbf{r} - \mathbf{w}(t_r) \Rightarrow \frac{\partial \mathbf{r}}{\partial t} = -\mathbf{v} \frac{\partial t_r}{\partial t}$ (as above), and

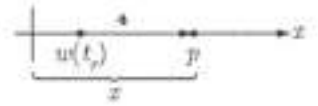
$$\begin{aligned} \frac{\partial \mathbf{v}}{\partial t} &= \frac{\partial \mathbf{v}}{\partial t_r} \frac{\partial t_r}{\partial t} = \mathbf{a} \frac{\partial t_r}{\partial t}. \\ &= \frac{q}{4\pi\epsilon_0 c (\mathbf{r} \cdot \mathbf{u})^2} \left\{ \mathbf{a} (\mathbf{r} \cdot \mathbf{u}) \frac{\partial t_r}{\partial t} - \mathbf{v} \left[c^2 \left(1 - \frac{\partial t_r}{\partial t}\right) + v^2 \frac{\partial t_r}{\partial t} - \mathbf{r} \cdot \mathbf{a} \frac{\partial t_r}{\partial t} \right] \right\} \\ &= \frac{q}{4\pi\epsilon_0 c (\mathbf{r} \cdot \mathbf{u})^2} \left\{ -c^2 \mathbf{v} + [(\mathbf{r} \cdot \mathbf{u}) \mathbf{a} + (c^2 - v^2 + \mathbf{r} \cdot \mathbf{a}) \mathbf{v}] \frac{\partial t_r}{\partial t} \right\} \\ &= \frac{q}{4\pi\epsilon_0 c (\mathbf{r} \cdot \mathbf{u})^2} \left\{ -c^2 \mathbf{v} + [(\mathbf{r} \cdot \mathbf{u}) \mathbf{a} + (c^2 - v^2 + \mathbf{r} \cdot \mathbf{a}) \mathbf{v}] \frac{c\mathbf{r}}{\mathbf{r} \cdot \mathbf{u}} \right\} \\ &= \frac{q}{4\pi\epsilon_0 c (\mathbf{r} \cdot \mathbf{u})^3} [-c^2 \mathbf{v} (\mathbf{r} \cdot \mathbf{u}) + c\mathbf{r} (\mathbf{r} \cdot \mathbf{u}) \mathbf{a} + c\mathbf{r} (c^2 - v^2 + \mathbf{r} \cdot \mathbf{a}) \mathbf{v}] \\ &= \frac{qc}{4\pi\epsilon_0} \frac{1}{(\mathbf{r} \cdot \mathbf{u})^3} \left[(\mathbf{r} \cdot \mathbf{u} - \mathbf{r} \cdot \mathbf{v}) \left(-\mathbf{v} + \frac{\mathbf{r}}{c} \mathbf{a} \right) + \frac{\mathbf{r}}{c} (c^2 - v^2 + \mathbf{r} \cdot \mathbf{a}) \mathbf{v} \right]. \quad \text{qed} \end{aligned}$$

Problem 10.20

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{(\mathbf{r} \cdot \mathbf{u})^3} [(c^2 - v^2) \mathbf{u} + \mathbf{r} \times (\mathbf{u} \times \mathbf{a})]. \text{ Here}$$

$\mathbf{v} = v \hat{\mathbf{x}}$, $\mathbf{a} = a \hat{\mathbf{x}}$, and, for points to the *right*, $\hat{\mathbf{r}} = \hat{\mathbf{x}}$.

So $\mathbf{u} = (c - v) \hat{\mathbf{x}}$, $\mathbf{u} \times \mathbf{a} = \mathbf{0}$, and $\mathbf{r} \cdot \mathbf{u} = \mathbf{r} \cdot (c - v) \hat{\mathbf{x}}$.



$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3 (c - v)^3} (c^2 - v^2)(c - v) \hat{\mathbf{x}} = \frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \frac{(c + v)(c - v)^2}{(c - v)^3} \hat{\mathbf{x}} = \frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \left(\frac{c + v}{c - v} \right) \hat{\mathbf{x}};$$

$$\mathbf{B} = \frac{1}{c} \hat{\mathbf{r}} \times \mathbf{E} = \mathbf{0}. \quad \text{qed}$$

For field points to the *left*, $\hat{\mathbf{r}} = -\hat{\mathbf{x}}$ and $\mathbf{u} = -(c + v) \hat{\mathbf{x}}$, so $\mathbf{r} \cdot \mathbf{u} = \mathbf{r} \cdot (c + v) \hat{\mathbf{x}}$, and

$$\mathbf{E} = -\frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3 (c + v)^3} (c^2 - v^2)(c + v) \hat{\mathbf{x}} = \boxed{-\frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \left(\frac{c - v}{c + v} \right) \hat{\mathbf{x}}; \quad \mathbf{B} = \mathbf{0}.}$$

Problem 10.21

By Gauss's law (in integral form) the answer has to be q/ϵ_0 .

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q(1 - v^2/c^2)}{(1 - \frac{v^2}{c^2} \sin^2 \theta)^{3/2}} \frac{\hat{\mathbf{R}}}{R^2} \quad (\text{Eq. 10.75}), \text{ so}$$

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{q(1 - v^2/c^2)}{4\pi\epsilon_0} \int \frac{R^2 \sin \theta \, d\theta \, d\phi}{R^2 (1 - \frac{v^2}{c^2} \sin^2 \theta)^{3/2}} = \frac{q(1 - v^2/c^2)}{4\pi\epsilon_0} 2\pi \int_0^\pi \frac{\sin \theta \, d\theta}{(1 - \frac{v^2}{c^2} \sin^2 \theta)^{3/2}}.$$

Let $u \equiv \cos \theta$, so $du = -\sin \theta \, d\theta$, $\sin^2 \theta = 1 - u^2$.

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{q(1 - v^2/c^2)}{2\epsilon_0} \int_{-1}^1 \frac{du}{[1 - \frac{v^2}{c^2} + \frac{v^2}{c^2} u^2]^{3/2}} = \frac{q(1 - v^2/c^2)}{2\epsilon_0} \left(\frac{c}{v}\right)^3 \int_{-1}^1 \frac{du}{(\frac{c^2}{v^2} - 1 + u^2)^{3/2}}.$$

The integral is: $\frac{u}{(\frac{c^2}{v^2} - 1) \sqrt{\frac{c^2}{v^2} - 1 + u^2}} \Big|_{-1}^{+1} = \frac{2}{(\frac{c^2}{v^2} - 1) \frac{c}{v}} = \left(\frac{v}{c}\right)^3 \frac{2}{(1 - v^2/c^2)}$. So

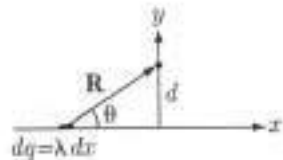
$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{q(1 - v^2/c^2)}{2\epsilon_0} \left(\frac{c}{v}\right)^3 \left(\frac{v}{c}\right)^3 \frac{2}{(1 - v^2/c^2)} = \boxed{\frac{q}{\epsilon_0}} \quad \checkmark$$

Problem 10.22

$$(a) \quad \mathbf{E} = \frac{\lambda}{4\pi\epsilon_0} (1 - v^2/c^2) \int \frac{\hat{\mathbf{R}}}{R^2} \frac{dx}{[1 - (v/c)^2 \sin^2 \theta]^{3/2}}.$$

The horizontal components cancel; the vertical component of $\hat{\mathbf{R}}$ is $\sin \theta$ (see diagram). Here $d = R \sin \theta$, so

$$\frac{1}{R^2} = \frac{\sin^2 \theta}{d^2}; \quad -\frac{x}{d} = \cot \theta, \text{ so } dx = -d(-\csc^2 \theta) d\theta = \frac{d}{\sin^2 \theta} d\theta;$$



$$\frac{1}{R^2} dx = \frac{d}{\sin^2 \theta} \frac{\sin^2 \theta}{d^2} d\theta = \frac{d\theta}{d}. \quad \text{Thus}$$

$$\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0} (1 - v^2/c^2) \left(\frac{\hat{\mathbf{y}}}{d}\right) \int_0^\pi \frac{\sin \theta}{[1 - (v/c)^2 \sin^2 \theta]^{3/2}} d\theta. \quad \text{Let } z \equiv \cos \theta, \text{ so } \sin^2 \theta = 1 - z^2.$$

$$= \frac{\lambda(1 - v^2/c^2) \hat{\mathbf{y}}}{4\pi\epsilon_0 d} \int_{-1}^1 \frac{1}{[1 - (v/c)^2 + (v/c)^2 z^2]^{3/2}} dz$$

$$= \frac{\lambda(1 - v^2/c^2) \hat{\mathbf{y}}}{4\pi\epsilon_0 d} \left[\frac{1}{(v/c)^3} \frac{z}{(c^2/v^2 - 1) \sqrt{(c/v)^2 - 1 + z^2}} \right] \Big|_{-1}^{+1}$$

$$= \frac{\lambda(1 - v^2/c^2) c}{4\pi\epsilon_0 d} \frac{1}{v(1 - v^2/c^2)} \frac{2}{\sqrt{(c/v)^2 - 1 + 1}} \hat{\mathbf{y}} = \boxed{\frac{1}{4\pi\epsilon_0} \frac{2\lambda}{d} \hat{\mathbf{y}}} \quad (\text{same as for a line charge at rest}).$$

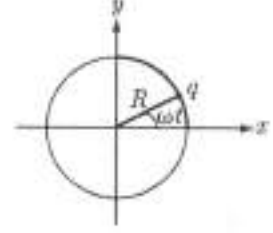
(b) $\mathbf{B} = \frac{1}{c^2} (\mathbf{v} \times \mathbf{E})$ for each segment $dq = \lambda dx$. Since $\mathbf{v}$ is constant, it comes outside the integral, and the same formula holds for the *total* field:

$$\mathbf{B} = \frac{1}{c^2} (\mathbf{v} \times \mathbf{E}) = \frac{1}{c^2} v \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{d} (\hat{\mathbf{x}} \times \hat{\mathbf{y}}) = \mu_0 \epsilon_0 v \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{d} \hat{\mathbf{z}} = \frac{\mu_0}{4\pi} \frac{2\lambda v}{d} \hat{\mathbf{z}}.$$

But $\lambda v = I$, so $\mathbf{B} = \frac{\mu_0}{4\pi} \frac{2I}{d} \hat{\phi}$ (the same as we got in magnetostatics, Eq. 5.39 and Ex. 5.7).

Problem 10.23

$$\begin{aligned}\mathbf{w}(t) &= R[\cos(\omega t) \hat{\mathbf{x}} + \sin(\omega t) \hat{\mathbf{y}}]; \\ \mathbf{v}(t) &= R\omega[-\sin(\omega t) \hat{\mathbf{x}} + \cos(\omega t) \hat{\mathbf{y}}]; \\ \mathbf{a}(t) &= -R\omega^2[\cos(\omega t) \hat{\mathbf{x}} + \sin(\omega t) \hat{\mathbf{y}}] = -\omega^2 \mathbf{w}(t); \\ \boldsymbol{\mathcal{Z}} &= -\mathbf{w}(t_r); \\ \mathcal{Z} &= R; \\ t_r &= t - R/c; \\ \hat{\boldsymbol{\mathcal{Z}}} &= -[\cos(\omega t_r) \hat{\mathbf{x}} + \sin(\omega t_r) \hat{\mathbf{y}}];\end{aligned}$$



$$\begin{aligned}\mathbf{u} &= c\hat{\boldsymbol{\mathcal{Z}}} - \mathbf{v}(t_r) = -c[\cos(\omega t_r) \hat{\mathbf{x}} + \sin(\omega t_r) \hat{\mathbf{y}}] - \omega R[-\sin(\omega t_r) \hat{\mathbf{x}} + \cos(\omega t_r) \hat{\mathbf{y}}] \\ &= -\{[c \cos(\omega t_r) - \omega R \sin(\omega t_r)] \hat{\mathbf{x}} + [c \sin(\omega t_r) + \omega R \cos(\omega t_r)] \hat{\mathbf{y}}\};\end{aligned}$$

$$\boldsymbol{\mathcal{Z}} \times (\mathbf{u} \times \mathbf{a}) = (\boldsymbol{\mathcal{Z}} \cdot \mathbf{a})\mathbf{u} - (\boldsymbol{\mathcal{Z}} \cdot \mathbf{u})\mathbf{a}; \quad \boldsymbol{\mathcal{Z}} \cdot \mathbf{a} = -\mathbf{w} \cdot (-\omega^2 \mathbf{w}) = \omega^2 R^2;$$

$$\boldsymbol{\mathcal{Z}} \cdot \mathbf{u} = R[c \cos^2(\omega t_r) - \omega R \sin(\omega t_r) \cos(\omega t_r) + c \sin^2(\omega t_r) + \omega R \sin(\omega t_r) \cos(\omega t_r)] = Rc;$$

$v^2 = (\omega R)^2$. So (Eq. 10.72):

$$\begin{aligned}\mathbf{E} &= \frac{q}{4\pi\epsilon_0} \frac{R}{(Rc)^3} [\mathbf{u}(c^2 - \omega^2 R^2) + \mathbf{u}(\omega R)^2 - \mathbf{a}(Rc)] = \frac{q}{4\pi\epsilon_0} \frac{c\mathbf{u} - R\mathbf{a}}{(Rc)^2} \\ &= \frac{q}{4\pi\epsilon_0} \frac{1}{(Rc)^2} \{-[c^2 \cos(\omega t_r) - \omega Rc \sin(\omega t_r)] \hat{\mathbf{x}} - [c^2 \sin(\omega t_r) + \omega Rc \cos(\omega t_r)] \hat{\mathbf{y}} \\ &\quad + R^2 \omega^2 \cos(\omega t_r) \hat{\mathbf{x}} + R^2 \omega^2 \sin(\omega t_r) \hat{\mathbf{y}}\} \\ &= \boxed{\frac{q}{4\pi\epsilon_0} \frac{1}{(Rc)^2} \{[(\omega^2 R^2 - c^2) \cos(\omega t_r) + \omega Rc \sin(\omega t_r)] \hat{\mathbf{x}} + [(\omega^2 R^2 - c^2) \sin(\omega t_r) - \omega Rc \cos(\omega t_r)] \hat{\mathbf{y}}\}}.\end{aligned}$$

$$\begin{aligned}\mathbf{B} &= \frac{1}{c} \hat{\boldsymbol{\mathcal{Z}}} \times \mathbf{E} = \frac{1}{c} (\hat{\boldsymbol{\mathcal{Z}}}_x E_y - \hat{\boldsymbol{\mathcal{Z}}}_y E_x) \hat{\mathbf{z}} \\ &= -\frac{1}{c} \frac{q}{4\pi\epsilon_0} \frac{1}{(Rc)^2} \{\cos(\omega t_r) [(\omega^2 R^2 - c^2) \sin(\omega t_r) - \omega Rc \cos(\omega t_r)] \\ &\quad - \sin(\omega t_r) [(\omega^2 R^2 - c^2) \cos(\omega t_r) + \omega Rc \sin(\omega t_r)]\} \hat{\mathbf{z}} \\ &= -\frac{q}{4\pi\epsilon_0} \frac{1}{R^2 c^3} [-\omega Rc \cos^2(\omega t_r) - \omega Rc \sin^2(\omega t_r)] \hat{\mathbf{z}} = \frac{q}{4\pi\epsilon_0} \frac{1}{R^2 c^3} \omega Rc \hat{\mathbf{z}} = \boxed{\frac{q}{4\pi\epsilon_0} \frac{\omega}{Rc^2} \hat{\mathbf{z}}}.\end{aligned}$$

Notice that $\mathbf{B}$ is constant in time.

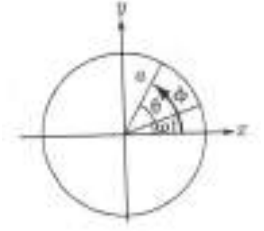
To obtain the field at the center of a circular *ring* of charge, let $q \rightarrow \lambda(2\pi R)$; for this ring to carry current I , we need $I = \lambda v = \lambda \omega R$, so $\lambda = I/\omega R$, and hence $q \rightarrow (I/\omega R)(2\pi R) = 2\pi I/\omega$. Thus $\mathbf{B} = \frac{2\pi I}{4\pi\epsilon_0} \frac{1}{Rc^2} \hat{\mathbf{z}}$, or,

since $1/c^2 = \epsilon_0 \mu_0$, $\mathbf{B} = \frac{\mu_0 I}{2R} \hat{\mathbf{z}}$, the same as Eq. 5.41, in the case $z = 0$.

Problem 10.24

$\lambda(\phi, t) = \lambda_0 |\sin(\theta/2)|$, where $\theta = \phi - \omega t$. So the (retarded) scalar potential at the center is (Eq. 10.26)

$$\begin{aligned} V(t) &= \frac{1}{4\pi\epsilon_0} \int \frac{\lambda}{r} dl' = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \frac{\lambda_0 |\sin[(\phi - \omega t_r)/2]|}{a} a d\phi \\ &= \frac{\lambda_0}{4\pi\epsilon_0} \int_0^{2\pi} \sin(\theta/2) d\theta = \frac{\lambda_0}{4\pi\epsilon_0} [-2 \cos(\theta/2)] \Big|_0^{2\pi} \\ &= \frac{\lambda_0}{4\pi\epsilon_0} [2 - (-2)] = \boxed{\frac{\lambda_0}{\pi\epsilon_0}}. \end{aligned}$$



(Note: at fixed t_r , $d\phi = d\theta$, and it goes through one full cycle of ϕ or θ .)

Meanwhile $\mathbf{I}(\phi, t) = \lambda \mathbf{v} = \lambda_0 \omega a |\sin[(\phi - \omega t)/2]| \hat{\phi}$. From Eq. 10.26 (again)

$$\mathbf{A}(t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I}}{r} dl' = \frac{\mu_0}{4\pi} \int_0^{2\pi} \frac{\lambda_0 \omega a |\sin[(\phi - \omega t_r)/2]| \hat{\phi}}{a} a d\phi.$$

But $t_r = t - a/c$ is again constant, for the ϕ integration, and $\hat{\phi} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}$.

$$= \frac{\mu_0 \lambda_0 \omega a}{4\pi} \int_0^{2\pi} |\sin[(\phi - \omega t_r)/2]| (-\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}) d\phi. \quad \text{Again, switch variables to } \theta = \phi - \omega t_r,$$

and integrate from $\theta = 0$ to $\theta = 2\pi$ (so we don't have to worry about the absolute value).

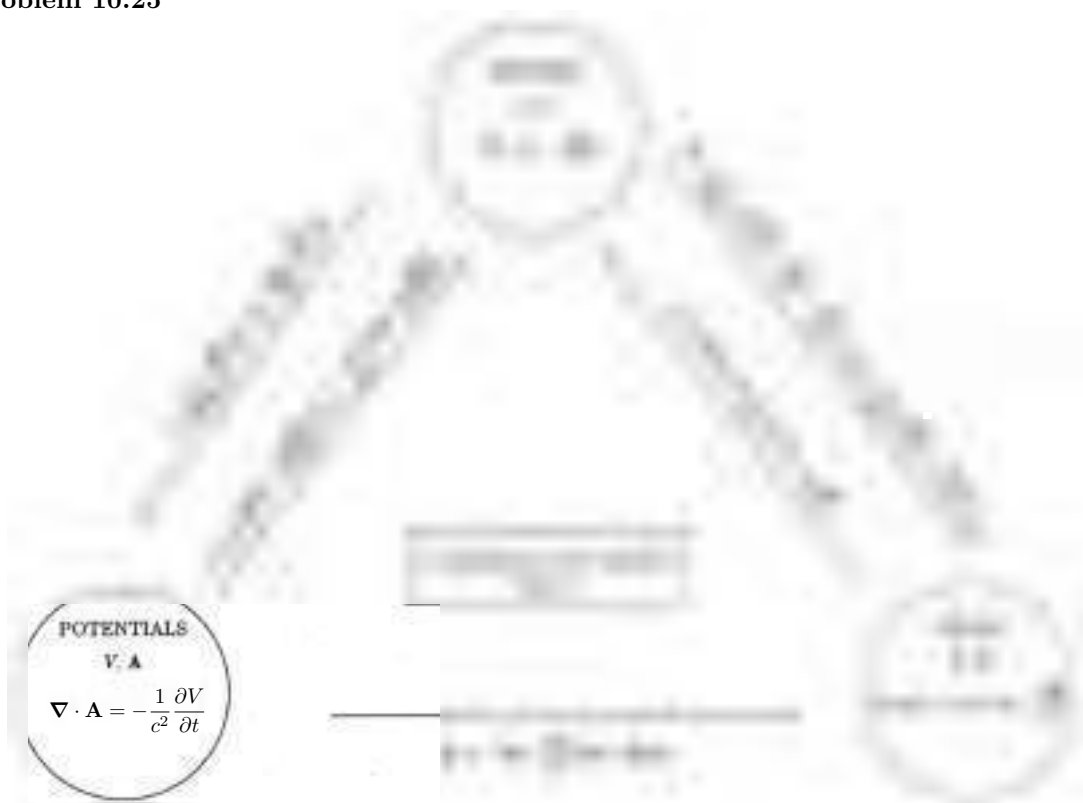
$$= \frac{\mu_0 \lambda_0 \omega a}{4\pi} \int_0^{2\pi} \sin(\theta/2) [-\sin(\theta + \omega t_r) \hat{\mathbf{x}} + \cos(\theta + \omega t_r) \hat{\mathbf{y}}] d\theta. \quad \text{Now}$$

$$\begin{aligned} \int_0^{2\pi} \sin(\theta/2) \sin(\theta + \omega t_r) d\theta &= \frac{1}{2} \int_0^{2\pi} [\cos(\theta/2 + \omega t_r) - \cos(3\theta/2 + \omega t_r)] d\theta \\ &= \frac{1}{2} \left[2 \sin(\theta/2 + \omega t_r) - \frac{2}{3} \sin(3\theta/2 + \omega t_r) \right] \Big|_0^{2\pi} \\ &= \sin(\pi + \omega t_r) - \sin(\omega t_r) - \frac{1}{3} \sin(3\pi + \omega t_r) + \frac{1}{3} \sin(\omega t_r) \\ &= -2 \sin(\omega t_r) + \frac{2}{3} \sin(\omega t_r) = -\frac{4}{3} \sin(\omega t_r). \end{aligned}$$

$$\begin{aligned} \int_0^{2\pi} \sin(\theta/2) \cos(\theta + \omega t_r) d\theta &= \frac{1}{2} \int_0^{2\pi} [-\sin(\theta/2 + \omega t_r) + \sin(3\theta/2 + \omega t_r)] d\theta \\ &= \frac{1}{2} \left[2 \cos(\theta/2 + \omega t_r) - \frac{2}{3} \cos(3\theta/2 + \omega t_r) \right] \Big|_0^{2\pi} \\ &= \cos(\pi + \omega t_r) - \cos(\omega t_r) - \frac{1}{3} \cos(3\pi + \omega t_r) + \frac{1}{3} \cos(\omega t_r) \\ &= -2 \cos(\omega t_r) + \frac{2}{3} \cos(\omega t_r) = -\frac{4}{3} \cos(\omega t_r). \quad \text{So} \end{aligned}$$

$$\mathbf{A}(t) = \frac{\mu_0 \lambda_0 \omega a}{4\pi} \left(\frac{4}{3} \right) [\sin(\omega t_r) \hat{\mathbf{x}} - \cos(\omega t_r) \hat{\mathbf{y}}] = \boxed{\frac{\mu_0 \lambda_0 \omega a}{3\pi} \{ \sin[\omega(t - a/c)] \hat{\mathbf{x}} - \cos[\omega(t - a/c)] \hat{\mathbf{y}} \}}.$$

Problem 10.25



Problem 10.26

$$\rho(\mathbf{r}, t) = \begin{cases} \frac{Q}{(4/3)\pi R^3} = \frac{3Q}{4\pi v^3 t^3}, & (r < R = vt), \\ 0, & (\text{otherwise}). \end{cases}$$

For a point at the center, $t_r = t - r/c$, so

$$\rho(\mathbf{r}, t_r) = \begin{cases} \frac{3Q}{4\pi v^3 (t - r/c)^3} = \frac{3Q}{4\pi (v/c)^3 (ct - r)^3}, & \left(r < v(t - r/c) \Rightarrow r < \frac{vt}{1 + v/c} \right), \\ 0, & (\text{otherwise}). \end{cases}$$

Let $a \equiv vt/(1 + v/c)$; then

$$\begin{aligned} Q_{\text{eff}} &= \frac{3Q}{4\pi (v/c)^3} \int_0^a \frac{1}{(ct - r)^3} 4\pi r^2 dr = \frac{3Q}{(v/c)^3} \int_0^a \frac{r^2}{(ct - r)^3} dr \\ &= -\frac{3Q}{(v/c)^3} \left[\ln(ct - r) + \frac{2ct}{(ct - r)} - \frac{(ct)^2}{2(ct - r)^2} \right] \Bigg|_0^a \\ &= -\frac{3Q}{(v/c)^3} \left[\ln(ct - a) + \frac{2ct}{(ct - a)} - \frac{(ct)^2}{2(ct - a)^2} - \ln(ct) - 2 + \frac{1}{2} \right] \\ &= \frac{3Q}{(v/c)^3} \left[\frac{3}{2} + \ln \left(\frac{ct}{ct - a} \right) - 2 \left(\frac{ct}{ct - a} \right) + \frac{1}{2} \left(\frac{ct}{ct - a} \right)^2 \right]. \end{aligned}$$

Now, $ct - a = ct - \frac{vt}{1+v/c} = \frac{ct}{1+v/c} \left(1 + \frac{v}{c} - \frac{v}{c}\right) = \frac{ct}{1+v/c}$, so $\frac{ct}{ct-a} = 1 + \frac{v}{c}$, and hence

$$\begin{aligned} Q_{\text{eff}} &= \frac{3Q}{(v/c)^3} \left[\frac{3}{2} + \ln \left(1 + \frac{v}{c}\right) - 2 \left(1 + \frac{v}{c}\right) + \frac{1}{2} \left(1 + \frac{v}{c}\right)^2 \right] \\ &= \frac{3Q}{(v/c)^3} \left[\frac{3}{2} + \ln \left(1 + \frac{v}{c}\right) - 2 - 2\frac{v}{c} + \frac{1}{2} + \frac{v}{c} + \frac{1}{2} \left(\frac{v}{c}\right)^2 \right] = \boxed{\frac{3Q}{(v/c)^3} \left[\ln \left(1 + \frac{v}{c}\right) - \frac{v}{c} + \frac{1}{2} \left(\frac{v}{c}\right)^2 \right]}. \end{aligned}$$

If $\epsilon \ll 1$, then $\ln(1 + \epsilon) = \epsilon - \frac{1}{2}\epsilon^2 + \frac{1}{3}\epsilon^3 - \frac{1}{4}\epsilon^4 + \dots$, so for $v \ll c$,

$$Q_{\text{eff}} \approx \frac{3Q}{(v/c)^3} \left[\frac{v}{c} - \frac{1}{2} \left(\frac{v}{c}\right)^2 + \frac{1}{3} \left(\frac{v}{c}\right)^3 - \frac{1}{4} \left(\frac{v}{c}\right)^4 - \frac{v}{c} + \frac{1}{2} \left(\frac{v}{c}\right)^2 \right] = Q \left(1 - \frac{3v}{4c}\right). \quad \checkmark$$

Problem 10.27

Using Product Rule #5, Eq. 10.50 $\Rightarrow$

$$\begin{aligned} \nabla \cdot \mathbf{A} &= \frac{\mu_0}{4\pi} q c \mathbf{v} \cdot \nabla [(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)]^{-1/2} \\ &= \frac{\mu_0 q c}{4\pi} \mathbf{v} \cdot \left\{ -\frac{1}{2} [(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)]^{-3/2} \nabla [(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)] \right\} \\ &= -\frac{\mu_0 q c}{8\pi} [(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)]^{-3/2} \mathbf{v} \cdot \{ -2(c^2 t - \mathbf{r} \cdot \mathbf{v}) \nabla(\mathbf{r} \cdot \mathbf{v}) + (c^2 - v^2) \nabla(r^2) \}. \end{aligned}$$

Product Rule #4 $\Rightarrow$

$$\nabla(\mathbf{r} \cdot \mathbf{v}) = \mathbf{v} \times (\nabla \times \mathbf{r}) + (\mathbf{v} \cdot \nabla) \mathbf{r}, \text{ but } \nabla \times \mathbf{r} = 0,$$

$$(\mathbf{v} \cdot \nabla) \mathbf{r} = \left(v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z} \right) (x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}) = v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}} = \mathbf{v}, \text{ and}$$

$$\nabla(r^2) = \nabla(\mathbf{r} \cdot \mathbf{r}) = 2\mathbf{r} \times (\nabla \times \mathbf{r}) + 2(\mathbf{r} \cdot \nabla) \mathbf{r} = 2\mathbf{r}. \text{ So}$$

$$\begin{aligned} \nabla \cdot \mathbf{A} &= -\frac{\mu_0 q c}{8\pi} [(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)]^{-3/2} \mathbf{v} \cdot [-2(c^2 t - \mathbf{r} \cdot \mathbf{v}) \mathbf{v} + (c^2 - v^2) 2\mathbf{r}] \\ &= \frac{\mu_0 q c}{4\pi} [(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)]^{-3/2} \{ (c^2 t - \mathbf{r} \cdot \mathbf{v}) v^2 - (c^2 - v^2)(\mathbf{r} \cdot \mathbf{v}) \}. \end{aligned}$$

But the term in curly brackets is: $c^2 t v^2 - v^2(\mathbf{r} \cdot \mathbf{v}) - c^2(\mathbf{r} \cdot \mathbf{v}) + v^2(\mathbf{r} \cdot \mathbf{v}) = c^2(v^2 t - \mathbf{r} \cdot \mathbf{v})$.

$$= \frac{\mu_0 q c^3}{4\pi} \frac{(v^2 t - \mathbf{r} \cdot \mathbf{v})}{[(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)]^{3/2}}.$$

Meanwhile, from Eq. 10.49,

$$\begin{aligned} -\mu_0 \epsilon_0 \frac{\partial V}{\partial t} &= -\mu_0 \epsilon_0 \frac{1}{4\pi \epsilon_0} q c \left(-\frac{1}{2} \right) [(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)]^{-3/2} \times \\ &\quad \frac{\partial}{\partial t} [(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)] \\ &= -\frac{\mu_0 q c}{8\pi} [(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)]^{-3/2} [2(c^2 t - \mathbf{r} \cdot \mathbf{v}) c^2 + (c^2 - v^2)(-2c^2 t)] \\ &= -\frac{\mu_0 q c^3}{4\pi} \frac{(c^2 t - \mathbf{r} \cdot \mathbf{v} - c^2 t + v^2 t)}{[(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)]^{3/2}} = \nabla \cdot \mathbf{A}. \quad \checkmark \end{aligned}$$

Problem 10.28

$$(a) \quad \mathbf{F}_2 = \frac{q_1 q_2}{4\pi\epsilon_0} \frac{1}{(b^2 + c^2 t^2)} \hat{\mathbf{x}}.$$

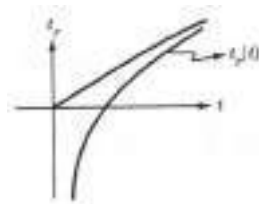


(This is just Coulomb's law, since q_1 is at rest.)

$$(b) \quad I_2 = \frac{q_1 q_2}{4\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{1}{(b^2 + c^2 t^2)} dt = \frac{q_1 q_2}{4\pi\epsilon_0} \left[\frac{1}{bc} \tan^{-1}(ct/b) \right] \Big|_{-\infty}^{\infty} = \frac{q_1 q_2}{4\pi\epsilon_0 bc} [\tan^{-1}(\infty) - \tan^{-1}(-\infty)]$$

$$= \frac{q_1 q_2}{4\pi\epsilon_0 bc} \left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right) \right] = \boxed{\frac{q_1 q_2}{4\pi\epsilon_0} \frac{\pi}{bc}}.$$

(c) From Prob. 10.20, $\mathbf{E} = -\frac{q_2}{4\pi\epsilon_0} \frac{1}{x^2} \left(\frac{c-v}{c+v} \right) \hat{\mathbf{x}}$. Here x and v are to be evaluated at the retarded time t_r , which is given by $c(t - t_r) = x(t_r) = \sqrt{b^2 + c^2 t_r^2} \Rightarrow c^2 t^2 - 2ctt_r + c^2 t_r^2 = b^2 + c^2 t_r^2 \Rightarrow t_r = \frac{c^2 t^2 - b^2}{2c^2 t}$. Note: As we found in Prob. 10.17, q_2 first “comes into view” (for q_1) at time $t = 0$. Before that it can exert no force on q_1 , and there is no retarded time. From the graph of t_r versus t we see that t_r ranges all the way from $-\infty$ to ∞ while $t > 0$.



$$x(t_r) = c(t - t_r) = \frac{2c^2 t^2 - c^2 t_r^2 + b^2}{2ct} = \frac{b^2 + c^2 t^2}{2ct} \quad (\text{for } t > 0). \quad v(t) = \frac{1}{2} \frac{2c^2 t}{\sqrt{b^2 + c^2 t^2}} = \frac{c^2 t}{x}, \text{ so}$$

$$v(t_r) = \left(\frac{c^2 t^2 - b^2}{2t} \right) \left(\frac{2ct}{b^2 + c^2 t^2} \right) = c \left(\frac{c^2 t^2 - b^2}{c^2 t^2 + b^2} \right) \quad (\text{for } t > 0). \quad \text{Therefore}$$

$$\frac{c-v}{c+v} = \frac{(c^2 t^2 + b^2) - (c^2 t^2 - b^2)}{(c^2 t^2 + b^2) + (c^2 t^2 - b^2)} = \frac{2b^2}{2c^2 t^2} = \frac{b^2}{c^2 t^2} \quad (\text{for } t > 0). \quad \mathbf{E} = -\frac{q_2}{4\pi\epsilon_0} \frac{4c^2 t^2}{(b^2 + c^2 t^2)^2} \frac{b^2}{c^2 t^2} \hat{\mathbf{x}} \Rightarrow$$

$$\mathbf{F}_1 = \begin{cases} 0, & t < 0; \\ -\frac{q_1 q_2}{4\pi\epsilon_0} \frac{4b^2}{(b^2 + c^2 t^2)^2} \hat{\mathbf{x}}, & t > 0. \end{cases}$$

$$(d) \quad I_1 = -\frac{q_1 q_2}{4\pi\epsilon_0} 4b^2 \int_0^{\infty} \frac{1}{(b^2 + c^2 t^2)^2} dt. \quad \text{The integral is}$$

$$\frac{1}{c^4} \int_0^{\infty} \frac{1}{[(b/c)^2 + t^2]^2} dt = \frac{1}{c^4} \left(\frac{c^2}{2b^2} \right) \left[\frac{t}{(b/c)^2 + t^2} \Big|_0^{\infty} + \int_0^{\infty} \frac{1}{[(b/c)^2 + t^2]} dt \right] = \frac{1}{2c^2 b^2} \left(\frac{\pi c}{2b} \right) = \frac{\pi}{4cb^3}.$$

$$\text{So } \boxed{I_1 = -\frac{q_1 q_2}{4\pi\epsilon_0} \frac{\pi}{bc}}.$$

(e) $\mathbf{F}_1 \neq -\mathbf{F}_2$, so Newton's third law is *not* obeyed. On the other hand, $I_1 = -I_2$ in this instance, which suggests that the *net* momentum delivered from (1) to (2) is equal and opposite to the net momentum delivered from (2) to (1), and hence that the total mechanical momentum is conserved. (In general the fields might carry off some momentum, leaving the mechanical momentum altered; but that doesn't happen in the present case.)

Problem 10.29

The electric field of q_1 at q_2 [Eq. 10.75, with $\theta = 45^\circ$ and $\mathbf{R} = (-vt \hat{\mathbf{x}} + vt \hat{\mathbf{y}})$] is

$$\mathbf{E}_1 = \frac{q_1}{4\pi\epsilon_0} \frac{1 - v^2/c^2}{(1 - v^2/2c^2)^{3/2}} \frac{1}{2\sqrt{2}(vt)^2} (-\hat{\mathbf{x}} + \hat{\mathbf{y}}).$$

The magnetic field [Eq. 10.76, with $\mathbf{v}_1 = -v \hat{\mathbf{x}}$] is

$$\mathbf{B}_1 = \frac{1}{c^2}(\mathbf{v}_1 \times \mathbf{E}) = -\frac{v}{c^2}(\hat{\mathbf{x}} \times \mathbf{E}) = -\frac{v}{c^2} \frac{q_1}{4\pi\epsilon_0} \frac{1 - v^2/c^2}{(1 - v^2/2c^2)^{3/2}} \frac{1}{2\sqrt{2}(vt)^2} \hat{\mathbf{z}}.$$

The force on q_2 is therefore [Lorentz force law with $\mathbf{v}_2 = -v \hat{\mathbf{y}}$]

$$\mathbf{F}_2 = q_2(\mathbf{E}_1 + \mathbf{v}_2 \times \mathbf{B}_1) = \boxed{\frac{q_1 q_2}{4\pi\epsilon_0} \frac{1 - v^2/c^2}{(1 - v^2/2c^2)^{3/2}} \frac{1}{2\sqrt{2}(vt)^2} \left(-\hat{\mathbf{x}} + \hat{\mathbf{y}} + \frac{v^2}{c^2} \hat{\mathbf{x}} \right)}.$$

The electric field of q_2 at q_1 is reversed, $\mathbf{E}_2 = -\mathbf{E}_1$; so is the magnetic field $\mathbf{B}_2 = -\mathbf{B}_1$. The electric force is also reversed, but the magnetic force now points in the y direction instead of the x direction. So the force on q_1 is

$$\mathbf{F}_1 = \boxed{\frac{q_1 q_2}{4\pi\epsilon_0} \frac{1 - v^2/c^2}{(1 - v^2/2c^2)^{3/2}} \frac{1}{2\sqrt{2}(vt)^2} \left(\hat{\mathbf{x}} - \hat{\mathbf{y}} + \frac{v^2}{c^2} \hat{\mathbf{y}} \right)}.$$

The forces are equal in magnitude, but not opposite in direction. No, Newton's third law is *not* obeyed.

Problem 10.30

$$c(t - t_1) = vt_1, \quad t = t_1(1 + v/c), \quad \boxed{t_1 = \frac{t}{1 + v/c}}, \quad c(t - t_2) = vt_2 + L, \quad t - L/c = t_2(1 + v/c), \quad \boxed{t_2 = \frac{t - L/c}{1 + v/c}}.$$

$$x_1 = vt_1 = \boxed{\frac{vt}{1 + v/c}}, \quad x_2 = vt_2 + L = \frac{v(t - L/c)}{1 + v/c} + L = \frac{vt - vL/c + L + vL/c}{1 + v/c} = \boxed{\frac{vt + L}{1 + v/c}}.$$

$$V(\mathbf{0}, t) = \frac{1}{4\pi\epsilon_0} \int_{x_1}^{x_2} \frac{\lambda}{x} dx = \frac{\lambda}{4\pi\epsilon_0} \ln \left(\frac{x_2}{x_1} \right) = \boxed{\frac{\lambda}{4\pi\epsilon_0} \ln \left(\frac{vt + L}{vt} \right)}.$$

If $L \ll vt$, $V = \frac{\lambda}{4\pi\epsilon_0} \ln \left(1 + \frac{L}{vt} \right) \approx \frac{\lambda}{4\pi\epsilon_0} \frac{L}{vt} = \boxed{\frac{q}{4\pi\epsilon_0} \left(\frac{1}{vt} \right)}$. The Liénard-Wiechert potential is

$$V = \frac{q}{4\pi\epsilon_0(z - \mathbf{z} \cdot \mathbf{v}/c)}. \quad \text{Here } (z - \mathbf{z} \cdot \mathbf{v}/c) = vt_r + v^2 t_r/c = v(1 + v/c)t_r = v(1 + v/c) \frac{t}{1 + v/c} = vt.$$

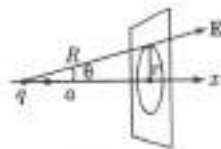
(In the limit $L \rightarrow 0$, $t_r = t_1 = t_2$.) So $V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{vt} \right)$. ✓

Problem 10.31

$$\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B}); \quad \mathbf{B} = \frac{1}{c^2}(\mathbf{v} \times \mathbf{E}) \quad (\text{Eq. 10.76}).$$

$$\text{So } \mathbf{S} = \frac{1}{\mu_0 c^2} [\mathbf{E} \times (\mathbf{v} \times \mathbf{E})] = \epsilon_0 [E^2 \mathbf{v} - (\mathbf{v} \cdot \mathbf{E}) \mathbf{E}].$$

The power crossing the plane is $P = \int \mathbf{S} \cdot d\mathbf{a}$,



and $d\mathbf{a} = 2\pi r dr \hat{\mathbf{x}}$ (see diagram). So

$$\begin{aligned}
 P &= \epsilon_0 \int (E^2 v - E_x^2 v) 2\pi r dr; \quad E_x = E \cos \theta, \text{ so } E^2 - E_x^2 = E^2 \sin^2 \theta. \\
 &= 2\pi\epsilon_0 v \int E^2 \sin^2 \theta r dr. \text{ From Eq. 10.75, } \mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{1}{\gamma^2} \frac{\hat{\mathbf{R}}}{R^2 [1 - (v/c)^2 \sin^2 \theta]^{3/2}} \text{ where } \gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}. \\
 &= 2\pi\epsilon_0 v \left(\frac{q}{4\pi\epsilon_0} \right)^2 \frac{1}{\gamma^2} \int_0^\infty \frac{r \sin^2 \theta}{R^4 [1 - (v/c)^2 \sin^2 \theta]^3} dr. \quad \text{Now } r = a \tan \theta \Rightarrow dr = a \frac{1}{\cos^2 \theta} d\theta; \quad \frac{1}{R} = \frac{\cos \theta}{a}. \\
 &= \frac{v}{2\gamma^4} \frac{q^2}{4\pi\epsilon_0} \frac{1}{a^2} \int_0^{\pi/2} \frac{\sin^3 \theta \cos \theta}{[1 - (v/c)^2 \sin^2 \theta]^3} d\theta. \quad \text{Let } u \equiv \sin^2 \theta, \text{ so } du = 2 \sin \theta \cos \theta d\theta. \\
 &= \frac{v q^2}{16\pi\epsilon_0 a^2 \gamma^4} \int_0^1 \frac{u}{[1 - (v/c)^2 u]^3} du = \frac{v q^2}{16\pi\epsilon_0 a^2 \gamma^4} \left(\frac{\gamma^4}{2} \right) = \boxed{\frac{v q^2}{32\pi\epsilon_0 a^2}}.
 \end{aligned}$$

Problem 10.32

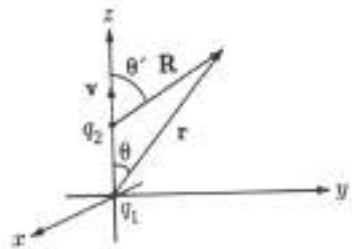
(a) $\boxed{\mathbf{F}_{12}(t) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{(vt)^2} \hat{\mathbf{z}}.}$

(b) From Eq. 10.75, with $\theta = 180^\circ$, $R = vt$, and $\hat{\mathbf{R}} = -\hat{\mathbf{z}}$:

$$\boxed{\mathbf{F}_{21}(t) = -\frac{1}{4\pi\epsilon_0} \frac{q_1 q_2 (1 - v^2/c^2)}{(vt)^2} \hat{\mathbf{z}}.}$$

$\boxed{\text{No,}}$ Newton's third law does *not* hold: $\mathbf{F}_{12} \neq \mathbf{F}_{21}$,

because of the extra factor $(1 - v^2/c^2)$.



(c) From Eq. 8.28, $\mathbf{p} = \epsilon_0 \int (\mathbf{E} \times \mathbf{B}) d\tau$. Here $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2$, whereas $\mathbf{B} = \mathbf{B}_2$, so $\mathbf{E} \times \mathbf{B} = (\mathbf{E}_1 \times \mathbf{B}_2) + (\mathbf{E}_2 \times \mathbf{B}_2)$. But the latter, when integrated over all space, is independent of time. We want only the time-dependent part:

$$\mathbf{p}(t) = \epsilon_0 \int (\mathbf{E}_1 \times \mathbf{B}_2) d\tau. \text{ Now } \mathbf{E}_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r^2} \hat{\mathbf{r}}, \text{ while, from Eq. 10.76, } \mathbf{B}_2 = \frac{1}{c^2} (\mathbf{v} \times \mathbf{E}_2), \text{ and (Eq. 10.75)}$$

$$\mathbf{E}_2 = \frac{q_2}{4\pi\epsilon_0} \frac{(1 - v^2/c^2)}{(1 - v^2 \sin^2 \theta'/c^2)^{3/2}} \frac{\hat{\mathbf{R}}}{R^2}. \text{ But } \mathbf{R} = \mathbf{r} - \mathbf{vt}; \quad R^2 = r^2 + v^2 t^2 - 2rvt \cos \theta; \quad \sin \theta' = \frac{r \sin \theta}{R}. \text{ So}$$

$$\mathbf{E}_2 = \frac{q_2}{4\pi\epsilon_0} \frac{(1 - v^2/c^2)}{[1 - (vr \sin \theta / Rc)^2]^{3/2}} \frac{(\mathbf{r} - \mathbf{vt})}{R^3}. \text{ Finally, noting that } \mathbf{v} \times (\mathbf{r} - \mathbf{vt}) = \mathbf{v} \times \mathbf{r} = vr \sin \theta \hat{\phi}, \text{ we get}$$

$$\mathbf{B}_2 = \frac{q_2(1 - v^2/c^2)}{4\pi\epsilon_0 c^2} \frac{vr \sin \theta}{[R^2 - (vr \sin \theta / c)^2]^{3/2}} \hat{\phi}. \text{ So } \mathbf{p}(t) = \epsilon_0 \frac{q_1}{4\pi\epsilon_0} \frac{q_2(1 - v^2/c^2)v}{4\pi\epsilon_0 c^2} \int \frac{1}{r^2} \frac{r \sin \theta (\hat{\mathbf{r}} \times \hat{\phi})}{[R^2 - (vr \sin \theta / c)^2]^{3/2}} d\tau.$$

But $\hat{\mathbf{r}} \times \hat{\phi} = -\hat{\theta} = -(\cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}})$, and the x and y components integrate to zero, so:

$$\begin{aligned}
 \mathbf{p}(t) &= \frac{q_1 q_2 v (1 - v^2/c^2)}{(4\pi c)^2 \epsilon_0} \hat{\mathbf{z}} \int \frac{\sin^2 \theta}{r [r^2 + (vt)^2 - 2rvt \cos \theta - (vr \sin \theta / c)^2]^{3/2}} r^2 \sin \theta dr d\theta d\phi \\
 &= \frac{q_1 q_2 v (1 - v^2/c^2)}{8\pi c^2 \epsilon_0} \hat{\mathbf{z}} \int \frac{r \sin^3 \theta}{[r^2 + (vt)^2 - 2rvt \cos \theta - (vr \sin \theta / c)^2]^{3/2}} dr d\theta.
 \end{aligned}$$

I'll do the r integral first. According to the CRC Tables,

$$\begin{aligned} \int_0^\infty \frac{x}{(a+bx+cx^2)^{3/2}} dx &= -\frac{2(bx+2a)}{(4ac-b^2)\sqrt{a+bx+cx^2}} \Big|_0^\infty = -\frac{2}{4ac-b^2} \left[\frac{b}{\sqrt{c}} - \frac{2a}{\sqrt{a}} \right] \\ &= -\frac{2}{\sqrt{c}(4ac-b^2)} (b-2\sqrt{ac}) = \frac{2}{\sqrt{c}} \frac{(2\sqrt{ac}-b)}{(2\sqrt{ac}-b)(2\sqrt{ac}+b)} = \frac{2}{\sqrt{c}} (2\sqrt{ac}+b)^{-1}. \end{aligned}$$

In this case $x = r$, $a = (vt)^2$, $b = -2vt \cos \theta$, and $c = 1 - (v/c)^2 \sin^2 \theta$. So the r integral is

$$\begin{aligned} &\frac{2}{\sqrt{1-(v/c)^2 \sin^2 \theta} \left[2vt\sqrt{1-(v/c)^2 \sin^2 \theta} - 2vt \cos \theta \right]} = \frac{1}{vt\sqrt{1-(v/c)^2 \sin^2 \theta} \left[\sqrt{1-(v/c)^2 \sin^2 \theta} - \cos \theta \right]} \\ &= \frac{\left[\sqrt{1-(v/c)^2 \sin^2 \theta} + \cos \theta \right]}{vt\sqrt{1-(v/c)^2 \sin^2 \theta} [1 - (v/c)^2 \sin^2 \theta - \cos^2 \theta]} = \frac{1}{vt \sin^2 \theta (1 - v^2/c^2)} \left[1 + \frac{\cos \theta}{\sqrt{1-(v/c)^2 \sin^2 \theta}} \right]. \end{aligned}$$

So

$$\begin{aligned} \mathbf{p}(t) &= \frac{q_1 q_2 v (1 - v^2/c^2)}{8\pi c^2 \epsilon_0} \frac{\hat{\mathbf{z}}}{vt(1 - v^2/c^2)} \int_0^\pi \frac{1}{\sin^2 \theta} \left[1 + \frac{\cos \theta}{\sqrt{1-(v/c)^2 \sin^2 \theta}} \right] \sin^3 \theta d\theta \\ &= \frac{q_1 q_2 \hat{\mathbf{z}}}{8\pi c^2 \epsilon_0 t} \left\{ \int_0^\pi \sin \theta d\theta + \frac{c}{v} \int_0^\pi \frac{\cos \theta \sin \theta}{\sqrt{(c/v)^2 - \sin^2 \theta}} d\theta \right\}. \end{aligned}$$

But $\int_0^\pi \sin \theta d\theta = 2$. In the second integral let $u \equiv \cos \theta$, so $du = -\sin \theta d\theta$:

$$\int_0^\pi \frac{\cos \theta \sin \theta}{\sqrt{(c/v)^2 - \sin^2 \theta}} d\theta = \int_{-1}^1 \frac{u}{\sqrt{(c/v)^2 - 1 + u^2}} du = 0 \quad (\text{the integrand is odd, and the interval is even}).$$

Conclusion: $\boxed{\mathbf{p}(t) = \frac{\mu_0 q_1 q_2}{4\pi t} \hat{\mathbf{z}}}$ (plus a term constant in time).

(d)

$$\begin{aligned} \mathbf{F}_{12} + \mathbf{F}_{21} &= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{v^2 t^2} \hat{\mathbf{z}} - \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2 (1 - v^2/c^2)}{v^2 t^2} \hat{\mathbf{z}} = \frac{q_1 q_2}{4\pi\epsilon_0 v^2 t^2} \left(1 - 1 + \frac{v^2}{c^2} \right) \hat{\mathbf{z}} = \frac{q_1 q_2}{4\pi\epsilon_0 c^2 t^2} \hat{\mathbf{z}} = \frac{\mu_0 q_1 q_2}{4\pi t^2} \hat{\mathbf{z}}. \\ -\frac{d\mathbf{p}}{dt} &= \frac{\mu_0 q_1 q_2}{4\pi t^2} \hat{\mathbf{z}} = \mathbf{F}_{12} + \mathbf{F}_{21}. \quad \text{qed} \end{aligned}$$

Since q_1 is at rest, and q_2 is moving at constant velocity, there must be another force ($\mathbf{F}_{\text{mech}}$) acting on them, to balance $\mathbf{F}_{12} + \mathbf{F}_{21}$; what we have found is that $\mathbf{F}_{\text{mech}} = d\mathbf{p}_{\text{em}}/dt$, which means that the impulse imparted to the system by the external force ends up as momentum in the fields. [For further discussion of this problem see J. J. G. Scanio, *Am. J. Phys.* **43**, 258 (1975).]

Problem 10.33

Maxwell's equations with magnetic charge read:

$$\begin{aligned} \text{(i)} \quad \nabla \cdot \mathbf{E} &= \frac{1}{\epsilon_0} \rho_e, & \text{(iii)} \quad \nabla \times \mathbf{E} &= -\mu_0 \mathbf{J}_m - \frac{\partial \mathbf{B}}{\partial t}, \\ \text{(ii)} \quad \nabla \cdot \mathbf{B} &= \mu_0 \rho_m, & \text{(iv)} \quad \nabla \times \mathbf{B} &= \mu_0 \mathbf{J}_e + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \end{aligned}$$

where ρ_e is the electric charge density, ρ_m is the magnetic charge density, $\mathbf{J}_e$ is the electric current density, and $\mathbf{J}_m$ is the magnetic current density.

If there are *only* electric charges and currents, then the usual potential formulation applies (Section 10.1.1), with $\mathbf{E} = -\nabla V_e - \partial \mathbf{A}_e / \partial t$, $\mathbf{B} = \nabla \times \mathbf{A}_e$. If there are *only* magnetic charges and currents,

$$\begin{aligned} \text{(i)} \quad \nabla \cdot \mathbf{E} &= 0, & \text{(iii)} \quad \nabla \times \mathbf{E} &= -\mu_0 \mathbf{J}_m - \frac{\partial \mathbf{B}}{\partial t}, \\ \text{(ii)} \quad \nabla \cdot \mathbf{B} &= \mu_0 \rho_m, & \text{(iv)} \quad \nabla \times \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \end{aligned}$$

This time equation (i) says that $\mathbf{E}$ can be expressed as the curl of a vector potential: $\mathbf{E} = -\nabla \times \mathbf{A}_m$, and plugging this into (iv) yields $\nabla \times (\mathbf{B} + \mu_0 \epsilon_0 \partial \mathbf{A}_m / \partial t) = \mathbf{0}$, which tells us that the quantity in parentheses can be represented as the gradient of a scalar: $\mathbf{B} = -\nabla V_m - \mu_0 \epsilon_0 \partial \mathbf{A}_m / \partial t$. [The *signs* of V_m and $\mathbf{A}_m$ are arbitrary, but I think this choice yields the most symmetrical formulation.] Putting these into (ii) and (iii) yields

$$\square^2 V_m = -\mu_0 \rho_m - \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(\nabla \cdot \mathbf{A}_m + \frac{\partial V_m}{\partial t} \right), \quad \square^2 \mathbf{A}_m = -\mu_0 \mathbf{J}_m + \nabla \left(\nabla \cdot \mathbf{A}_m + \frac{\partial V_m}{\partial t} \right).$$

The “Lorenz gauge” for the magnetic potentials, $\nabla \cdot \mathbf{A}_m = -\partial V_m / \partial t$, kills the terms in parentheses.

In general, if there are both electric *and* magnetic charges and currents present, the total fields are the sums (by the superposition principle):

$$\boxed{\mathbf{E} = -\nabla V_e - \frac{\partial \mathbf{A}_e}{\partial t} - \nabla \times \mathbf{A}_m,} \quad \boxed{\mathbf{B} = -\nabla V_m - \mu_0 \epsilon_0 \frac{\partial \mathbf{A}_m}{\partial t} + \nabla \times \mathbf{A}_e.}$$

If we choose to work in the Lorenz gauge:

$$\boxed{\nabla \cdot \mathbf{A}_e = -\mu_0 \epsilon_0 \frac{\partial V_e}{\partial t}, \quad \nabla \cdot \mathbf{A}_m = -\frac{\partial V_m}{\partial t},}$$

Maxwell's equations become (in terms of the potentials)

$$\boxed{\square^2 V_e = -\frac{1}{\epsilon_0} \rho_e, \quad \square^2 V_m = -\mu_0 \rho_m, \quad \square^2 \mathbf{A}_e = -\mu_0 \mathbf{J}_e, \quad \square^2 \mathbf{A}_m = -\mu_0 \mathbf{J}_m,}$$

and the retarded solutions are

$$\boxed{V_e(\mathbf{r}, t) = \frac{1}{4\pi \epsilon_0} \int \frac{\rho_e(\mathbf{r}', t_r)}{r} d\tau', \quad V_m(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\rho_m(\mathbf{r}', t_r)}{r} d\tau',}$$

$$\boxed{\mathbf{A}_e(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}_e(\mathbf{r}', t_r)}{r} d\tau', \quad \mathbf{A}_m(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}_m(\mathbf{r}', t_r)}{r} d\tau' .}$$

Problem 10.34

The retarded potentials are

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{r} d\tau', \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{r} d\tau'.$$

We want a source localized at the origin, so we expand in powers of $\mathbf{r}'$, keeping terms up to first order:

$$\begin{aligned} r^2 &= (\mathbf{r} - \mathbf{r}') \cdot (\mathbf{r} - \mathbf{r}') = r^2 - 2\mathbf{r} \cdot \mathbf{r}' + (r')^2, \\ r &\approx r \left(1 - \frac{\mathbf{r} \cdot \mathbf{r}'}{r^2} \right), \\ \frac{1}{r} &\approx \frac{1}{r} \left(1 + \frac{\mathbf{r} \cdot \mathbf{r}'}{r^2} \right), \\ t - \frac{r}{c} &\approx t - \frac{r}{c} + \frac{\mathbf{r} \cdot \mathbf{r}'}{rc} = t_0 + \frac{\mathbf{r} \cdot \mathbf{r}'}{rc}, \end{aligned}$$

where $t_0 \equiv t - r/c$ is the retarded time for a source at the origin. Thus (Taylor expanding about t_0)

$$\rho(\mathbf{r}', t_r) = \rho\left(\mathbf{r}', t - \frac{r}{c}\right) \approx \rho(\mathbf{r}', t_0) + \frac{\mathbf{r} \cdot \mathbf{r}'}{rc} \dot{\rho}(\mathbf{r}', t_0),$$

$$\frac{\rho(\mathbf{r}', t_r)}{r} = \frac{1}{r} \left(1 + \frac{\mathbf{r} \cdot \mathbf{r}'}{r^2} \right) \left[\rho(\mathbf{r}', t_0) + \frac{\mathbf{r} \cdot \mathbf{r}'}{rc} \dot{\rho}(\mathbf{r}', t_0) \right] \approx \frac{1}{r} \rho(\mathbf{r}', t_0) + \frac{\mathbf{r} \cdot \mathbf{r}'}{r^3} \rho(\mathbf{r}', t_0) + \frac{\mathbf{r} \cdot \mathbf{r}'}{r^2 c} \dot{\rho}(\mathbf{r}', t_0),$$

and hence

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \left[\frac{1}{r} \int \rho(\mathbf{r}', t_0) d\tau' + \frac{\mathbf{r}}{r^3} \cdot \int \mathbf{r}' \rho(\mathbf{r}', t_0) d\tau' + \frac{\mathbf{r}}{r^2 c} \cdot \int \mathbf{r}' \dot{\rho}(\mathbf{r}', t_0) d\tau' \right].$$

The first integral is the total charge of the dipole, which is zero; the second integral is the dipole moment, and the third is the time derivative of the dipole moment:

$$V(\mathbf{r}, t) = \boxed{\frac{1}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}}}{r^2} \cdot \left[\mathbf{p}(t_0) + \frac{r}{c} \dot{\mathbf{p}}(t_0) \right]}.$$

By the same reasoning, the vector potential is

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \left[\frac{1}{r} \int \mathbf{J}(\mathbf{r}', t_0) d\tau' + \frac{1}{r^3} \int (\mathbf{r} \cdot \mathbf{r}') \mathbf{J}(\mathbf{r}', t_0) d\tau' + \frac{1}{r^2 c} \int (\mathbf{r} \cdot \mathbf{r}') \dot{\mathbf{J}}(\mathbf{r}', t_0) d\tau' \right].$$

From Eq. 5.31 $\int \mathbf{J}(\mathbf{r}', t) d\tau' = \dot{\mathbf{p}}(t)$, and $\mathbf{J} = \rho \mathbf{v}' = \rho(d\mathbf{r}'/dt)$ is already first order in r' , so the second and third integrals are second order, and we drop them.

$$\mathbf{A}(\mathbf{r}, t) = \boxed{\frac{\mu_0}{4\pi} \left[\frac{\dot{\mathbf{p}}(t_0)}{r} \right]}.$$

In calculating the fields, remember that the dipole moments themselves depend on r , through their argument ($t_0 = t - r/c$). For a function of t_0 alone,

$$\nabla f(t_0) = \frac{df}{dt_0} \nabla t_0 = \dot{f} \left(-\frac{1}{c} \nabla r \right) = -\frac{1}{c} \dot{f} \hat{\mathbf{r}},$$

so you simply replace ∇ by $-\frac{\hat{\mathbf{r}}}{c} \frac{d}{dt}$. For a structure of the form $\left[f(t_0) + \frac{r}{c} \dot{f}(t_0) \right]$

$$\nabla \left[f(t_0) + \frac{r}{c} \dot{f}(t_0) \right] = -\frac{\hat{\mathbf{r}}}{c} \dot{f} + \frac{r}{c} \left(-\frac{\hat{\mathbf{r}}}{c} \ddot{f} \right) + \frac{\dot{f}}{c} \hat{\mathbf{r}} = -\frac{\mathbf{r}}{c^2} \ddot{f}.$$

Using the product rule for $\nabla(\mathbf{A} \cdot \mathbf{B})$:

$$\begin{aligned} \nabla V &= \frac{1}{4\pi\epsilon_0} \left\{ \frac{\hat{\mathbf{r}}}{r^2} \times \left(\nabla \times \left[\mathbf{p} + \frac{r}{c} \dot{\mathbf{p}} \right] \right) + \left[\mathbf{p} + \frac{r}{c} \dot{\mathbf{p}} \right] \times \left(\nabla \times \frac{\hat{\mathbf{r}}}{r^2} \right) + \left(\frac{\hat{\mathbf{r}}}{r^2} \cdot \nabla \right) \left[\mathbf{p} + \frac{r}{c} \dot{\mathbf{p}} \right] + \left(\left[\mathbf{p} + \frac{r}{c} \dot{\mathbf{p}} \right] \cdot \nabla \right) \frac{\hat{\mathbf{r}}}{r^2} \right\} \\ &= \frac{1}{4\pi\epsilon_0} \left\{ \frac{\hat{\mathbf{r}}}{r^2} \times \left(-\frac{\mathbf{r}}{c^2} \times \ddot{\mathbf{p}} \right) + \mathbf{0} + \left[\frac{\hat{\mathbf{r}}}{r^2} \cdot \left(-\frac{\mathbf{r}}{c^2} \right) \right] \ddot{\mathbf{p}} + \frac{1}{r^3} \left(\left[\mathbf{p} + \frac{r}{c} \dot{\mathbf{p}} \right] - 3\hat{\mathbf{r}} \left(\hat{\mathbf{r}} \cdot \left[\mathbf{p} + \frac{r}{c} \dot{\mathbf{p}} \right] \right) \right) \right\} \\ &= \frac{1}{4\pi\epsilon_0} \left\{ -\frac{\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}})}{rc^2} + \frac{[\mathbf{p} + (r/c)\dot{\mathbf{p}}] - 3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot [\mathbf{p} + (r/c)\dot{\mathbf{p}}])}{r^3} \right\}. \end{aligned}$$

The others are easier:

$$\begin{aligned} \nabla \times \mathbf{A} &= \frac{\mu_0}{4\pi} \left[\frac{1}{r} (\nabla \times \dot{\mathbf{p}}) - \dot{\mathbf{p}} \times \left(\nabla \frac{1}{r} \right) \right] = \frac{\mu_0}{4\pi} \left[\frac{1}{r} \left(-\frac{\hat{\mathbf{r}}}{c} \times \ddot{\mathbf{p}} \right) + \left(\dot{\mathbf{p}} \times \frac{\hat{\mathbf{r}}}{r^2} \right) \right] = -\frac{\mu_0}{4\pi} \left\{ \frac{\hat{\mathbf{r}} \times [(\dot{\mathbf{p}} + (r/c)\ddot{\mathbf{p}})]}{r^2} \right\}, \\ \frac{\partial \mathbf{A}}{\partial t} &= \frac{\mu_0}{4\pi} \left(\frac{\ddot{\mathbf{p}}}{r} \right). \end{aligned}$$

The fields are therefore

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= -\nabla V - \frac{\partial \mathbf{A}}{\partial t} = \boxed{-\frac{\mu_0}{4\pi} \left\{ \frac{\ddot{\mathbf{p}} - \hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}})}{r} + c^2 \frac{[\mathbf{p} + (r/c)\dot{\mathbf{p}}] - 3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot [\mathbf{p} + (r/c)\dot{\mathbf{p}}])}{r^3} \right\}}, \\ \mathbf{B}(\mathbf{r}, t) &= \nabla \times \mathbf{A} = \boxed{-\frac{\mu_0}{4\pi} \left\{ \frac{\hat{\mathbf{r}} \times [\dot{\mathbf{p}} + (r/c)\ddot{\mathbf{p}}]}{r^2} \right\}} \end{aligned}$$

(with all dipole moments evaluated at the retarded time $t_0 = t - r/c$).

Chapter 11

Radiation

Problem 11.1

From Eq. 11.17, $\mathbf{A} = -\frac{\mu_0 p_0 \omega}{4\pi} \frac{1}{r} \sin[\omega(t - r/c)] (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}})$, so

$$\begin{aligned} \nabla \cdot \mathbf{A} &= -\frac{\mu_0 p_0 \omega}{4\pi} \left\{ \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{1}{r} \sin[\omega(t - r/c)] \cos \theta \right] + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left[-\sin^2 \theta \frac{1}{r} \sin[\omega(t - r/c)] \right] \right\} \\ &= -\frac{\mu_0 p_0 \omega}{4\pi} \left\{ \frac{1}{r^2} \left(\sin[\omega(t - r/c)] - \frac{\omega r}{c} \cos[\omega(t - r/c)] \right) \cos \theta - \frac{2 \sin \theta \cos \theta}{r^2 \sin \theta} \sin[\omega(t - r/c)] \right\} \\ &= \mu_0 \epsilon_0 \left\{ \frac{p_0 \omega}{4\pi \epsilon_0} \left(\frac{1}{r^2} \sin[\omega(t - r/c)] + \frac{\omega}{rc} \cos[\omega(t - r/c)] \right) \cos \theta \right\}. \end{aligned}$$

Meanwhile, from Eq. 11.12,

$$\begin{aligned} \frac{\partial V}{\partial t} &= \frac{p_0 \cos \theta}{4\pi \epsilon_0 r} \left\{ -\frac{\omega^2}{c} \cos[\omega(t - r/c)] - \frac{\omega}{r} \sin[\omega(t - r/c)] \right\} \\ &= -\frac{p_0 \omega}{4\pi \epsilon_0} \left\{ \frac{1}{r^2} \sin[\omega(t - r/c)] + \frac{\omega}{rc} \cos[\omega(t - r/c)] \right\} \cos \theta. \quad \text{So } \nabla \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}. \quad \text{qed} \end{aligned}$$

Problem 11.2

$$\text{Eq. 11.14: } V(\mathbf{r}, t) = -\frac{\omega}{4\pi \epsilon_0 c} \frac{\mathbf{p}_0 \cdot \hat{\mathbf{r}}}{r} \sin[\omega(t - r/c)]. \quad \text{Eq. 11.17: } \mathbf{A}(\mathbf{r}, t) = -\frac{\mu_0 \omega}{4\pi} \frac{\mathbf{p}_0}{r} \sin[\omega(t - r/c)].$$

Now $\mathbf{p}_0 \times \hat{\mathbf{r}} = p_0 \sin \theta \hat{\boldsymbol{\phi}}$ and $\hat{\mathbf{r}} \times (\mathbf{p}_0 \times \hat{\mathbf{r}}) = p_0 \sin \theta (\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}}) = -p_0 \sin \theta \hat{\boldsymbol{\theta}}$, so

$$\text{Eq. 11.18: } \mathbf{E}(\mathbf{r}, t) = \frac{\mu_0 \omega^2}{4\pi} \frac{\hat{\mathbf{r}} \times (\mathbf{p}_0 \times \hat{\mathbf{r}})}{r} \cos[\omega(t - r/c)]. \quad \text{Eq. 11.19: } \mathbf{B}(\mathbf{r}, t) = -\frac{\mu_0 \omega^2}{4\pi c} \frac{(\mathbf{p}_0 \times \hat{\mathbf{r}})}{r} \cos[\omega(t - r/c)].$$

$$\text{Eq. 11.21: } \langle \mathbf{S} \rangle = \frac{\mu_0 \omega^4}{32\pi^2 c} \frac{(\mathbf{p}_0 \times \hat{\mathbf{r}})^2}{r^2} \hat{\mathbf{r}}.$$

Problem 11.3

$P = I^2 R = q_0^2 \omega^2 \sin^2(\omega t) R$ (Eq. 11.15) $\Rightarrow \langle P \rangle = \frac{1}{2} q_0^2 \omega^2 R$. Equate this to Eq. 11.22:

$$\frac{1}{2} q_0^2 \omega^2 R = \frac{\mu_0 q_0^2 d^2 \omega^4}{12\pi c} \Rightarrow R = \frac{\mu_0 d^2 \omega^2}{6\pi c}; \quad \text{or, since } \omega = \frac{2\pi c}{\lambda},$$

$$R = \frac{\mu_0 d^2}{6\pi c} \frac{4\pi^2 c^2}{\lambda^2} = \frac{2}{3} \pi \mu_0 c \left(\frac{d}{\lambda} \right)^2 = \frac{2}{3} \pi (4\pi \times 10^{-7}) (3 \times 10^8) \left(\frac{d}{\lambda} \right)^2 = 80\pi^2 \left(\frac{d}{\lambda} \right)^2 \Omega = \boxed{789.6(d/\lambda)^2 \Omega}.$$

For the wires in an ordinary radio, with $d = 5 \times 10^{-2}$ m and (say) $\lambda = 10^3$ m, $R = 790(5 \times 10^{-5})^2 = 2 \times 10^{-6} \Omega$, which is negligible compared to the Ohmic resistance.

Problem 11.4

By the superposition principle, we can *add* the potentials of the two dipoles. Let's first express V (Eq. 11.14) in Cartesian coordinates: $V(x, y, z, t) = -\frac{p_0\omega}{4\pi\epsilon_0 c} \left(\frac{z}{x^2 + y^2 + z^2} \right) \sin[\omega(t - r/c)]$. That's for an oscillating dipole along the z axis. For one along x or y , we just change z to x or y . In the present case,

$\mathbf{p} = p_0[\cos(\omega t)\hat{\mathbf{x}} + \cos(\omega t - \pi/2)\hat{\mathbf{y}}]$, so the one along y is delayed by a phase angle $\pi/2$:

$\sin[\omega(t - r/c)] \rightarrow \sin[\omega(t - r/c) - \pi/2] = -\cos[\omega(t - r/c)]$ (just let $\omega t \rightarrow \omega t - \pi/2$). Thus

$$\begin{aligned} V &= -\frac{p_0\omega}{4\pi\epsilon_0 c} \left\{ \frac{x}{x^2 + y^2 + z^2} \sin[\omega(t - r/c)] - \frac{y}{x^2 + y^2 + z^2} \cos[\omega(t - r/c)] \right\} \\ &= -\frac{p_0\omega}{4\pi\epsilon_0 c} \frac{\sin\theta}{r} \{ \cos\phi \sin[\omega(t - r/c)] - \sin\phi \cos[\omega(t - r/c)] \}. \quad \text{Similarly,} \\ \mathbf{A} &= -\frac{\mu_0 p_0 \omega}{4\pi r} \{ \sin[\omega(t - r/c)] \hat{\mathbf{x}} - \cos[\omega(t - r/c)] \hat{\mathbf{y}} \}. \end{aligned}$$

We *could* get the fields by differentiating these potentials, but I prefer to work with Eqs. 11.18 and 11.19, using superposition. Since $\hat{\mathbf{z}} = \cos\theta\hat{\mathbf{r}} - \sin\theta\hat{\boldsymbol{\theta}}$, and $\cos\theta = z/r$, Eq. 11.18 can be written

$\mathbf{E} = \frac{\mu_0 p_0 \omega^2}{4\pi r} \cos[\omega(t - r/c)] \left(\hat{\mathbf{z}} - \frac{z}{r} \hat{\mathbf{r}} \right)$. In the case of the rotating dipole, therefore,

$$\begin{aligned} \mathbf{E} &= \frac{\mu_0 p_0 \omega^2}{4\pi r} \left\{ \cos[\omega(t - r/c)] \left(\hat{\mathbf{x}} - \frac{x}{r} \hat{\mathbf{r}} \right) + \sin[\omega(t - r/c)] \left(\hat{\mathbf{y}} - \frac{y}{r} \hat{\mathbf{r}} \right) \right\}, \\ \mathbf{B} &= \frac{1}{c} (\hat{\mathbf{r}} \times \mathbf{E}). \end{aligned}$$

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0 c} [\mathbf{E} \times (\hat{\mathbf{r}} \times \mathbf{E})] = \frac{1}{\mu_0 c} [E^2 \hat{\mathbf{r}} - (\mathbf{E} \cdot \hat{\mathbf{r}}) \mathbf{E}] = \frac{E^2}{\mu_0 c} \hat{\mathbf{r}} \quad (\text{notice that } \mathbf{E} \cdot \hat{\mathbf{r}} = 0). \quad \text{Now}$$

$$E^2 = \left(\frac{\mu_0 p_0 \omega^2}{4\pi r} \right)^2 \{ a^2 \cos^2[\omega(t - r/c)] + b^2 \sin^2[\omega(t - r/c)] + 2(\mathbf{a} \cdot \mathbf{b}) \sin[\omega(t - r/c)] \cos[\omega(t - r/c)] \},$$

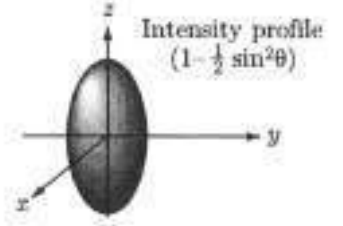
where $\mathbf{a} \equiv \hat{\mathbf{x}} - (x/r)\hat{\mathbf{r}}$ and $\mathbf{b} \equiv \hat{\mathbf{y}} - (y/r)\hat{\mathbf{r}}$. Noting that $\hat{\mathbf{x}} \cdot \mathbf{r} = x$ and $\hat{\mathbf{y}} \cdot \mathbf{r} = y$, we have

$$a^2 = 1 + \frac{x^2}{r^2} - 2\frac{x^2}{r^2} = 1 - \frac{x^2}{r^2}; \quad b^2 = 1 - \frac{y^2}{r^2}; \quad \mathbf{a} \cdot \mathbf{b} = -\frac{y}{r} \frac{x}{r} - \frac{x}{r} \frac{y}{r} + \frac{xy}{r^2} = -\frac{xy}{r^2}.$$

$$\begin{aligned} E^2 &= \left(\frac{\mu_0 p_0 \omega^2}{4\pi r} \right)^2 \left\{ \left(1 - \frac{x^2}{r^2} \right) \cos^2[\omega(t - r/c)] + \left(1 - \frac{y^2}{r^2} \right) \sin^2[\omega(t - r/c)] \right. \\ &\quad \left. - 2\frac{xy}{r^2} \sin[\omega(t - r/c)] \cos[\omega(t - r/c)] \right\} \\ &= \left(\frac{\mu_0 p_0 \omega^2}{4\pi r} \right)^2 \left\{ 1 - \frac{1}{r^2} (x^2 \cos^2[\omega(t - r/c)] + 2xy \sin[\omega(t - r/c)] \cos[\omega(t - r/c)] + y^2 \sin^2[\omega(t - r/c)]) \right\} \\ &= \left(\frac{\mu_0 p_0 \omega^2}{4\pi r} \right)^2 \left\{ 1 - \frac{1}{r^2} (x \cos[\omega(t - r/c)] + y \sin[\omega(t - r/c)])^2 \right\} \end{aligned}$$

But $x = r \sin \theta \cos \phi$ and $y = r \sin \theta \sin \phi$.

$$\begin{aligned} &= \left(\frac{\mu_0 p_0 \omega^2}{4\pi r} \right)^2 \left\{ 1 - \sin^2 \theta (\cos \phi \cos[\omega(t - r/c)] + \sin \phi \sin[\omega(t - r/c)])^2 \right\} \\ &= \left(\frac{\mu_0 p_0 \omega^2}{4\pi r} \right)^2 \left\{ 1 - (\sin \theta \cos[\omega(t - r/c) - \phi])^2 \right\}. \end{aligned}$$



$$\mathbf{S} = \boxed{\frac{\mu_0}{c} \left(\frac{p_0 \omega^2}{4\pi r} \right)^2 \left\{ 1 - (\sin \theta \cos[\omega(t - r/c) - \phi])^2 \right\} \hat{\mathbf{r}}}.$$

$$\langle \mathbf{S} \rangle = \frac{\mu_0}{c} \left(\frac{p_0 \omega^2}{4\pi r} \right)^2 \left[1 - \frac{1}{2} \sin^2 \theta \right] \hat{\mathbf{r}}.$$

$$\begin{aligned} P &= \int \langle \mathbf{S} \rangle \cdot d\mathbf{a} = \frac{\mu_0}{c} \left(\frac{p_0 \omega^2}{4\pi} \right)^2 \int \frac{1}{r^2} \left(1 - \frac{1}{2} \sin^2 \theta \right) r^2 \sin \theta d\theta d\phi \\ &= \frac{\mu_0 p_0^2 \omega^4}{16\pi^2 c} 2\pi \left[\int_0^\pi \sin \theta d\theta - \frac{1}{2} \int_0^\pi \sin^3 \theta d\theta \right] = \frac{\mu_0 p_0^2 \omega^4}{8\pi c} \left(2 - \frac{1}{2} \cdot \frac{4}{3} \right) = \boxed{\frac{\mu_0 p_0^2 \omega^4}{6\pi c}}. \end{aligned}$$

This is *twice* the power radiated by either oscillating dipole alone (Eq. 11.22). In general, $\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0} [(\mathbf{E}_1 + \mathbf{E}_2) \times (\mathbf{B}_1 + \mathbf{B}_2)] = \frac{1}{\mu_0} [(\mathbf{E}_1 \times \mathbf{B}_1) + (\mathbf{E}_2 \times \mathbf{B}_2) + (\mathbf{E}_1 \times \mathbf{B}_2) + (\mathbf{E}_2 \times \mathbf{B}_1)] = \mathbf{S}_1 + \mathbf{S}_2 + \text{cross terms}$. In this particular case the fields of 1 and 2 are 90° out of phase, so the cross terms go to zero in the time averaging, and the total power radiated is just the sum of the two individual powers.

Problem 11.5

Go back to Eq. 11.33:

$$\mathbf{A} = \frac{\mu_0 m_0}{4\pi} \left(\frac{\sin \theta}{r} \right) \left\{ \frac{1}{r} \cos[\omega(t - r/c)] - \frac{\omega}{c} \sin[\omega(t - r/c)] \right\} \hat{\phi}.$$

Since $V = 0$ here,

$$\begin{aligned}
\mathbf{E} &= -\frac{\partial \mathbf{A}}{\partial t} = -\frac{\mu_0 m_0}{4\pi} \left(\frac{\sin \theta}{r} \right) \left\{ \frac{1}{r} (-\omega) \sin[\omega(t - r/c)] - \frac{\omega}{c} \cos[\omega(t - r/c)] \right\} \hat{\phi} \\
&= \boxed{\frac{\mu_0 m_0 \omega}{4\pi} \left(\frac{\sin \theta}{r} \right) \left\{ \frac{1}{r} \sin[\omega(t - r/c)] + \frac{\omega}{c} \cos[\omega(t - r/c)] \right\} \hat{\phi}} \\
\mathbf{B} &= \nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\phi \sin \theta) \hat{\mathbf{r}} - \frac{1}{r} \frac{\partial}{\partial r} (r A_\phi) \hat{\boldsymbol{\theta}} \\
&= \frac{\mu_0 m_0}{4\pi} \left\{ \frac{1}{r \sin \theta} \frac{2 \sin \theta \cos \theta}{r} \left[\frac{1}{r} \cos[\omega(t - r/c)] - \frac{\omega}{c} \sin[\omega(t - r/c)] \right] \hat{\mathbf{r}} \right. \\
&\quad \left. - \frac{\sin \theta}{r} \left[-\frac{1}{r^2} \cos[\omega(t - r/c)] + \frac{\omega}{rc} \sin[\omega(t - r/c)] - \frac{\omega}{c} \left(-\frac{\omega}{c} \right) \cos[\omega(t - r/c)] \right] \hat{\boldsymbol{\theta}} \right\} \\
&= \boxed{\frac{\mu_0 m_0}{4\pi} \left\{ \frac{2 \cos \theta}{r^2} \left[\frac{1}{r} \cos[\omega(t - r/c)] - \frac{\omega}{c} \sin[\omega(t - r/c)] \right] \hat{\mathbf{r}} \right. \\
&\quad \left. - \frac{\sin \theta}{r} \left[-\frac{1}{r^2} \cos[\omega(t - r/c)] + \frac{\omega}{rc} \sin[\omega(t - r/c)] + \left(\frac{\omega}{c} \right)^2 \cos[\omega(t - r/c)] \right] \hat{\boldsymbol{\theta}} \right\}}.
\end{aligned}$$

These are precisely the fields we studied in Prob. 9.35, with $A \rightarrow \frac{\mu_0 m_0 \omega^2}{4\pi c}$. The Poynting vector (quoting the solution to that problem) is

$$\boxed{\mathbf{S} = \frac{\mu_0 m_0^2 \omega^3}{16\pi^2 c^2} \left(\frac{\sin \theta}{r^2} \right) \left\{ \frac{2 \cos \theta}{r} \left[\left(1 - \frac{c^2}{\omega^2 r^2} \right) \sin u \cos u + \frac{c}{\omega r} (\cos^2 u - \sin^2 u) \right] \hat{\boldsymbol{\theta}} \right.} \\
\left. \sin \theta \left[\left(-\frac{2}{r} + \frac{c^2}{\omega^2 r^3} \right) \sin u \cos u + \frac{\omega}{c} \cos^2 u + \frac{c}{\omega r^2} (\sin^2 u - \cos^2 u) \right] \hat{\mathbf{r}} \right\},}$$

where $u \equiv -\omega(t - r/c)$. The intensity is $\langle \mathbf{S} \rangle = \frac{\mu_0 m_0^2 \omega^4}{32\pi^2 c^3} \frac{\sin^2 \theta}{r^2} \hat{\mathbf{r}}$, the same as Eq. 11.39.

Problem 11.6

$$I^2 R = I_0^2 R \cos^2(\omega t) \Rightarrow \langle P \rangle = \frac{1}{2} I_0^2 R = \frac{\mu_0 m_0^2 \omega^4}{12\pi c^3} = \frac{\mu_0 \pi^2 b^4 I_0^2 \omega^4}{12\pi c^3}, \text{ so } \boxed{R = \frac{\mu_0 \pi b^4 \omega^4}{6c^3}}; \text{ or, since } \omega = \frac{2\pi c}{\lambda},$$

$$R = \frac{\mu_0 \pi b^4}{6c^3} \frac{16\pi^4 c^4}{\lambda^4} = \boxed{\frac{8}{3} \pi^5 \mu_0 c \left(\frac{b}{\lambda} \right)^4} = \frac{8}{3} (\pi^5) (4\pi \times 10^{-7}) (3 \times 10^8) (b/\lambda)^4 = \boxed{3.08 \times 10^5 (b/\lambda)^4 \Omega}.$$

Because $b \ll \lambda$, and R goes like the *fourth* power of this small number, R is typically much smaller than the electric radiative resistance (Prob. 11.3). For the dimensions we used in Prob. 11.3 ($b = 5 \text{ cm}$ and $\lambda = 10^3 \text{ m}$), $R = 3 \times 10^5 (5 \times 10^{-5})^4 = 2 \times 10^{-12} \Omega$, which is a millionth of the comparable electrical radiative resistance.

Problem 11.7

With $\alpha = 90^\circ$, Eq. 7.68 $\Rightarrow \mathbf{E}' = c\mathbf{B}$, $\mathbf{B}' = -\mathbf{E}/c$, $q'_m = -cq_e \Rightarrow m_0 \equiv q'_m d = -cq_e d = -cp_0$. So

$$\mathbf{E}' = c \left\{ -\frac{\mu_0 (-m_0/c) \omega^2}{4\pi c} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\phi} \right\} = \boxed{\frac{\mu_0 m_0 \omega^2}{4\pi c} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\phi}}.$$

$$\mathbf{B}' = -\frac{1}{c} \left\{ -\frac{\mu_0 (-m_0/c) \omega^2}{4\pi} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\boldsymbol{\theta}} \right\} = \boxed{-\frac{\mu_0 m_0 \omega^2}{4\pi c^2} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\boldsymbol{\theta}}}.$$

These are *identical* to the fields of an Ampère dipole (Eqs. 11.36 and 11.37), which is consistent with our general experience that the two models generate identical fields *except right at* the dipole (not relevant here, since we're in the radiation zone).

Problem 11.8

(a) The power radiated (Eq. 11.60) is $\frac{dW_r}{dt} = \frac{\mu_0}{6\pi c} \ddot{p}^2$. In this case $p = Qd$, so $\ddot{p} = \ddot{Q}d = Q_0 \left(\frac{1}{RC} \right)^2 e^{-t/RC} d$, so $\frac{dW_r}{dt} = \frac{\mu_0}{6\pi c} \frac{(Q_0 d)^2}{(RC)^4} e^{-2t/RC}$, and the total energy radiated is

$$W_r = \frac{\mu_0}{6\pi c} \frac{(Q_0 d)^2}{(RC)^4} \int_0^\infty e^{-2t/RC} dt = \frac{\mu_0}{6\pi c} \frac{(Q_0 d)^2}{(RC)^4} \left[-\frac{RC}{2} e^{-2t/RC} \right] \bigg|_0^\infty = \frac{\mu_0}{6\pi c} \frac{(Q_0 d)^2}{(RC)^4} \frac{RC}{2} = \frac{\mu_0}{12\pi c} \frac{(Q_0 d)^2}{(RC)^3}.$$

The fraction of the original energy that is radiated is therefore

$$f = \frac{W_r}{W_0} = \frac{\mu_0}{12\pi c} \frac{(Q_0 d)^2}{(RC)^3} \frac{2C}{Q_0^2} = \boxed{\frac{\mu_0}{6\pi c} \frac{d^2}{R^3 C^2}}.$$

[Technically, $\dot{Q}(t)$ is discontinuous at $t = 0$, and $\ddot{Q}$ picks up a delta function. But any *real* circuit has some (self-)inductance, which smoothes out the sudden change in $\dot{Q}$.]

(b) With the parameters given,

$$f = \frac{4\pi \times 10^{-7}}{6\pi(3 \times 10^8)} \frac{10^{-8}}{(10^6)(10^{-24})} = \boxed{2.22 \times 10^{-6}}.$$

About a millionth of the total energy is radiated away, so yes, this is negligible.

Problem 11.9

$\mathbf{p}(t) = p_0[\cos(\omega t) \hat{\mathbf{x}} + \sin(\omega t) \hat{\mathbf{y}}] \Rightarrow \ddot{\mathbf{p}}(t) = -\omega^2 p_0[\cos(\omega t) \hat{\mathbf{x}} + \sin(\omega t) \hat{\mathbf{y}}] \Rightarrow$
 $[\ddot{\mathbf{p}}(t)]^2 = \omega^4 p_0^2[\cos^2(\omega t) + \sin^2(\omega t)] = p_0^2 \omega^4$. So Eq. 11.59 says $\mathbf{S} = \frac{\mu_0 p_0^2 \omega^4}{16\pi^2 c} \frac{\sin^2 \theta}{r^2} \hat{\mathbf{r}}$. (This appears to disagree with the answer to Prob. 11.4. The reason is that in Eq. 11.59 the polar axis is along the direction of $\ddot{\mathbf{p}}(t_0)$; as the dipole rotates, so do the axes. Thus the angle θ here is not the same as in Prob. 11.4.) Meanwhile, Eq. 11.60 says $P = \frac{\mu_0 p_0^2 \omega^4}{6\pi c}$. (This *does* agree with Prob. 11.4, because we have now integrated over all angles, and the orientation of the polar axis irrelevant.)

Problem 11.10

At $t = 0$ the dipole moment of the ring is

$$\begin{aligned} \mathbf{p}_0 &= \int \lambda \mathbf{r} dl = \int (\lambda_0 \sin \phi)(b \sin \phi \hat{\mathbf{y}} + b \cos \phi \hat{\mathbf{x}}) b d\phi = \lambda_0 b^2 \left(\hat{\mathbf{y}} \int_0^{2\pi} \sin^2 \phi d\phi + \hat{\mathbf{x}} \int_0^{2\pi} \sin \phi \cos \phi d\phi \right) \\ &= \lambda b^2 (\pi \hat{\mathbf{y}} + 0 \hat{\mathbf{x}}) = \pi b^2 \lambda_0 \hat{\mathbf{y}}. \end{aligned}$$

As it rotates (counterclockwise, say) $\mathbf{p}(t) = p_0[\cos(\omega t) \hat{\mathbf{y}} - \sin(\omega t) \hat{\mathbf{x}}]$, so $\ddot{\mathbf{p}} = -\omega^2 \mathbf{p}$, and hence $(\ddot{\mathbf{p}})^2 = \omega^4 p_0^2$.

Therefore (Eq. 11.60) $P = \frac{\mu_0}{6\pi c} \omega^4 (\pi b^2 \lambda_0)^2 = \boxed{\frac{\pi \mu_0 \omega^4 b^4 \lambda_0^2}{6c}}$.

Problem 11.11

Here $V = 0$ (since the ring is neutral), and the current depends only on t (not on position), so the retarded vector potential (Eq. 11.52) is $\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \oint \frac{I(t - \mathbf{r} \cdot \mathbf{z} / c)}{\mathbf{z}} d\mathbf{l}'$. But in this case it does *not* suffice to replace $\mathbf{z}$ by r in the denominator—that would lead to Eq. 11.54, and hence to $\mathbf{A} = \mathbf{0}$ (since $\mathbf{p} = \mathbf{0}$). Instead, use Eq. 11.30: $\frac{1}{\mathbf{z}} \cong \frac{1}{r} \left(1 + \frac{b}{r} \sin \theta \cos \phi' \right)$. Meanwhile, $d\mathbf{l}' = b d\phi' \hat{\phi} = b(-\sin \phi' \hat{\mathbf{x}} + \cos \phi' \hat{\mathbf{y}}) d\phi'$, and

$$I(t - \mathbf{r} \cdot \mathbf{z} / c) \cong I(t - r/c + (b/c) \sin \theta \cos \phi') = I(t_0 + (b/c) \sin \theta \cos \phi') \cong I(t_0) + \dot{I}(t_0) \frac{b}{c} \sin \theta \cos \phi'$$

(carrying all terms to first order in b). As always, $t_0 = t - r/c$. (From now on I'll suppress the argument: I , $\dot{I}$, etc. are all to be evaluated at t_0 .) Then

$$\begin{aligned} \mathbf{A}(\mathbf{r}, t) &= \frac{\mu_0}{4\pi} \oint \frac{1}{r} \left(1 + \frac{b}{r} \sin \theta \cos \phi' \right) \left(I + \dot{I} \frac{b}{c} \sin \theta \cos \phi' \right) b(-\sin \phi' \hat{\mathbf{x}} + \cos \phi' \hat{\mathbf{y}}) d\phi' \\ &\cong \frac{\mu_0 b}{4\pi r} \int_0^{2\pi} \left[I + \dot{I} \frac{b}{c} \sin \theta \cos \phi' + I \frac{b}{r} \sin \theta \cos \phi' \right] (-\sin \phi' \hat{\mathbf{x}} + \cos \phi' \hat{\mathbf{y}}) d\phi'. \\ \text{But } \int_0^{2\pi} \sin \phi' d\phi' &= \int_0^{2\pi} \cos \phi' d\phi' = \int_0^{2\pi} \sin \phi' \cos \phi' d\phi' = 0, \text{ while } \int_0^{2\pi} \cos^2 \phi' d\phi' = \pi. \\ &= \frac{\mu_0 b}{4\pi r} (\pi \hat{\mathbf{y}}) \left[\dot{I} \frac{b}{c} \sin \theta + I \frac{b}{r} \sin \theta \right] = \frac{\mu_0 b^2}{4r^2} \sin \theta \left(I + \frac{r}{c} \dot{I} \right) \hat{\mathbf{y}}. \end{aligned}$$

In general (i.e. for points *not* on the xz plane) $\hat{\mathbf{y}} \rightarrow \hat{\phi}$; moreover, in the radiation zone we are not interested

in terms that go like $1/r^2$, so $\boxed{\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0 b^2}{4c} \left[\dot{I}(t - r/c) \right] \frac{\sin \theta}{r} \hat{\phi}.}$

$$\mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{A}}{\partial t} = \boxed{-\frac{\mu_0 b^2}{4c} \left[\ddot{I}(t - r/c) \right] \frac{\sin \theta}{r} \hat{\phi}.}$$

$$\begin{aligned} \mathbf{B}(\mathbf{r}, t) &= \nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\phi \sin \theta) \hat{\mathbf{r}} - \frac{1}{r} \frac{\partial}{\partial r} (r A_\phi) \hat{\theta} \\ &= \frac{\mu_0 b^2}{4c} \left[\frac{\dot{I}}{r \sin \theta} \frac{1}{r} 2 \sin \theta \cos \theta \hat{\mathbf{r}} - \frac{1}{r} \ddot{I} \left(-\frac{1}{c} \right) \sin \theta \hat{\theta} \right] = \boxed{\frac{\mu_0 b^2}{4c^2} \ddot{I} \frac{\sin \theta}{r} \hat{\theta}.} \end{aligned}$$

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0 c} \left(\frac{\mu_0 b^2}{4c} \ddot{I} \frac{\sin \theta}{r} \right)^2 (-\hat{\phi} \times \hat{\theta}) = \boxed{\frac{\mu_0}{16c^3} \left(b^2 \ddot{I} \right)^2 \frac{\sin^2 \theta}{r^2} \hat{\mathbf{r}}.}$$

$$\begin{aligned} P &= \int \mathbf{S} \cdot d\mathbf{a} = \frac{\mu_0}{16c^3} \left(b^2 \ddot{I} \right)^2 \int \frac{\sin^2 \theta}{r^2} r^2 \sin \theta d\theta d\phi = \frac{\mu_0}{16c^3} \left(b^2 \ddot{I} \right)^2 (2\pi) \left(\frac{4}{3} \right) = \frac{\mu_0 \pi}{6c^3} \left(b^2 \ddot{I} \right)^2 \\ &= \boxed{\frac{\mu_0 \ddot{m}^2}{6\pi c^3}.} \quad (\text{Note that } m = I\pi b^2, \text{ so } \ddot{m} = \ddot{I}\pi b^2.) \end{aligned}$$

Problem 11.12

$\mathbf{p} = -ey \hat{\mathbf{y}}$, $y = \frac{1}{2}gt^2$, so $\mathbf{p} = -\frac{1}{2}get^2 \hat{\mathbf{y}}$; $\ddot{\mathbf{p}} = -ge \hat{\mathbf{y}}$. Therefore (Eq. 11.60) $P = \frac{\mu_0}{6\pi c} (ge)^2$. Now, the time it takes to fall a distance h is given by $h = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{2h/g}$, so the energy radiated in falling a distance h

is $U_{\text{rad}} = Pt = \frac{\mu_0 (ge)^2}{6\pi c} \sqrt{2h/g}$. Meanwhile, the potential energy lost is $U_{\text{pot}} = mgh$. So the *fraction* is

$$f = \frac{U_{\text{rad}}}{U_{\text{pot}}} = \frac{\mu_0 g^2 e^2}{6\pi c} \sqrt{\frac{2h}{g}} \frac{1}{mgh} = \boxed{\frac{\mu_0 e^2}{6\pi mc} \sqrt{\frac{2g}{h}}} = \frac{(4\pi \times 10^{-7})(1.6 \times 10^{-19})^2}{6\pi(9.11 \times 10^{-31})(3 \times 10^8)} \sqrt{\frac{(2)(9.8)}{(0.01)}} = \boxed{2.76 \times 10^{-22}}.$$

Evidently *almost* all the energy goes into kinetic form (as indeed I *assumed* in saying $y = \frac{1}{2}gt^2$).

Problem 11.13

The power radiated (Eq. 11.70) is $\frac{dW_r}{dt} = \frac{\mu_0 q^2 a^2}{6\pi c}$. Here $F = \frac{1}{4\pi\epsilon_0} \frac{qQ}{x^2} = ma$, so $a = \frac{k}{x^2}$, where $k \equiv \frac{qQ}{4\pi\epsilon_0 m}$. The energy radiated (twice that radiated on the way *out*) is

$$W_r = \frac{\mu_0 q^2}{6\pi c} \int \frac{k^2}{x^4} dt = \frac{\mu_0 q^2}{6\pi c} 2k^2 \int_{x_0}^{\infty} \frac{1}{x^4} \frac{1}{v} dx,$$

where x_0 is the distance of closest approach. Conservation of energy (ignoring radiative losses, as suggested) says

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 + \frac{1}{4\pi\epsilon_0} \frac{qQ}{x} \Rightarrow v^2 = v_0^2 - 2\frac{qQ}{4\pi\epsilon_0 m} \frac{1}{x} = v_0^2 - \frac{2k}{x}, \text{ and } x_0 = \frac{2k}{v_0^2}$$

(note that q is at x_0 when $v = 0$). So

$$W_r = \frac{\mu_0 q^2}{6\pi c} 2k^2 \int_{x_0}^{\infty} \frac{1}{x^4 \sqrt{v_0^2 - 2k/x}} dx = \frac{\mu_0 q^2}{6\pi c} \frac{2k^2}{\sqrt{2k}} \int_{x_0}^{\infty} \frac{1}{x^4 \sqrt{(1/x_0) - (1/x)}} dx = \frac{\mu_0 q^2}{6\pi c} \frac{2k^2}{\sqrt{2k}} \frac{16}{15x_0^{5/2}}.$$

(I used Mathematica to do the integral.) Simplifying,

$$W_r = \frac{2\mu_0 q^2 v_0^5}{45\pi c k} = \frac{2\mu_0 q^2 v_0^5}{45\pi c} \frac{4\pi\epsilon_0 m}{qQ} = \frac{8qm v_0^5}{45c^3 Q} \Rightarrow f = \frac{W_r}{W_0} = \frac{8qm v_0^5}{45c^3 Q} \frac{2}{mv_0^2} = \boxed{\frac{16q}{45Q} \left(\frac{v_0}{c}\right)^3}.$$

Problem 11.14

$$F = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} = ma = m \frac{v^2}{r} \Rightarrow v = \sqrt{\frac{1}{4\pi\epsilon_0} \frac{q^2}{mr}}. \text{ At the beginning } (r_0 = 0.5 \text{ \AA}),$$

$$\frac{v}{c} = \left[\frac{(1.6 \times 10^{-19})^2}{4\pi(8.85 \times 10^{-12})(9.11 \times 10^{-31})(5 \times 10^{-11})} \right]^{-1/2} \frac{1}{3 \times 10^8} = 0.0075,$$

and when the radius is one hundredth of this v/c is only 10 times greater (0.075), so for *most* of the trip the velocity is safely nonrelativistic.

From the Larmor formula, $P = \frac{\mu_0 q^2}{6\pi c} \left(\frac{v^2}{r}\right)^2 = \frac{\mu_0 q^2}{6\pi c} \left(\frac{1}{4\pi\epsilon_0} \frac{q^2}{mr^2}\right)^2$ (since $a = v^2/r$), and $P = -dU/dt$, where U is the (total) energy of the electron:

$$U = U_{\text{kin}} + U_{\text{pot}} = \frac{1}{2}mv^2 - \frac{1}{4\pi\epsilon_0} \frac{q^2}{r} = \frac{1}{2} \left(\frac{1}{4\pi\epsilon_0} \frac{q^2}{r} \right) - \frac{1}{4\pi\epsilon_0} \frac{q^2}{r} = -\frac{1}{8\pi\epsilon_0} \frac{q^2}{r}.$$

$$\text{So } -\frac{dU}{dt} = -\frac{1}{8\pi\epsilon_0} \frac{q^2}{r^2} \frac{dr}{dt} = P = \frac{q^2}{6\pi\epsilon_0 c^3} \left(\frac{1}{4\pi\epsilon_0} \frac{q^2}{mr^2} \right)^2, \text{ and hence } \frac{dr}{dt} = -\frac{1}{3c} \left(\frac{q^2}{2\pi\epsilon_0 mc} \right)^2 \frac{1}{r^2}, \text{ or}$$

$$\begin{aligned}
dt &= -3c \left(\frac{2\pi\epsilon_0 mc}{q^2} \right)^2 r^2 dr \Rightarrow t = -3c \left(\frac{2\pi\epsilon_0 mc}{q^2} \right)^2 \int_{r_0}^0 r^2 dr = \boxed{c \left(\frac{2\pi\epsilon_0 mc}{q^2} \right)^2 r_0^3} \\
&= (3 \times 10^8) \left[\frac{2\pi(8.85 \times 10^{-12})(9.11 \times 10^{-31})(3 \times 10^8)}{(1.6 \times 10^{-19})^2} \right]^2 (5 \times 10^{-11})^3 = \boxed{1.3 \times 10^{-11} \text{ s.}} \text{ (Not very long!)}
\end{aligned}$$

Problem 11.15

According to Eq. 11.74, the maximum occurs at $\frac{d}{d\theta} \left[\frac{\sin^2 \theta}{(1 - \beta \cos \theta)^5} \right] = 0$. Thus

$$\frac{2 \sin \theta \cos \theta}{(1 - \beta \cos \theta)^5} - \frac{5 \sin^2 \theta (\beta \sin \theta)}{(1 - \beta \cos \theta)^6} = 0 \Rightarrow 2 \cos \theta (1 - \beta \cos \theta) = 5 \beta \sin^2 \theta = 5 \beta (1 - \cos^2 \theta);$$

$$2 \cos \theta - 2 \beta \cos^2 \theta = 5 \beta - 5 \beta \cos^2 \theta, \text{ or } 3 \beta \cos^2 \theta + 2 \cos \theta - 5 \beta = 0. \text{ So}$$

$$\cos \theta = \frac{-2 \pm \sqrt{4 + 60\beta^2}}{6\beta} = \frac{1}{3\beta} \left(\pm \sqrt{1 + 15\beta^2} - 1 \right). \text{ We want the plus sign, since } \theta_m \rightarrow 90^\circ (\cos \theta_m = 0) \text{ when}$$

$$\beta \rightarrow 0 \text{ (Fig. 11.11): } \boxed{\theta_{\max} = \cos^{-1} \left(\frac{\sqrt{1 + 15\beta^2} - 1}{3\beta} \right)}.$$

For $v \approx c$, $\beta \approx 1$; write $\beta = 1 - \epsilon$ (where $\epsilon \ll 1$), and expand to first order in ϵ :

$$\begin{aligned}
\left(\frac{\sqrt{1 + 15\beta^2} - 1}{3\beta} \right) &= \frac{1}{3(1 - \epsilon)} \left[\sqrt{1 + 15(1 - \epsilon)^2} - 1 \right] \cong \frac{1}{3}(1 + \epsilon) \left[\sqrt{1 + 15(1 - 2\epsilon)} - 1 \right] \\
&= \frac{1}{3}(1 + \epsilon) \left[\sqrt{16 - 30\epsilon} - 1 \right] = \frac{1}{3}(1 + \epsilon) \left[4\sqrt{1 - (15\epsilon/8)} - 1 \right] = \frac{1}{3}(1 + \epsilon) \left[4 \left(1 - \frac{15\epsilon}{16} \right) - 1 \right] \\
&= \frac{1}{3}(1 + \epsilon) \left(3 - \frac{15\epsilon}{4} \right) = (1 + \epsilon) \left(1 - \frac{5\epsilon}{4} \right) \cong 1 + \epsilon - \frac{5}{4}\epsilon = 1 - \frac{1}{4}\epsilon.
\end{aligned}$$

$$\text{Evidently } \theta_{\max} \approx 0, \text{ so } \cos \theta_{\max} \cong 1 - \frac{1}{2}\theta_{\max}^2 = 1 - \frac{1}{4}\epsilon \Rightarrow \theta_{\max}^2 = \frac{1}{2}\epsilon, \text{ or } \theta_{\max} \cong \sqrt{\epsilon/2} = \boxed{\sqrt{(1 - \beta)/2}}.$$

$$\text{Let } f \equiv \frac{(dP/d\Omega|_{\theta_m})_{\text{ur}}}{(dP/d\Omega|_{\theta_m})_{\text{rest}}} = \left[\frac{\sin^2 \theta_{\max}}{(1 - \beta \cos \theta_{\max})^5} \right]_{\text{ur}}. \text{ Now } \sin^2 \theta_{\max} \cong \epsilon/2, \text{ and}$$

$$(1 - \beta \cos \theta_{\max}) \cong 1 - (1 - \epsilon)(1 - \frac{1}{4}\epsilon) \cong 1 - (1 - \epsilon - \frac{1}{4}\epsilon) = \frac{5}{4}\epsilon. \text{ So } f = \frac{\epsilon/2}{(5\epsilon/4)^5} = \left(\frac{4}{5} \right)^5 \frac{1}{2\epsilon^4}. \text{ But}$$

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} = \frac{1}{\sqrt{1 - (1 - \epsilon)^2}} \cong \frac{1}{\sqrt{1 - (1 - 2\epsilon)}} = \frac{1}{\sqrt{2\epsilon}} \Rightarrow \epsilon = \frac{1}{2\gamma^2}. \text{ Therefore}$$

$$f = \left(\frac{4}{5} \right)^5 \frac{1}{2} (2\gamma^2)^4 = \boxed{\frac{1}{4} \left(\frac{8}{5} \right)^5 \gamma^8 = 2.62\gamma^8}.$$

Problem 11.16

$$\text{Equation 11.72 says } \frac{dP}{d\Omega} = \frac{q^2}{16\pi^2\epsilon_0} \frac{|\hat{\mathbf{z}} \times (\mathbf{u} \times \mathbf{a})|^2}{(\hat{\mathbf{z}} \cdot \mathbf{u})^5}. \text{ Let } \beta \equiv v/c.$$

$$\mathbf{u} = c\hat{\mathbf{z}} - \mathbf{v} = c\hat{\mathbf{z}} - v\hat{\mathbf{z}} \Rightarrow \hat{\mathbf{z}} \cdot \mathbf{u} = c - v(\hat{\mathbf{z}} \cdot \hat{\mathbf{z}}) = c - v \cos \theta = c \left(1 - \frac{v}{c} \cos \theta \right) = c(1 - \beta \cos \theta);$$

$$\mathbf{a} \cdot \mathbf{u} = ac(\hat{\mathbf{x}} \cdot \hat{\mathbf{z}}) - av(\hat{\mathbf{x}} \cdot \hat{\mathbf{z}}) = ac \sin \theta \cos \phi; \quad u^2 = \mathbf{u} \cdot \mathbf{u} = c^2 - 2cv(\hat{\mathbf{z}} \cdot \hat{\mathbf{z}}) + v^2 = c^2 + v^2 - 2cv \cos \theta.$$

$$\begin{aligned} \hat{\mathbf{z}} \times (\mathbf{u} \times \mathbf{a}) &= (\hat{\mathbf{z}} \cdot \mathbf{a})\mathbf{u} - (\hat{\mathbf{z}} \cdot \mathbf{u})\mathbf{a}; \\ |\hat{\mathbf{z}} \times (\mathbf{u} \times \mathbf{a})|^2 &= (\hat{\mathbf{z}} \cdot \mathbf{a})^2 u^2 - 2(\mathbf{u} \cdot \mathbf{a})(\hat{\mathbf{z}} \cdot \mathbf{a})(\hat{\mathbf{z}} \cdot \mathbf{u}) + (\hat{\mathbf{z}} \cdot \mathbf{u})^2 a^2 \\ &= (c^2 + v^2 - 2cv \cos \theta)(a \sin \theta \cos \phi)^2 - 2(ac \sin \theta \cos \phi)(a \sin \theta \cos \phi)(c - v \cos \theta) + a^2 c^2 (1 - \beta \cos \theta) \\ &= a^2 [c^2 (1 - \beta \cos \theta)^2 + (\sin^2 \theta \cos^2 \phi)(c^2 + v^2 - 2cv \cos \theta - 2c^2 + 2cv \cos \theta)] \\ &= a^2 c^2 [(1 - \beta \cos \theta)^2 - (1 - \beta^2)(\sin \theta \cos \phi)^2]. \\ \frac{dP}{d\Omega} &= \boxed{\frac{\mu_0 q^2 a^2 [(1 - \beta \cos \theta)^2 - (1 - \beta^2) \sin^2 \theta \cos^2 \phi]}{16\pi^2 c (1 - \beta \cos \theta)^5}}. \end{aligned}$$

The total power radiated (in all directions) is:

$$\begin{aligned} P &= \int \frac{dP}{d\Omega} d\Omega = \int \frac{dP}{d\Omega} \sin \theta d\theta d\phi = \frac{\mu_0 q^2 a^2}{16\pi^2 c} \int \int \frac{[(1 - \beta \cos \theta)^2 - (1 - \beta^2) \sin^2 \theta \cos^2 \phi]}{(1 - \beta \cos \theta)^5} \sin \theta d\theta d\phi. \\ \text{But } \int_0^{2\pi} d\phi &= 2\pi \quad \text{and} \quad \int_0^{2\pi} \cos^2 \phi d\phi = \pi. \\ &= \frac{\mu_0 q^2 a^2}{16\pi^2 c} \pi \int_0^\pi \frac{[2(1 - \beta \cos \theta)^2 - (1 - \beta^2) \sin^2 \theta]}{(1 - \beta \cos \theta)^5} \sin \theta d\theta. \end{aligned}$$

Let $w \equiv (1 - \beta \cos \theta)$. Then $(1 - w)/\beta = \cos \theta$; $\sin^2 \theta = [\beta^2 - (1 - w)^2] / \beta^2$, and the numerator becomes

$$\begin{aligned} 2w^2 - \frac{(1 - \beta^2)}{\beta^2}(\beta^2 - 1 + 2w - w^2) &= \frac{1}{\beta^2} [2w^2 \beta^2 + (1 - \beta^2)^2 - 2(1 - \beta^2)w + w^2(1 - \beta^2)] \\ &= \frac{1}{\beta^2} [(1 - \beta^2)^2 - 2(1 - \beta^2)w + (1 + \beta^2)w^2]; \end{aligned}$$

$$dw = \beta \sin \theta d\theta \Rightarrow \sin \theta d\theta = \frac{1}{\beta} dw. \quad \text{When } \theta = 0, w = (1 - \beta); \text{ when } \theta = \pi, w = (1 + \beta).$$

$$\begin{aligned} P &= \frac{\mu_0 q^2 a^2}{16\pi c} \frac{1}{\beta^3} \int_{(1-\beta)}^{(1+\beta)} \frac{1}{w^5} [(1 - \beta^2)^2 - 2(1 - \beta^2)w + (1 + \beta^2)w^2] dw. \quad \text{The integral is} \\ \text{Int} &= (1 - \beta^2)^2 \int \frac{1}{w^5} dw - 2(1 - \beta^2) \int \frac{1}{w^4} dw + (1 + \beta^2) \int \frac{1}{w^3} dw \\ &= \left[(1 - \beta^2)^2 \left(-\frac{1}{4w^4} \right) - 2(1 - \beta^2) \left(-\frac{1}{3w^3} \right) + (1 + \beta^2) \left(-\frac{1}{2w^2} \right) \right] \Big|_{1-\beta}^{1+\beta}. \end{aligned}$$

$$\begin{aligned} \frac{1}{w^2} \Big|_{1-\beta}^{1+\beta} &= \frac{1}{(1 + \beta)^2} - \frac{1}{(1 - \beta)^2} = \frac{(1 - 2\beta + \beta^2) - (1 + 2\beta + \beta^2)}{(1 + \beta)^2(1 - \beta)^2} = -\frac{4\beta}{(1 - \beta^2)^2}. \\ \frac{1}{w^3} \Big|_{1-\beta}^{1+\beta} &= \frac{1}{(1 + \beta)^3} - \frac{1}{(1 - \beta)^3} = \frac{(1 - 3\beta + 3\beta^2 - \beta^3) - (1 + 3\beta + 3\beta^2 + \beta^3)}{(1 + \beta)^3(1 - \beta)^3} = -\frac{2\beta(3 + \beta^2)}{(1 - \beta^2)^3}. \\ \frac{1}{w^4} \Big|_{1-\beta}^{1+\beta} &= \frac{1}{(1 + \beta)^4} - \frac{1}{(1 - \beta)^4} = \frac{(1 - 4\beta + 6\beta^2 - 4\beta^3 + \beta^4) - (1 + 4\beta + 6\beta^2 + 4\beta^3 + \beta^4)}{(1 + \beta)^4(1 - \beta)^4} = -\frac{8\beta(1 + \beta^2)}{(1 - \beta^2)^4}. \end{aligned}$$

$$\begin{aligned}\text{Int} &= (1 - \beta^2)^2 \left(-\frac{1}{4} \right) \frac{-8\beta(1 + \beta^2)}{(1 - \beta^2)^4} - 2(1 - \beta^2) \left(-\frac{1}{3} \right) \frac{-2\beta(3 + \beta^2)}{(1 - \beta^2)^3} + (1 + \beta^2) \left(-\frac{1}{2} \right) \frac{-4\beta}{(1 - \beta^2)^2} \\ &= \frac{2\beta}{(1 - \beta^2)^2} \left[(1 + \beta^2) - \frac{2}{3}(3 + \beta^2) + (1 + \beta^2) \right] = \frac{8}{3} \frac{\beta^3}{(1 - \beta^2)^2} \cdot \\ P &= \frac{\mu_0 q^2 a^2}{16\pi c} \frac{1}{\beta^3} \frac{8}{3} \frac{\beta^3}{(1 - \beta^2)^2} = \boxed{\frac{\mu_0 q^2 a^2 \gamma^4}{6\pi c}}, \quad \text{where } \gamma = \frac{1}{\sqrt{1 - \beta^2}}.\end{aligned}$$

Is this consistent with the Liénard formula (Eq. 11.73)? Here $\mathbf{v} \times \mathbf{a} = v a (\hat{\mathbf{z}} \times \hat{\mathbf{x}}) = v a \hat{\mathbf{y}}$, so $a^2 - \left(\frac{\mathbf{v} \times \mathbf{a}}{c} \right)^2 = a^2 \left(1 - \frac{v^2}{c^2} \right) = (1 - \beta^2) a^2 = \frac{1}{\gamma^2} a^2$, so the Liénard formula says $P = \frac{\mu_0 q^2 \gamma^6 a^2}{6\pi c} \frac{1}{\gamma^2}$. ✓

Problem 11.17

(a) To counteract the radiation reaction (Eq. 11.80), you must exert a force $\mathbf{F}_e = -\frac{\mu_0 q^2}{6\pi c} \dot{\mathbf{a}}$.

For circular motion, $\mathbf{r}(t) = R [\cos(\omega t) \hat{\mathbf{x}} + \sin(\omega t) \hat{\mathbf{y}}]$, $\mathbf{v}(t) = \dot{\mathbf{r}} = R\omega [-\sin(\omega t) \hat{\mathbf{x}} + \cos(\omega t) \hat{\mathbf{y}}]$;

$$\mathbf{a}(t) = \dot{\mathbf{v}} = -R\omega^2 [\cos(\omega t) \hat{\mathbf{x}} + \sin(\omega t) \hat{\mathbf{y}}] = -\omega^2 \mathbf{r}; \quad \dot{\mathbf{a}} = -\omega^2 \dot{\mathbf{r}} = -\omega^2 \mathbf{v}. \quad \text{So } \boxed{\mathbf{F}_e = \frac{\mu_0 q^2}{6\pi c} \omega^2 \mathbf{v}.}$$

$$\boxed{P_e = \mathbf{F}_e \cdot \mathbf{v} = \frac{\mu_0 q^2}{6\pi c} \omega^2 v^2.} \quad \text{This is the power you must supply.}$$

Meanwhile, the power radiated is (Eq. 11.70) $P_{\text{rad}} = \frac{\mu_0 q^2 a^2}{6\pi c}$, and $a^2 = \omega^4 r^2 = \omega^4 R^2 = \omega^2 v^2$, so

$$P_{\text{rad}} = \frac{\mu_0 q^2}{6\pi c} \omega^2 v^2, \text{ and the two expressions agree.}$$

(b) For simple harmonic motion, $\mathbf{r}(t) = A \cos(\omega t) \hat{\mathbf{z}}$; $\mathbf{v} = \dot{\mathbf{r}} = -A\omega \sin(\omega t) \hat{\mathbf{z}}$; $\mathbf{a} = \dot{\mathbf{v}} = -A\omega^2 \cos(\omega t) \hat{\mathbf{z}} = -\omega^2 \mathbf{r}$; $\dot{\mathbf{a}} = -\omega^2 \dot{\mathbf{r}} = -\omega^2 \mathbf{v}$. So $\boxed{\mathbf{F}_e = \frac{\mu_0 q^2}{6\pi c} \omega^2 \mathbf{v}; P_e = \frac{\mu_0 q^2}{6\pi c} \omega^2 v^2.}$ But this time $a^2 = \omega^4 r^2 = \omega^4 A^2 \cos^2(\omega t)$, whereas $\omega^2 v^2 = \omega^4 A^2 \sin^2(\omega t)$, so

$$P_{\text{rad}} = \frac{\mu_0 q^2}{6\pi c} \omega^4 A^2 \cos^2(\omega t) \neq P_e = \frac{\mu_0 q^2}{6\pi c} \omega^4 A^2 \sin^2(\omega t);$$

the power you deliver is *not* equal to the power radiated. However, since the time *averages* of $\sin^2(\omega t)$ and $\cos^2(\omega t)$ are equal (to wit: 1/2), *over a full cycle* the energy radiated is the same as the energy input. (In the mean time energy is evidently being stored temporarily in the nearby fields.)

(c) In free fall, $\mathbf{v}(t) = \frac{1}{2} g t^2 \hat{\mathbf{y}}$; $\mathbf{v} = g t \hat{\mathbf{y}}$; $\mathbf{a} = g \hat{\mathbf{y}}$; $\dot{\mathbf{a}} = 0$. So $\boxed{\mathbf{F}_e = \mathbf{0}}$; the radiation reaction is zero, and

hence $\boxed{P_e = 0.}$ But there *is* radiation: $\boxed{P_{\text{rad}} = \frac{\mu_0 q^2}{6\pi c} g^2.}$ Evidently energy is being continuously extracted from the nearby fields. This paradox persists even in the *exact* solution (where we do *not* assume $v \ll c$, as in the Larmor formula and the Abraham-Lorentz formula)—see Prob. 11.34.

Problem 11.18

(a) From Eq. 11.80, $F_{\text{rad}} = \frac{\mu_0 q^2}{6\pi c} \dot{a} = m\tau \ddot{x}$ (Eq. 11.82). The equation of motion is

$$F = m\ddot{x} = F_{\text{spring}} + F_{\text{rad}} = -kx + m\tau \ddot{x}, \text{ or } \ddot{x} + \omega_0^2 x - \tau \ddot{x} = 0, \text{ with } \omega_0 = \sqrt{k/m}.$$

Since the damping is small, it oscillates at the natural frequency ω_0 , and hence $\ddot{x} = -\omega_0^2 \dot{x}$, so $\ddot{x} + \omega_0^2 \tau \dot{x} + \omega_0^2 x = 0$, or $\ddot{x} + \gamma \dot{x} + \omega_0^2 x = 0$, with $\gamma = \omega_0^2 \tau$. The solution with $x(0) = 0$ is (for $\gamma \ll \omega_0$)

$$x(t) = A e^{-\gamma t/2} \sin(\omega_0 t); \quad v(t) = -\frac{\gamma}{2} A e^{-\gamma t/2} \sin(\omega_0 t) + \omega_0 A e^{-\gamma t/2} \cos(\omega_0 t), \quad \text{so } v(0) = A \omega_0 = v_0 \Rightarrow A = \frac{v_0}{\omega_0},$$

and
$$x(t) = \frac{v_0}{\omega_0} e^{-\gamma t/2} \sin(\omega_0 t).$$

According to the Larmor formula (Eq. 11.70), the power radiated is $P = \frac{\mu_0 q^2 a^2}{6\pi c}$. In this case (still assuming $\gamma \ll \omega_0$)

$$a = -\omega_0^2 x = -\omega_0^2 \frac{v_0}{\omega_0} e^{-\gamma t/2} \sin(\omega_0 t), \quad P = \frac{\mu_0 q^2}{6\pi c} (\omega_0 v_0)^2 e^{-\gamma t} \sin^2(\omega_0 t).$$

Averaging over a full cycle (holding $e^{-\gamma t}$ constant) $\langle P \rangle = \frac{\mu_0 q^2 \omega_0^2 v_0^2}{6\pi c} e^{-\gamma t} \frac{1}{2}$, and the total energy radiated is

$$\int_0^\infty \langle P \rangle dt = \frac{\mu_0 q^2 \omega_0^2 v_0^2}{12\pi c} \left[-\frac{1}{\gamma} e^{-\gamma t} \right]_0^\infty = \frac{\mu_0 q^2 \omega_0^2 v_0^2}{12\pi c \gamma} = \frac{\mu_0 q^2 \omega_0^2 v_0^2}{12\pi c \omega_0^2 \tau} = \frac{\mu_0 q^2 v_0^2}{12\pi c} \left(\frac{6\pi m c}{\mu_0 q^2} \right) = \frac{1}{2} m v_0^2. \quad \checkmark$$

(b) This “equivalent” single oscillator has twice the charge, and twice the mass, so τ (and hence γ) is doubled. Since $\int \langle P \rangle dt$ goes like q^2/γ , it also doubles. The power radiated is indeed four times as great, but the oscillations die away faster, and the total energy radiated is just twice as much as for one oscillator.

Problem 11.19

(a) $a = \tau \dot{a} + \frac{F}{m} \Rightarrow \frac{dv}{dt} = \tau \frac{da}{dt} + \frac{F}{m} \Rightarrow \int \frac{dv}{dt} dt = \tau \int \frac{da}{dt} dt + \frac{1}{m} \int F dt.$

$[v(t_0 + \epsilon) - v(t_0 - \epsilon)] = \tau [a(t_0 + \epsilon) - a(t_0 - \epsilon)] + \frac{2\epsilon}{m} F_{\text{ave}}$, where F_{ave} is the average force during the interval. But v is continuous, so as long as F is not a delta function, we are left (in the limit $\epsilon \rightarrow 0$) with $[a(t_0 + \epsilon) - a(t_0 - \epsilon)] = 0$. Thus a , too, is continuous. qed

(b) (i) $a = \tau \dot{a} \Rightarrow \frac{da}{a} = \frac{1}{\tau} dt \Rightarrow \int \frac{da}{a} = \frac{1}{\tau} \int dt \Rightarrow \ln a = \frac{t}{\tau} + \text{constant} \Rightarrow \boxed{a(t) = A e^{t/\tau}}$, where A is a constant.

(ii) $a = \tau \dot{a} + \frac{F}{m} \Rightarrow \tau \frac{da}{dt} = a - \frac{F}{m} \Rightarrow \frac{da}{a - F/m} = \frac{1}{\tau} dt \Rightarrow \ln(a - F/m) = \frac{t}{\tau} + \text{constant} \Rightarrow a - \frac{F}{m} = B e^{t/\tau} \Rightarrow$

$\boxed{a(t) = \frac{F}{m} + B e^{t/\tau}}$, where B is some other constant.

(iii) Same as (i): $\boxed{a(t) = C e^{t/\tau}}$, where C is a third constant.

(c) At $t = 0$, $A = F/m + B$; at $t = T$, $F/m + B e^{T/\tau} = C e^{T/\tau} \Rightarrow C = (F/m) e^{-T/\tau} + B$. So

$$a(t) = \begin{cases} [(F/m) + B] e^{t/\tau}, & t \leq 0; \\ [(F/m) + B e^{t/\tau}], & 0 \leq t \leq T; \\ [(F/m) e^{-T/\tau} + B] e^{t/\tau}, & t \geq T. \end{cases}$$

To eliminate the runaway in region (iii), we'd need $B = -(F/m) e^{-T/\tau}$; to avoid preacceleration in region (i), we'd need $B = -(F/m)$. Obviously, we cannot do both at once.

(d) If we choose to eliminate the runaway, then

$$a(t) = \begin{cases} (F/m) [1 - e^{-T/\tau}] e^{t/\tau}, & t \leq 0; \\ (F/m) [1 - e^{(t-T)/\tau}], & 0 \leq t \leq T; \\ 0, & t \geq T. \end{cases}$$

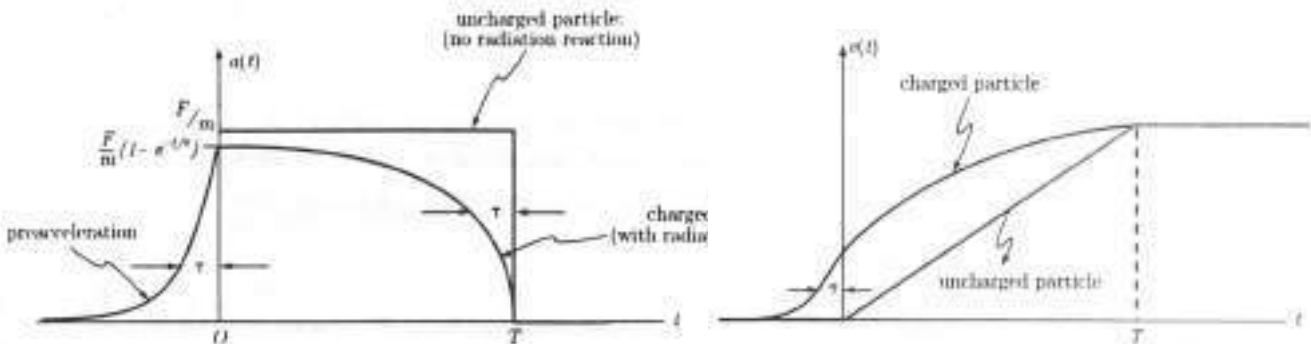
(i) $v = (F/m) [1 - e^{-T/\tau}] \int e^{t/\tau} dt = (F\tau/m) [1 - e^{-T/\tau}] e^{t/\tau} + D$, where D is a constant determined by the condition $v(-\infty) = 0 \Rightarrow D = 0$.

(ii) $v = (F/m) [t - \tau e^{(t-T)/\tau}] + E$, where E is a constant determined by the continuity of v at $t = 0$: $(F\tau/m) [1 - e^{-T/\tau}] = (F/m) [-\tau e^{-T/\tau}] + E \Rightarrow E = (F\tau/m)$.

(iii) v is a constant determined by the continuity of v at $t = T$: $v = (F/m)[T + \tau - \tau] = (F/m)T$.

$$v(t) = \begin{cases} (F\tau/m) [1 - e^{-T/\tau}] e^{t/\tau}, & t \leq 0; \\ (F/m) [t + \tau - \tau e^{(t-T)/\tau}], & 0 \leq t \leq T; \\ (F/m)T, & t \geq T. \end{cases}$$

(e)



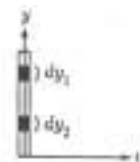
Problem 11.20

(a) From Eq. 11.80, $F_{\text{rad}}^{\text{end}} = \frac{\mu_0(q/2)^2}{6\pi c} \dot{a}$, so $F_{\text{rad}} = F_{\text{rad}}^{\text{int}} + 2F_{\text{rad}}^{\text{end}} = \frac{\mu_0 q^2}{6\pi c} \dot{a} \left[\frac{1}{2} + 2 \left(\frac{1}{4} \right) \right] = \frac{\mu_0 q^2}{6\pi c} \dot{a}$. ✓

(b) Following the suggested method:

$$F(q) = F^{\text{int}}(q) + 2 \left(\frac{1}{4} \right) F(q) \Rightarrow \frac{1}{2} F(q) = F^{\text{int}}(q) \Rightarrow F(q) = 2F^{\text{int}}(q) = \frac{\mu_0 q^2 \dot{a}}{6\pi c}. \quad \checkmark$$

(c) $F_{\text{rad}} = \frac{\mu_0}{12\pi c} \dot{a} \int_0^L \left\{ \int_0^{y_1} 2\lambda dy_2 \right\} 2\lambda dy_1$. (Running the y_2 integral up to y_1 insures that $y_1 \geq y_2$, so we don't count the same pair twice. Alternatively, run *both* integrals from 0 to L —intentionally double-counting—and divide the result by 2.)

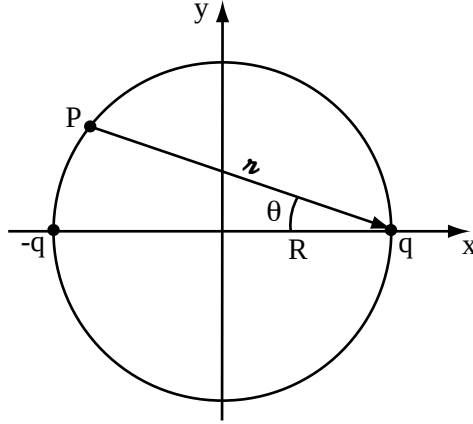


$$F_{\text{rad}} = \frac{\mu_0 \dot{a}}{12\pi c} (4\lambda^2) \int_0^L y_1 dy_1 = \frac{\mu_0 \dot{a}}{12\pi c} (4\lambda^2) \frac{L^2}{2} = \frac{\mu_0}{6\pi c} (\lambda L)^2 \dot{a} = \frac{\mu_0 q^2}{6\pi c} \dot{a}. \quad \checkmark$$

Problem 11.21

(a) The total torque is twice the torque on $+q$; we might as well calculate it at time $t = 0$. First we need the electric field at $+q$, due to $-q$ when it was at the retarded point P (Eq. 10.72). From the figure, $\hat{\mathbf{z}} = \cos \theta \hat{\mathbf{x}} - \sin \theta \hat{\mathbf{y}}$, $z = 2R \cos \theta$. The velocity of $-q$ (at P) was $\mathbf{v} = -\omega R (\sin 2\theta \hat{\mathbf{x}} + \cos 2\theta \hat{\mathbf{y}})$, and its acceleration was $\mathbf{a} = \omega^2 R (\cos 2\theta \hat{\mathbf{x}} - \sin 2\theta \hat{\mathbf{y}})$. Quantities we will need in Eq. 10.72 are:

$$\begin{aligned} \mathbf{u} &= c\hat{\mathbf{z}} - \mathbf{v} = (c \cos \theta + \omega R \sin 2\theta) \hat{\mathbf{x}} - (c \sin \theta - \omega R \cos 2\theta) \hat{\mathbf{y}}, \\ \hat{\mathbf{z}} \cdot \mathbf{u} &= 2R \cos \theta (c + \omega R \sin \theta), \quad \hat{\mathbf{z}} \cdot \mathbf{a} = 2(\omega R \cos \theta)^2. \end{aligned}$$



$$\begin{aligned} \mathbf{E} &= \frac{-q}{4\pi\epsilon_0} \frac{2R \cos \theta}{(2R \cos \theta)^3 (c + \omega R \sin \theta)^3} \left\{ [c^2 - (\omega R)^2 + 2(\omega R \cos \theta)^2] [(c \cos \theta + \omega R \sin 2\theta) \hat{\mathbf{x}} - (c \sin \theta - \omega R \cos 2\theta) \hat{\mathbf{y}}] \right. \\ &\quad \left. - 2R \cos \theta (c + \omega R \sin \theta) \omega^2 R (\cos 2\theta \hat{\mathbf{x}} - \sin 2\theta \hat{\mathbf{y}}) \right\} \end{aligned}$$

The total torque (about the origin) is

$$\begin{aligned} \mathbf{N} &= 2(R \hat{\mathbf{x}}) \times (q\mathbf{E}) = \frac{-2q^2 R}{4\pi\epsilon_0} \frac{\hat{\mathbf{z}}}{(2R \cos \theta)^2 (c + \omega R \sin \theta)^3} \left\{ -[c^2 - (\omega R)^2 + 2(\omega R \cos \theta)^2] (c \sin \theta - \omega R \cos 2\theta) \right. \\ &\quad \left. + 2(\omega R)^2 \cos \theta (c + \omega R \sin \theta) \sin 2\theta \right\} \\ &= -\frac{q^2}{4\pi\epsilon_0} \frac{\hat{\mathbf{z}}}{2R \cos^2 \theta (c + \omega R \sin \theta)^3} [-c^3 \sin \theta + c^2 \omega R (2 \cos^2 \theta - 1) + c(\omega R)^2 (2 \cos^2 \theta + 1) \sin \theta + (\omega R)^3] \\ &= -\frac{q^2}{4\pi\epsilon_0} \frac{1}{2R \cos^2 \theta (1 + \beta \sin \theta)^3} [-\sin \theta + \beta (2 \cos^2 \theta - 1) + \beta^2 (2 \cos^2 \theta + 1) \sin \theta + \beta^3] \hat{\mathbf{z}}, \end{aligned}$$

where $\beta \equiv \omega R/c$. [Since $\mathbf{E}$ and $\hat{\mathbf{z}}$ both lie in the xy plane, $\mathbf{B} = (1/c)\hat{\mathbf{z}} \times \mathbf{E}$ is along the z direction, $\mathbf{v} \times \mathbf{B}$ is radial, and hence the magnetic contribution to the torque is zero.]

The angle θ is determined by the retarded time condition, $z = -ct_r$ (note that t_r is negative, here), and 2θ is the angle through which the dipole rotates in time $-t_r$, so $2R \cos \theta = -ct_r = c(2\theta/\omega)$, or $\theta = \beta \cos \theta$. We

can use this to eliminate the trig functions:

$$\mathbf{N} = -\frac{q^2}{4\pi\epsilon_0} \frac{\beta}{2R\theta^2 \left(1 + \sqrt{\beta^2 - \theta^2}\right)^3} \left[(\beta^4 - \beta^2 + 2\theta^2) + \sqrt{\beta^2 - \theta^2}(\beta^2 + 2\theta^2 - 1) \right] \hat{\mathbf{z}}.$$

Meanwhile, expanding in powers of θ :

$$\beta = \theta \sec \theta = \theta + \frac{1}{2}\theta^3 + \frac{5}{24}\theta^5 + \frac{61}{720}\theta^7 + \dots$$

This can be “solved” (for θ as a function of β) by reverting the series:

$$\theta = \beta - \frac{1}{2}\beta^3 + \frac{13}{24}\beta^5 - \frac{541}{720}\beta^7 + \dots$$

Then

$$\begin{aligned} \theta^2 &= \beta^2 \left(1 - \beta^2 + \frac{4}{3}\beta^4 + \dots \right), \quad \sqrt{\beta^2 - \theta^2} = \beta^2 \left(1 - \frac{2}{3}\beta^2 + \frac{4}{5}\beta^4 + \dots \right), \\ \frac{1}{\theta^2} &= \frac{1}{\beta^2} (1 + \beta^2 + \dots), \quad \frac{1}{\left(1 + \sqrt{\beta^2 - \theta^2}\right)^3} = 1 - 3\beta^2 + \dots, \\ \left[(\beta^4 - \beta^2 + 2\theta^2) + \sqrt{\beta^2 - \theta^2}(\beta^2 + 2\theta^2 - 1) \right] &= \frac{8}{3}\beta^4 \left(1 - \frac{4}{5}\beta^2 + \dots \right). \end{aligned}$$

To leading order in β , then,

$$\mathbf{N} = -\frac{q^2}{4\pi\epsilon_0} \frac{\beta}{2R} \frac{1}{\beta^2} \frac{8\beta^4}{3} \hat{\mathbf{z}} = \boxed{-\frac{q^2}{4\pi\epsilon_0} \frac{4\beta^3}{3R} \hat{\mathbf{z}}}.$$

(b) The radiation reaction force on $+q$ is (Eq. 11.80) $\mathbf{F} = \frac{\mu_0 q^2}{6\pi c} \dot{\mathbf{a}}$. In this case $\dot{\mathbf{a}} = -\omega^2 \mathbf{v} = -\omega^3 R \hat{\mathbf{y}}$, so the net *torque* (counting both ends) is

$$\mathbf{N} = -2R \frac{\mu_0 q^2}{6\pi c} \omega^3 R \hat{\mathbf{z}} = -\frac{q^2}{4\pi\epsilon_0} \frac{4\beta^3}{3R} \hat{\mathbf{z}}.$$

Adding this to the interaction torque from (a), the total is

$$\mathbf{N} = -\frac{q^2}{4\pi\epsilon_0} \frac{8\beta^3}{3R} \hat{\mathbf{z}} = \boxed{-\frac{\mu_0 p^2 \omega^3}{6\pi c} \hat{\mathbf{z}}}.$$

(c) $\ddot{\mathbf{p}} = 2q\ddot{\mathbf{r}} = 2q(-\omega^2)\mathbf{r} = -\omega^2\mathbf{p}$, so Eq. 11.60 says the power radiated is $P = \frac{\mu_0}{6\pi c} \omega^4 p^2$. The power associated with the torque in (b) is $N\omega = -\frac{\mu_0 p^2 \omega^3}{6\pi c} \omega$, so they are in agreement. ✓

Problem 11.22

(a) This is an oscillating electric dipole, with amplitude $p_0 = qd$ and frequency $\omega = \sqrt{k/m}$. The (averaged) Poynting vector is given by Eq. 11.21: $\langle \mathbf{S} \rangle = \left(\frac{\mu_0 p_0^2 \omega^4}{32\pi^2 c} \right) \frac{\sin^2 \theta}{r^2} \hat{\mathbf{r}}$, so the power per unit area of *floor* is

$$\begin{aligned} I_f &= \langle \mathbf{S} \rangle \cdot \hat{\mathbf{z}} = \left(\frac{\mu_0 p_0^2 \omega^4}{32\pi^2 c} \right) \frac{\sin^2 \theta \cos \theta}{r^2}. \quad \text{But } \sin \theta = \frac{R}{r}, \quad \cos \theta = \frac{h}{r}, \quad \text{and } r^2 = R^2 + h^2. \\ &= \boxed{\left(\frac{\mu_0 q^2 d^2 \omega^4}{32\pi^2 c} \right) \frac{R^2 h}{(R^2 + h^2)^{5/2}}}. \end{aligned}$$

$$\frac{dI_f}{dR} = 0 \Rightarrow \frac{d}{dR} \left[\frac{R^2}{(R^2 + h^2)^{5/2}} \right] = 0 \Rightarrow \frac{2R}{(R^2 + h^2)^{5/2}} - \frac{5}{2} \frac{R^2}{(R^2 + h^2)^{7/2}} 2R = 0 \Rightarrow$$

$$(R^2 + h^2) - \frac{5}{2} R^2 = 0 \Rightarrow h^2 = \frac{3}{2} R^2 \Rightarrow \boxed{R = \sqrt{2/3} h}, \text{ for maximum intensity.}$$

(b)

$$P = \int I_f(R) da = \int I_f(R) 2\pi R dR = 2\pi \left(\frac{\mu_0 (qd)^2 \omega^4}{32\pi^2 c} \right) h \int_0^\infty \frac{R^3}{(R^2 + h^2)^{5/2}} dR. \quad \text{Let } x \equiv R^2 :$$

$$\int_0^\infty \frac{R^3}{(R^2 + h^2)^{5/2}} dR = \frac{1}{2} \int_0^\infty \frac{x}{(x + h^2)^{5/2}} dx = \frac{1}{2h} \frac{\Gamma(2)\Gamma(1/2)}{\Gamma(5/2)} = \frac{2}{3h}.$$

$$= 2\pi \left(\frac{\mu_0 q^2 d^2 \omega^4}{32\pi^2 c} \right) h \frac{2}{3h} = \boxed{\frac{\mu_0 q^2 d^2 \omega^4}{24\pi c}},$$

which should be (and *is*) half the *total* radiated power (Eq. 11.22)—the rest hits the ceiling, of course.

(c) The amplitude is $x_0(t)$, so $U = \frac{1}{2} k x_0^2$ is the energy, at time t , and $dU/dt = -2P$ is the power radiated:

$$\frac{1}{2} k \frac{d}{dt} (x_0^2) = -\frac{\mu_0 \omega^4}{12\pi c} q^2 x_0^2 \Rightarrow \frac{d}{dt} (x_0^2) = -\frac{\mu_0 \omega^4 q^2}{6\pi k c} (x_0^2) = -\kappa x_0^2 \Rightarrow x_0^2 = d^2 e^{-\kappa t} \text{ or } x_0(t) = d e^{-\kappa t/2}.$$

$$\tau = \frac{2}{\kappa} = \frac{12\pi k c}{\mu_0 q^2 k^2} m^2 = \boxed{\frac{12\pi c m^2}{\mu_0 q^2 k}}.$$

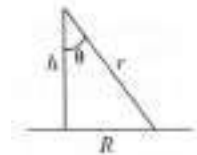
Problem 11.23

(a) From Eq. 11.39, $\langle \mathbf{S} \rangle = \left(\frac{\mu_0 m_0^2 \omega^4}{32\pi^2 c^3} \right) \frac{\sin^2 \theta}{r^2} \hat{\mathbf{r}}$. Here $\sin \theta =$

R/r , $r = \sqrt{R^2 + h^2}$, and the total radiated power (Eq. 11.40) is

$$P = \frac{\mu_0 m_0^2 \omega^4}{12\pi c^3}. \text{ So the intensity is } I(R) = \left(\frac{12P}{32\pi} \right) \frac{R^2}{(R^2 + h^2)^2} =$$

$$\boxed{\frac{3P}{8\pi} \frac{R^2}{(R^2 + h^2)^2}}.$$



(b) The intensity *directly* below the antenna ($R = 0$) would (ideally) have been *zero*. The engineer *should* have measured it at the position of *maximum* intensity:

$$\frac{dI}{dR} = \frac{3P}{8\pi} \left[\frac{2R}{(R^2 + h^2)^2} - \frac{2R^2}{(R^2 + h^2)^3} 2R \right] = \frac{3P}{8\pi} \frac{2R}{(R^2 + h^2)^3} (R^2 + h^2 - 2R^2) = 0 \Rightarrow \boxed{R = h}.$$

$$\text{At this location the intensity is } I(h) = \frac{3P}{8\pi} \frac{h^2}{(2h^2)^2} = \boxed{\frac{3P}{32\pi h^2}}.$$

$$(c) I_{\max} = \frac{3(35 \times 10^3)}{32\pi(200)^2} = 0.026 \text{ W/m}^2 = \boxed{2.6 \mu\text{W/cm}^2}. \quad \boxed{\text{Yes, KRUD is in compliance.}}$$

Problem 11.24

(a)

$$V_{\pm} = \mp \frac{p_0 \omega}{4\pi \epsilon_0 c} \left(\frac{\cos \theta_{\pm}}{r_{\pm}} \right) \sin[\omega(t - r_{\pm}/c)]. \quad V_{\text{tot}} = V_+ + V_-.$$

$$r_{\pm} = \sqrt{r^2 + (d/2)^2 \mp 2r(d/2) \cos \theta} \cong r \sqrt{1 \mp (d/r) \cos \theta} \cong r \left(1 \mp \frac{d}{2r} \cos \theta \right).$$

$$\frac{1}{r_{\pm}} \cong \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right).$$

$$\begin{aligned} \cos \theta_{\pm} &= \frac{r \cos \theta \mp (d/2)}{r_{\pm}} = r \left(\cos \theta \mp \frac{d}{2r} \right) \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right) = \cos \theta \pm \frac{d}{2r} \cos^2 \theta \mp \frac{d}{2r} \\ &= \cos \theta \mp \frac{d}{2r} (1 - \cos^2 \theta) = \cos \theta \mp \frac{d}{2r} \sin^2 \theta. \end{aligned}$$

$$\begin{aligned} \sin[\omega(t - r_{\pm}/c)] &= \sin \left\{ \omega \left[t - \frac{r}{c} \left(1 \mp \frac{d}{2r} \cos \theta \right) \right] \right\} = \sin \left(\omega t_0 \pm \frac{\omega d}{2c} \cos \theta \right), \text{ where } t_0 \equiv t - r/c. \\ &= \sin(\omega t_0) \cos \left(\frac{\omega d}{2c} \cos \theta \right) \pm \cos(\omega t_0) \sin \left(\frac{\omega d}{2c} \cos \theta \right) \cong \sin(\omega t_0) \pm \frac{\omega d}{2c} \cos \theta \cos(\omega t_0). \end{aligned}$$

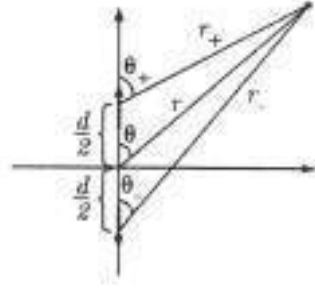
$$\begin{aligned} V_{\pm} &= \mp \frac{p_0 \omega}{4\pi \epsilon_0 c r} \left\{ \left(1 \pm \frac{d}{2r} \cos \theta \right) \left(\cos \theta \mp \frac{d}{2r} \sin^2 \theta \right) \left[\sin(\omega t_0) \pm \frac{\omega d}{2c} \cos \theta \cos(\omega t_0) \right] \right\} \\ &= \mp \frac{p_0 \omega}{4\pi \epsilon_0 c r} \left\{ \left(\cos \theta \mp \frac{d}{2r} \sin^2 \theta \pm \frac{d}{2r} \cos^2 \theta \right) \left[\sin(\omega t_0) \pm \frac{\omega d}{2c} \cos \theta \cos(\omega t_0) \right] \right\} \\ &= \mp \frac{p_0 \omega}{4\pi \epsilon_0 c r} \left[\cos \theta \sin(\omega t_0) \pm \frac{\omega d}{2c} \cos^2 \theta \cos(\omega t_0) \pm \frac{d}{2r} (\cos^2 \theta - \sin^2 \theta) \sin(\omega t_0) \right]. \\ V_{\text{tot}} &= -\frac{p_0 \omega}{4\pi \epsilon_0 c r} \left[\frac{\omega d}{c} \cos^2 \theta \cos(\omega t_0) + \frac{d}{r} (\cos^2 \theta - \sin^2 \theta) \sin(\omega t_0) \right] \\ &= -\frac{p_0 \omega^2 d}{4\pi \epsilon_0 c^2 r} \left[\cos^2 \theta \cos(\omega t_0) + \frac{c}{\omega r} (\cos^2 \theta - \sin^2 \theta) \sin(\omega t_0) \right]. \end{aligned}$$

In the radiation zone ($r \gg \omega/c$) the second term is negligible, so

$$V = -\frac{p_0 \omega^2 d}{4\pi \epsilon_0 c^2 r} \cos^2 \theta \cos[\omega(t - r/c)].$$

Meanwhile

$$\begin{aligned} \mathbf{A}_{\pm} &= \mp \frac{\mu_0 p_0 \omega}{4\pi r_{\pm}} \sin[\omega(t - r_{\pm}/c)] \hat{\mathbf{z}} \\ &= \mp \frac{\mu_0 p_0 \omega}{4\pi r} \left\{ \left(1 \pm \frac{d}{2r} \cos \theta \right) \left[\sin(\omega t_0) \pm \frac{\omega d}{2c} \cos \theta \cos(\omega t_0) \right] \right\} \hat{\mathbf{z}} \\ &= \mp \frac{\mu_0 p_0 \omega}{4\pi r} \left[\sin(\omega t_0) \pm \frac{\omega d}{2c} \cos \theta \cos(\omega t_0) \pm \frac{d}{2r} \cos \theta \sin(\omega t_0) \right] \hat{\mathbf{z}}. \\ \mathbf{A}_{\text{tot}} &= \mathbf{A}_+ + \mathbf{A}_- = -\frac{\mu_0 p_0 \omega}{4\pi r} \left[\frac{\omega d}{c} \cos \theta \cos(\omega t_0) + \frac{d}{r} \cos \theta \sin(\omega t_0) \right] \hat{\mathbf{z}} \\ &= -\frac{\mu_0 p_0 \omega^2 d}{4\pi c r} \cos \theta \left[\cos(\omega t_0) + \frac{c}{\omega r} \sin(\omega t_0) \right] \hat{\mathbf{z}}. \end{aligned}$$



In the radiation zone, $\mathbf{A} = -\frac{\mu_0 p_0 \omega^2 d}{4\pi cr} \cos \theta \cos[\omega(t - r/c)] \hat{\mathbf{z}}.$

(b) To simplify the notation, let $\alpha \equiv -\frac{\mu_0 p_0 \omega^2 d}{4\pi}$. Then

$$\begin{aligned} V &= \alpha \frac{\cos^2 \theta}{r} \cos[\omega(t - r/c)]; \\ \nabla V &= \frac{\partial V}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\boldsymbol{\theta}} = \alpha \cos^2 \theta \left\{ -\frac{1}{r^2} \cos[\omega(t - r/c)] + \frac{\omega}{rc} \sin[\omega(t - r/c)] \right\} \hat{\mathbf{r}} \\ &\quad + \alpha \frac{-2 \cos \theta \sin \theta}{r^2} \cos[\omega(t - r/c)] \hat{\boldsymbol{\theta}} = \alpha \frac{\omega \cos^2 \theta}{c r} \sin[\omega(t - r/c)] \hat{\mathbf{r}} \quad (\text{in the radiation zone}). \\ \mathbf{A} &= \frac{\alpha \cos \theta}{c r} \cos[\omega(t - r/c)] (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}). \quad \frac{\partial \mathbf{A}}{\partial t} = -\frac{\alpha \omega \cos \theta}{c r} \sin[\omega(t - r/c)] (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}). \end{aligned}$$

$$\begin{aligned} \mathbf{E} &= -\nabla V - \frac{\partial \mathbf{A}}{\partial t} = -\frac{\alpha \omega}{cr} \sin[\omega(t - r/c)] (\cos^2 \theta \hat{\mathbf{r}} - \cos^2 \theta \hat{\mathbf{r}} + \sin \theta \cos \theta \hat{\boldsymbol{\theta}}) \\ &= \boxed{-\frac{\alpha \omega}{cr} \sin \theta \cos \theta \sin[\omega(t - r/c)] \hat{\boldsymbol{\theta}}}. \end{aligned}$$

$$\begin{aligned} \mathbf{B} &= \nabla \times \mathbf{A} = \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{\boldsymbol{\phi}} \\ &= \frac{\alpha}{cr} \left\{ \frac{\partial}{\partial r} (\cos \theta \cos[\omega(t - r/c)](-\sin \theta)) - \frac{\partial}{\partial \theta} \left[\frac{\cos^2 \theta}{r} \cos[\omega(t - r/c)] \right] \right\} \hat{\boldsymbol{\phi}} \\ &= \frac{\alpha}{cr} (-\sin \theta \cos \theta) \frac{\omega}{c} \sin[\omega(t - r/c)] \hat{\boldsymbol{\phi}} \quad (\text{in the radiation zone}) = \boxed{-\frac{\alpha \omega}{c^2 r} \sin \theta \cos \theta \sin[\omega(t - r/c)] \hat{\boldsymbol{\phi}}}. \end{aligned}$$

Notice that $\mathbf{B} = \frac{1}{c}(\hat{\mathbf{r}} \times \mathbf{E})$ and $\mathbf{E} \cdot \hat{\mathbf{r}} = 0$.

$$\begin{aligned} \mathbf{S} &= \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0 c} \mathbf{E} \times (\hat{\mathbf{r}} \times \mathbf{E}) = \frac{1}{\mu_0 c} [E^2 \hat{\mathbf{r}} - (\mathbf{E} \cdot \hat{\mathbf{r}}) \mathbf{E}] = \frac{E^2}{\mu_0 c} \hat{\mathbf{r}} \\ &= \boxed{\frac{1}{\mu_0 c} \left\{ \frac{\alpha \omega}{rc} \sin \theta \cos \theta \sin[\omega(t - r/c)] \right\}^2 \hat{\mathbf{r}}}. \quad \boxed{I = \frac{1}{2\mu_0 c} \left(\frac{\alpha \omega}{rc} \sin \theta \cos \theta \right)^2}. \\ P &= \int \langle \mathbf{S} \rangle \cdot d\mathbf{a} = \frac{1}{\mu_0 c} \left(\frac{\alpha \omega}{c} \right)^2 \int \sin^2 \theta \cos^2 \theta \sin \theta d\theta d\phi = \frac{1}{2\mu_0 c} \left(\frac{\alpha \omega}{c} \right)^2 2\pi \int_0^\pi (1 - \cos^2 \theta) \cos^2 \theta \sin \theta d\theta. \\ \text{The integral is : } &-\frac{\cos^3 \theta}{3} \Big|_0^\pi + \frac{\cos^5 \theta}{5} \Big|_0^\pi = \frac{2}{3} - \frac{2}{5} = \frac{4}{15}. \\ &= \frac{1}{2\mu_0 c} \frac{\omega^2}{c^2} \frac{\mu_0^2}{16\pi^2} (p_0 d)^2 \omega^4 2\pi \frac{4}{15} = \boxed{\frac{\mu_0}{60\pi c^3} (p_0 d)^2 \omega^6}. \end{aligned}$$

Notice that it goes like ω^6 , whereas *dipole* radiation goes like ω^4 .

Problem 11.25

(a) $\mathbf{m}(t) = M \cos \psi \hat{\mathbf{z}} + M \sin \psi [\cos(\omega t) \hat{\mathbf{x}} + \sin(\omega t) \hat{\mathbf{y}}]$. As in Prob. 11.4, the power radiated will be twice that of an oscillating magnetic dipole with dipole moment of amplitude $m_0 = M \sin \psi$. Therefore (quoting

Eq. 11.40): $\boxed{P = \frac{\mu_0 M^2 \omega^4 \sin^2 \psi}{6\pi c^3}}.$ (Alternatively, you can get this from the answer to Prob. 11.11.)

(b) From Eq. 5.88, with $r \rightarrow R$, $m \rightarrow M$, and $\theta = \pi/2$: $B = \frac{\mu_0}{4\pi} \frac{M}{R^3}$, so

$$\boxed{M = \frac{4\pi R^3}{\mu_0} B} = \frac{4\pi(6.4 \times 10^6)^3(5 \times 10^{-5})}{4\pi \times 10^{-7}} = \boxed{1.3 \times 10^{23} \text{ A m}^2}.$$

$$(c) P = \frac{(4\pi \times 10^{-7})(1.3 \times 10^{23})^2 \sin^2(11^\circ)}{6\pi(3 \times 10^8)^3} \left(\frac{2\pi}{24 \times 60 \times 60} \right)^4 = \boxed{4 \times 10^{-5} \text{ W}} \text{ (not much).}$$

$$(d) P = \frac{\mu_0(4\pi R^3 B/\mu_0)^2 \omega^4 \sin^2 \psi}{6\pi c^3} = \frac{8\pi}{3\mu_0 c^3} (\omega^2 R^3 B \sin \psi)^2. \text{ Using the average value } (1/2) \text{ for } \sin^2 \psi,$$

$$P = \frac{8\pi}{3(4\pi \times 10^{-7})(3 \times 10^8)^3} \left[\left(\frac{2\pi}{10^{-3}} \right)^2 (10^4)^3 (10^8) \right] \frac{1}{2} = \boxed{2 \times 10^{36} \text{ W}} \text{ (a lot).}$$

Problem 11.26

(a) Write $\mathbf{p}(t) = q(t)d\hat{\mathbf{z}}$, with $q(t) = kt^2$, where $kd = (1/2)\ddot{p}_0$. As in Eq. 11.5,

$$\begin{aligned} V(\mathbf{r}, t) &= \frac{1}{4\pi\epsilon_0} \left[\frac{k(t - z_+/c)^2}{z_+} - \frac{k(t - z_-/c)^2}{z_-} \right] \\ &= \frac{k}{4\pi\epsilon_0} \left[\frac{(t^2 - (2t/c)z_+ + z_+^2/c^2)}{z_+} - \frac{(t^2 - (2t/c)z_- + z_-^2/c^2)}{z_-} \right] \\ &= \frac{k}{4\pi\epsilon_0} \left[t^2 \left(\frac{1}{z_+} - \frac{1}{z_-} \right) + \frac{1}{c^2} (z_+ - z_-) \right]. \end{aligned}$$

From Eqs. 11.8 and 11.9,

$$z_{\pm} = r \left(1 \mp \frac{d}{2r} \cos \theta \right), \quad \frac{1}{z_{\pm}} = \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right),$$

so

$$V(\mathbf{r}, t) = \frac{k}{4\pi\epsilon_0} \left[\frac{t^2}{r} \left(\frac{d}{r} \cos \theta \right) - \frac{r}{c^2} \left(\frac{d}{r} \cos \theta \right) \right] = \frac{k}{4\pi\epsilon_0 c^2} d \cos \theta \left[\left(\frac{ct}{r} \right)^2 - 1 \right] = \frac{\mu_0 \ddot{p}_0}{8\pi} \cos \theta \left[\left(\frac{ct}{r} \right)^2 - 1 \right].$$

As in Eq. 11.15, $\mathbf{I}(t) = (dq/dt) \hat{\mathbf{z}} = 2kt \hat{\mathbf{z}}$, so (following Eqs. 11.16, and 11.17),

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \hat{\mathbf{z}} \int_{-d/2}^{d/2} \frac{2k(t - z/c)}{z} dz = \frac{\mu_0}{4\pi} 2k \frac{(t - r/c)}{r} d \hat{\mathbf{z}} = \frac{\mu_0 \ddot{p}_0}{4\pi c} \left[\left(\frac{ct}{r} \right) - 1 \right] \hat{\mathbf{z}}.$$

$$\begin{aligned} \mathbf{E} &= -\nabla V - \frac{\partial \mathbf{A}}{\partial t} = -\frac{\mu_0 \ddot{p}_0}{8\pi} \left\{ \cos \theta \left[-2 \frac{(ct)^2}{r^3} \right] \hat{\mathbf{r}} - \frac{1}{r} \sin \theta \left[\left(\frac{ct}{r} \right)^2 - 1 \right] \hat{\boldsymbol{\theta}} \right\} - \frac{\mu_0 \ddot{p}_0}{4\pi c} \left(\frac{c}{r} \right) \hat{\mathbf{z}} \\ &= \frac{\mu_0 \ddot{p}_0}{4\pi r} \left[\cos \theta \left(\frac{ct}{r} \right)^2 \hat{\mathbf{r}} + \frac{1}{2} \sin \theta \left(\frac{ct}{r} \right)^2 \hat{\boldsymbol{\theta}} - \frac{1}{2} \sin \theta \hat{\boldsymbol{\theta}} - (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}) \right] \\ &= \boxed{\frac{\mu_0 \ddot{p}_0}{4\pi r} \left\{ \left[\left(\frac{ct}{r} \right)^2 - 1 \right] \cos \theta \hat{\mathbf{r}} + \frac{1}{2} \left[\left(\frac{ct}{r} \right)^2 + 1 \right] \sin \theta \hat{\boldsymbol{\theta}} \right\}}. \end{aligned}$$

$$\begin{aligned}
\mathbf{B} &= -\nabla \times \mathbf{A} = \frac{\mu_0 \ddot{p}_0}{4\pi c} \nabla \times \left\{ \left[\left(\frac{ct}{r} \right) - 1 \right] (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}) \right\} \\
&= \frac{\mu_0 \ddot{p}_0}{4\pi cr} \left\{ -\frac{\partial}{\partial r} \left(r \left[\left(\frac{ct}{r} \right) - 1 \right] \sin \theta \right) - \frac{\partial}{\partial \theta} \left(\left[\left(\frac{ct}{r} \right) - 1 \right] \cos \theta \right) \right\} \hat{\boldsymbol{\phi}} \\
&= \frac{\mu_0 \ddot{p}_0}{4\pi cr} \left\{ \sin \theta + \left[\left(\frac{ct}{r} \right) - 1 \right] \sin \theta \right\} \hat{\boldsymbol{\phi}} = \boxed{\frac{\mu_0 \ddot{p}_0 t}{4\pi r^2} \sin \theta \hat{\boldsymbol{\phi}}}.
\end{aligned}$$

(b) The Poynting vector is

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{\mu_0 \ddot{p}_0^2 t}{16\pi^2 r^3} \sin \theta \left\{ \left[\left(\frac{ct}{r} \right)^2 - 1 \right] \cos \theta (-\hat{\boldsymbol{\theta}}) + \frac{1}{2} \left[\left(\frac{ct}{r} \right)^2 + 1 \right] \sin \theta \hat{\mathbf{r}} \right\}.$$

$$\mathbf{S} \cdot d\mathbf{a} = \frac{\mu_0 \ddot{p}_0^2 t}{32\pi^2} \left[\left(\frac{ct}{r} \right)^2 + 1 \right] \frac{\sin^2 \theta}{r^3} (r^2 \sin \theta d\theta d\phi).$$

$$P(r, t) = \frac{\mu_0 \ddot{p}_0^2 t}{32\pi^2 r} \left[\left(\frac{ct}{r} \right)^2 + 1 \right] 2\pi \int_0^\pi \sin^3 \theta d\theta = \boxed{\frac{\mu_0 \ddot{p}_0^2 t}{12\pi r} \left[\left(\frac{ct}{r} \right)^2 + 1 \right]}.$$

$$(c) P(r, t_0 + r/c) = \frac{\mu_0 \ddot{p}_0^2}{12\pi r} \left(t_0 + \frac{r}{c} \right) \left[\frac{c^2}{r^2} \left(t_0^2 + 2t_0 \frac{r}{c} + \frac{r^2}{c^2} \right) + 1 \right] = \frac{\mu_0 \ddot{p}_0^2}{12\pi c} \left(1 + \frac{ct_0}{r} \right) \left[2 + 2\frac{ct_0}{r} + \left(\frac{ct_0}{r} \right)^2 \right].$$

$$P_{\text{rad}}(t_0) = \lim_{r \rightarrow \infty} P(r, t_0 + r/c) = \boxed{\frac{\mu_0 \ddot{p}_0^2}{6\pi c}},$$

in agreement with Eq. 11.60.

Problem 11.27

The momentum flux density is (minus) the Maxwell stress tensor (Section 8.2.3),

$$T_{ij} = \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right),$$

(Eq. 8.17) so the momentum radiated per unit (retarded) time is

$$\frac{d\mathbf{p}}{dt_r} = - \oint \frac{1}{(\partial t_r / \partial t)} \overleftrightarrow{\mathbf{T}} \cdot d\mathbf{A},$$

where (Eq. 10.78)

$$\frac{\partial t_r}{\partial t} = \frac{z}{\boldsymbol{z} \cdot \mathbf{u}} = \left(1 - \frac{\hat{\boldsymbol{z}} \cdot \mathbf{v}}{c} \right)^{-1}.$$

(This factor is 1, when $\mathbf{v} = \mathbf{0}$; it can be ignored in deriving the Larmor formula, but it does contribute to the momentum radiated.) The integration is over a large spherical surface centered on the charge:

$$d\mathbf{A} = z^2 \sin \theta d\theta d\phi \hat{\boldsymbol{z}}.$$

As in the case of the power radiated, only the radiation fields contribute (Eq. 11.66):

$$\mathbf{E}_{\text{rad}} = \frac{q}{4\pi\epsilon_0} \frac{z}{(\boldsymbol{z} \cdot \mathbf{u})^3} [\boldsymbol{z} \times (\mathbf{u} \times \mathbf{a})], \quad \mathbf{B}_{\text{rad}} = \frac{1}{c} \hat{\boldsymbol{z}} \times \mathbf{E}_{\text{rad}}.$$

Thus

$$\overleftrightarrow{T} \cdot d\mathbf{A} = \epsilon_0 \left[\mathbf{E}(\mathbf{E} \cdot d\mathbf{A}) - \frac{1}{2} E^2 d\mathbf{A} \right] + \frac{1}{\mu_0} \left[\mathbf{B}(\mathbf{B} \cdot d\mathbf{A}) - \frac{1}{2} B^2 d\mathbf{A} \right] = -\frac{1}{2} \left[\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right] d\mathbf{A} = -\epsilon_0 E^2 d\mathbf{A}$$

(note that $\mathbf{B} \cdot d\mathbf{A} = 0$, and, for radiation fields, $\mathbf{E} \cdot d\mathbf{A} = 0$, $(1/\mu_0)B^2 = (1/\mu_0 c^2)(\hat{\mathbf{r}} \times \mathbf{E}) \cdot (\hat{\mathbf{r}} \times \mathbf{E}) = \epsilon_0 \hat{\mathbf{r}} \cdot [\mathbf{E} \times (\hat{\mathbf{r}} \times \mathbf{E})] = \epsilon_0 \hat{\mathbf{r}} \cdot [\hat{\mathbf{r}} E^2 - \mathbf{E}(\hat{\mathbf{r}} \cdot \mathbf{E})] = \epsilon_0 E^2$). So

$$\frac{d\mathbf{p}}{dt_r} = \epsilon_0 \oint \left(\frac{\mathbf{r} \cdot \mathbf{u}}{r c} \right) E^2 d\mathbf{A} = \frac{q^2}{16\pi^2 \epsilon_0 c} \oint \frac{r}{(\mathbf{r} \cdot \mathbf{u})^5} [\mathbf{r} \times (\mathbf{u} \times \mathbf{a})]^2 d\mathbf{A}.$$

We expand the integrand to first order in $\beta \equiv \mathbf{v}/c$: $\mathbf{u} = c\hat{\mathbf{r}} - \mathbf{v} = c(\hat{\mathbf{r}} - \beta)$, $\mathbf{r} \cdot \mathbf{u} = cr(1 - \hat{\mathbf{r}} \cdot \beta)$,

$$\frac{1}{(\mathbf{r} \cdot \mathbf{u})^5} = \frac{1}{r^5 c^5} [1 + 5(\hat{\mathbf{r}} \cdot \beta)],$$

$$\begin{aligned} \mathbf{r} \times (\mathbf{u} \times \mathbf{a}) &= \mathbf{u}(\mathbf{r} \cdot \mathbf{a}) - \mathbf{a}(\mathbf{r} \cdot \mathbf{u}) = r c (\hat{\mathbf{r}} - \beta)(\hat{\mathbf{r}} \cdot \mathbf{a}) - a r c (1 - \hat{\mathbf{r}} \cdot \beta) \\ &= r c [\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{a}) - \mathbf{a} + \mathbf{a}(\hat{\mathbf{r}} \cdot \beta) - \beta(\hat{\mathbf{r}} \cdot \mathbf{a})]. \end{aligned}$$

$$\begin{aligned} [\mathbf{r} \times (\mathbf{u} \times \mathbf{a})]^2 &= r^2 c^2 [(\hat{\mathbf{r}} \cdot \mathbf{a})^2 - 2(\hat{\mathbf{r}} \cdot \mathbf{a})^2 + a^2 + 2(\hat{\mathbf{r}} \cdot \mathbf{a})^2(\hat{\mathbf{r}} \cdot \beta) - 2(\hat{\mathbf{r}} \cdot \beta)(\hat{\mathbf{r}} \cdot \mathbf{a})^2 - 2a^2(\hat{\mathbf{r}} \cdot \beta) \\ &\quad + 2(\mathbf{a} \cdot \beta)(\hat{\mathbf{r}} \cdot \mathbf{a})] = r^2 c^2 [a^2 - (\hat{\mathbf{r}} \cdot \mathbf{a})^2 - 2a^2(\hat{\mathbf{r}} \cdot \beta) + 2(\hat{\mathbf{r}} \cdot \mathbf{a})(\mathbf{a} \cdot \beta)]. \end{aligned}$$

$$\begin{aligned} \frac{r}{(\mathbf{r} \cdot \mathbf{u})^5} [\mathbf{r} \times (\mathbf{u} \times \mathbf{a})]^2 &= \frac{1}{r^2 c^3} [1 + 5(\hat{\mathbf{r}} \cdot \beta)] [a^2 - (\hat{\mathbf{r}} \cdot \mathbf{a})^2 - 2a^2(\hat{\mathbf{r}} \cdot \beta) + 2(\hat{\mathbf{r}} \cdot \mathbf{a})(\mathbf{a} \cdot \beta)] \\ &= \frac{1}{r^2 c^3} [a^2 - (\hat{\mathbf{r}} \cdot \mathbf{a})^2 - 2a^2(\hat{\mathbf{r}} \cdot \beta) + 2(\hat{\mathbf{r}} \cdot \mathbf{a})(\mathbf{a} \cdot \beta) + 5(\hat{\mathbf{r}} \cdot \beta)a^2 - 5(\hat{\mathbf{r}} \cdot \beta)(\hat{\mathbf{r}} \cdot \mathbf{a})^2] \\ &= \frac{1}{r^2 c^3} [a^2 - (\hat{\mathbf{r}} \cdot \mathbf{a})^2 + 3a^2(\hat{\mathbf{r}} \cdot \beta) + 2(\hat{\mathbf{r}} \cdot \mathbf{a})(\mathbf{a} \cdot \beta) - 5(\hat{\mathbf{r}} \cdot \beta)(\hat{\mathbf{r}} \cdot \mathbf{a})^2] \\ &= \frac{1}{r^2 c^3} [(\hat{\mathbf{r}} \times \mathbf{a})^2 + \hat{\mathbf{r}} \cdot [3a^2\beta + 2(\mathbf{a} \cdot \beta)\mathbf{a}] - 5(\hat{\mathbf{r}} \cdot \beta)(\hat{\mathbf{r}} \cdot \mathbf{a})^2]. \end{aligned}$$

To integrate the first term we set the polar axis along $\mathbf{a}$, so $(\hat{\mathbf{r}} \times \mathbf{a})^2 = a^2 \sin^2 \theta$, while $\hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}$. The ϕ integral kills the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ components, leaving

$$\oint \frac{1}{r^2} (\hat{\mathbf{r}} \times \mathbf{a})^2 d\mathbf{A} = 2\pi a^2 \hat{\mathbf{z}} \int_0^\pi \sin^3 \theta \cos \theta d\theta = 0.$$

[Note that if $\mathbf{v} = \mathbf{0}$ then $d\mathbf{p}/dt_r = 0$ —a particle instantaneously at rest radiates *no* momentum. That's why we had to carry the expansion to first order in β , whereas in deriving the Larmor formula we could afford to set $v = 0$.] The second term is of the form $(\hat{\mathbf{r}} \cdot \mathbf{g})$, for a constant vector $\mathbf{g}$. Setting the polar axis along $\mathbf{g}$, so $\hat{\mathbf{r}} \cdot \mathbf{g} = g \cos \theta$, the x and y components again vanish, leaving

$$\oint \frac{1}{r^2} (\hat{\mathbf{r}} \cdot \mathbf{g}) d\mathbf{A} = 2\pi g \hat{\mathbf{z}} \int_0^\pi \cos^2 \theta \sin \theta d\theta = \frac{4\pi}{3} \mathbf{g}.$$

The last term involves $(\hat{\mathbf{r}} \cdot \beta)(\hat{\mathbf{r}} \cdot \mathbf{a})^2$; this time we orient the polar axis along $\mathbf{a}$ and let $\mathbf{v}$ lie in the xz plane: $\beta = \beta_x \hat{\mathbf{x}} + \beta_z \hat{\mathbf{z}}$, so $(\hat{\mathbf{r}} \cdot \beta) = \beta_x \sin \theta \cos \phi + \beta_z \cos \theta$. Then

$$\oint \frac{1}{r^2} (\hat{\mathbf{r}} \cdot \beta)(\hat{\mathbf{r}} \cdot \mathbf{a})^2 d\mathbf{A} = a^2 \int (\beta_x \sin \theta \cos \phi + \beta_z \cos \theta) \cos^2 \theta [\sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}] \sin \theta d\theta d\phi$$

$$= a^2 \left(\beta_x \hat{\mathbf{x}} \int \sin^3 \theta \cos^2 \theta \cos^2 \phi d\theta d\phi + \beta_z \hat{\mathbf{z}} \int \cos^4 \theta \sin \theta d\theta d\phi \right) = \frac{4\pi}{15} a^2 (\beta_x \hat{\mathbf{x}} + 3\beta_z \hat{\mathbf{z}}) = \frac{4\pi}{15} (a^2 \boldsymbol{\beta} + 2(\mathbf{a} \cdot \boldsymbol{\beta}) \mathbf{a})$$

Putting all this together,

$$\frac{d\mathbf{p}}{dt_r} = \frac{q^2}{16\pi^2 \epsilon_0} \frac{1}{c^4} \left\{ \frac{4\pi}{3} [2\mathbf{a}(\mathbf{a} \cdot \boldsymbol{\beta}) + 3a^2 \boldsymbol{\beta}] - 5 \frac{4\pi}{15} [a^2 \boldsymbol{\beta} + 2(\mathbf{a} \cdot \boldsymbol{\beta}) \mathbf{a}] \right\} = \boxed{\frac{\mu_0 q^2}{6\pi c^3} a^2 \mathbf{v}.}$$

The *angular* momentum radiated is

$$\frac{d\mathbf{L}}{dt_r} = - \oint \frac{1}{(\partial t_r / \partial t)} (\boldsymbol{\mathcal{Z}} \times \overleftrightarrow{T}) \cdot d\mathbf{A}.$$

Because of the “extra” $\boldsymbol{\mathcal{Z}}$ in the integrand, it seems at first glance that the radiation fields alone will produce a result that grows without limit (as $\boldsymbol{\mathcal{Z}} \rightarrow \infty$); however, the coefficient of this term is precisely zero: $(\boldsymbol{\mathcal{Z}} \times \overleftrightarrow{T}) \cdot d\mathbf{A} = -\epsilon_0 E^2 (\boldsymbol{\mathcal{Z}} \times d\mathbf{A}) = 0$ (for radiation fields). The finite contribution comes from the cross terms, in which one field (in $\overleftrightarrow{T}$) is a radiation field and the other a Coulomb field $\left(\mathbf{E}_{\text{coul}} \equiv \frac{q}{4\pi\epsilon_0} \frac{\boldsymbol{\mathcal{Z}}}{(\boldsymbol{\mathcal{Z}} \cdot \mathbf{u})^3} (c^2 - v^2) \mathbf{u} \right)$:

$$T_{ij} = \epsilon_0 \left(E_i^{(r)} E_j^{(c)} + E_i^{(c)} E_j^{(r)} - \delta_{ij} \mathbf{E}^{(r)} \cdot \mathbf{E}^{(c)} \right) + \frac{1}{\mu_0} \left(B_i^{(r)} B_j^{(c)} + B_i^{(c)} B_j^{(r)} - \delta_{ij} \mathbf{B}^{(r)} \cdot \mathbf{B}^{(c)} \right).$$

This time

$$\overleftrightarrow{T} \cdot d\mathbf{A} = \epsilon_0 \left[\mathbf{E}^{(c)} (\mathbf{E}^{(c)} \cdot d\mathbf{A}) - (\mathbf{E}^{(r)} \cdot \mathbf{E}^{(c)}) d\mathbf{A} \right] - \frac{1}{\mu_0} (\mathbf{B}^{(r)} \cdot \mathbf{B}^{(c)}) d\mathbf{A},$$

so

$$(\boldsymbol{\mathcal{Z}} \times \overleftrightarrow{T}) \cdot d\mathbf{A} = \epsilon_0 (\boldsymbol{\mathcal{Z}} \times \mathbf{E}^{(r)}) (\mathbf{E}^{(c)} \cdot d\mathbf{A}).$$

Thus

$$\frac{d\mathbf{L}}{dt_r} = - \frac{q^2 c}{16\pi^2 \epsilon_0 \gamma^2} \oint \frac{\boldsymbol{\mathcal{Z}}}{(\boldsymbol{\mathcal{Z}} \cdot \mathbf{u})^5} \left\{ \boldsymbol{\mathcal{Z}} \times [\boldsymbol{\mathcal{Z}} \times (\mathbf{u} \times \mathbf{a})] \right\} \mathbf{u} \cdot d\mathbf{A}, \text{ where } \gamma \equiv \frac{1}{\sqrt{1 - (v/c)^2}}.$$

Now, $\{\boldsymbol{\mathcal{Z}} \times [\boldsymbol{\mathcal{Z}} \times (\mathbf{u} \times \mathbf{a})]\} = \boldsymbol{\mathcal{Z}} [\boldsymbol{\mathcal{Z}} \cdot (\mathbf{u} \times \mathbf{a})] - (\mathbf{u} \times \mathbf{a}) \boldsymbol{\mathcal{Z}}^2$, and $(\mathbf{u} \times \mathbf{a}) = c(\hat{\boldsymbol{\mathcal{Z}}} - \boldsymbol{\beta}) \times \mathbf{a} = c[(\hat{\boldsymbol{\mathcal{Z}}} \times \mathbf{a}) - (\boldsymbol{\beta} \times \mathbf{a})]$, $\hat{\boldsymbol{\mathcal{Z}}} \cdot (\mathbf{u} \times \mathbf{a}) = -c \hat{\boldsymbol{\mathcal{Z}}} \cdot (\boldsymbol{\beta} \times \mathbf{a})$, so $\{\boldsymbol{\mathcal{Z}} \times [\boldsymbol{\mathcal{Z}} \times (\mathbf{u} \times \mathbf{a})]\} = -\boldsymbol{\mathcal{Z}}^2 c \{(\hat{\boldsymbol{\mathcal{Z}}} \times \mathbf{a}) - (\boldsymbol{\beta} \times \mathbf{a}) + \hat{\boldsymbol{\mathcal{Z}}} [\hat{\boldsymbol{\mathcal{Z}}} \cdot (\boldsymbol{\beta} \times \mathbf{a})]\}$. Meanwhile $\mathbf{u} \cdot d\mathbf{A} = (\mathbf{u} \cdot \hat{\boldsymbol{\mathcal{Z}}}) \boldsymbol{\mathcal{Z}}^2 d\Omega$, where $d\Omega \equiv \sin \theta d\theta d\phi$. Expanding to first order in $\boldsymbol{\beta}$,

$$\begin{aligned} \frac{d\mathbf{L}}{dt_r} &= - \frac{\mu_0 q^2 c^3}{16\pi^2} \oint \frac{\boldsymbol{\mathcal{Z}}}{(\boldsymbol{\mathcal{Z}} \cdot \mathbf{u})^5} (-\boldsymbol{\mathcal{Z}}^2 c) \{(\hat{\boldsymbol{\mathcal{Z}}} \times \mathbf{a}) - (\boldsymbol{\beta} \times \mathbf{a}) + \hat{\boldsymbol{\mathcal{Z}}} [\hat{\boldsymbol{\mathcal{Z}}} \cdot (\boldsymbol{\beta} \times \mathbf{a})]\} \boldsymbol{\mathcal{Z}} (\boldsymbol{\mathcal{Z}} \cdot \mathbf{u}) d\Omega \\ &= \frac{\mu_0 q^2}{16\pi^2} \int [1 + 4(\hat{\boldsymbol{\mathcal{Z}}} \cdot \boldsymbol{\beta})] \{(\hat{\boldsymbol{\mathcal{Z}}} \times \mathbf{a}) - (\boldsymbol{\beta} \times \mathbf{a}) + \hat{\boldsymbol{\mathcal{Z}}} [\hat{\boldsymbol{\mathcal{Z}}} \cdot (\boldsymbol{\beta} \times \mathbf{a})]\} d\Omega \\ &= \frac{\mu_0 q^2}{16\pi^2} \int \left\{ (\hat{\boldsymbol{\mathcal{Z}}} \times \mathbf{a}) - (\boldsymbol{\beta} \times \mathbf{a}) + \hat{\boldsymbol{\mathcal{Z}}} [\hat{\boldsymbol{\mathcal{Z}}} \cdot (\boldsymbol{\beta} \times \mathbf{a})] + 4(\hat{\boldsymbol{\mathcal{Z}}} \cdot \boldsymbol{\beta})(\hat{\boldsymbol{\mathcal{Z}}} \times \mathbf{a}) \right\} d\Omega \end{aligned}$$

The first integral is

$$-\mathbf{a} \times \int \hat{\boldsymbol{\mathcal{Z}}} \sin \theta d\theta d\phi = -\mathbf{a} \times \int (\sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}) \sin \theta d\theta d\phi = \mathbf{0},$$

the second is

$$-(\boldsymbol{\beta} \times \mathbf{a}) \int \sin \theta d\theta d\phi = -4\pi(\boldsymbol{\beta} \times \mathbf{a}),$$

for the third we set the polar axis along $\mathbf{g} \equiv \boldsymbol{\beta} \times \mathbf{a}$ and get

$$g \int \hat{\mathbf{z}} \cos \theta \sin \theta d\theta d\phi = 2\pi g \hat{\mathbf{z}} \int \cos^2 \theta \sin \theta d\theta = \frac{4\pi}{3} \mathbf{g} = \frac{4\pi}{3} (\boldsymbol{\beta} \times \mathbf{a}),$$

and for the last we put the polar axis along $\mathbf{a}$ and let $\boldsymbol{\beta} = \beta_x \hat{\mathbf{x}} + \beta_z \hat{\mathbf{z}}$ lie in the xz plane:

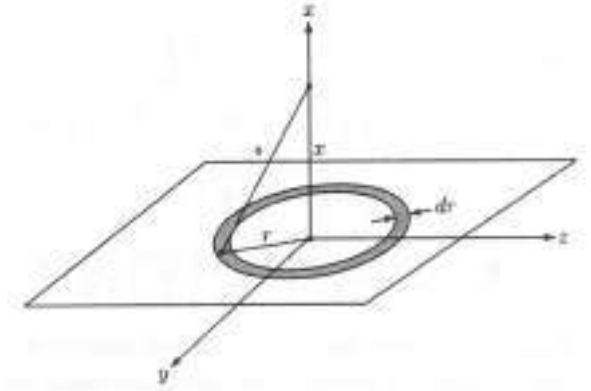
$$\begin{aligned} & 4a \int (\beta_x \sin \theta \cos \phi + \beta_z \cos \theta) \sin \theta (-\hat{\phi}) \sin \theta d\theta d\phi \\ &= 4a \int (\beta_x \sin \theta \cos \phi + \beta_z \cos \theta) (\sin \phi \hat{\mathbf{x}} - \cos \phi \hat{\mathbf{y}}) \sin^2 \theta d\theta d\phi \\ &= -4a\beta_x \hat{\mathbf{y}} \int \cos^2 \phi \sin^3 \theta d\theta d\phi = -\frac{16\pi}{3} a\beta_x \hat{\mathbf{y}} = \frac{16\pi}{3} (\boldsymbol{\beta} \times \mathbf{a}). \end{aligned}$$

Putting this all together, we conclude

$$\frac{d\mathbf{L}}{dt_r} = \frac{\mu_0 q^2}{16\pi^2} \left[-4\pi(\boldsymbol{\beta} \times \mathbf{a}) + \frac{4\pi}{3}(\boldsymbol{\beta} \times \mathbf{a}) + \frac{16\pi}{3}(\boldsymbol{\beta} \times \mathbf{a}) \right] = \boxed{\frac{\mu_0 q^2}{6\pi c} (\mathbf{v} \times \mathbf{a})}.$$

Problem 11.28

$$\begin{aligned} \text{(a)} \quad \mathbf{A}(x, t) &= \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}(t_r)}{r} da \\ &= \frac{\mu_0 \hat{\mathbf{z}}}{4\pi} \int \frac{K(t_r)}{\sqrt{r^2 + x^2}} 2\pi r dr \\ &= \frac{\mu_0 \hat{\mathbf{z}}}{2} \int \frac{K(t - \sqrt{r^2 + x^2}/c)}{\sqrt{r^2 + x^2}} r dr. \end{aligned}$$



The maximum r is given by $t - \sqrt{r^2 + x^2}/c = 0$;

$$r_{\max} = \sqrt{c^2 t^2 - x^2} \quad (\text{since } K(t) = 0 \text{ for } t < 0).$$

(i)

$$\mathbf{A}(x, t) = \frac{\mu_0 K_0 \hat{\mathbf{z}}}{2} \int_0^{r_m} \frac{r}{\sqrt{r^2 + x^2}} dr = \frac{\mu_0 K_0 \hat{\mathbf{z}}}{2} \sqrt{r^2 + x^2} \Big|_0^{r_m} = \frac{\mu_0 K_0 \hat{\mathbf{z}}}{2} (\sqrt{r_m^2 - x^2} - x) = \frac{\mu_0 K_0 (ct - x)}{2} \hat{\mathbf{z}}.$$

$$\mathbf{E}(x, t) = -\frac{\partial \mathbf{A}}{\partial t} = \boxed{-\frac{\mu_0 K_0 c}{2} \hat{\mathbf{z}}}, \text{ for } ct > x, \text{ and } 0, \text{ for } ct < x.$$

$$\mathbf{B}(x, t) = \boldsymbol{\nabla} \times \mathbf{A} = -\frac{\partial A_z}{\partial x} \hat{\mathbf{y}} = \boxed{\frac{\mu_0 K_0}{2} \hat{\mathbf{y}}}, \text{ for } ct > x, \text{ and } 0, \text{ for } ct < x.$$

(ii)

$$\begin{aligned}\mathbf{A}(x, t) &= \frac{\mu_0 \alpha \hat{\mathbf{z}}}{2} \int_0^{r_m} \frac{(t - \sqrt{r^2 + x^2}/c)}{\sqrt{r^2 + x^2}} r dr = \frac{\mu_0 \alpha \hat{\mathbf{z}}}{2} \left[t \int_0^{r_m} \frac{r}{\sqrt{r^2 + x^2}} dr - \frac{1}{c} \int_0^{r_m} r dr \right] \\ &= \frac{\mu_0 \alpha \hat{\mathbf{z}}}{2} \left[t(ct - x) - \frac{1}{2c}(c^2 t^2 - x^2) \right] = \frac{\mu_0 \alpha \hat{\mathbf{z}}}{4c}(x^2 - 2ctx + c^2 t^2) = \frac{\mu_0 \alpha (x - ct)^2}{4c} \hat{\mathbf{z}}.\end{aligned}$$

$$\mathbf{E}(x, t) = -\frac{\partial \mathbf{A}}{\partial t} = \boxed{\frac{\mu_0 \alpha (x - ct)}{2} \hat{\mathbf{z}}}, \text{ for } ct > x, \text{ and } 0, \text{ for } ct < x.$$

$$\mathbf{B}(x, t) = \nabla \times \mathbf{A} = -\frac{\partial A_z}{\partial x} \hat{\mathbf{y}} = \boxed{-\frac{\mu_0 \alpha}{2c}(x - ct) \hat{\mathbf{y}}}, \text{ for } ct > x, \text{ and } 0, \text{ for } ct < x.$$

$$(b) \text{ Let } u \equiv \frac{1}{c} \left(\sqrt{r^2 + x^2} - x \right), \text{ so } du = \frac{1}{c} \left[\frac{1}{2} \frac{1}{\sqrt{r^2 + x^2}} 2r dr \right] = \frac{1}{c} \frac{r}{\sqrt{r^2 + x^2}} dr, \text{ and}$$

$$t - \frac{\sqrt{r^2 + x^2}}{c} = t - \frac{x}{c} - u, \text{ and as } r : 0 \rightarrow \infty, u : 0 \rightarrow \infty. \text{ Then } \mathbf{A}(x, t) = \frac{\mu_0 c \hat{\mathbf{z}}}{2} \int_0^\infty K\left(t - \frac{x}{c} - u\right) du. \text{ qed}$$

$$\begin{aligned}\mathbf{E}(x, t) &= -\frac{\partial \mathbf{A}}{\partial t} = -\frac{\mu_0 c \hat{\mathbf{z}}}{2} \int_0^\infty \frac{\partial}{\partial t} K\left(t - \frac{x}{c} - u\right) du. \text{ But } \frac{\partial}{\partial t} K\left(t - \frac{x}{c} - u\right) = -\frac{\partial}{\partial u} K\left(t - \frac{x}{c} - u\right). \\ &= \frac{\mu_0 c}{2} \hat{\mathbf{z}} \int_0^\infty \frac{\partial}{\partial u} K\left(t - \frac{x}{c} - u\right) du = \frac{\mu_0 c}{2} \hat{\mathbf{z}} \left[K\left(t - \frac{x}{c} - u\right) \right]_0^\infty = -\frac{\mu_0 c}{2} [K(t - x/c) - K(-\infty)] \hat{\mathbf{z}} \\ &= \boxed{-\frac{\mu_0 c}{2} K(t - x/c) \hat{\mathbf{z}}}, [\text{if } K(-\infty) = 0].\end{aligned}$$

Note that (i) and (ii) are consistent with this result. Meanwhile

$$\begin{aligned}\mathbf{B}(x, t) &= -\frac{\partial A_z}{\partial x} \hat{\mathbf{y}} = -\frac{\mu_0 c}{c} \hat{\mathbf{y}} \int_0^\infty \frac{\partial}{\partial x} K\left(t - \frac{x}{c} - u\right) du. \text{ But } \frac{\partial}{\partial x} K\left(t - \frac{x}{c} - u\right) = \frac{1}{c} \frac{\partial}{\partial u} K\left(t - \frac{x}{c} - u\right). \\ &= -\frac{\mu_0}{2} \hat{\mathbf{y}} \int_0^\infty \frac{\partial}{\partial u} K\left(t - \frac{x}{c} - u\right) du = -\frac{\mu_0}{2} \hat{\mathbf{y}} \left[K\left(t - \frac{x}{c} - u\right) \right]_0^\infty = \frac{\mu_0}{2} [K(t - x/c) - K(-\infty)] \hat{\mathbf{y}} \\ &= \boxed{\frac{\mu_0}{2} K(t - x/c) \hat{\mathbf{y}}}, [\text{if } K(-\infty) = 0]. \\ \mathbf{S} &= \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0} \left(\frac{\mu_0 c}{2} \right) \left(\frac{\mu_0}{2} \right) K(t - x/c) [-\hat{\mathbf{z}} \times \hat{\mathbf{y}}] = \frac{\mu_0 c}{4} [K(t - x/c)]^2 \hat{\mathbf{x}}.\end{aligned}$$

This is the power per unit area that reaches x at time t ; it left the surface at time $(t - x/c)$. Moreover, an equal amount of energy is radiated *downward*, so the total power leaving the surface at time t is $\frac{\mu_0 c}{2} [K(t)]^2$.

Problem 11.29

With $\alpha = 90^\circ$, Eq. 7.68 gives $\mathbf{E}' = c\mathbf{B}$, $\mathbf{B}' = -\frac{1}{c}\mathbf{E}$, $q'_m = -cq_e$. Use this to “translate” Eqs. 10.72, 10.73,

and 11.70:

$$\mathbf{E}' = c \left(\frac{1}{c} \hat{\mathbf{z}} \times \mathbf{E} \right) = \hat{\mathbf{z}} \times (-c\mathbf{B}') = -c(\hat{\mathbf{z}} \times \mathbf{B}').$$

$$\begin{aligned} \mathbf{B}' &= -\frac{1}{c} \mathbf{E} = -\frac{1}{c} \frac{q_e}{4\pi\epsilon_0} \frac{\mathbf{z}}{(\hat{\mathbf{z}} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + \hat{\mathbf{z}} \times (\mathbf{u} \times \mathbf{a})] \\ &= -\frac{1}{c} \frac{(-q'_m/c)}{4\pi\epsilon_0} \frac{\mathbf{z}}{(\hat{\mathbf{z}} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + \hat{\mathbf{z}} \times (\mathbf{u} \times \mathbf{a})] = \frac{\mu_0 q'_m}{4\pi} \frac{\mathbf{z}}{(\hat{\mathbf{z}} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + \hat{\mathbf{z}} \times (\mathbf{u} \times \mathbf{a})]. \end{aligned}$$

$$P = \frac{\mu_0 a^2}{6\pi c} q_e^2 = \frac{\mu_0 a^2}{6\pi c} \left(-\frac{1}{c} q'_m \right)^2 = \frac{\mu_0 a^2}{6\pi c^3} (q'_m)^2.$$

Or, dropping the primes,

$$\begin{aligned} \mathbf{B}(\mathbf{r}, t) &= \frac{\mu_0 q_m}{4\pi} \frac{\mathbf{z}}{(\hat{\mathbf{z}} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + \hat{\mathbf{z}} \times (\mathbf{u} \times \mathbf{a})]. \\ \mathbf{E}(\mathbf{r}, t) &= -c(\hat{\mathbf{z}} \times \mathbf{B}). \\ P &= \frac{\mu_0 q_m^2 a^2}{6\pi c^3}. \end{aligned}$$

Problem 11.30

(a) $W_{\text{ext}} = \int F dx = F \int_0^T v(t) dt$. From Prob. 11.19, $v(t) = \frac{F}{m} \left[t + \tau - \tau e^{(t-T)/\tau} \right]$. So

$$\begin{aligned} W_{\text{ext}} &= \frac{F^2}{m} \left[\int_0^T t dt + \tau \int_0^T dt - \tau e^{-T/\tau} \int_0^T e^{t/\tau} dt \right] = \frac{F^2}{m} \left[\frac{t^2}{2} + \tau t - \tau e^{-T/\tau} \tau e^{t/\tau} \right] \Big|_0^T \\ &= \frac{F^2}{m} \left[\frac{1}{2} T^2 + \tau T - \tau^2 e^{-T/\tau} (e^{T/\tau} - 1) \right] = \boxed{\frac{F^2}{m} \left(\frac{1}{2} T^2 + \tau T - \tau^2 + \tau^2 e^{-T/\tau} \right)}. \end{aligned}$$

(b) From Prob. 11.19, the final velocity is $v_f = (F/m)T$, so $W_{\text{kin}} = \frac{1}{2} m v_f^2 = \frac{1}{2} m \frac{F^2}{m^2} T^2 = \boxed{\frac{F^2 T^2}{2m}}$.

(c) $W_{\text{rad}} = \int P dt$. According to the Larmor formula, $P = \frac{\mu_0 q^2 a^2}{6\pi c}$, and (again from Prob. 11.19)

$$a(t) = \begin{cases} (F/m) [1 - e^{-T/\tau}] e^{t/\tau}, & (t \leq 0); \\ (F/m) [1 - e^{(t-T)/\tau}], & (0 \leq t \leq T). \end{cases}$$

$$\begin{aligned}
W_{\text{rad}} &= \frac{\mu_0 q^2}{6\pi c} \frac{F^2}{m^2} \left\{ \left(1 - e^{-T/\tau}\right)^2 \int_{-\infty}^0 e^{2t/\tau} dt + \int_0^T \left[1 - e^{(t-T)/\tau}\right]^2 dt \right\} \\
&= \tau \frac{F^2}{m} \left\{ \left(1 - e^{-T/\tau}\right)^2 \left(\frac{\tau}{2} e^{2t/\tau}\right) \Big|_{-\infty}^0 + \int_0^T dt - 2e^{-T/\tau} \int_0^T e^{t/\tau} dt + e^{-2T/\tau} \int_0^T e^{2t/\tau} dt \right\} \\
&= \frac{\tau F^2}{m} \left[\frac{\tau}{2} \left(1 - e^{-T/\tau}\right)^2 + T - 2e^{-T/\tau} \left(\tau e^{t/\tau}\right) \Big|_0^T + e^{-2T/\tau} \left(\frac{\tau}{2} e^{2t/\tau}\right) \Big|_0^T \right] \\
&= \frac{\tau F^2}{m} \left[\frac{\tau}{2} \left(1 - 2e^{-T/\tau} + e^{-2T/\tau}\right) + T - 2\tau e^{-T/\tau} \left(e^{T/\tau} - 1\right) + \frac{\tau}{2} e^{-2T/\tau} \left(e^{2T/\tau} - 1\right) \right] \\
&= \frac{\tau F^2}{m} \left[\frac{\tau}{2} - \tau e^{-T/\tau} + \frac{\tau}{2} e^{-2T/\tau} + T - 2\tau + 2\tau e^{-T/\tau} + \frac{\tau}{2} - \frac{\tau}{2} e^{-2T/\tau} \right] = \boxed{\frac{\tau F^2}{m} \left(T - \tau + \tau e^{-T/\tau}\right)}.
\end{aligned}$$

Energy conservation requires that the work done by the external force equal the final kinetic energy plus the energy radiated:

$$W_{\text{kin}} + W_{\text{rad}} = \frac{F^2 T^2}{2m} + \frac{\tau F^2}{m} \left(T - \tau + \tau e^{-T/\tau}\right) = \frac{F^2}{m} \left(\frac{1}{2} T^2 + \tau T - \tau^2 + \tau^2 e^{-T/\tau}\right) = W_{\text{ext}}. \checkmark$$

Problem 11.31

$$(a) \quad a = \tau \dot{a} + \frac{k}{m} \delta(t) \Rightarrow \int_{-\epsilon}^{\epsilon} a(t) dt = v(\epsilon) - v(-\epsilon) = \tau \int_{-\epsilon}^{\epsilon} \frac{da}{dt} dt + \frac{k}{m} \int_{-\epsilon}^{\epsilon} \delta(t) dt = \tau [a(\epsilon) - a(-\epsilon)] + \frac{k}{m}.$$

If the velocity is continuous, so $v(\epsilon) = v(-\epsilon)$, then $\boxed{a(\epsilon) - a(-\epsilon) = -\frac{k}{m\tau}}.$

When $t < 0$, $a = \tau \dot{a} \Rightarrow a(t) = A e^{t/\tau}$; when $t > 0$, $a = \tau \dot{a} \Rightarrow a(t) = B e^{t/\tau}$; $\Delta a = B - A = -\frac{k}{m\tau}$

$\Rightarrow B = A - \frac{k}{m\tau}$, so the general solution is $\boxed{a(t) = \begin{cases} A e^{t/\tau}, & (t < 0); \\ [A - (k/m\tau)] e^{t/\tau}, & (t > 0). \end{cases}}$

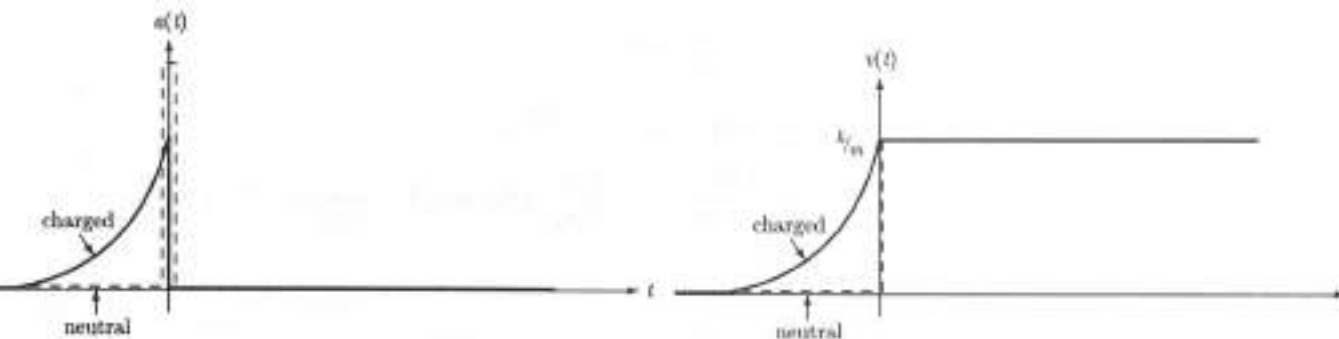
To eliminate the runaway we'd need $A = k/m\tau$; to eliminate preacceleration we'd need $A = 0$. Obviously, you can't do both. If you choose to eliminate the runaway, then $\boxed{a(t) = \begin{cases} (k/m\tau) e^{t/\tau}, & (t < 0); \\ 0, & (t > 0). \end{cases}}$

$$v(t) = \int_{-\infty}^t a(t) dt = \frac{k}{m\tau} \int_{-\infty}^t e^{t/\tau} dt = \frac{k}{m\tau} \left(\tau e^{t/\tau}\right) \Big|_{-\infty}^t = \frac{k}{m} e^{t/\tau} \text{ (for } t < 0);$$

$$\text{for } t > 0, v(t) = v(0) + \int_0^t a(t) dt = v(0) = \frac{k}{m}. \text{ So } \boxed{v(t) = \begin{cases} (k/m) e^{t/\tau}, & (t < 0); \\ (k/m), & (t > 0). \end{cases}}$$

For an *uncharged* particle we would have $a(t) = \frac{k}{m} \delta(t)$, $v(t) = \int_{-\infty}^t a(t) dt = \begin{cases} 0, & (t < 0); \\ (k/m), & (t > 0). \end{cases}$

The graphs:



(b)

$$W_{\text{ext}} = \int F dx = \int F v dt = k \int \delta(t) v(t) dt = k v(0) = \frac{k^2}{m}.$$

$$W_{\text{kin}} = \frac{1}{2} m v_f^2 = \frac{1}{2} m \left(\frac{k}{m} \right)^2 = \frac{k^2}{2m}.$$

$$W_{\text{ext}} = \int P_{\text{rad}} dt = \frac{\mu_0 q^2}{6\pi c} \int [a(t)]^2 dt = \tau m \left(\frac{k}{m\tau} \right)^2 \int_{-\infty}^0 e^{2t/\tau} dt = \frac{k^2}{m\tau} \left(\frac{\tau}{2} e^{2t/\tau} \right) \Big|_{-\infty}^0 = \frac{k^2}{m\tau} \frac{\tau}{2} = \frac{k^2}{2m}.$$

Clearly, $W_{\text{ext}} = W_{\text{kin}} + W_{\text{rad}}$. ✓

Problem 11.32

Our task is to solve the equation $a = \tau \dot{a} + \frac{U_0}{m} [-\delta(x) + \delta(x - L)]$, subject to the boundary conditions

- (1) x continuous at $x = 0$ and $x = L$;
- (2) v continuous at $x = 0$ and $x = L$;
- (3) $\Delta a = \pm U_0/m\tau v$ (plus at $x = 0$, minus at $x = L$).

The third of these follows from integrating the equation of motion:

$$\begin{aligned} \int \frac{dv}{dt} dt &= \tau \int \frac{da}{dt} dt + \frac{U_0}{m} \int [-\delta(x) + \delta(x - L)] dt, \\ \Delta v &= \tau \Delta a + \frac{U_0}{m} \int [-\delta(x) + \delta(x - L)] \frac{dt}{dx} dx = 0, \\ \Delta a &= -\frac{U_0}{m\tau} \int \frac{1}{v} [-\delta(x) + \delta(x - L)] dx = \pm \frac{U_0}{m\tau v}. \end{aligned}$$

In each of the three regions the force is zero (it acts only at $x = 0$ and $x = L$), and the general solution is

$$a(t) = A e^{t/\tau}; \quad v(t) = A \tau e^{t/\tau} + B; \quad x(t) = A \tau^2 e^{t/\tau} + B t + C.$$

(I'll put subscripts on the constants A , B , and C , to distinguish the three regions.)

Region iii ($x > L$): To avoid the runaway we pick $A_3 = 0$; then $a(t) = 0$, $v(t) = B_3$, $x(t) = B_3 t + C_3$. Let the final velocity be $v_f (= B_3)$, set the clock so that $t = 0$ when the particle is at $x = 0$, and let T be the time it takes to traverse the barrier, so $x(T) = L = v_f T + C_3$, and hence $C_3 = L - v_f T$. Then

$$\boxed{a(t) = 0; \quad v(t) = v_f, \quad x(t) = L + v_f(t - T),} \quad (t < T).$$

Region ii ($0 < x < L$): $a = A_2 e^{t/\tau}$, $v = A_2 \tau e^{t/\tau} + B_2$, $x = A_2 \tau^2 e^{t/\tau} + B_2 t + C_2$.

$$(3) \Rightarrow 0 - A_2 e^{T/\tau} = -\frac{U_0}{m\tau v_f} \Rightarrow A_2 = \frac{U_0}{m\tau v_f} e^{-T/\tau}.$$

$$(2) \Rightarrow v_f = A_2 \tau e^{T/\tau} + B_2 = \frac{U_0}{mv_f} + B_2 \Rightarrow B_2 = v_f - \frac{U_0}{mv_f}.$$

$$(1) \Rightarrow L = A_2 \tau^2 e^{T/\tau} + B_2 T + C_2 = \frac{U_0 \tau}{mv_f} + v_f T - \frac{U_0 T}{mv_f} + C_2 = v_f T + \frac{U_0}{mv_f} (\tau - T) + C_2 \Rightarrow$$

$$C_2 = L - v_f T + \frac{U_0}{mv_f} (T - \tau).$$

$$\boxed{\begin{aligned} a(t) &= \frac{U_0}{m\tau v_f} e^{(t-T)/\tau}; \\ v(t) &= v_f + \frac{U_0}{mv_f} \left[e^{(t-T)/\tau} - 1 \right]; \\ x(t) &= L + v_f (t - T) + \frac{U_0}{mv_f} \left[\tau e^{(t-T)/\tau} - t + T - \tau \right]; \end{aligned}} \quad (0 < t < T).$$

[Note: if the barrier is sufficiently wide (or high) the particle may turn around before reaching L , but we're interested here in the régime where it *does* tunnel through.]

In particular, for $t = 0$ (when $x = 0$):

$$0 = L - v_f T + \frac{U_0}{mv_f} \left[\tau e^{-T/\tau} + T - \tau \right] \Rightarrow L = v_f T - \frac{U_0}{mv_f} \left[\tau e^{-T/\tau} + T - \tau \right]. \quad \text{qed}$$

Region i ($x < 0$): $a = A_1 e^{t/\tau}$, $v = A_1 \tau e^{t/\tau} + B_1$, $x = A_1 \tau^2 e^{t/\tau} + B_1 t + C_1$. Let v_i be the incident velocity (at $t \rightarrow -\infty$); then $B_1 = v_i$. Condition (3) says

$$\frac{U_0}{m\tau v_f} e^{-T/\tau} - A_1 = \frac{U_0}{m\tau v_0},$$

where v_0 is the speed of the particle as it passes $x = 0$. From the solution in region (ii) it follows that $v_0 = v_f + \frac{U_0}{mv_f} (e^{-T/\tau} - 1)$. But we can also express it in terms of the solution in region (i): $v_0 = A_1 \tau + v_i$. Therefore

$$\begin{aligned} v_i &= v_f + \frac{U_0}{mv_f} (e^{-T/\tau} - 1) - A_1 \tau = v_f + \frac{U_0}{mv_f} (e^{-T/\tau} - 1) + \frac{U_0}{mv_0} - \frac{U_0}{mv_f} e^{-T/\tau} \\ &= v_f - \frac{U_0}{mv_f} + \frac{U_0}{mv_0} = v_f - \frac{U_0}{mv_f} \left(1 - \frac{v_f}{v_0} \right) = v_f - \frac{U_0}{mv_f} \left\{ 1 - \frac{v_f}{v_f + (U_0/mv_f) [e^{-T/\tau} - 1]} \right\} \\ &= v_f - \frac{U_0}{mv_f} \left\{ 1 - \frac{1}{1 + (U_0/mv_f^2) [e^{-T/\tau} - 1]} \right\}. \quad \text{qed} \end{aligned}$$

If $\frac{1}{2}mv_f^2 = \frac{1}{2}U_0$, then

$$L = v_f T - v_f \left[\tau e^{-T/\tau} + T - \tau \right] = v_f \left[T - \tau e^{-T/\tau} - T + \tau \right] = \tau v_f (1 - e^{-T/\tau});$$

$$v_i = v_f - v_f \left[1 - \frac{1}{1 + e^{-T/\tau} - 1} \right] = v_f (1 - 1 + e^{T/\tau}) = v_f e^{T/\tau}.$$

Putting these together,

$$\frac{L}{\tau v_f} = 1 - e^{-T/\tau} \Rightarrow e^{-T/\tau} = 1 - \frac{L}{\tau v_f} \Rightarrow e^{T/\tau} = \frac{1}{1 - (L/\tau v_f)} \Rightarrow v_i = \frac{v_f}{1 - (L/v_f \tau)}. \quad \text{qed}$$

In particular, for $L = v_f \tau / 4$, $v_i = \frac{v_f}{1 - 1/4} = \frac{4}{3} v_f$, so $\frac{KE_i}{KE_f} = \frac{\frac{1}{2} m v_i^2}{\frac{1}{2} m v_f^2} = \left(\frac{v_i}{v_f} \right)^2 = \frac{16}{9} \Rightarrow$

$$KE_i = \frac{16}{9} KE_f = \frac{16}{9} \frac{1}{2} U_0 = \frac{8}{9} U_0.$$

Problem 11.33

(a) From Eq. 10.72, $\mathbf{E}_1 = \frac{(q/2)}{4\pi\epsilon_0 (\mathbf{r} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + (\mathbf{r} \cdot \mathbf{a})\mathbf{u} - (\mathbf{r} \cdot \mathbf{u})\mathbf{a}]$. Here $\mathbf{u} = c\hat{\mathbf{r}} - \mathbf{v}$, $\mathbf{r} = l\hat{\mathbf{x}} + d\hat{\mathbf{y}}$, $\mathbf{v} = v\hat{\mathbf{x}}$, $\mathbf{a} = a\hat{\mathbf{x}}$, so $\mathbf{r} \cdot \mathbf{v} = lv$, $\mathbf{r} \cdot \mathbf{a} = la$, $\mathbf{r} \cdot \mathbf{u} = c\mathbf{r} \cdot \hat{\mathbf{r}} - \mathbf{r} \cdot \mathbf{v} = c\mathbf{r} \cdot \hat{\mathbf{r}} - lv$. We want only the x component. Noting that $u_x = (c/\mathbf{r})l - v = (cl - v\mathbf{r})/\mathbf{r}$, we have:

$$\begin{aligned} E_{1x} &= \frac{q}{8\pi\epsilon_0} \frac{\mathbf{r}}{(c\mathbf{r} - lv)^3} \left[\frac{1}{\mathbf{r}} (cl - v\mathbf{r})(c^2 - v^2 + la) - a(c\mathbf{r} - lv) \right] \\ &= \frac{q}{8\pi\epsilon_0} \frac{1}{(c\mathbf{r} - lv)^3} [(cl - v\mathbf{r})(c^2 - v^2) + cl^2a - v\mathbf{r}la - ac\mathbf{r}^2 + alv\mathbf{r}]. \text{ But } \mathbf{r}^2 = l^2 + d^2. \\ &= \frac{q}{8\pi\epsilon_0} \frac{1}{(c\mathbf{r} - lv)^3} [(cl - v\mathbf{r})(c^2 - v^2) - acd^2]. \\ \mathbf{F}_{\text{self}} &= \frac{q^2}{8\pi\epsilon_0} \frac{1}{(c\mathbf{r} - lv)^3} [(cl - v\mathbf{r})(c^2 - v^2) - acd^2] \hat{\mathbf{x}}. \quad (\text{This generalizes Eq. 11.90}) \end{aligned}$$

Now $x(t) - x(t_r) = l = vT + \frac{1}{2}aT^2 + \frac{1}{6}\dot{a}T^3 + \dots$, where $T = t - t_r$, and v , a , and $\dot{a}$ are all evaluated at the retarded time t_r .

$$(cT)^2 = \mathbf{r}^2 = l^2 + d^2 = d^2 + (vT + \frac{1}{2}aT^2 + \frac{1}{6}\dot{a}T^3)^2 = d^2 + v^2T^2 + vaT^3 + \frac{1}{3}v\dot{a}T^4 + \frac{1}{4}a^2T^4;$$

$$c^2T^2(1 - v^2/c^2) = c^2T^2/\gamma^2 = d^2 + vaT^3 + \left(\frac{1}{3}v\dot{a} + \frac{1}{4}a^2 \right) T^4. \text{ Solve for } T \text{ as a power series in } d:$$

$$T = \frac{\gamma d}{c} (1 + Ad + Bd^2 + \dots) \Rightarrow \frac{c^2}{\gamma^2} \frac{\gamma^2 d^2}{c^2} (1 + 2Ad + 2Bd^2 + A^2d^2) = d^2 + va\frac{\gamma^3 d^3}{c^3} (1 + 3Ad) + \left(\frac{v\dot{a}}{3} + \frac{a^2}{4} \right) \frac{\gamma^4}{c^4} d^4.$$

$$\text{Comparing like powers of } d: A = \frac{1}{2}va\frac{\gamma^3}{c^3}; \quad 2B + A^2 = \frac{3va\gamma^3}{c^3}A + \left(\frac{v\dot{a}}{3} + \frac{a^2}{4} \right) \frac{\gamma^4}{c^4}.$$

$$\begin{aligned} 2B &= \frac{3va\gamma^3}{c^3} \frac{1}{2}va\frac{\gamma^3}{c^3} - \frac{1}{4}v^2a^2\frac{\gamma^6}{c^6} + \frac{v\dot{a}}{3}\frac{\gamma^4}{c^4} + \frac{a^2\gamma^4}{4c^4} = \frac{v\dot{a}}{3}\frac{\gamma^4}{c^4} + \frac{\gamma^6a^2}{4c^4} \left(\frac{1}{\gamma^2} - \frac{v^2}{c^2} \right) + \frac{3}{2}\frac{v^2a^2\gamma^6}{c^6} \\ &= \frac{\gamma^4}{c^4} \left[\frac{v\dot{a}}{3} + \frac{a^2\gamma^2}{4} \left(1 - \frac{v^2}{c^2} - \frac{v^2}{c^2} + 6\frac{v^2}{c^2} \right) \right] \Rightarrow B = \frac{\gamma^4}{2c^4} \left[\frac{v\dot{a}}{3} + \frac{\gamma^2a^2}{4} \left(1 + 4\frac{v^2}{c^2} \right) \right]. \end{aligned}$$

$$T = \frac{\gamma d}{c} \left\{ 1 + \frac{va\gamma^3}{2c^3}d + \frac{\gamma^4}{2c^4} \left[\frac{v\dot{a}}{3} + \frac{\gamma^2a^2}{4} \left(1 + 4\frac{v^2}{c^2} \right) \right] d^2 \right\} + (\dots) d^4 + \dots \quad (\text{generalizing Eq. 11.93}).$$

$$\begin{aligned}
l &= vT + \frac{1}{2}aT^2 + \frac{1}{6}\dot{a}T^3 + \dots \\
&= \frac{v\gamma d}{c} \left\{ 1 + \frac{va}{2} \frac{\gamma^3}{c^3} d + \frac{\gamma^4}{2c^4} \left[\frac{v\dot{a}}{3} + \frac{\gamma^2 a^2}{4} \left(1 + 4 \frac{v^2}{c^2} \right) \right] d^2 \right\} + \frac{1}{2} a \frac{\gamma^2 d^2}{c^2} \left[1 + va \frac{\gamma^3}{c^3} d \right] + \frac{1}{6} \dot{a} \frac{\gamma^3}{c^3} d^3 \\
&= \left(\frac{v\gamma}{c} \right) d + \frac{a}{2} \frac{\gamma^4}{c^2} \left(1 - \frac{v^2}{c^2} + \frac{v^2}{c^2} \right) d^2 + \left\{ \frac{v\gamma}{2c} \frac{\gamma^4}{c^4} \left[\frac{v\dot{a}}{3} + \frac{\gamma^2 a^2}{4} \left(1 + 4 \frac{v^2}{c^2} \right) \right] + \frac{1}{2} a \frac{\gamma^2}{c^2} va \frac{\gamma^3}{c^3} + \frac{1}{6} \dot{a} \frac{\gamma^3}{c^3} \right\} d^3 \\
&= \left(\frac{v\gamma}{c} \right) d + \left(\frac{a\gamma^4}{2c^2} \right) d^2 + \frac{\gamma^3}{2c^3} \left[\frac{\dot{a}}{3} \left(1 + \gamma^2 \frac{v^2}{c^2} \right) + \frac{v\gamma^4 a^2}{c^2} \left(\frac{1}{4} + \frac{v^2}{c^2} + 1 - \frac{v^2}{c^2} \right) \right] d^3 \\
&= \left(\frac{v\gamma}{c} \right) d + \left(\frac{a\gamma^4}{2c^2} \right) d^2 + \frac{\gamma^5}{2c^3} \left[\frac{\dot{a}}{3} + \frac{5}{4} \frac{v\gamma^2 a^2}{c^2} \right] d^3 + () d^4 + \dots
\end{aligned}$$

$$\begin{aligned}
\mathbf{r} &= cT = \gamma d \left\{ 1 + \frac{va}{2} \frac{\gamma^3}{c^3} d + \frac{\gamma^4}{2c^4} \left[\frac{v\dot{a}}{3} + \gamma^2 a^2 \left(\frac{1}{4} + \frac{v^2}{c^2} \right) \right] d^2 \right\} + () d^4 + \dots \\
c\mathbf{r} - lv &= c\gamma d + \frac{va\gamma^4}{2c^2} d^2 + \frac{\gamma^5}{2c^3} \left[\frac{v\dot{a}}{3} + \gamma^2 a^2 \left(\frac{1}{4} + \frac{v^2}{c^2} \right) \right] d^3 - \frac{v^2\gamma}{c} d - \frac{av\gamma^4}{2c^2} d^2 - \frac{\gamma^5 v}{2c^3} \left[\frac{\dot{a}}{3} + \frac{5}{4} \frac{v\gamma^2 a^2}{c^2} \right] d^3 + \dots \\
&= c\gamma d \left(1 - \frac{v^2}{c^2} \right) + \frac{\gamma^5}{2c^3} \left[\frac{v\dot{a}}{3} + \gamma^2 a^2 \left(\frac{1}{4} + \frac{v^2}{c^2} \right) - \frac{v\dot{a}}{3} - \frac{5}{4} \frac{v^2\gamma^2 a^2}{c^2} \right] d^3 + \dots \\
&= \frac{c}{\gamma} d + \frac{\gamma^5 a^2}{8c^3} d^3 + () d^4 + \dots
\end{aligned}$$

$$\begin{aligned}
cl - v\mathbf{r} &= v\gamma d + \frac{a\gamma^4}{2c} d^2 + \frac{\gamma^5}{2c^2} \left(\frac{\dot{a}}{3} + \frac{5}{4} \frac{v\gamma^2 a^2}{c^2} \right) d^3 - v\gamma d - \frac{v^2 a}{2} \frac{\gamma^4}{c^3} d^2 - \frac{v\gamma^5}{2c^4} \left[\frac{v\dot{a}}{3} + \gamma^2 a^2 \left(\frac{1}{4} + \frac{v^2}{c^2} \right) \right] d^3 \\
&= \frac{a\gamma^4}{2c} \left(1 - \frac{v^2}{c^2} \right) d^2 + \frac{\gamma^5}{2c^2} \left[\frac{\dot{a}}{3} + \frac{5}{4} \frac{v\gamma^2 a^2}{c^2} - \frac{v^2}{c^2} \frac{\dot{a}}{3} - \frac{v\gamma^2 a^2}{c^2} \left(\frac{1}{4} + \frac{v^2}{c^2} \right) \right] d^3 + () d^4 + \dots \\
&= \left(\frac{a\gamma^2}{2c} \right) d^2 + \frac{\gamma^5}{2c^2} \left[\frac{\dot{a}}{3\gamma^2} + \frac{v\gamma^2 a^2}{c^2} \left(\frac{5}{4} - \frac{1}{4} - \frac{v^2}{c^2} \right) \right] d^3 + () d^4 + \dots \\
&= \left(\frac{a\gamma^2}{2c} \right) d^2 + \frac{\gamma^3}{2c^2} \left(\frac{\dot{a}}{3} + \frac{v\gamma^2 a^2}{c^2} \right) d^3 + () d^4 + \dots \\
(c\mathbf{r} - lv)^{-3} &= \left[\frac{cd}{\gamma} \left(1 + \frac{\gamma^6 a^2}{8c^4} d^2 \right) \right]^{-3} = \left(\frac{\gamma}{cd} \right)^3 \left(1 - 3 \frac{\gamma^6 a^2}{8c^4} d^2 \right) + \dots
\end{aligned}$$

$$\begin{aligned}
\mathbf{F}_{\text{self}} &= \frac{q^2}{8\pi\epsilon_0} \left(\frac{\gamma}{cd} \right)^3 \left(1 - 3 \frac{\gamma^6 a^2}{8c^4} d^2 \right) \left\{ \left[\left(\frac{a\gamma^2}{2c} \right) d^2 + \frac{\gamma^3}{2c^2} \left(\frac{\dot{a}}{3} + \frac{v\gamma^2 a^2}{c^2} \right) d^3 \right] \frac{c^2}{\gamma^2} - acd^2 \right\} \hat{\mathbf{x}} \\
&= \frac{q^2}{8\pi\epsilon_0} \frac{\gamma^3}{c^3 d} \left(1 - \frac{3}{8} \frac{\gamma^6 a^2}{c^4} d^2 \right) \left[-\frac{ac}{2} + \frac{\gamma}{2} \left(\frac{\dot{a}}{3} + \frac{v\gamma^2 a^2}{c^2} \right) d \right] \hat{\mathbf{x}} \\
&= \frac{q^2}{8\pi\epsilon_0} \frac{\gamma^3}{c^3 d} \frac{1}{2} \left[-ac + \gamma \left(\frac{\dot{a}}{3} + \frac{v\gamma^2 a^2}{c^2} \right) d + () d^2 + \dots \right] \hat{\mathbf{x}} \\
&= \frac{q^2}{4\pi\epsilon_0} \left[-\gamma^3 \frac{a}{4c^2 d} + \frac{\gamma^4}{4c^3} \left(\frac{\dot{a}}{3} + \frac{v\gamma^2 a^2}{c^2} \right) + () d + \dots \right] \hat{\mathbf{x}} \text{ (generalizing Eq. 11.95).}
\end{aligned}$$

Switching to t : $v(t_r) = v(t) + \dot{v}(t)(t_r - t) + \dots = v(t) - a(t)T = v(t) - a\gamma d/c$. (When multiplied by d , it doesn't matter—to this order—whether we evaluate at t or at t_r .)

$$1 - \left[\frac{v(t_r)}{c} \right]^2 = 1 - \frac{[v(t)^2 - 2va\gamma d/c]}{c^2} = \left[1 - \frac{v(t)^2}{c^2} \right] \left(1 + \frac{2av\gamma^3 d}{c^3} \right), \text{ so}$$

$$\gamma = \left[1 - \left(\frac{v(t_r)}{c} \right)^2 \right]^{-1/2} = \gamma(t) \left(1 - \frac{va\gamma^3 d}{c^3} \right); \quad a(t_r) = a(t) - T\dot{a} = a(t) - \frac{\dot{a}\gamma}{c}d.$$

Evaluating everything now at time t :

$$\begin{aligned} \mathbf{F}_{\text{self}} &= \frac{q^2}{4\pi\epsilon_0} \left[-\gamma^3 \frac{(1 - 3va\gamma^3 d/c^3)(a - \dot{a}\gamma d/c)}{4c^2 d} + \frac{\gamma^4}{4c^3} \left(\frac{\dot{a}}{3} + \frac{v\gamma^2 a^2}{c^2} \right) + () d^2 + \dots \right] \hat{\mathbf{x}} \\ &= \frac{q^2}{4\pi\epsilon_0} \left[-\frac{\gamma^3 a}{4c^2 d} + \frac{\gamma^3}{4c^2} \left(\frac{\dot{a}\gamma}{c} + 3\frac{va^2\gamma^2}{c^3} \right) + \frac{\gamma^4}{4c^3} \left(\frac{\dot{a}}{3} + \frac{v\gamma^2 a^2}{c^2} \right) + () d + \dots \right] \hat{\mathbf{x}} \\ &= \frac{q^2}{4\pi\epsilon_0} \left[-\frac{\gamma^3 a}{4c^2 d} + \frac{\gamma^4}{4c^3} \left(\dot{a} + \frac{\dot{a}}{3} + 3\frac{va^2\gamma^2}{c^2} + \frac{v\gamma^2 a^2}{c^2} \right) + () d + \dots \right] \hat{\mathbf{x}} \\ &= \frac{q^2}{4\pi\epsilon_0} \left[-\frac{\gamma^3 a}{4c^2 d} + \frac{\gamma^4}{3c^3} \left(\dot{a} + 3\frac{va^2\gamma^2}{c^2} \right) + () d + \dots \right] \hat{\mathbf{x}} \text{ (generalizing Eq. 11.96).} \end{aligned}$$

The first term is the electromagnetic mass; the radiation reaction itself is the second term:

$$F_{\text{rad}}^{\text{int}} = \frac{\mu_0 q^2}{12\pi c} \gamma^4 \left(\dot{a} + 3\frac{va^2\gamma^2}{c^2} \right) \text{ (generalizing Eq. 11.99), so the generalization of Eq. 11.100 is}$$

$$F_{\text{rad}} = \frac{\mu_0 q^2}{6\pi c} \gamma^4 \left(\dot{a} + 3\frac{va^2\gamma^2}{c^2} \right).$$

(b) $F_{\text{rad}} = A\gamma^4 \left(\dot{a} + \frac{3\gamma^2 a^2 v}{c^2} \right)$, where $A \equiv \frac{\mu_0 q^2}{6\pi c}$. $P = Aa^2\gamma^6$ (Eq. 11.75). What we must show is that

$$\int_{t_1}^{t_2} F_{\text{rad}} v dt = - \int_{t_1}^{t_2} P dt, \quad \text{or} \quad \int_{t_1}^{t_2} \gamma^4 \left(\dot{a} v + 3\frac{v^2 a^2 \gamma^2}{c^2} \right) dt = - \int_{t_1}^{t_2} a^2 \gamma^6 dt$$

(except for boundary terms—see Sect. 11.2.2).

Rewrite the first term: $\int_{t_1}^{t_2} \gamma^4 \dot{a} v dt = \int_{t_1}^{t_2} (\gamma^4 v) \frac{da}{dt} dt = \gamma^4 v a \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} (\gamma^4 v) a dt.$

Now $\frac{d}{dt} (\gamma^4 v) = 4\gamma^3 \frac{d\gamma}{dt} v + \gamma^4 \dot{a}$; $\frac{d\gamma}{dt} = \frac{d}{dt} \left(\frac{1}{\sqrt{1 - v^2/c^2}} \right) = -\frac{1}{2} \frac{1}{(1 - v^2/c^2)^{3/2}} \left(-\frac{2va}{c^2} \right) = \frac{va\gamma^3}{c^2}$. So

$$\frac{d}{dt} (\gamma^4 v) = 4\gamma^3 v \frac{va\gamma^3}{c^2} + \gamma^4 \dot{a} = \gamma^6 a \left(1 - \frac{v^2}{c^2} + 4\frac{v^2}{c^2} \right) = \gamma^6 a \left(1 + 3\frac{v^2}{c^2} \right).$$

$$\int_{t_1}^{t_2} \gamma^4 \dot{a} v dt = \gamma^4 v a \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} \gamma^6 a^2 \left(1 + 3\frac{v^2}{c^2} \right) dt, \text{ and hence}$$

$$\int_{t_1}^{t_2} \gamma^4 \left(\dot{a} v + \frac{3\gamma^2 a^2 v^2}{c^2} \right) dt = \gamma^4 v a \Big|_{t_1}^{t_2} + \int_{t_1}^{t_2} \left[-\gamma^6 a^2 \left(1 + 3\frac{v^2}{c^2} \right) + 3\gamma^6 \frac{a^2 v^2}{c^2} \right] dt = \gamma^4 v a \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} \gamma^6 a^2 dt. \quad \text{qed}$$

Problem 11.34

(a) $P = \frac{\mu_0 q^2 a^2 \gamma^6}{6\pi c}$ (Eq. 11.75). $w = \sqrt{b^2 + c^2 t^2}$ (Eq. 10.52); $v = \dot{w} = \frac{c^2 t}{\sqrt{b^2 + c^2 t^2}}$;

$$a = \dot{v} = \frac{c^2}{\sqrt{b^2 + c^2 t^2}} - \frac{c^2 t (c^2 t)}{(b^2 + c^2 t^2)^{3/2}} = \frac{c^2}{(b^2 + c^2 t^2)^{3/2}} (b^2 + c^2 t^2 - c^2 t^2) = \frac{b^2 c^2}{(b^2 + c^2 t^2)^{3/2}};$$

$$\gamma^2 = \frac{1}{1 - v^2/c^2} = \frac{1}{1 - [c^2 t^2 / (b^2 + c^2 t^2)]} = \frac{b^2 + c^2 t^2}{b^2 + c^2 t^2 - c^2 t^2} = \frac{1}{b^2} (b^2 + c^2 t^2). \quad \text{So}$$

$$P = \frac{\mu_0 q^2}{6\pi c} \frac{b^4 c^4}{(b^2 + c^2 t^2)^3} \frac{(b^2 + c^2 t^2)^3}{b^6} = \boxed{\frac{q^2 c}{6\pi \epsilon_0 b^2}}. \quad \boxed{\text{Yes, it radiates}} \quad (\text{in fact, at a constant rate}).$$

$$\begin{aligned} \text{(b) } F_{\text{rad}} &= \frac{\mu_0 q^2 \gamma^4}{6\pi c} \left(\dot{a} + \frac{3\gamma^2 a^2 v}{c^2} \right); \quad \dot{a} = -\frac{3}{2} \frac{b^2 c^2 (2c^2 t)}{(b^2 + c^2 t^2)^{5/2}} = -\frac{3b^2 c^4 t}{(b^2 + c^2 t^2)^{5/2}}; \quad \left(\dot{a} + \frac{3\gamma^2 a^2 v}{c^2} \right) = \\ &= -\frac{3b^2 c^4 t}{(b^2 + c^2 t^2)^{5/2}} + \frac{3}{c^2} \frac{(b^2 + c^2 t^2)}{b^2} \frac{b^4 c^4}{(b^2 + c^2 t^2)^3} \frac{c^2 t}{\sqrt{b^2 + c^2 t^2}} = 0. \quad \boxed{F_{\text{rad}} = 0}. \quad \boxed{\text{No, the radiation reaction is zero.}} \end{aligned}$$

Problem 11.35

The fields of a nonstatic ideal dipole (Problem 10.34) contain terms that go like $1/r$, $1/r^2$, and $1/r^3$, for fixed $t_0 \equiv t - r/c$; radiation comes from the first of these:

$$\mathbf{E}_r = -\frac{\mu_0}{4\pi r} [\ddot{\mathbf{p}} - \hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}})], \quad \mathbf{B}_r = -\frac{\mu_0}{4\pi r c} (\hat{\mathbf{r}} \times \ddot{\mathbf{p}})$$

where the dipole moments are evaluated at time t_0 . The $1/r^2$ term in the Poynting vector is

$$\begin{aligned} \mathbf{S}_r &= \frac{1}{\mu_0} (\mathbf{E}_r \times \mathbf{B}_r) = \frac{\mu_0}{16\pi^2 c r^2} [\ddot{\mathbf{p}} - \hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}})] \times (\hat{\mathbf{r}} \times \ddot{\mathbf{p}}) = \frac{\mu_0}{16\pi^2 c r^2} \{ \ddot{\mathbf{p}} \times (\hat{\mathbf{r}} \times \ddot{\mathbf{p}}) - (\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}) [\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \ddot{\mathbf{p}})] \} \\ &= \frac{\mu_0}{16\pi^2 c r^2} \{ \hat{\mathbf{r}} \ddot{p}^2 - \ddot{\mathbf{p}}(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}) - (\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}) [\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}) - \ddot{\mathbf{p}}] \} = \frac{\mu_0}{16\pi^2 c r^2} [\ddot{p}^2 - (\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}})^2] \hat{\mathbf{r}}. \end{aligned}$$

Setting the z axis along $\ddot{\mathbf{p}}$, $[\ddot{p}^2 - (\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}})^2] = \ddot{p}^2 \sin^2 \theta$, and the power radiated is

$$P = \oint \mathbf{S}_r \cdot d\mathbf{a} = \frac{\mu_0}{16\pi^2 c} \ddot{p}^2 \oint \frac{1}{r^2} \sin^2 \theta r^2 \sin \theta d\theta d\phi = \frac{\mu_0}{8\pi c} \ddot{p}^2 \int_0^\pi \sin^3 \theta d\theta = \boxed{\frac{\mu_0}{6\pi c} \ddot{p}^2}.$$

For the sinusoidal case, $\mathbf{p} = \mathbf{p}_0 \cos \omega t$, we have $\ddot{\mathbf{p}} = -\omega^2 \mathbf{p}_0 \cos \omega t$, $\ddot{p}^2 = \omega^4 p_0^2 \cos^2 \omega t$, and its time average is $(1/2)p_0^2 \omega^4$, so

$$\langle P \rangle = \frac{\mu_0}{12\pi c} \ddot{p}_0^2 \omega^4,$$

in agreement with Eq. 11.22. And in the case of quadratic time dependence our result agrees with the answer to Problem 11.26.

Chapter 12

Electrodynamics and Relativity

Problem 12.1

Let $\mathbf{u}$ be the velocity of a particle in $\mathcal{S}$, $\bar{\mathbf{u}}$ its velocity in $\bar{\mathcal{S}}$, and $\mathbf{v}$ the velocity of $\bar{\mathcal{S}}$ with respect to $\mathcal{S}$. Galileo's velocity addition rule says that $\mathbf{u} = \bar{\mathbf{u}} + \mathbf{v}$. For a free particle, $\mathbf{u}$ is constant (that's Newton's first law in $\mathcal{S}$).

- (a) If $\mathbf{v}$ is constant, then $\bar{\mathbf{u}} = \mathbf{u} - \mathbf{v}$ is also constant, so Newton's first law holds in $\bar{\mathcal{S}}$, and hence $\bar{\mathcal{S}}$ is inertial.
(b) If $\bar{\mathcal{S}}$ is inertial, then $\bar{\mathbf{u}}$ is also constant, so $\mathbf{v} = \mathbf{u} - \bar{\mathbf{u}}$ is constant.

Problem 12.2

$$\begin{aligned} \text{(a)} \quad m_A \mathbf{u}_A + m_B \mathbf{u}_B &= m_C \mathbf{u}_C + m_D \mathbf{u}_D; & \mathbf{u}_i &= \bar{\mathbf{u}}_i + \mathbf{v}. \\ m_A(\bar{\mathbf{u}}_A + \mathbf{v}) + m_B(\bar{\mathbf{u}}_B + \mathbf{v}) &= m_C(\bar{\mathbf{u}}_C + \mathbf{v}) + m_D(\bar{\mathbf{u}}_D + \mathbf{v}), \\ m_A \bar{\mathbf{u}}_A + m_B \bar{\mathbf{u}}_B + (m_A + m_B)\mathbf{v} &= m_C \bar{\mathbf{u}}_C + m_D \bar{\mathbf{u}}_D + (m_C + m_D)\mathbf{v}. \end{aligned}$$

Assuming *mass* is conserved, $(m_A + m_B) = (m_C + m_D)$, it follows that

$$m_A \bar{\mathbf{u}}_A + m_B \bar{\mathbf{u}}_B = m_C \bar{\mathbf{u}}_C + m_D \bar{\mathbf{u}}_D, \text{ so momentum is conserved in } \bar{\mathcal{S}}.$$

$$\begin{aligned} \text{(b)} \quad \frac{1}{2} m_A u_A^2 + \frac{1}{2} m_B u_B^2 &= \frac{1}{2} m_C u_C^2 + \frac{1}{2} m_D u_D^2 \quad \Rightarrow \\ \frac{1}{2} m_A (\bar{u}_A^2 + 2\bar{\mathbf{u}}_A \cdot \mathbf{v} + v^2) + \frac{1}{2} m_B (\bar{u}_B^2 + 2\bar{\mathbf{u}}_B \cdot \mathbf{v} + v^2) &= \frac{1}{2} m_C (\bar{u}_C^2 + 2\bar{\mathbf{u}}_C \cdot \mathbf{v} + v^2) + \frac{1}{2} m_D (\bar{u}_D^2 + 2\bar{\mathbf{u}}_D \cdot \mathbf{v} + v^2) \\ \frac{1}{2} m_A \bar{u}_A^2 + \frac{1}{2} m_B \bar{u}_B^2 + \mathbf{v} \cdot (m_A \bar{\mathbf{u}}_A + m_B \bar{\mathbf{u}}_B) + \frac{1}{2} v^2 (m_A + m_B) &= \frac{1}{2} m_C \bar{u}_C^2 + \frac{1}{2} m_D \bar{u}_D^2 + \mathbf{v} \cdot (m_C \bar{\mathbf{u}}_C + m_D \bar{\mathbf{u}}_D) + \frac{1}{2} v^2 (m_C + m_D). \end{aligned}$$

But the middle terms are equal by conservation of momentum, and the last terms are equal by conservation of mass, so $\frac{1}{2} m_A \bar{u}_A^2 + \frac{1}{2} m_B \bar{u}_B^2 = \frac{1}{2} m_C \bar{u}_C^2 + \frac{1}{2} m_D \bar{u}_D^2$. *qed*

Problem 12.3

$$\text{(a)} \quad v_G = v_{AB} + v_{BC}; \quad v_E = \frac{v_{AB} + v_{BC}}{1 + v_{AB}v_{BC}/c^2} \approx v_G \left(1 - \frac{v_{AB}v_{BC}}{c^2}\right); \quad \therefore \frac{v_G - v_E}{v_G} = \frac{v_{AB}v_{BC}}{c^2}.$$

$$\text{In mi/h, } c = (186,000 \text{ mi/s}) \times (3600 \text{ sec/hr}) = 6.7 \times 10^8 \text{ mi/hr.}$$

$$\therefore \frac{v_G - v_E}{v_G} = \frac{(5)(60)}{(6.7 \times 10^8)^2} = 6.7 \times 10^{-16}. \quad \therefore \boxed{6.7 \times 10^{-14}\% \text{ error}} \quad (\text{pretty small!})$$

$$\text{(b)} \quad \left(\frac{1}{2}c + \frac{3}{4}c\right) / \left(1 + \frac{1}{2} \cdot \frac{3}{4}\right) = \left(\frac{5}{4}c\right) / \left(\frac{11}{8}\right) = \boxed{\frac{10}{11}c} \quad (\text{still less than } c)$$

(c) To simplify notation, let $\beta = v_{AC}/c$, $\beta_1 = v_{AB}/c$, $\beta_2 = v_{BC}/c$. Then Eq. 12.3 says: $\beta = \frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}$, or:

$$\beta^2 = \frac{\beta_1^2 + 2\beta_1\beta_2 + 2\beta_1\beta_2 + \beta_2^2}{(1 + 2\beta_1\beta_2 + \beta_1^2\beta_2^2)} = \frac{1 + 2\beta_1\beta_2 + \beta_1^2\beta_2^2}{(1 + 2\beta_1\beta_2 + \beta_1^2\beta_2^2)} - \frac{(1 + \beta_1^2\beta_2^2 - \beta_1^2 - \beta_2^2)}{(1 + 2\beta_1\beta_2 + \beta_1^2\beta_2^2)} = 1 - \frac{(1 - \beta_1^2)(1 - \beta_2^2)}{(1 + \beta_1\beta_2)^2} = 1 - \Delta,$$

where $\Delta \equiv (1 - \beta_1^2)(1 - \beta_2^2)/(1 + \beta_1\beta_2)^2$ is clearly a *positive* number. So $\beta_2 < 1$, and hence $|v_{AC}| < c$. qed.

Problem 12.4

(a) Velocity of bullet relative to ground is $\frac{1}{2}c + \frac{1}{3}c = \frac{5}{6}c = \frac{10}{12}c$.

Velocity of getaway car is $\frac{3}{4}c = \frac{9}{12}c$. Since $v_b > v_g$, bullet *does* reach target.

(b) Velocity of bullet relative to ground is $\frac{\frac{1}{2}c + \frac{1}{3}c}{1 + \frac{1}{2} \cdot \frac{1}{3}} = \frac{\frac{5}{6}c}{\frac{7}{6}} = \frac{5}{7}c = \frac{20}{28}c$.

Velocity of getaway car is $\frac{3}{4}c = \frac{21}{28}c$. Since $v_g > v_b$, bullet *does not* reach target.

Problem 12.5

(a) Light from 90th clock took $\frac{90 \times 10^9 \text{ m}}{3 \times 10^8 \text{ m/s}} = 300 \text{ sec} = 5 \text{ min}$ to reach me, so the time I *see* on the clock is 11:55 am.

(b) I *observe* 12 noon.

Problem 12.6

$$\begin{cases} \text{light signal leaves } a \text{ at time } t'_a; \text{ arrives at earth at time } t_a = t'_a + \frac{d_a}{c} \\ \text{light signal leaves } b \text{ at time } t'_b; \text{ arrives at earth at time } t_b = t'_b + \frac{d_b}{c} \end{cases}$$

$$\therefore \Delta t = t_b - t_a = t'_b - t'_a + \frac{(d_b - d_a)}{c} = \Delta t' + \frac{(-v\Delta t' \cos \theta)}{c} = \Delta t' \left[1 - \frac{v}{c} \cos \theta \right]$$

(Here d_a is the distance from a to earth, and d_b is the distance from b to earth.)

$$\Delta s = v\Delta t' \sin \theta = \frac{v \sin \theta \Delta t}{(1 - \frac{v}{c} \cos \theta)} \quad \therefore \quad u = \frac{v \sin \theta}{(1 - \frac{v}{c} \cos \theta)} \quad \text{is the the apparent velocity.}$$

$$\begin{aligned} \frac{du}{d\theta} &= \frac{v \left[(1 - \frac{v}{c} \cos \theta)(\cos \theta) - \sin \theta (\frac{v}{c} \sin \theta) \right]}{(1 - \frac{v}{c} \cos \theta)^2} = 0 \Rightarrow (1 - \frac{v}{c} \cos \theta) \cos \theta = \frac{v}{c} \sin^2 \theta \\ &\Rightarrow \cos \theta = \frac{v}{c} (\sin^2 \theta + \cos^2 \theta) = \frac{v}{c} \end{aligned}$$

$$\theta_{\max} = \cos^{-1}(v/c) \quad \text{At this maximal angle, } u = \frac{v\sqrt{1-v^2/c^2}}{1-v^2/c^2} = \frac{v}{\sqrt{1-v^2/c^2}}$$

As $v \rightarrow c$, $u \rightarrow \infty$, because the denominator $\rightarrow 0$ — even though $v < c$.

Problem 12.7

The student has not taken into account time dilation of the muon's “internal clock”. In the *laboratory*, the muon lasts $\gamma\tau = \frac{\tau}{\sqrt{1-v^2/c^2}}$, where τ is the “proper” lifetime, 2×10^{-6} sec. Thus

$$v = \frac{d}{t/\sqrt{1-v^2/c^2}} = \frac{d}{\tau} \sqrt{1 - \frac{v^2}{c^2}}, \quad \text{where } d = 800 \text{ meters.}$$

$$\left(\frac{\tau}{d}\right)^2 v^2 = 1 - \frac{v^2}{c^2}; \quad v^2 \left(\left(\frac{\tau}{d}\right)^2 + \frac{1}{c^2} \right) = 1; \quad v^2 = \frac{1}{(\tau/d)^2 + (1/c^2)}.$$

$$\frac{v^2}{c^2} = \frac{1}{1 + (\tau c/d)^2}; \quad \frac{\tau c}{d} = \frac{(2 \times 10^{-6})(3 \times 10^8)}{800} = \frac{6}{8} = \frac{3}{4}; \quad \frac{v^2}{c^2} = \frac{1}{1 + 9/16} = \frac{16}{25}; \quad \boxed{v = \frac{4}{5}c}.$$

Problem 12.8

(a) Rocket clock runs slow; so earth clock reads $\gamma t = \frac{1}{\sqrt{1-v^2/c^2}} \cdot 1 \text{ hr}$. Here $\gamma = \frac{1}{\sqrt{1-v^2/c^2}} = \frac{1}{\sqrt{1-9/25}} = \frac{5}{4}$.
 $\therefore$ According to earth clocks signal was sent 1 hr, 15 min after take-off.

(b) By earth observer, rocket is now a distance $(\frac{3}{5}c)(\frac{5}{4})(1 \text{ hr}) = \frac{3}{4}c \text{ hr}$ (three-quarters of a light hour) away. Light signal will therefore take $\frac{3}{4} \text{ hr}$ to return to earth. Since it *left* 1 hr and 15 min after departure, light signal reaches earth 2 hrs after takeoff

(c) Earth clocks run slow: $t_{\text{rocket}} = \gamma \cdot (2 \text{ hrs}) = \frac{5}{4} \cdot (2 \text{ hrs}) = \text{span style="border: 1px solid black; padding: 2px;">2.5 hrs$

Problem 12.9

$$L_c = 2L_v; \frac{L_c}{\gamma_c} = \frac{L_v}{\gamma_v}; \text{ so } \frac{2}{\gamma_c} = \frac{1}{\gamma_v} = \sqrt{1 - \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4}}; \frac{1}{\gamma_c^2} = 1 - \frac{v^2}{c^2} = \frac{v^2}{c^2} = \frac{3}{16}; \frac{v^2}{c^2} = 1 - \frac{3}{16} = \frac{13}{16}; \quad v = \frac{\sqrt{13}}{4}c$$

Problem 12.10

Say length of mast (at rest) is $\bar{l}$. To an observer on the boat, height of mast is $\bar{l} \sin \bar{\theta}$, horizontal projection is $\bar{l} \cos \bar{\theta}$. To observer on dock, the former is unaffected, but the latter is Lorentz contracted to $\frac{1}{\gamma} \bar{l} \cos \bar{\theta}$. Therefore:

$$\tan \theta = \frac{\bar{l} \sin \bar{\theta}}{\frac{1}{\gamma} \bar{l} \cos \bar{\theta}} = \gamma \tan \bar{\theta}, \quad \text{or} \quad \tan \theta = \frac{\tan \bar{\theta}}{\sqrt{1 - v^2/c^2}}$$

Problem 12.11

Naively, circumference/diameter $= \frac{1}{\gamma}(2\pi R)/(2R) = \pi/\gamma = \pi\sqrt{1 - (\omega R/c)^2}$ — but this is nonsense. Point is: an accelerating object cannot remain rigid, in relativity. To decide what *actually* happens here, you need a specific model for the internal forces holding the disc together.

Problem 12.12

(iv) $\Rightarrow t = \frac{\bar{t}}{\gamma} + \frac{vx}{c^2}$. Put this into (i), and solve for x :

$$\bar{x} = \gamma x - \gamma v \left(\frac{\bar{t}}{\gamma} + \frac{vx}{c^2} \right) = \gamma x \left(1 - \frac{v^2}{c^2} \right) - v\bar{t} = \gamma x \frac{1}{\gamma^2} - v\bar{t} = \frac{x}{\gamma} - v\bar{t}; \quad \boxed{x = \gamma(\bar{x} + v\bar{t})} \quad \checkmark$$

Similarly, (i) $\Rightarrow x = \frac{\bar{x}}{\gamma} + vt$. Put this into (iv) and solve for t :

$$\bar{t} = \gamma t - \frac{\gamma v}{c^2} \left(\frac{\bar{x}}{\gamma} + vt \right) = \gamma t \left(1 - \frac{v^2}{c^2} \right) - \frac{v}{c^2} \bar{x} \frac{t}{\gamma} - \frac{v}{c^2} \bar{x}; \quad \boxed{t = \gamma \left(\bar{t} + \frac{v}{c^2} \bar{x} \right)} \quad \checkmark$$

Problem 12.13

Let brother's accident occur at origin, time zero, in both frames. In system $\mathcal{S}$ (Sophie's), the coordinates of Sophie's cry are $x = 5 \times 10^5 \text{ m}$, $t = 0$. In system $\bar{\mathcal{S}}$ (scientist's), $\bar{t} = \gamma(t - \frac{v}{c^2}x) = -\gamma vx/c^2$. Since this is *negative*, Sophie's cry occurred before the accident, in $\bar{\mathcal{S}}$. $\gamma = \frac{1}{\sqrt{1-(12/13)^2}} = \frac{13}{\sqrt{169-144}} = \frac{13}{5}$. So

$$\bar{t} = -\left(\frac{13}{5}\right)\left(\frac{12}{13}c\right)(5 \times 10^5)/c^2 = -12 \times 10^5/3 \times 10^8 = 10^{-3}. \quad \boxed{4 \times 10^{-3} \text{ seconds earlier}}$$

Problem 12.14

(a) In $\mathcal{S}$ it moves a distance dy in time dt . In $\bar{\mathcal{S}}$, meanwhile, it moves a distance $d\bar{y} = dy$ in time $d\bar{t} = \gamma(dt - \frac{v}{c^2}dx)$.

$$\therefore \frac{d\bar{y}}{d\bar{t}} = \frac{dy}{\gamma(dt - \frac{v}{c^2}dx)} = \frac{(dy/dt)}{\gamma(1 - \frac{v}{c^2} \frac{dx}{dt})}; \quad \text{or} \quad \boxed{\bar{u}_y = \frac{u_y}{\gamma(1 - \frac{vu_x}{c^2})}; \quad \bar{u}_z = \frac{u_z}{\gamma(1 - \frac{vu_x}{c^2})}}$$

(b) $\mathcal{S}$ = dock frame; $\mathcal{S}'$ = boat frame; we need reverse transformations ($v \rightarrow -v$):

$$\tan \theta = -\frac{u_y}{u_x} = -\frac{\bar{u}_y/\gamma \left(1 + \frac{v\bar{u}_x}{c^2}\right)}{(\bar{u}_x + v)/(1 + \frac{v\bar{u}_x}{c^2})} = -\frac{1}{\gamma} \frac{\bar{u}_y}{(\bar{u}_x + v)}. \text{ In this case } \bar{u}_x = -c \cos \bar{\theta}; \bar{u}_y = c \sin \bar{\theta}, \text{ so}$$

$$\tan \theta = -\frac{1}{\gamma} \frac{c \sin \bar{\theta}}{(-c \cos \bar{\theta} + v)} = \frac{1}{\gamma} \left(\frac{\sin \bar{\theta}}{\cos \bar{\theta} - v/c} \right)$$

[Contrast $\tan \theta = \gamma \frac{\sin \bar{\theta}}{\cos \bar{\theta}}$ in Prob. 12.10. The point is that *velocities* are sensitive not only to the transformation of *distances*, but also of *times*. That's why there is no *universal* rule for translating angles — you have to know whether it's an angle made by a *velocity* vector or a *position* vector.]

That's how the velocity vector of an *individual photon* transforms. But the beam as a whole is a snapshot of many *different* photons at one instant of time, and *it* transforms the *same* way the mast does.

Problem 12.15

Bullet relative to ground: $\frac{5}{7}c$. Outlaws relative to police: $\frac{\frac{3}{4}c - \frac{1}{2}c}{1 - \frac{3}{4} \cdot \frac{1}{2}} = \frac{\frac{1}{4}c}{\frac{5}{8}} = \frac{2}{5}c$.

Bullet relative to outlaws: $\frac{\frac{5}{7}c - \frac{3}{4}c}{1 - \frac{5}{7} \cdot \frac{3}{4}} = \frac{-\frac{1}{28}c}{\frac{13}{28}} = -\frac{1}{13}c$. [Velocity of A relative to B is minus the velocity of B relative to A , so all entries below the diagonal are trivial. Note that in every case $v_{\text{bullet}} < v_{\text{outlaws}}$, so no matter how you look at it, the bad guys get away.]

speed of $\rightarrow$	Ground	Police	Outlaws	Bullet	Do they escape?
Ground $\downarrow$	0	$\frac{1}{2}c$	$\frac{3}{4}c$	$\frac{5}{7}c$	Yes
Police	$-\frac{1}{2}$	0	$\frac{2}{5}c$	$\frac{1}{3}c$	Yes
Outlaws	$-\frac{3}{4}c$	$-\frac{2}{5}c$	0	$-\frac{1}{13}c$	Yes
Bullet	$-\frac{5}{7}c$	$-\frac{1}{3}c$	$\frac{1}{13}c$	0	Yes

Problem 12.16

(a) Moving clock runs slow, by a factor $\gamma = \frac{1}{\sqrt{1-(4/5)^2}} = \frac{5}{3}$. Since 18 years elapsed on the moving clock, $\frac{5}{3} \times 18 = 30$ years elapsed on the stationary clock. 51 years old

(b) By earth clock, it took 15 years to get there, at $\frac{4}{5}c$, so $d = \frac{4}{5}c \times 15 \text{ yrs} = \text{span style="border: 1px solid black; padding: 2px;">12c years (12 light years)$

(c) $t = 15 \text{ yrs}$, $x = 12c \text{ yrs}$

(d) $\bar{t} = 9 \text{ yrs}$, $\bar{x} = 0$. [She got *on* at the origin in $\bar{\mathcal{S}}$, and rode along on $\bar{\mathcal{S}}$, so she's *still* at the origin. If you doubt these values, use the Lorentz Transformations, with x and t in (c).]

(e) Lorentz Transformations: $\left\{ \begin{array}{l} \tilde{x} = \gamma(x + vt) \\ \tilde{t} = \gamma(t + \frac{v}{c^2}x) \end{array} \right\}$ [note that v is negative, since $\tilde{\mathcal{S}}$ is going to the *left*]

$$\therefore \tilde{x} = \frac{5}{3}(12c \text{ yrs} + \frac{4}{5}c \cdot 15 \text{ yrs}) = \frac{5}{3} \cdot 24c \text{ yrs} = \text{span style="border: 1px solid black; padding: 2px;">40c years.$$

$$\tilde{t} = \frac{5}{3}(15 \text{ yrs} + \frac{4}{5} \frac{c}{c^2} \cdot 12c \text{ yrs}) = \frac{5}{3}(15 + \frac{48}{5}) \text{ yrs} = (25 + 16) \text{ yrs} = \text{span style="border: 1px solid black; padding: 2px;">41 years.$$

(f) Set her clock ahead 32 years, from 9 to 41 ($\bar{t} \rightarrow \tilde{t}$). Return trip takes 9 years (moving time), so her clock will now read 50 years at her arrival. Note that this is $\frac{5}{3} \cdot 30$ years—precisely what she would calculate if the *stay-at-home* had been the traveler, for 30 years of his own time.

(g) (i) $\bar{t} = 9$ yrs, $x = 0$. What is t ? $t = \frac{v}{c^2}x + \frac{\bar{t}}{\gamma} = \frac{3}{5} \cdot 9 \text{ yrs} = \frac{27}{5} = 5.4$ years, and he started at age 21, so he's 26.4 years old (*Younger* than traveler (!) because to the traveller it's the stay-at-home who's moving.)

(ii) $\tilde{t} = 41$ yrs, $x = 0$. What is t ? $t = \frac{\tilde{t}}{\gamma} = \frac{3}{5} \cdot 41 \text{ yrs}$, or $\frac{123}{5}$ yrs, or 24.6 yrs, and he started at 21, so he's 45.6 years old.

(h) It will take another 5.4 years of earth time for the return, so when she gets back, she will say her twin's age is $45.6 + 5.4 = 51$ years—which is what we found in (a). But note that to make it work from traveler's point of view you *must* take into account the jump in *perceived* age of the stay-at-home when she changes coordinates from $\bar{S}$ to $\tilde{S}$.)

Problem 12.17

$$\begin{aligned}
 -\bar{a}^0\bar{b}^0 + \bar{a}^1\bar{b}^1 + \bar{a}^2\bar{b}^2 + \bar{a}^3\bar{b}^3 &= -\gamma^2(a^0 - \beta a^1)(b^0 - \beta b^1) + \gamma^2(a^1 - \beta a^0)(b^1 - \beta a^0) + a^2b^2 + a^3b^3 \\
 &= -\gamma^2(a^0b^0 - \cancel{\beta a^0b^1} - \cancel{\beta a^1b^0} + \beta^2 a^1b^1 - a^1b^1 + \cancel{\beta a^1b^0} + \cancel{\beta a^0b^1} - \beta^2 a^0b^0) + a^2b^2 + a^3b^3 \\
 &= -\gamma^2 a^0b^0(1 - \beta^2) + \gamma^2 a^1b^1(1 - \beta^2) + a^2b^2 + a^3b^3 \\
 &= -a^0b^0 + a^1b^1 + a^2b^2 + a^3b^3. \text{ qed [Note: } \gamma^2(1 - \beta^2) = 1.]
 \end{aligned}$$

Problem 12.18

(a)

$$\begin{pmatrix} c\bar{t} \\ \bar{x} \\ \bar{y} \\ \bar{z} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -\beta & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$
 (using the notation of Eq. 12.24, for best comparison)

(b)

$$\Lambda = \begin{pmatrix} \gamma & 0 & -\gamma\beta & 0 \\ 0 & 1 & 0 & 0 \\ -\gamma\beta & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(c) Multiply the matrices: $\Lambda = \begin{pmatrix} \bar{\gamma} & 0 & -\bar{\gamma}\bar{\beta} & 0 \\ 0 & 1 & 0 & 0 \\ -\bar{\gamma}\bar{\beta} & 0 & \bar{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \gamma\bar{\gamma} & -\gamma\bar{\gamma}\beta & -\bar{\gamma}\bar{\beta} & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ -\bar{\gamma}\gamma\bar{\beta} & \gamma\bar{\gamma}\beta\bar{\beta} & \bar{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

Yes, the order *does* matter. In the other order “bars” and “no-bars” would be switched, and this would yield a *different matrix*.

Problem 12.19

(a) Since $\tanh \theta = \frac{\sinh \theta}{\cosh \theta}$, and $\cosh^2 \theta - \sinh^2 \theta = 1$, we have:

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} = \frac{1}{\sqrt{1 - \tanh^2 \theta}} = \frac{\cosh \theta}{\sqrt{\cosh^2 \theta - \sinh^2 \theta}} = \cosh \theta; \quad \gamma\beta = \cosh \theta \tanh \theta = \sinh \theta.$$

$\therefore \Lambda = \begin{pmatrix} \cosh \theta & -\sinh \theta & 0 & 0 \\ -\sinh \theta & \cosh \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ Compare: $R = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(b) $\bar{u} = \frac{u-v}{1-\frac{uv}{c^2}} \Rightarrow \frac{\bar{u}}{c} = \frac{\left(\frac{u}{c}\right) - \left(\frac{v}{c}\right)}{1 - \left(\frac{u}{c}\right)\left(\frac{v}{c}\right)} \Rightarrow \tanh \bar{\phi} = \frac{\tanh \phi - \tanh \theta}{1 - \tanh \phi \tanh \theta}$, where $\tanh \phi = u/c$, $\tanh \theta = v/c$; $\tanh \bar{\phi} = \bar{u}/c$. But a “trig” formula for hyperbolic functions (CRC Handbook, 18th Ed., p. 204) says:

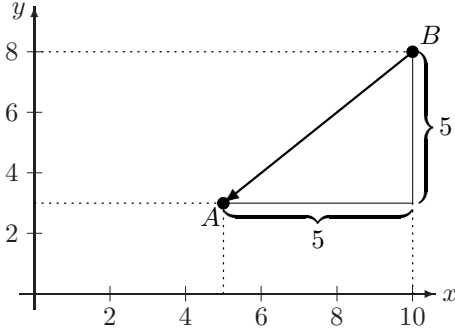
$$\frac{\tanh \phi - \tanh \theta}{1 - \tanh \phi \tanh \theta} = \tanh(\phi - \theta). \quad \therefore \tanh \bar{\phi} = \tanh(\phi - \theta), \text{ or: } \boxed{\bar{\phi} = \phi - \theta}$$

Problem 12.20

(a) (i) $I = -c^2 \Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2 = -(5-15)^2 + (10-5)^2 + (8-3)^2 + (0-0)^2 = -100 + 25 + 25 = \boxed{-50}$

(ii) No. (In such a system $\Delta \bar{t} = 0$, so I would have to be *positive*, which it *isn't*.)

(iii) Yes.



$\bar{S}$ travels in the direction from B toward A , making the trip in time $10/c$.

$$\therefore \mathbf{v} = \frac{-5\hat{\mathbf{x}} - 5\hat{\mathbf{y}}}{10/c} = \boxed{-\frac{c}{2}\hat{\mathbf{x}} - \frac{c}{2}\hat{\mathbf{y}}}$$

Note that $\frac{v^2}{c^2} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$, so $v = \frac{1}{\sqrt{2}}c$, safely less than c .

(b) (i) $I = -(3-1)^2 + (5-2)^2 + 0 + 0 = -4 + 9 = \boxed{5}$

(ii) Yes. By Lorentz Transformation: $\Delta(ct) = \gamma(\Delta(ct) - \beta(\Delta x))$. We want $\Delta \bar{t} = 0$, so $\Delta(ct) = \beta(\Delta x)$; or

$$\frac{v}{c} = \frac{\Delta(ct)}{(\Delta x)} = \frac{(3-1)}{(5-2)} = \frac{2}{3}. \text{ So } \boxed{v = \frac{2}{3}c} \text{ in the } +x \text{ direction.}$$

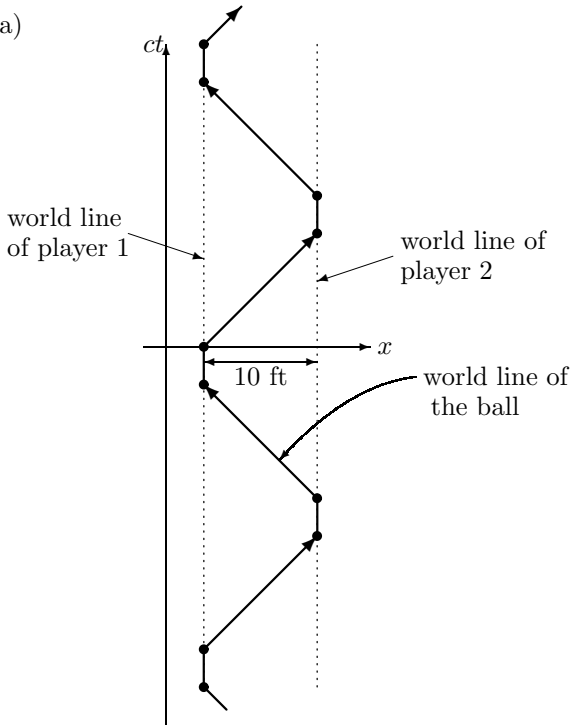
(iii) No. (In such a system $\Delta x = \Delta y = \Delta z = 0$ so I would be negative, which it *isn't*.)

Problem 12.21

Using Eq. 12.18 (iv): $\Delta \bar{t} = \gamma(\Delta t - \frac{v}{c^2} \Delta x) = 0 \Rightarrow \Delta t = \frac{v}{c^2} \Delta x$, or $v = \frac{\Delta t}{\Delta x} c^2 = \boxed{\frac{t_B - t_A}{x_B - x_A} c^2}$

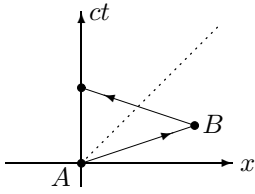
Problem 12.22

(a)



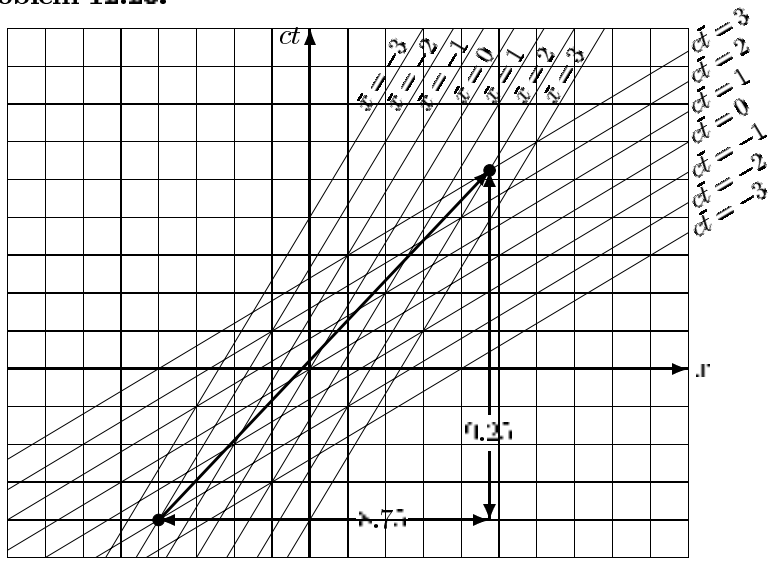
Truth is, you never *do* communicate with the other person *right now*—you communicate with the person he/she *will be* when the message gets there; and the response comes back to an older and wiser you.

(b) No way It is true that a moving observer might say she arrived at B before she left A , but for the *round trip* everyone must agree that she arrives back after she set out.



Problem 12.23.

(a)



(b) $\frac{c}{v} = \text{slope} = \frac{9.25}{8.75}$

$\Rightarrow v = \frac{8.75}{9.25}c = \frac{35}{37}c$

(c) $v' = \frac{4}{5}c$, so $v = \frac{\frac{4}{5}c + \frac{3}{5}c}{1 + \frac{4}{5} \cdot \frac{3}{5}}$

$= \frac{(7/5)c}{(37/25)} = \frac{35}{37}c$

Problem 12.24

(a) $(1 - \frac{u^2}{c^2})\eta^2 = u^2; u^2(1 + \frac{\eta^2}{c^2}) = \eta^2; \mathbf{u} = \frac{1}{\sqrt{1 + \eta^2/c^2}}\boldsymbol{\eta}.$

$$(b) \frac{1}{\sqrt{1-u^2/c^2}} = \frac{1}{\sqrt{1-\tanh^2 \theta}} = \frac{\cosh \theta}{\sqrt{\cosh^2 \theta - \sinh^2 \theta}} = \cosh \theta \quad \therefore \eta = \frac{1}{\sqrt{1-u^2/c^2}} u = \cosh \theta c \tanh \theta = \boxed{c \sinh \theta}.$$

Problem 12.25

$$(a) u_x = u_y = u \cos 45^\circ = \frac{1}{\sqrt{2}} \frac{2}{\sqrt{5}} c = \boxed{\sqrt{\frac{2}{5}} c}.$$

$$(b) \frac{1}{\sqrt{1-u^2/c^2}} = \frac{1}{\sqrt{1-4/5}} = \frac{\sqrt{5}}{\sqrt{5-4}} = \sqrt{5}. \quad \therefore \boldsymbol{\eta} = \frac{\mathbf{u}}{\sqrt{1-u^2/c^2}} \Rightarrow \boxed{\eta_x = \eta_y = \sqrt{2}}$$

$$(c) \eta_0 = \gamma c = \boxed{\sqrt{5} c}.$$

$$(d) \text{ Eq. 12.45} \Rightarrow \begin{cases} \bar{u}_x = \frac{u_x - v}{1 - \frac{u_x v}{c^2}} = \frac{\frac{\sqrt{2}}{5} c - \frac{\sqrt{2}}{5} c}{1 - \frac{2}{5}} = \boxed{0} \\ \bar{u}_y = \frac{1}{\gamma} \left(\frac{u_y}{1 - \frac{u_x v}{c^2}} \right) = \sqrt{1 - \frac{2}{5}} \frac{\frac{\sqrt{2}}{5} c}{1 - \frac{2}{5}} = \frac{\sqrt{2/5}}{\sqrt{3/5}} c = \boxed{\sqrt{\frac{2}{3}} c}. \end{cases}$$

$$(e) \bar{\eta}_x = \gamma(\eta_x - \beta \eta^0) = \sqrt{1 - \frac{2}{5}} (\sqrt{2} c - \sqrt{\frac{2}{5}} \sqrt{5} c) = \boxed{0}. \quad \boxed{\bar{\eta}_y = \eta_y = \sqrt{2} c}.$$

$$(f) \frac{1}{\sqrt{1-\bar{u}^2/c^2}} = \frac{1}{\sqrt{1-2/3}} = \sqrt{3}; \quad \therefore \bar{\boldsymbol{\eta}} = \sqrt{3} \bar{\mathbf{u}} \Rightarrow \left\{ \begin{array}{l} \bar{\eta}_x = \sqrt{3} \bar{u}_x = 0 \quad \checkmark \\ \bar{\eta}_y = \sqrt{3} \bar{u}_y = \sqrt{2} c \quad \checkmark \end{array} \right\}$$

Problem 12.26

$$\eta^\mu \eta_\mu = -(\eta^0)^2 + \eta^2 = \frac{1}{(1-u^2/c^2)} (-c^2 + u^2) = -c^2 \frac{(1-u^2/c^2)}{(1-u^2/c^2)} = \boxed{-c^2}. \quad \boxed{\text{Timelike.}}$$

Problem 12.27

Use the result of Problem 12.24(a): $u = \frac{1}{\sqrt{1+\eta^2/c^2}} \eta$. Here $\frac{\eta}{c} = \frac{4}{3}$, so $\frac{1}{\sqrt{1+16/9}} = \frac{3}{5}$, and hence $u = \frac{3}{5} (4 \times 10^8) = 2.4 \times 10^8 \text{ m/s}$. Innocent.

Problem 12.28

$$(a) \text{ From Prob. 11.34 we have } \gamma = \frac{1}{b} \sqrt{b^2 + c^2 t^2}. \quad \therefore \tau = \int \frac{1}{\gamma} dt = b \int \frac{dt}{\sqrt{b^2 + c^2 t^2}} = \frac{b}{c} \ln(ct + \sqrt{b^2 + c^2 t^2}) + k; \text{ at } t = 0 \text{ we want } \tau = 0: 0 = \frac{b}{c} \ln b + k, \text{ so } k = -\frac{b}{c} \ln b; \quad \boxed{\tau = \frac{b}{c} \ln \left[\frac{1}{b} (ct + \sqrt{b^2 + c^2 t^2}) \right]}$$

$$(b) \sqrt{x^2 - b^2} + x = b e^{c\tau/b}; \quad \sqrt{x^2 - b^2} = b e^{c\tau/b} - x; \quad x^2 - b^2 = b^2 e^{2c\tau/b} - 2x b e^{c\tau/b} + x^2; \quad 2x b e^{c\tau/b} = b^2 (1 + e^{2c\tau/b}); \quad x = b \left(\frac{e^{c\tau/b} + e^{-c\tau/b}}{2} \right) = \boxed{b \cosh(c\tau/b)}. \text{ Also from Prob. 11.34: } v = c^2 t / \sqrt{b^2 + c^2 t^2}.$$

$$v = \frac{c}{x} \sqrt{x^2 - b^2} = \frac{c}{b \cosh(c\tau/b)} \sqrt{b^2 \cosh^2(c\tau/b) - b^2} = c \frac{\sqrt{\cosh^2(c\tau/b) - 1}}{\cosh(c\tau/b)} = c \frac{\sinh(c\tau/b)}{\cosh(c\tau/b)} = \boxed{c \tanh \left(\frac{c\tau}{b} \right)}.$$

$$(c) \eta^\mu = \gamma(c, v, 0, 0). \quad \gamma = \frac{x}{b} = \cosh \frac{c\tau}{b}, \text{ so } \eta^\mu = \cosh \frac{c\tau}{b} \left(c, c \tanh \frac{c\tau}{b}, 0, 0 \right) = \boxed{c \left(\cosh \frac{c\tau}{b}, \sinh \frac{c\tau}{b}, 0, 0 \right)}.$$

Problem 12.29

$$(a) m_A u_A + m_B u_B = m_C u_C + m_D u_D; \quad u_i = \frac{\bar{u}_i + v}{1 + (\bar{u}_i v / c^2)}.$$

$$m_A \frac{\bar{u}_A + v}{1 + (\bar{u}_A v / c^2)} + m_B \frac{\bar{u}_B + v}{1 + (\bar{u}_B v / c^2)} = m_C \frac{\bar{u}_C + v}{1 + (\bar{u}_C v / c^2)} + m_D \frac{\bar{u}_D + v}{1 + (\bar{u}_D v / c^2)}.$$

This time, because the denominators are all different, we *cannot* conclude that

$$m_A \bar{u}_A + m_B \bar{u}_B = m_C \bar{u}_C + m_D \bar{u}_D.$$

As an explicit counterexample, suppose all the masses are equal, and $u_A = -u_B = v$, $u_C = u_D = 0$. This is a symmetric “completely inelastic” collision in $\mathcal{S}$, and momentum is clearly conserved ($0=0$). But the Einstein

velocity addition rule gives $\bar{u}_A = 0$, $\bar{u}_B = -2v/(1 + v^2/c^2)$, $\bar{u}_C = \bar{u}_D = -v$, so in $\bar{S}$ the (incorrectly defined) momentum is *not* conserved:

$$m \left(\frac{-2v}{1 + v^2/c^2} \right) \neq -2mv.$$

(b) $m_A \eta_A + m_B \eta_B = m_C \eta_C + m_D \eta_D$; $\eta_i = \gamma(\bar{\eta}_i + \beta \bar{\eta}_i^0)$. (The inverse Lorentz transformation.)
 $m_A \gamma(\bar{\eta}_A + \beta \bar{\eta}_A^0) + m_B \gamma(\bar{\eta}_B + \beta \bar{\eta}_B^0) = m_C \gamma(\bar{\eta}_C + \beta \bar{\eta}_C^0) + m_D \gamma(\bar{\eta}_D + \beta \bar{\eta}_D^0)$. The gamma's cancel:
 $m_A \bar{\eta}_A + m_B \bar{\eta}_B + \beta(m_A \bar{\eta}_A^0 + m_B \bar{\eta}_B^0) = m_C \bar{\eta}_C + m_D \bar{\eta}_D + \beta(m_C \bar{\eta}_C^0 + m_D \bar{\eta}_D^0)$.

But $m_i \eta_i^0 = p_i^0 = E_i/c$, so if energy is conserved in $\bar{S}$ ($\bar{E}_A + \bar{E}_B = \bar{E}_C + \bar{E}_D$), then so too is the momentum (correctly defined):

$$m_A \bar{\eta}_A + m_B \bar{\eta}_B = m_C \bar{\eta}_C + m_D \bar{\eta}_D. \quad \text{qed}$$

Problem 12.30

$$\gamma mc^2 - mc^2 = nmc^2 \Rightarrow \gamma = n + 1 = \frac{1}{1 - \sqrt{u^2/c^2}} \Rightarrow 1 - \frac{u^2}{c^2} = \frac{1}{(n+1)^2}$$

$$\therefore \frac{u^2}{c^2} = 1 - \frac{1}{(n+1)^2} = \frac{n^2 + 2n + 1 - 1}{(n+1)^2} = \frac{n(n+2)}{(n+1)^2}; \quad u = \frac{\sqrt{n(n+2)}}{n+1} c$$

Problem 12.31

$$E_T = E_1 + E_2 + \cdots; p_T = p_1 + p_2 + \cdots; \bar{p}_T = \gamma(p_T - \beta E_T/c) = 0 \Rightarrow \beta = v/c = p_T c / E_T$$

$$v = c^2 p_T / E_T = \boxed{c^2 (p_1 + p_2 + \cdots) / (E_1 + E_2 + \cdots)}$$

Problem 12.32

$$E_\mu = \frac{(m_\pi^2 + m_\mu^2)}{2m_\pi} c^2 = \gamma m_\mu c^2 \Rightarrow \gamma = \frac{(m_\pi^2 + m_\mu^2)}{2m_\pi m_\mu} = \frac{1}{\sqrt{1 - v^2/c^2}}; \quad 1 - \frac{v^2}{c^2} = \frac{1}{\gamma^2};$$

$$\frac{v^2}{c^2} = 1 - \frac{1}{\gamma^2} = 1 - \frac{4m_\pi^2 m_\mu^2}{(m_\pi^2 + m_\mu^2)^2} = \frac{m_\pi^4 + 2m_\pi^2 m_\mu^2 + m_\mu^4 - 4m_\pi^2 m_\mu^2}{(m_\pi^2 + m_\mu^2)^2} = \frac{(m_\pi^2 - m_\mu^2)^2}{(m_\pi^2 + m_\mu^2)^2}; v = \boxed{\left(\frac{(m_\pi^2 - m_\mu^2)}{(m_\pi^2 + m_\mu^2)} \right) c}.$$

Problem 12.33

$$\text{Initial momentum: } E^2 - p^2 c^2 = m^2 c^4 \Rightarrow p^2 c^2 = (2mc^2)^2 - m^2 c^4 = 3m^2 c^4 \Rightarrow p = \sqrt{3} mc.$$

$$\text{Initial energy: } 2mc^2 + mc^2 = 3mc^2.$$

Each is conserved, so final energy is $3mc^2$, final momentum is $\sqrt{3} mc$.

$$E^2 - p^2 c^2 = (3mc^2)^2 - (\sqrt{3} mc)^2 c^2 = 6m^2 c^4 = M^2 c^4. \quad \therefore \boxed{M = \sqrt{6} m} \approx 2.5m$$

(In this process some kinetic energy was converted into rest energy, so $M > 2m$.)

$$v = \frac{pc^2}{E} = \frac{\sqrt{3} mc c^2}{3mc^2} = \boxed{\frac{c}{\sqrt{3}} = v}.$$

Problem 12.34

$$\text{First calculate pion's energy: } E^2 = p^2 c^2 + m^2 c^4 = \frac{9}{16} m^2 c^4 + m^2 c^4 = \frac{25}{16} m^2 c^4 \Rightarrow E = \frac{5}{4} mc^2.$$

$$\left. \begin{array}{l} \text{Conservation of energy: } \frac{5}{4} mc^2 = E_A + E_B \\ \text{Conservation of momentum: } \frac{3}{4} mc = p_A + p_B = \frac{E_A}{c} - \frac{E_B}{c} \Rightarrow \frac{3}{4} mc^2 = E_A - E_B \end{array} \right\} 2E_A = 2mc^2$$

$$\Rightarrow \boxed{E_A = mc^2}; \quad \boxed{E_B = \frac{1}{4} mc^2}.$$

Problem 12.35

Classically, $E = \frac{1}{2}mv^2$. In a colliding beam experiment, the relative velocity (classically) is *twice* the velocity of either one, so the relative energy is $4E$.



Let $\bar{S}$ be the system in which ① is at rest. Its speed v , relative to S , is just the speed of ① in S .

$$\bar{p}^0 = \gamma(p^0 - \beta p^1) \Rightarrow \frac{\bar{E}}{c} = \gamma \left(\frac{E}{c} - \beta p \right), \text{ where } p \text{ is the momentum of ② in } S.$$

$$E = \gamma M c^2, \text{ so } \gamma = \frac{E}{M c^2}; p = -\gamma M V = -\gamma M \beta c. \therefore \bar{E} = \gamma \left(\frac{E}{c} + \beta \gamma M \beta c \right) c = \gamma(E + \gamma M c^2 \beta^2)$$

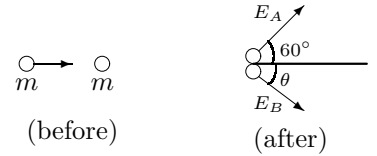
$$\gamma^2 = \frac{1}{1-\beta^2} \Rightarrow 1 - \beta^2 = \frac{1}{\gamma^2} \Rightarrow \beta^2 = 1 - \frac{1}{\gamma^2} = \frac{\gamma^2 - 1}{\gamma^2}. \therefore \bar{E} = \frac{E}{M c^2} E + \left[\left(\frac{E}{M c^2} \right)^2 - 1 \right] M c^2$$

$$\bar{E} = \frac{E^2}{M c^2} + \frac{E^2}{M c^2} - M c^2; \quad \boxed{\bar{E} = \frac{2E^2}{M c^2} - M c^2.}$$

$$\text{For } E = 30 \text{ GeV and } M c^2 = 1 \text{ GeV, we have } \bar{E} = \frac{(2)(900)}{1} - 1 = 1800 - 1 = \boxed{1799 \text{ GeV}} = \boxed{60E}.$$

Problem 12.36

One photon is impossible, because in the “center of momentum” frame (Prob. 12.31) we’d be left with a photon *at rest*, whereas photons *have* to travel at speed c .



$$\left\{ \begin{array}{l} \text{Cons. of energy: } \sqrt{p_0^2 c^2 + m^2 c^4} + m c^2 = E_A + E_B \\ \text{Cons. of mom.: } \left\{ \begin{array}{l} \text{horizontal: } p_0 = \frac{E_A}{c} \cos 60^\circ + \frac{E_B}{c} \cos \theta \Rightarrow E_B \cos \theta = p_0 c - \frac{1}{2} E_A \\ \text{vertical: } 0 = \frac{E_A}{c} \sin 60^\circ - \frac{E_B}{c} \sin \theta \Rightarrow E_B \sin \theta = \frac{\sqrt{3}}{2} E_A \end{array} \right\} \text{square and add:} \end{array} \right.$$

$$E_B^2 (\cos^2 \theta + \sin^2 \theta) = p_0^2 c^2 - p_0 c E_A + \frac{1}{4} E_A^2 + \frac{3}{4} E_A^2$$

$$\Rightarrow E_B^2 = p_0^2 c^2 - p_0 c E_A + E_A^2 = \left[\sqrt{p_0^2 c^2 + m^2 c^4} + m c^2 - E_A \right]^2$$

$$= p_0^2 c^2 + m^2 c^4 + 2 \sqrt{p_0^2 c^2 + m^2 c^4} (m c^2 - E_A) + m^2 c^4 - 2 E_A m c^2 + E_A^2. \quad \text{Or:}$$

$$-p_0 c E_A = 2 m^2 c^4 + 2 m c^2 \sqrt{p_0^2 c^2 + m^2 c^4} - 2 E_A \sqrt{p_0^2 c^2 + m^2 c^4} - 2 E_A m c^2;$$

$$\Rightarrow E_A (m c^2 + \sqrt{p_0^2 c^2 + m^2 c^4} - p_0 c / 2) = m^2 c^4 + m c^2 \sqrt{p_0^2 c^2 + m^2 c^4};$$

$$E_A = m c^2 \frac{(m c^2 + \sqrt{p_0^2 c^2 + m^2 c^4})}{(m c^2 + \sqrt{p_0^2 c^2 + m^2 c^4} - p_0 c / 2)} \cdot \frac{(m c^2 - \sqrt{p_0^2 c^2 + m^2 c^4} - p_0 c / 2)}{(m c^2 - \sqrt{p_0^2 c^2 + m^2 c^4} - p_0 c / 2)}$$

$$= m c^2 \frac{(\cancel{m^2 c^4} - p_0^2 c^2 - \cancel{m^2 c^4} - \frac{1}{2} p_0 m c^3 - \frac{p_0 c}{2} \sqrt{p_0^2 c^2 + m^2 c^4})}{(\cancel{m^2 c^4} - p_0 m c^3 + \frac{p_0 c^2}{4} - p_0 c^2 - \cancel{m^2 c^4})} = \boxed{\frac{m c^2}{2} \frac{(m c + 2 p_0 + \sqrt{p_0^2 + m^2 c^2})}{(m c + \frac{3}{4} p_0)}}.$$

Problem 12.40

$K_\mu K^\mu = -(K^0)^2 + \mathbf{K} \cdot \mathbf{K}$. From Eq. 12.69, $\mathbf{K} \cdot \mathbf{K} = \frac{F^2}{(1-u^2/c^2)}$. From Eq. 12.70:

$$K^0 = \frac{1}{c} \frac{dE}{d\tau} = \frac{1}{c\sqrt{1-u^2/c^2}} \frac{d}{dt} \left(\frac{mc^2}{\sqrt{1-u^2/c^2}} \right) = \frac{mc}{\sqrt{1-u^2/c^2}} \left[-\frac{1}{2} \frac{(-1/c^2)}{(1-u^2/c^2)^{3/2}} 2\mathbf{u} \cdot \mathbf{a} \right] = \frac{m}{c} \frac{(\mathbf{u} \cdot \mathbf{a})}{(1-u^2/c^2)^2}$$

But Eq. 12.74: $\mathbf{u} \cdot \mathbf{F} = uF \cos \theta = \frac{m}{\sqrt{1-u^2/c^2}} \left[(\mathbf{u} \cdot \mathbf{a}) + \frac{u^2(\mathbf{u} \cdot \mathbf{a})}{c^2(1-u^2/c^2)} \right] = \frac{m(\mathbf{u} \cdot \mathbf{a})}{(1-u^2/c^2)^{3/2}}$, so:

$$K^0 = \frac{uF \cos \theta}{c\sqrt{1-u^2/c^2}}. \quad \therefore K_\mu K^\mu = \frac{F^2}{(1-u^2/c^2)} - \frac{u^2 F^2 \cos^2 \theta}{c^2(1-u^2/c^2)} = \left[\frac{1 - (u^2/c^2) \cos^2 \theta}{(1-u^2/c^2)} \right] F^2. \quad \text{qed}$$

Problem 12.41

$$\mathbf{F} = \frac{m}{\sqrt{1-u^2/c^2}} \left[\mathbf{a} + \frac{\mathbf{u}(\mathbf{u} \cdot \mathbf{a})}{c^2 - u^2} \right] = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}) \Rightarrow \mathbf{a} + \frac{\mathbf{u}(\mathbf{u} \cdot \mathbf{a})}{(c^2 - u^2)} = \frac{q}{m} \sqrt{1-u^2/c^2} (\mathbf{E} + \mathbf{u} \times \mathbf{B}).$$

$$\text{Dot in } \mathbf{u}: (\mathbf{u} \cdot \mathbf{a}) + \frac{u^2(\mathbf{u} \cdot \mathbf{a})}{c^2(1-u^2/c^2)} = \frac{\mathbf{u} \cdot \mathbf{a}}{(1-u^2/c^2)} = \frac{q}{m} \sqrt{1-u^2/c^2} [\mathbf{u} \cdot \mathbf{E} + \underbrace{\mathbf{u} \cdot (\mathbf{u} \times \mathbf{B})}_{=0}];$$

$$\therefore \frac{\mathbf{u}(\mathbf{u} \cdot \mathbf{a})}{(c^2 - u^2)} = \frac{q}{m} \sqrt{1 - \frac{u^2}{c^2}} \frac{\mathbf{u}(\mathbf{u} \cdot \mathbf{E})}{c^2}. \text{ So } \mathbf{a} = \frac{q}{m} \sqrt{1 - \frac{u^2}{c^2}} (\mathbf{E} + \mathbf{u} \times \mathbf{B} - \frac{1}{c^2} \mathbf{u}(\mathbf{u} \cdot \mathbf{E})). \quad \text{qed}$$

Problem 12.42

One way to see it is to look back at the general formula for $\mathbf{E}$ (Eq. 10.36). For a uniform infinite plane of charge, moving at constant velocity in the plane, $\dot{\mathbf{J}} = 0$ and $\dot{\rho} = 0$, while ρ (or rather, σ) is independent of t (so retardation does nothing). Therefore the field is exactly the same as it would be for a plane at rest (except that σ itself is altered by Lorentz contraction).

A more elegant argument exploits the fact that $\mathbf{E}$ is a *vector* (whereas $\mathbf{B}$ is a *pseudovector*). This means that any given component changes sign if the configuration is reflected in a plane perpendicular to that direction. But in Fig. 12.35(b), if we reflect in the xy plane the configuration is unaltered, so the z component of $\mathbf{E}$ would have to stay the *same*. Therefore it must in fact be zero. (By contrast, if you reflect in a plane perpendicular to the y direction the charges trade places, so it is perfectly appropriate that the y component of $\mathbf{E}$ should reverse its sign.)

Problem 12.43

(a) Field is σ_0/ϵ_0 , and it points perpendicular to the positive plate, so:

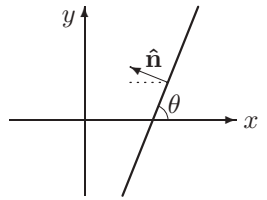
$$\mathbf{E}_0 = \frac{\sigma_0}{\epsilon_0} (\cos 45^\circ \hat{\mathbf{x}} + \sin 45^\circ \hat{\mathbf{y}}) = \boxed{\frac{\sigma_0}{\sqrt{2} \epsilon_0} (-\hat{\mathbf{x}} + \hat{\mathbf{y}})}.$$

(b) From Eq. 12.109, $E_x = E_{x_0} = -\frac{\sigma_0}{\sqrt{2} \epsilon_0}$; $E_y = \gamma E_{y_0} = \gamma \frac{\sigma_0}{\sqrt{2} E_0}$. So $\mathbf{E} = \boxed{\frac{\sigma_0}{\sqrt{2} \epsilon_0} (-\hat{\mathbf{x}} + \gamma \hat{\mathbf{y}})}.$

(c) From Prob. 12.10: $\tan \theta = \gamma$, so $\boxed{\theta = \tan^{-1} \gamma}.$

(d) Let $\hat{\mathbf{n}}$ be a unit vector perpendicular to the plates in $\mathcal{S}$ — evidently

$$\hat{\mathbf{n}} = -\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{y}}; |E| = \frac{\sigma_0}{\sqrt{2} \epsilon_0} \sqrt{1 + \gamma^2}.$$



So the angle ϕ between $\hat{\mathbf{n}}$ and $\mathbf{E}$ is:

$$\frac{\mathbf{E} \cdot \hat{\mathbf{n}}}{|E|} = \cos \phi = \frac{1}{\sqrt{1+\gamma^2}} (\sin \theta + \gamma \cos \theta) = \frac{\cos \theta}{\sqrt{1+\gamma^2}} (\tan \theta + \gamma) = \frac{2\gamma}{\sqrt{1+\gamma^2}} \cos \theta$$

But $\gamma = \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sqrt{1-\cos^2 \theta}}{\cos \theta} = \sqrt{\frac{1}{\cos^2 \theta} - 1} \Rightarrow \frac{1}{\cos^2 \theta} = \gamma^2 + 1 \Rightarrow \cos \theta = \frac{1}{\sqrt{1+\gamma^2}}$. So $\cos \phi = \left(\frac{2\gamma}{1+\gamma^2} \right)$.

Evidently the field is not perpendicular to the plates in $\mathcal{S}$.

Problem 12.44

(a) $\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{\mathbf{s}} = \frac{\lambda}{2\pi\epsilon_0} \frac{x_0 \hat{\mathbf{x}} + y_0 \hat{\mathbf{y}}}{(x_0^2 + y_0^2)}$.

(b) $\bar{E}_x = E_x = \frac{\lambda}{2\pi\epsilon_0} \frac{x_0}{(x_0^2 + y_0^2)}$, $\bar{E}_y = \gamma E_y = \gamma \frac{\lambda}{2\pi\epsilon_0} \frac{y_0}{(x_0^2 + y_0^2)}$, $\bar{E}_z = \gamma E_z = 0$, $\bar{\mathbf{E}} = \frac{\lambda}{2\pi\epsilon_0} \frac{(x_0 \hat{\mathbf{x}} + \gamma y_0 \hat{\mathbf{y}})}{(x_0^2 + y_0^2)}$.

Using the inverse Lorentz transformations (Eq. 12.19), $x_0 = \gamma(x + vt)$, $y_0 = y$,

$$\bar{\mathbf{E}} = \frac{\lambda}{2\pi\epsilon_0} \frac{\gamma(x + vt) \hat{\mathbf{x}} + \gamma y \hat{\mathbf{y}}}{[\gamma^2(x + vt)^2 + y^2]} = \frac{\lambda}{2\pi\epsilon_0} \frac{1}{\gamma} \frac{(x + vt) \hat{\mathbf{x}} + y \hat{\mathbf{y}}}{[(x + vt)^2 + y^2/\gamma^2]}.$$

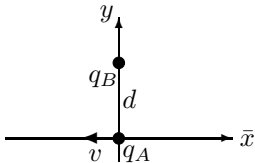
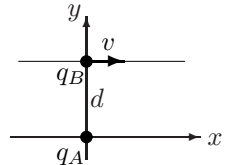
Now $\mathbf{S} = (x + vt) \hat{\mathbf{x}} + y \hat{\mathbf{y}}$, and $y = S \sin \theta$, so $[(x + vt)^2 + y^2/\gamma^2] = [(x + vt)^2 + y^2(1 - v^2/c^2)] = S^2 - (v/c)^2 S^2 \sin^2 \theta = S^2[1 - (v/c)^2 \sin^2 \theta]$, so

$$\bar{\mathbf{E}} = \frac{\lambda}{2\pi\epsilon_0} \frac{\sqrt{1 - (v/c)^2}}{(1 - v^2 \sin^2 \theta/c^2)} \frac{\hat{\mathbf{S}}}{S}.$$

This is reminiscent of Eq. 10.75. Yes, the field *does* point away from the present location of the wire.

Problem 12.45

(a) Fields of A at B : $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q_A}{d^2} \hat{\mathbf{y}}$; $\mathbf{B} = \mathbf{0}$. So force on q_B is $\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{q_A q_B}{d^2} \hat{\mathbf{y}}$.



(b) (i) From Eq. 12.67: $\bar{\mathbf{F}} = \frac{\gamma}{4\pi\epsilon_0} \frac{q_A q_B}{d^2} \hat{\mathbf{y}}$. (Note: here the particle is at rest in $\bar{\mathcal{S}}$.)

(ii) From Eq. 12.93, with $\theta = 90^\circ$: $\bar{E} = \frac{1}{4\pi\epsilon_0} \frac{q_A(1 - v^2/c^2)}{(1 - v^2/c^2)^{3/2}} \frac{1}{d^2} \hat{\mathbf{y}} = \frac{\gamma}{4\pi\epsilon_0} \frac{q_A}{d^2} \hat{\mathbf{y}}$
(this also follows from Eq. 12.109).

$\bar{\mathbf{B}} \neq 0$, but since $v_B = 0$ in $\bar{\mathcal{S}}$, there is no magnetic force anyway, and $\bar{\mathbf{F}} = \frac{\gamma}{4\pi\epsilon_0} \frac{q_A q_B}{d^2} \hat{\mathbf{y}}$ (as before).

Problem 12.46

System A: Use Eqs. 12.93 and 12.112, with $\theta = 90^\circ$, $\mathbf{R} = d\hat{\mathbf{y}}$, and $\hat{\phi} = \hat{\mathbf{z}}$:

$$\mathbf{E} = -\frac{q}{4\pi\epsilon_0} \frac{\gamma}{d^2} \hat{\mathbf{y}}; \quad \mathbf{B} = -\frac{q}{4\pi\epsilon_0} \frac{v}{c^2} \frac{\gamma}{d^2} \hat{\mathbf{z}}; \quad \text{where} \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

[Note that $(E^2 - B^2c^2) = \left(\frac{q}{4\pi\epsilon_0 d^2}\right)^2 \gamma^2 (1 - \frac{v^2}{c^2}) = \left(\frac{q}{4\pi\epsilon_0 d^2}\right)^2$ is invariant, since it doesn't depend on v (see Prob. 12.47b for the general proof). We'll use this as a check.]

$$\mathbf{F} = q(\mathbf{E} + (-v\hat{\mathbf{x}}) \times \mathbf{B}) = -\frac{q^2}{4\pi\epsilon_0} \frac{\gamma}{d^2} \left(\hat{\mathbf{y}} - \frac{v^2}{c^2} (\hat{\mathbf{x}} \times \hat{\mathbf{z}}) \right) = -\frac{q^2}{4\pi\epsilon_0} \frac{\gamma}{d^2} \left(1 + \frac{v^2}{c^2} \right) \hat{\mathbf{y}}.$$

System B: The speed of $-q$ is $v_B = \frac{v+v}{1+v^2/c^2} = \frac{2v}{(1+v^2/c^2)}$

$$\gamma_B = \frac{1}{\sqrt{1 - \frac{4v^2/c^2}{(1+v^2/c^2)^2}}} = \frac{(1+v^2/c^2)}{\sqrt{1 - 2\frac{v^2}{c^2} + \frac{v^4}{c^4}}} = \frac{(1+v^2/c^2)}{(1-v^2/c^2)} = \gamma^2 \left(1 + \frac{v^2}{c^2} \right); \quad v_B \gamma_B = 2v\gamma^2.$$

$$\therefore \mathbf{E} = -\frac{q}{4\pi\epsilon_0} \frac{1}{d^2} \gamma^2 \left(1 + \frac{v^2}{c^2} \right) \hat{\mathbf{y}}; \quad \mathbf{B} = -\frac{q}{4\pi\epsilon_0} \frac{2v}{c^2} \frac{\gamma^2}{d^2} \hat{\mathbf{z}}.$$

$$[Check: (E^2 - B^2c^2) = \left(\frac{q}{4\pi\epsilon_0 d^2}\right)^2 \gamma^4 \left(1 + \frac{2v^2}{c^2} + \frac{v^4}{c^4} - \frac{4v^2}{c^2} \right) = \left(\frac{q}{4\pi\epsilon_0 d^2}\right)^2 \gamma^4 \frac{1}{\gamma^4} = \left(\frac{q}{4\pi\epsilon_0 d^2}\right)^2 \checkmark]$$

$$\mathbf{F} = q\mathbf{E} = -\frac{q^2}{4\pi\epsilon_0} \frac{\gamma^2}{d^2} \left(1 + \frac{v^2}{c^2} \right) \hat{\mathbf{y}} \quad (+q \text{ at rest} \Rightarrow \text{no magnetic force}). \quad [Check: \text{Eq. 12.67} \Rightarrow F_A = \frac{1}{\gamma} F_B. \quad \checkmark]$$

$$\text{System C:} \quad v_C = 0. \quad \mathbf{E} = -\frac{q}{4\pi\epsilon_0} \frac{1}{d^2} \hat{\mathbf{y}}; \quad \mathbf{B} = \mathbf{0}. \quad \mathbf{F} = q\mathbf{E} = -\frac{q^2}{4\pi\epsilon_0} \frac{1}{d^2} \hat{\mathbf{y}}.$$

[The relative velocity of B and C is $2v/(1+v^2/c^2)$, and corresponding γ is $\gamma^2(1+v^2/c^2)$. So Eq. 12.67 $\Rightarrow F_C = \frac{1}{\gamma^2(1+v^2/c^2)} F_B$. $\checkmark$]

Summary:

$\left(-\frac{q}{4\pi\epsilon_0 d^2}\right) \gamma \hat{\mathbf{y}}$	$\left(-\frac{q}{4\pi\epsilon_0 d^2}\right) \gamma^2 (1+v^2/c^2) \hat{\mathbf{y}}$	$\left(-\frac{q}{4\pi\epsilon_0 d^2}\right) \hat{\mathbf{y}}$
$\left(-\frac{q}{4\pi\epsilon_0 d^2}\right) \frac{v}{c^2} \gamma \hat{\mathbf{z}}$	$\left(-\frac{q}{4\pi\epsilon_0 d^2}\right) \frac{2v}{c^2} \gamma^2 \hat{\mathbf{z}}$	$\mathbf{0}$
$\left(-\frac{q^2}{4\pi\epsilon_0 d^2}\right) \gamma (1+v^2/c^2) \hat{\mathbf{y}}$	$\left(-\frac{q^2}{4\pi\epsilon_0 d^2}\right) \gamma^2 (1+v^2/c^2) \hat{\mathbf{y}}$	$\left(-\frac{q^2}{4\pi\epsilon_0 d^2}\right) \hat{\mathbf{y}}$

Problem 12.47

(a) From Eq. 12.109:

$$\begin{aligned} \bar{\mathbf{E}} \cdot \bar{\mathbf{B}} &= \bar{E}_x \bar{B}_x + \bar{E}_y \bar{B}_y + \bar{E}_z \bar{B}_z = E_x B_x + \gamma^2 (E_y - vB_z)(B_y + \frac{v}{c^2} E_z) + \gamma (E_z + vB_y)(B_z - \frac{v}{c^2} E_y) \\ &= E_x B_x + \gamma^2 \{ E_y B_y + \cancel{\frac{v}{c^2} E_y E_z} - \cancel{v B_y B_z} - \frac{v^2}{c^2} E_z B_z + E_z B_z - \cancel{\frac{v}{c^2} E_y E_z} + \cancel{v B_y B_z} - \frac{v^2}{c^2} E_y B_y \} \\ &= E_x B_x + \gamma^2 \left(E_y B_y \left(1 - \frac{v^2}{c^2} \right) + E_z B_z \left(1 - \frac{v^2}{c^2} \right) \right) = E_x B_x + E_y B_y + E_z B_z = \mathbf{E} \cdot \mathbf{B}. \quad \text{qed} \end{aligned}$$

$$\begin{aligned}
\text{(b)} \quad \bar{E}^2 - c^2 \bar{B}^2 &= [E_x^2 + \gamma^2(E_y - vB_z)^2 + \gamma^2(E_z + vB_y)^2] - c^2[B_x^2 + \gamma^2(B_y + \frac{v}{c^2}E_z)^2 + \gamma^2(B_z - \frac{v}{c^2}E_y)] \\
&= E_x^2 + \gamma^2(E_y^2 - 2E_y v B_z + v^2 B_z^2 + E_z^2 + 2E_z v B_y + v^2 B_y^2 - c^2 B_y^2 - c^2 2 \cancel{B_y E_z} \\
&\quad - c^2 \frac{v^2}{c^4} E_z^2 - c^2 B_z^2 + c^2 2 \cancel{B_z E_y} - c^2 \frac{v^2}{c^4} E_y^2) - c^2 B_x^2 \\
&= E_x^2 - c^2 B_x^2 + \gamma^2 \left(E_y^2 \left(1 - \frac{v^2}{c^2}\right) + E_z^2 \left(1 - \frac{v^2}{c^2}\right) - c^2 (B_y^2) \left(1 - \frac{v^2}{c^2}\right) - c^2 B_z^2 \left(1 - \frac{v^2}{c^2}\right) \right) \\
&= (E_x^2 + E_y^2 + E_z^2) - c^2 (B_x^2 + B_y^2 + B_z^2) = E^2 - B^2 c^2. \quad \text{qed}
\end{aligned}$$

(c) No. For if $\mathbf{B} = \mathbf{0}$ in one system, then $(E^2 - c^2 B^2)$ is *positive*. Since it is invariant, it must be positive in *any* system. $\therefore \mathbf{E} \neq \mathbf{0}$ in *all* systems.

Problem 12.48

(a) Making the appropriate modifications in Eq. 9.48 (and picking $\delta = 0$ for convenience),

$$\mathbf{E}(x, y, z, t) = E_0 \cos(kx - \omega t) \hat{\mathbf{y}}, \quad \mathbf{B}(x, y, z, t) = \frac{E_0}{c} \cos(kx - \omega t) \hat{\mathbf{z}}, \quad \text{where } k \equiv \frac{\omega}{c}.$$

(b) Using Eq. 12.109 to transform the fields:

$$\bar{E}_x = \bar{E}_z = 0, \quad \bar{E}_y = \gamma(E_y - vB_z) = \gamma E_0 \left[\cos(kx - \omega t) - \frac{v}{c} \cos(kx - \omega t) \right] = \alpha E_0 \cos(kx - \omega t),$$

$$\bar{B}_x = \bar{B}_y = 0, \quad \bar{B}_z = \gamma(B_z - \frac{v}{c^2} E_y) = \gamma E_0 \left[\frac{1}{c} \cos(kx - \omega t) - \frac{v}{c^2} \cos(kx - \omega t) \right] = \alpha \frac{E_0}{c} \cos(kx - \omega t),$$

where $\alpha \equiv \gamma \left(1 - \frac{v}{c}\right) = \sqrt{\frac{1 - v/c}{1 + v/c}}.$

Now the inverse Lorentz transformations (Eq. 12.19) $\Rightarrow x = \gamma(\bar{x} + v\bar{t})$ and $t = \gamma\left(\bar{t} + \frac{v}{c^2}\bar{x}\right)$, so

$$kx - \omega t = \gamma \left[k(\bar{x} + v\bar{t}) - \omega \left(\bar{t} + \frac{v}{c^2} \bar{x} \right) \right] = \gamma \left[\left(k - \frac{\omega v}{c^2} \right) \bar{x} - (\omega - kv) \bar{t} \right] = \bar{k} \bar{x} - \bar{\omega} \bar{t},$$

where, recalling that $k = \omega/c$: $\bar{k} \equiv \gamma \left(k - \frac{\omega v}{c^2} \right) = \gamma k (1 - v/c) = \alpha k$ and $\bar{\omega} \equiv \gamma \omega (1 - v/c) = \alpha \omega$.

Conclusion:

$$\begin{aligned}
\bar{\mathbf{E}}(\bar{x}, \bar{y}, \bar{z}, \bar{t}) &= \bar{E}_0 \cos(\bar{k}\bar{x} - \bar{\omega}\bar{t}) \hat{\mathbf{y}}, \quad \bar{\mathbf{B}}(\bar{x}, \bar{y}, \bar{z}, \bar{t}) = \frac{\bar{E}_0}{c} \cos(\bar{k}\bar{x} - \bar{\omega}\bar{t}) \hat{\mathbf{z}}, \\
\text{where } \bar{E}_0 &= \alpha E_0, \quad \bar{k} = \alpha k, \quad \bar{\omega} = \alpha \omega, \quad \text{and} \quad \alpha \equiv \sqrt{\frac{1 - v/c}{1 + v/c}}.
\end{aligned}$$

(c) $\bar{\omega} = \omega \sqrt{\frac{1 - v/c}{1 + v/c}}.$ This is the Doppler shift for light. $\bar{\lambda} = \frac{2\pi}{\bar{k}} = \frac{2\pi}{\alpha k} = \frac{\lambda}{\alpha}.$ The velocity of the wave

in $\bar{\mathcal{S}}$ is $\bar{v} = \frac{\bar{\omega}}{2\pi} \bar{\lambda} = \frac{\omega}{2\pi} \lambda = \text{c.}$ Yup, this is exactly what I expected (the velocity of a light wave is the same in any inertial system).

(d) Since the intensity goes like E^2 , the ratio is $\frac{\bar{I}}{I} = \frac{\bar{E}_0^2}{E_0^2} = \alpha^2 = \boxed{\frac{1 - v/c}{1 + v/c}}.$

Dear Al,

The amplitude, frequency, and intensity of the light will all decrease to zero as you run faster and faster. It'll get so faint you won't be able to see it, and so red-shifted even your night-vision goggles won't help. But it'll still be going 3×10^8 m/s relative to you. Sorry about that.

Sincerely,

David

Problem 12.49

$$\begin{aligned}\bar{t}^{02} &= \Lambda_\lambda^0 \Lambda_\sigma^2 t^{\lambda\sigma} = \Lambda_0^0 \Lambda_2^2 t^{02} + \Lambda_1^0 \Lambda_2^2 t^{12} = \gamma t^{02} + (-\gamma\beta) t^{12} = \gamma(t^{02} - \beta t^{12}). \\ \bar{t}^{03} &= \Lambda_\lambda^0 \Lambda_\sigma^3 t^{\lambda\sigma} = \Lambda_0^0 \Lambda_3^3 t^{03} + \Lambda_1^0 \Lambda_3^3 t^{13} = \gamma t^{03} + (-\gamma\beta) t^{13} = \gamma(t^{03} - \beta t^{13}) = \gamma(t^{03} + \beta t^{31}). \\ \bar{t}_{23} &= \Lambda_\lambda^2 \Lambda_\sigma^3 t^{\lambda\sigma} = \Lambda_2^2 \Lambda_3^3 t^{23} = t^{23}. \\ \bar{t}_{31} &= \Lambda_\lambda^3 \Lambda_\sigma^1 t^{\lambda\sigma} = \Lambda_3^3 \Lambda_0^1 t^{30} + \Lambda_3^3 \Lambda_1^1 t^{31} = (-\gamma\beta) t^{30} + \gamma t^{31} = \gamma(t^{31} - \beta t^{03}). \\ \bar{t}_{12} &= \Lambda_\lambda^1 \Lambda_\sigma^2 t^{\lambda\sigma} = \Lambda_0^1 \Lambda_2^2 t^{02} + \Lambda_1^1 \Lambda_2^2 t^{12} = (-\gamma\beta) t^{02} + \gamma t^{12} = \gamma(t^{12} - \beta t^{02}).\end{aligned}$$

Problem 12.50

Suppose $t^{\nu\mu} = \pm t^{\mu\nu}$ (+ for symmetric, - for antisymmetric).

$$\begin{aligned}\bar{t}^{\kappa\lambda} &= \Lambda_\mu^\kappa \Lambda_\nu^\lambda t^{\mu\nu} \\ \bar{t}^{\lambda\kappa} &= \Lambda_\mu^\lambda \Lambda_\nu^\kappa t^{\mu\nu} = \Lambda_\nu^\lambda \Lambda_\mu^\kappa t^{\nu\mu} \quad [\text{Because } \mu \text{ and } \nu \text{ are both summed from } 0 \rightarrow 3, \\ &\quad \text{it doesn't matter which we call } \mu \text{ and which call } \nu.] \\ &= \Lambda_\mu^\kappa \Lambda_\mu^\lambda (\pm t^{\mu\nu}) \quad [\text{Using symmetry of } t^{\mu\nu}, \text{ and writing the } \Lambda\text{'s in the other order.}] \\ &= \pm \bar{t}^{\kappa\lambda}. \quad \text{qed}\end{aligned}$$

Problem 12.51

$$\begin{aligned}F^{\mu\nu} F_{\mu\nu} &= F^{00} F^{00} - F^{01} F^{01} - F^{02} F^{02} - F^{03} F^{03} - F^{10} F^{10} - F^{20} F^{20} - F^{30} F^{30} \\ &\quad + F^{11} F^{11} + F^{12} F^{12} + F^{13} F^{13} + F^{21} F^{21} + F^{22} F^{22} + F^{23} F^{23} + F^{31} F^{31} + F^{32} F^{32} + F^{33} F^{33} \\ &= -(E_x/c)^2 - (E_y/c)^2 - (E_z/c)^2 - (E_x/c)^2 - (E_y/c)^2 - (E_z/c)^2 + B_z^2 + B_y^2 + B_z^2 + B_x^2 + B_y^2 + B_x^2 \\ &= 2B^2 - 2E^2/c^2 = \boxed{2\left(B^2 - \frac{E^2}{c^2}\right)},\end{aligned}$$

which, apart from the constant factor $-2/c^2$, is the invariant we found in Prob. 12.47(b).

$$\boxed{G^{\mu\nu} G_{\mu\nu} = 2(E^2/c^2 - B^2)} \quad (\text{the same invariant}).$$

$$\begin{aligned}F^{\mu\nu} G_{\mu\nu} &= -2(F^{01} G^{01} + F^{02} G^{02} + F^{03} G^{03}) + 2(F^{12} G^{12} + F^{13} G^{13} + F^{23} G^{23}) \\ &= -2\left(\frac{1}{c} E_x B_x + \frac{1}{c} E_y B_y + \frac{1}{c} E_z B_z\right) + 2[B_z(-E_z/c) + (-B_y)(E_y/c) + B_x(-E_x/c)] \\ &= -\frac{2}{c}(\mathbf{E} \cdot \mathbf{B}) - \frac{2}{c}(\mathbf{E} \cdot \mathbf{B}) = \boxed{-\frac{4}{c}(\mathbf{E} \cdot \mathbf{B})},\end{aligned}$$

which, apart from the factor $-4/c$, is the invariant we found of Prob. 12.47(a). [These are, incidentally, the *only* fundamental invariants you can construct from $\mathbf{E}$ and $\mathbf{B}$.]

Problem 12.52

$$\left. \begin{aligned} \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{x} \hat{\mathbf{x}} = \frac{\mu_0}{2\pi} \frac{\lambda c^2}{x} \hat{\mathbf{x}} \\ \mathbf{B} &= \frac{\mu_0}{4\pi} \frac{2\lambda v}{x} \hat{\mathbf{y}} = \frac{\mu_0}{2\pi} \frac{\lambda v}{x} \hat{\mathbf{y}} \end{aligned} \right\} \quad \boxed{F^{\mu\nu} = \frac{\mu_0\lambda}{2\pi x} \begin{pmatrix} 0 & c & 0 & 0 \\ -c & 0 & 0 & -v \\ 0 & 0 & 0 & 0 \\ 0 & v & 0 & 0 \end{pmatrix}} \quad \boxed{G^{\mu\nu} = \frac{\mu_0\lambda}{2\pi x} \begin{pmatrix} 0 & 0 & v & 0 \\ 0 & 0 & 0 & 0 \\ -v & 0 & 0 & -c \\ 0 & 0 & c & 0 \end{pmatrix}}$$

Problem 12.53

$\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$. Differentiate: $\partial_\mu \partial_\nu F^{\mu\nu} = \mu_0 \partial_\mu J^\mu$.

But $\partial_\mu \partial_\nu = \partial_\nu \partial_\mu$ (the combination is *symmetric*) while $F^{\nu\mu} = -F^{\mu\nu}$ (*antisymmetric*).

$\therefore \partial_\mu \partial_\nu F^{\mu\nu} = 0$. [Why? Well, these indices are both summed from $0 \rightarrow 3$, so it doesn't matter which we call μ , which ν : $\partial_\mu \partial_\nu F^{\mu\nu} = \partial_\nu \partial_\mu F^{\nu\mu} = \partial_\mu \partial_\nu (-F^{\mu\nu}) = -\partial_\mu \partial_\nu F^{\mu\nu}$. But if a quantity is equal to minus itself, it must be zero.] *Conclusion:* $\partial_\mu J^\mu = 0$. qed

Problem 12.54

We know that $\partial_\nu G^{\mu\nu} = 0$ is equivalent to the two homogeneous Maxwell equations, $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$. All we have to show, then, is that $\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0$ is *also* equivalent to them. Now this equation stands for 64 separate equations ($\mu = 0 \rightarrow 3$, $\nu = 0 \rightarrow 3$, $\lambda = 0 \rightarrow 3$, and $4 \times 4 \times 4 = 64$). But many of them are redundant, or trivial.

Suppose two indices are the same (say, $\mu = \nu$). Then $\partial_\lambda F_{\mu\mu} + \partial_\mu F_{\mu\lambda} = \partial_\mu F_{\lambda\mu} = 0$. But $F_{\mu\mu} = 0$ and $F_{\mu\lambda} = -F_{\lambda\mu}$, so this is trivial: $0 = 0$. To get anything significant, then, μ, ν, λ must all be *different*. They could be all *spatial* ($\mu, \nu, \lambda = 1, 2, 3 = x, y, z$ — or some permutation thereof), or *one temporal and two spatial* ($\mu = 0, \nu, \lambda = 1, 2$ or $2, 3$, or $1, 3$ — or some permutation). Let's examine these two cases separately.

All spatial: say, $\mu = 1, \nu = 2, \lambda = 3$ (other permutations yield the same equation, or minus it.)

$$\partial_3 F_{12} + \partial_1 F_{23} + \partial_2 F_{31} = 0 \Rightarrow \frac{\partial}{\partial z}(B_z) + \frac{\partial}{\partial x}(B_x) + \frac{\partial}{\partial y}(B_y) = 0 \Rightarrow \nabla \cdot \mathbf{B} = 0.$$

One temporal: say, $\mu = 0, \nu = 1, \lambda = 2$ (other permutations of these indices yield same result, or minus it).

$$\partial_2 F_{01} + \partial_0 F_{12} + \partial_1 F_{31} = 0 \Rightarrow \frac{\partial}{\partial y}\left(-\frac{E_x}{c}\right) = \frac{\partial}{\partial(ct)}(B_z) + \frac{\partial}{\partial x}\left(+\frac{E_y}{c}\right) = 0.$$

or: $-\frac{\partial B_z}{\partial t} + \left(\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x}\right) = 0$, which is the z -component of $-\frac{\partial \mathbf{B}}{\partial t} = \nabla \times \mathbf{E}$. (If $\mu = 0, \nu = 1, \lambda = 2$, we get the y component; for $\nu = 2, \lambda = 3$ we get the x component.)

Conclusion: $\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0$ is equivalent to $\nabla \cdot \mathbf{B} = 0$ and $\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}$, and hence to $\partial_\nu G^{\mu\nu} = 0$. qed

Problem 12.55

$K^0 = q\eta_\nu F^{0\nu} - q(\eta_1 F^{01} + \eta_2 F^{02} + \eta_3 F^{03}) = q(\boldsymbol{\eta} \cdot \mathbf{E})/c = \boxed{\frac{q}{c} \boldsymbol{\eta} \cdot \mathbf{E}}$. Now from Eq. 12.70 we know that $K^0 = \frac{1}{c} \frac{dW}{d\tau}$, where W is the energy of the particle. Since $d\tau = \frac{1}{\gamma} dt$, we have:

$$\frac{1}{c} \gamma \frac{dW}{dt} = \frac{q}{c} \boldsymbol{\eta} \cdot \mathbf{E} \Rightarrow \boxed{\frac{dW}{dt} = q(\mathbf{u} \cdot \mathbf{E})}$$

This says *the power delivered to the particle is force ($q\mathbf{E}$) times velocity ($\mathbf{u}$)* — which is as it *should* be.

Problem 12.56

$$\overline{\partial^0 \phi} = \frac{\partial}{\partial \bar{x}_0} \phi = -\frac{1}{c} \frac{\partial}{\partial \bar{t}} \phi = \frac{1}{c} \left(\frac{\partial \phi}{\partial t} \frac{\partial t}{\partial \bar{t}} + \frac{\partial \phi}{\partial x} \frac{\partial x}{\partial \bar{t}} + \frac{\partial \phi}{\partial y} \frac{\partial y}{\partial \bar{t}} + \frac{\partial \phi}{\partial z} \frac{\partial z}{\partial \bar{t}} \right)$$

From Eq. 12.19, we have: $\frac{\partial t}{\partial \bar{t}} = \gamma$, $\frac{\partial x}{\partial \bar{t}} = \gamma v$, $\frac{\partial y}{\partial \bar{t}} = \frac{\partial z}{\partial \bar{t}} = 0$.

$$\text{So } \overline{\partial^0 \phi} = -\frac{1}{c} \gamma \left(\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} \right) \text{ or (since } ct = x^0 = -x_0): \overline{\partial^0 \phi} = \gamma \left(\frac{\partial \phi}{\partial x_0} - \frac{v}{c} \frac{\partial \phi}{\partial x^1} \right) = \gamma [(\partial^0 \phi) - \beta(\partial^1 \phi)].$$

$$\overline{\partial^1 \phi} = \frac{\partial}{\partial \bar{x}} \phi = \frac{\partial \phi}{\partial t} \frac{\partial t}{\partial \bar{x}} + \frac{\partial \phi}{\partial x} \frac{\partial x}{\partial \bar{x}} + \frac{\partial \phi}{\partial y} \frac{\partial y}{\partial \bar{x}} + \frac{\partial \phi}{\partial z} \frac{\partial z}{\partial \bar{x}} = \gamma \frac{v}{c^2} \frac{\partial \phi}{\partial t} + \gamma \frac{\partial \phi}{\partial x} = \gamma \left(\frac{\partial \phi}{\partial x_1} - \frac{v}{c} \frac{\partial \phi}{\partial x_0} \right) = \gamma [(\partial^1 \phi) - \beta(\partial^0 \phi)].$$

$$\overline{\partial^2 \phi} = \frac{\partial \phi}{\partial \bar{y}} = \frac{\partial \phi}{\partial t} \frac{\partial t}{\partial \bar{y}} + \frac{\partial \phi}{\partial x} \frac{\partial x}{\partial \bar{y}} + \frac{\partial \phi}{\partial y} \frac{\partial y}{\partial \bar{y}} + \frac{\partial \phi}{\partial z} \frac{\partial z}{\partial \bar{y}} = \frac{\partial \phi}{\partial y} = \partial^2 \phi.$$

$$\overline{\partial^3 \phi} = \frac{\partial \phi}{\partial \bar{z}} = \frac{\partial \phi}{\partial t} \frac{\partial t}{\partial \bar{z}} + \frac{\partial \phi}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial \phi}{\partial y} \frac{\partial y}{\partial \bar{z}} + \frac{\partial \phi}{\partial z} \frac{\partial z}{\partial \bar{z}} = \frac{\partial \phi}{\partial z} = \partial^3 \phi.$$

Conclusion: $\partial^\mu \phi$ transforms in the same way as a^μ (Eq. 12.27)—and hence is a contravariant 4-vector. qed

Problem 12.57

According to Prob. 12.54, $\frac{\partial G^{\mu\nu}}{\partial x^\nu} = 0$ is equivalent to Eq. 12.130. Using Eq. 12.133, we find (in the notation of Prob. 12.56):

$$\begin{aligned} \frac{\partial F_{\mu\nu}}{\partial x^\lambda} + \frac{\partial F_{\nu\lambda}}{\partial x^\mu} + \frac{\partial F_{\lambda\mu}}{\partial x^\nu} &= \partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} \\ &= \partial_\lambda (\partial_\mu A_\nu - \partial_\nu A_\mu) + \partial_\mu (\partial_\nu A_\lambda - \partial_\lambda A_\nu) + \partial_\nu (\partial_\lambda A_\mu - \partial_\mu A_\lambda) \\ &= (\partial_\lambda \partial_\mu A_\nu - \partial_\mu \partial_\lambda A_\nu) + (\partial_\mu \partial_\nu A_\lambda - \partial_\nu \partial_\mu A_\lambda) + (\partial_\nu \partial_\lambda A_\mu - \partial_\lambda \partial_\nu A_\mu) = 0. \quad \text{qed} \end{aligned}$$

[Note that $\partial_\lambda \partial_\mu A_\nu = \frac{\partial^2 A_\nu}{\partial x^\lambda \partial x^\mu} = \frac{\partial^2 A_\nu}{\partial x^\mu \partial x^\lambda} = \partial_\nu \partial_\lambda A_\mu$, by equality of cross-derivatives.]

Problem 12.58

From Eqs. 12.40 and 12.42, $\eta^\mu = \gamma(c, \mathbf{v})$, while $\mathbf{z}^\mu = (ct - ct_r, \mathbf{r} - \mathbf{w}(t_r)) = (\mathbf{z}^0, \mathbf{z})$, so $\eta^\nu \mathbf{z}_\nu = -\gamma c \mathbf{z}^0 + \gamma \mathbf{v} \cdot \mathbf{z} = -\gamma(\mathbf{z}^0 c - \mathbf{z} \cdot \mathbf{v})$.

$$-\frac{q}{4\pi\epsilon_0 c} \frac{\eta^0}{(\eta^\nu \mathbf{z}_\nu)} = \frac{q}{4\pi\epsilon_0 c} \frac{\gamma c}{\gamma(\mathbf{z}^0 c - \mathbf{z} \cdot \mathbf{v})} = \frac{1}{4\pi\epsilon_0 c} \frac{qc}{(\mathbf{z}^0 c - \mathbf{z} \cdot \mathbf{v})} = \frac{1}{c} V$$

(Eq. 10.46),

$$-\frac{q}{4\pi\epsilon_0 c} \frac{\boldsymbol{\eta}}{(\eta^\nu \mathbf{z}_\nu)} = \frac{q}{4\pi\epsilon_0 c} \frac{\gamma \mathbf{v}}{\gamma(\mathbf{z}^0 c - \mathbf{z} \cdot \mathbf{v})} = \frac{1}{4\pi\epsilon_0 c} \frac{q \mathbf{v}}{(\mathbf{z}^0 c - \mathbf{z} \cdot \mathbf{v})} = \mathbf{A}$$

(Eq. 10.47), so (Eq. 12.132)

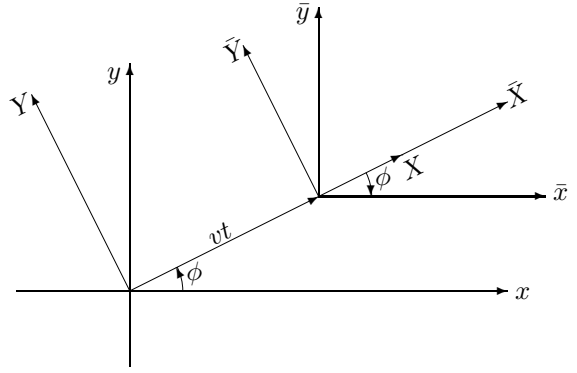
$$-\frac{q}{4\pi\epsilon_0 c} \frac{\eta^\mu}{(\eta^\nu \mathbf{z}_\nu)} = A^\mu. \quad \checkmark$$

Problem 12.59

Step 1: rotate from xy to XY , using Eq. 1.29:

$$X = \cos \phi x + \sin \phi y$$

$$Y = -\sin \phi x + \cos \phi y$$



Step 2: Lorentz-transform from XY to $\bar{X}\bar{Y}$, using Eq. 12.18:

$$\bar{X} = \gamma(X - vt) = \gamma[\cos \phi x + \sin \phi y - \beta ct]$$

$$\bar{Y} = Y = -\sin \phi x + \cos \phi y$$

$$\bar{Z} = Z = z$$

$$c\bar{t} = \gamma(ct - \beta X) = \gamma[ct - \beta(\cos \phi x + \sin \phi y)]$$

Step 3: Rotate from $\bar{X}\bar{Y}$ to $\bar{x}\bar{y}$, using Eq. 1.29 with negative ϕ :

$$\bar{x} = \cos \phi \bar{X} - \sin \phi \bar{Y} = \gamma \cos \phi [\cos \phi x + \sin \phi y - \beta ct] - \sin \phi [-\sin \phi x + \cos \phi y]$$

$$= (\gamma \cos^2 \phi + \sin^2 \phi)x + (\gamma - 1) \sin \phi \cos \phi y - \gamma \beta \cos \phi (ct)$$

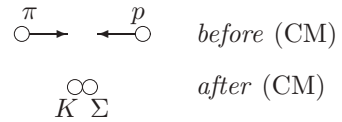
$$\bar{y} = \sin \phi \bar{X} + \cos \phi \bar{Y} = \gamma \sin \phi (\cos \phi x + \sin \phi y - \beta ct) + \cos \phi (-\sin \phi x + \cos \phi y)$$

$$= (\gamma - 1) \sin \phi \cos \phi x + (\gamma \sin^2 \phi + \cos^2 \phi)y - \gamma \beta \sin \phi (ct)$$

In matrix form:
$$\begin{pmatrix} c\bar{t} \\ \bar{x} \\ \bar{y} \\ \bar{z} \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta \cos \phi & -\gamma\beta \sin \phi & 0 \\ -\gamma\beta \cos \phi & (\gamma \cos^2 \phi + \sin^2 \phi) & (\gamma - 1) \sin \phi \cos \phi & 0 \\ -\gamma\beta \sin \phi & (\gamma - 1) \sin \phi \cos \phi & (\gamma \sin^2 \phi + \cos^2 \phi) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

Problem 12.60

In center-of-momentum system, threshold occurs when incident energy is *just* sufficient to cover the *rest* energy of the resulting particles, with none “wasted” as kinetic energy. Thus, in lab system, we want the outgoing K and Σ to have the *same velocity*, at threshold:



Initial momentum: p_π ; Initial energy of π : $E^2 - p^2 c^2 = m^2 c^4 \Rightarrow E_\pi^2 = m_\pi^2 c^4 + p_\pi^2 c^2$.

Total initial energy: $m_p c^2 = \sqrt{m_\pi^2 c^4 + p_\pi^2 c^2}$. These are also the final energy and momentum: $E^2 - p^2 c^2 = (m_K + m_\Sigma)^2 c^4$.

$$\left(m_p c^2 + \sqrt{m_\pi^2 c^4 + p_\pi^2 c^2} \right)^2 - p_\pi^2 c^2 = (m_K + m_\Sigma)^2 c^4$$

$$\frac{m_p^2 c^4 + 2m_p c^2 \sqrt{m_\pi^2 c^4 + p_\pi^2 c^2} + m_\pi^2 c^4 + p_\pi^2 c^2 - p_\pi^2 c^2}{c^4} = (m_K + m_\Sigma)^2 c^4$$

$$\frac{2m_p}{c} \sqrt{m_\pi^2 c^4 + p_\pi^2 c^2} = (m_K + m_\Sigma)^2 - m_p^2 - m_\pi^2$$

$$(m_\pi^2 c^2 + p_\pi^2) \frac{4m_p^2}{c^2} = (m_K + m_\Sigma)^4 - 2(m_p^2 + m_\pi^2)(m_K + m_\Sigma)^2 + m_p^4 + m_\pi^4 + 2m_p^2 m_\pi^2$$

$$\frac{4m_p^2}{c^2} p_\pi^2 = (m_K + m_\Sigma)^4 - 2(m_p^2 + m_\pi^2)(m_K + m_\Sigma)^2 + (m_p^2 - m_\pi^2)^2$$

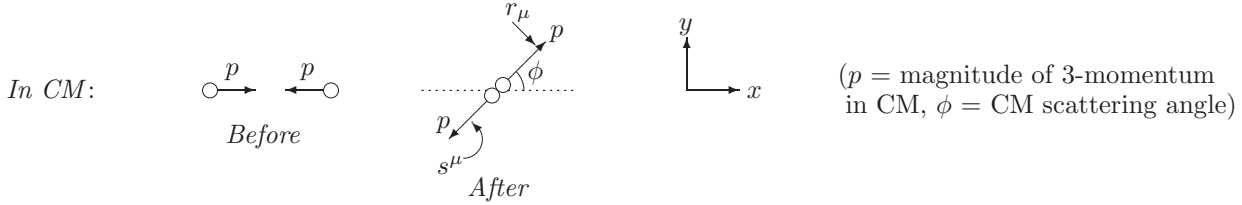
$$p_\pi = \frac{c}{2m_p} \sqrt{(m_K + m_\Sigma)^4 - 2(m_p^2 + m_\pi^2)(m_K + m_\Sigma)^2 + (m_p^2 - m_\pi^2)^2}$$

$$= \frac{1}{(2m_p c^2)c} \sqrt{(m_K c^2 + m_\Sigma c^2)^4 - 2((m_p c^2)^2 + (m_\pi c^2)^2)(m_K c^2 + m_\Sigma c^2)^2 + ((m_p c^2)^2 - (m_\pi c^2)^2)^2}$$

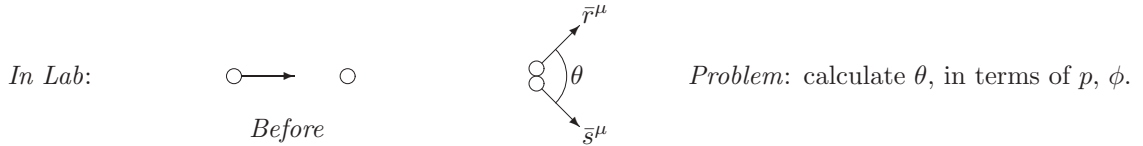
$$= \frac{1}{2c(900)} \sqrt{(1700)^4 - 2((900)^2 + (150)^2)(1700)^2 + ((900)^2 - (150)^2)^2}$$

$$= \frac{1}{1800c} \sqrt{(8.35 \times 10^{12}) - (4.81 \times 10^{12}) + (0.62 \times 10^{12})} = \frac{1}{1800c} (2.04 \times 10^6) = \boxed{1133 \text{ MeV}/c}$$

Problem 12.61



Outgoing 4-momentua: $r^\mu = (\frac{E}{c}, p \cos \phi, p \sin \phi, 0)$; $s^\mu = (\frac{E}{c}, -p \cos \phi, -p \sin \phi, 0)$.



Lorentz transformation: $\bar{r}_x = \gamma(r_x - \beta r^0)$; $\bar{r}_y = r_y$; $\bar{s}_x = \gamma(s_x - \beta s^0)$; $\bar{s}_y = s_y$.

Now $E = \gamma m c^2$; $p = -\gamma m v$ (v here is to the left; $E^2 - p^2 c^2 = m^2 c^4$, so $\beta = -\frac{pc}{E}$).

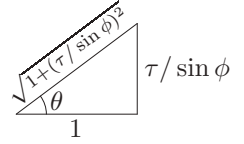
$\therefore \bar{r}_x = \gamma(p \cos \phi + \frac{pc}{E} \frac{E}{c}) = \gamma p(1 + \cos \phi)$; $\bar{r}_y = p \sin \phi$; $\bar{s}_x = \gamma p(1 - \cos \phi)$; $\bar{s}_y = -p \sin \phi$.

$$\begin{aligned} \cos \theta &= \frac{\bar{\mathbf{r}} \cdot \bar{\mathbf{s}}}{\bar{r} \bar{s}} = \frac{\gamma^2 p^2 (1 - \cos^2 \phi) - p^2 \sin^2 \phi}{\sqrt{[\gamma^2 p^2 (1 + \cos \phi)^2 + p^2 \sin^2 \phi] [\gamma^2 p^2 (1 - \cos \phi)^2 + p^2 \sin^2 \phi]}} \\ &= \frac{(\gamma^2 - 1) \sin^2 \phi}{\sqrt{[\gamma^2 (1 + \cos \phi)^2 + \sin^2 \phi] [\gamma^2 (1 - \cos \phi)^2 + \sin^2 \phi]}} \\ &= \frac{(\gamma^2 - 1)}{\sqrt{[\gamma^2 (\frac{1 + \cos \phi}{\sin \phi})^2 + 1] [\gamma^2 (\frac{1 - \cos \phi}{\sin \phi})^2 + 1]}} = \frac{(\gamma^2 - 1)}{\sqrt{(\gamma^2 \cot^2 \frac{\phi}{2} + 1)(\tan^2 \frac{\phi}{2} + 1)}} \end{aligned}$$

$$\begin{aligned}
\cos \theta &= \frac{\omega}{\sqrt{(1 + \cot^2 \frac{\phi}{2} + \omega \cot^2 \frac{\phi}{2})(1 + \tan^2 \frac{\phi}{2} + \omega \tan^2 \frac{\phi}{2})}} \quad (\text{where } \omega \equiv \gamma^2 - 1) \\
&= \frac{\omega}{\sqrt{(\csc^2 \frac{\phi}{2} + \omega \cot^2 \frac{\phi}{2})(\sec^2 \frac{\phi}{2} + \omega \tan^2 \frac{\phi}{2})}} = \frac{\omega \sin \frac{\phi}{2} \cos \frac{\phi}{2}}{\sqrt{(1 + \omega \cos^2 \frac{\phi}{2})(1 + \omega \sin^2 \frac{\phi}{2})}} \\
&= \frac{\frac{1}{2} \omega \sin \phi}{\sqrt{(1 + \omega \frac{1}{2}(1 + \cos \phi))(1 + \omega \frac{1}{2}(1 - \cos \phi))}} = \frac{\sin \phi}{\sqrt{[(\frac{2}{\omega} + 1) + \cos \phi][(\frac{2}{\omega} + 1) - \cos \phi]}} \\
&= \frac{\sin \phi}{\sqrt{(\frac{2}{\omega} + 1)^2 - \cos^2 \phi}} = \frac{\sin \phi}{\sqrt{\frac{4}{\omega^2} + \frac{4}{\omega} + \sin^2 \phi}} = \frac{1}{\sqrt{1 + (\frac{\tau^2}{\sin^2 \phi})}}, \text{ where } \tau^2 = \frac{4}{\omega^2} + \frac{4}{\omega}.
\end{aligned}$$

$$\sin \theta = \frac{\tau}{\sin \phi}. \quad \tau^2 = \frac{4}{\omega^2}(1 + \omega) = \frac{4}{(\gamma^2 - 1)^2} \gamma^2, \text{ so } \tan \theta = \frac{2\gamma}{(\gamma^2 - 1) \sin \phi}.$$

$$\text{Or, since } (\gamma^2 - 1) = \gamma^2 \left(1 - \frac{1}{\gamma^2}\right) = \gamma^2 \frac{v^2}{c^2}, \quad \boxed{\tan \theta = \frac{2c^2}{\gamma v^2 \sin \phi}}$$



Problem 12.62

$$\frac{dp}{d\tau} = K \text{ (a constant)} \Rightarrow \frac{dp}{dt} \frac{dt}{d\tau} = K. \text{ But } \frac{dt}{d\tau} = \frac{1}{\sqrt{1 - u^2/c^2}}; p = \frac{mu}{\sqrt{1 - u^2/c^2}}.$$

$$\therefore \frac{d}{dt} \left(\frac{u}{\sqrt{1 - u^2/c^2}} \right) = \frac{K}{m} \sqrt{1 - u^2/c^2}. \text{ Multiply by } \frac{dt}{dx} = \frac{1}{u}:$$

$$\frac{dt}{dx} \frac{d}{dt} \left(\frac{u}{\sqrt{1 - u^2/c^2}} \right) = \frac{d}{dx} \left(\frac{u}{\sqrt{1 - u^2/c^2}} \right) = \frac{K}{m} \frac{\sqrt{1 - u^2/c^2}}{u}. \text{ Let } w = \frac{u}{\sqrt{1 - u^2/c^2}}.$$

$$\frac{dw}{dx} = \frac{K}{m} \frac{1}{w}; \quad w \frac{dw}{dx} = \frac{1}{2} \frac{d}{dx} w^2 = \frac{k}{m}; \quad \frac{d(w^2)}{dx} = \frac{2K}{m} \Rightarrow d(w^2) = \frac{2K}{m} (dx).$$

$$\therefore w^2 = \frac{2K}{m} x + \text{constant}. \text{ But at } t = 0, x = 0 \text{ and } u = 0 \text{ (so } w = 0), \text{ and hence the constant is 0.}$$

$$w^2 = \frac{2K}{m} x = \frac{u^2}{1 - u^2/c^2}; \quad u^2 = \frac{2Kx}{m} - \frac{2Kx}{mc^2} u^2; \quad u^2 \left(1 + \frac{2Kx}{mc^2}\right) = \frac{2Kx}{m}.$$

$$u^2 = \frac{2Kx/m}{1 + \frac{2Kx}{mc^2}} = \frac{c^2}{1 + (\frac{mc^2}{2Kx})}; \quad \frac{dx}{dt} = \frac{c}{\sqrt{1 + (\frac{mc^2}{2Kx})}}; \quad ct = \int \sqrt{1 + (\frac{mc^2}{2Kx})} dx$$

$$\text{Let } \frac{mc^2}{2K} = a^2; \quad ct = \int \frac{\sqrt{x + a^2}}{\sqrt{x}} dx. \text{ Let } x = y^2; dx = 2y dy; \sqrt{x} = y.$$

$$ct = \int \frac{\sqrt{y^2 + a^2}}{y} 2y dy = 2 \int \sqrt{y^2 + a^2} dy = \left[y \sqrt{y^2 + a^2} + a^2 \ln(y + \sqrt{y^2 + a^2}) \right] + \text{constant}.$$

$$\text{At } t = 0, x = 0 \Rightarrow y = 0. \therefore 0 = a^2 \ln a + \text{constant}, \text{ so constant} = -a^2 \ln a.$$

$$\therefore ct = y \sqrt{y^2 + a^2} + a^2 \ln(y/a + \sqrt{(y/a)^2 + 1}) = a^2 \left[\left(\frac{y}{a}\right) \sqrt{\left(\frac{y}{a}\right)^2 + 1} + \ln\left(\frac{y}{a} + \sqrt{\left(\frac{y}{a}\right)^2 + 1}\right) \right]$$

$$\text{Let: } z = \frac{y}{a} = \sqrt{x} \sqrt{\frac{2K}{mc^2}} = \sqrt{\frac{2Kx}{mc^2}} = z. \text{ Then } \boxed{\frac{2Kt}{mc} = z \sqrt{1 + z^2} + \ln(z + \sqrt{1 + z^2}).}$$

Problem 12.63

(a) $x(t) = \frac{c}{\alpha} \left[\sqrt{1 + (\alpha t)^2} - 1 \right]$, where $\alpha = \frac{F}{mc}$. The force of $+q$ on $-q$ will be the mirror image of the force of $-q$ on $+q$ (in the x -axis), so the *net* force is in the x direction (the net *magnetic* force is zero). So all we need is the x -component of $\mathbf{E}$.

The field at $+q$ due to $-q$ is: (Eq. 10.72)

$$\mathbf{E} = -\frac{q}{4\pi\epsilon_0} \frac{\mathbf{z}}{(\mathbf{z} \cdot \mathbf{u})^3} [\mathbf{u}(c^2 - v^2) + \mathbf{u}(\mathbf{z} \cdot \mathbf{a}) - \mathbf{a}(\mathbf{z} \cdot \mathbf{u})].$$

$\mathbf{u} = c\mathbf{z} - \mathbf{v} \Rightarrow u_x = c\frac{l}{z} - v = \frac{1}{z}(cl - vz)$; $\mathbf{z} \cdot \mathbf{u} = c\mathbf{z} \cdot \mathbf{z} - \mathbf{z} \cdot \mathbf{v} = (c\mathbf{z} \cdot \mathbf{z} - lv)$; $\mathbf{z} \cdot \mathbf{a} = la$. So:

$$\begin{aligned} E_x &= -\frac{q}{4\pi\epsilon_0} \frac{z}{(c\mathbf{z} \cdot \mathbf{z} - vl)^3} \left[\frac{1}{z}(cl - vz)(c^2 - v^2) = \frac{1}{z}(cl - vz)la - a(c\mathbf{z} \cdot \mathbf{z} - lv) \right] \\ &= -\frac{q}{4\pi\epsilon_0} \frac{1}{(c\mathbf{z} \cdot \mathbf{z} - vl)^3} [(cl - vz)(c^2 - v^2) - cad^2]. \end{aligned}$$

The *force* on $+q$ is qE_x , and there is an equal force on $-q$, so the net force on the dipole is:

$$\mathbf{F} = -\frac{2q^2}{4\pi\epsilon_0} \frac{1}{(c\mathbf{z} \cdot \mathbf{z} - lv)^3} [(cl - vz)(c^2 - v^2) - cad^2] \hat{\mathbf{x}}$$

It remains to determine $\mathbf{z}$, l , v , and a , and plug these in.

$$v(t) = \frac{dx}{dt} = \frac{c}{\alpha} \frac{1}{2} \frac{1}{\sqrt{1 + (\alpha t)^2}} 2\alpha^2 t = \frac{c\alpha t}{\sqrt{1 + (\alpha t)^2}}; \quad v = v(t_r) = \frac{c\alpha t_r}{T}, \quad \text{where } T = \sqrt{1 + (\alpha t_r)^2}.$$

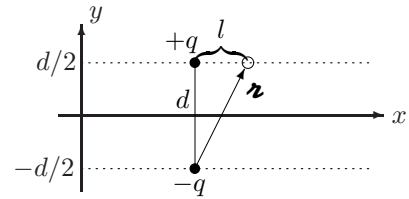
$$a(t_r) = \frac{dv}{dt_r} = \frac{c\alpha}{T} + c\alpha t_r \left(-\frac{1}{2} \right) \frac{2\alpha^2 t_r}{T^3} = \frac{c\alpha}{T^3} (1 + (\alpha t_r)^2 - (\alpha t_r)^2) = \frac{c\alpha}{T^3}$$

Now calculate t_r : $c^2(t - t_r)^2 = \mathbf{z} \cdot \mathbf{z} = l^2 + d^2$; $l = x(t) - x(t_r) = \frac{c}{\alpha} [\sqrt{1 + (\alpha t)^2} - \sqrt{1 + (\alpha t_r)^2}]$, so
 $t^2 - 2tt_r + t_r^2 = \frac{1}{\alpha^2} [1 + (\alpha t)^2 + 1 + (\alpha t_r)^2 - 2\sqrt{1 + (\alpha t)^2}\sqrt{1 + (\alpha t_r)^2}] + (d/c)^2$
 (★) $\sqrt{1 + (\alpha t)^2}\sqrt{1 + (\alpha t_r)^2} = 1 + \alpha^2 tt_r + \frac{1}{2} \left(\frac{\alpha d}{c} \right)^2$. Square both sides:

$$\begin{aligned} \cancel{t^2} + (\alpha t)^2 + (\alpha t_r)^2 + \cancel{\alpha^4 t_r^2} &= \cancel{t^2} + \cancel{\alpha^4 t_r^2} + \frac{1}{4} \left(\frac{\alpha d}{c} \right)^4 + 2\alpha^2 tt_r + \left(\frac{\alpha d}{c} \right)^2 + \alpha^2 tt_r \left(\frac{\alpha d}{c} \right)^2 \\ t^2 + t_r^2 - 2tt_r - tt_r \left(\frac{\alpha d}{c} \right)^2 - \left(\frac{d}{c} \right)^2 - \frac{\alpha^2}{4} \left(\frac{d}{c} \right)^4 &= 0 \end{aligned}$$

At this point we *could* solve for t_r (in terms of t), but since v and a are already expressed in terms of t_r , it is simpler to solve for t (in terms of t_r), and express everything in terms of t_r :

$$\begin{aligned} t^2 - tt_r \left[2 + \left(\frac{\alpha d}{c} \right)^2 \right] + \left[t_r^2 - \left(\frac{d}{c} \right)^2 - \frac{\alpha^2}{4} \left(\frac{d}{c} \right)^4 \right] &= 0 \Rightarrow \\ t &= \frac{1}{2} \left\{ t_r \left[2 + \left(\frac{\alpha d}{c} \right)^2 \right] \pm \sqrt{t_r^2 \left[4 + 4 \left(\frac{\alpha d}{c} \right)^2 + \left(\frac{\alpha d}{c} \right)^4 \right] - 4 \left(\frac{d}{c} \right)^2 + \alpha^2 \left(\frac{d}{c} \right)^4} \right\} \\ &= t_r \left[1 + \frac{1}{2} \left(\frac{\alpha d}{c} \right)^2 \right] \pm \sqrt{[1 + (\alpha t_r)^2] \left(\frac{d}{c} \right)^2 \left[1 + \left(\frac{\alpha d}{2c} \right)^2 \right]} \end{aligned}$$



Which sign? For small α we want $t \approx t_r + d/c$, so we need the + sign:

$$t = t_r \left[1 + \frac{1}{2} \left(\frac{\alpha d}{c} \right)^2 \right] + \frac{d}{c} TD, \text{ where } D = \sqrt{1 + \left(\frac{\alpha d}{2c} \right)^2}$$

So $\mathcal{Z} = c(t - t_r) \Rightarrow \mathcal{Z} = \frac{ct_r}{2} \left(\frac{\alpha d}{c} \right)^2 + dTD$. Now go back to Eq. (★) and solve for $\sqrt{1 + (\alpha t)^2}$:

$$\begin{aligned} \sqrt{1 + (\alpha t)^2} &= \frac{1}{T} \left\{ 1 + \frac{1}{2} \left(\frac{\alpha d}{c} \right)^2 + \alpha^2 t_r \left[t_r \left(1 + \frac{1}{2} \left(\frac{\alpha d}{c} \right)^2 \right) + \frac{d}{c} TD \right] \right\} \\ &= \frac{1}{T} \left\{ \underbrace{\left[1 + (\alpha t_r)^2 \right]}_{T^2} \left[1 + \frac{1}{2} \left(\frac{\alpha d}{c} \right)^2 \right] + \frac{\alpha^2 t_r d}{c} TD \right\} = \left[1 + \frac{1}{2} \left(\frac{\alpha d}{c} \right)^2 \right] T + \frac{\alpha^2 t_r d}{c} D \end{aligned}$$

$$l = \frac{c}{\alpha} \left[\sqrt{1 + (\alpha t)^2} - \sqrt{1 + (\alpha t_r)^2} \right] = \frac{c}{\alpha} \left\{ \left[\mathcal{X} + \frac{1}{2} \left(\frac{\alpha d}{c} \right)^2 \right] T + \frac{\alpha^2 t_r d}{c} D - \mathcal{X} \right\} = \alpha d \left(\frac{d}{2c} T + t_r D \right)$$

Putting all this in, the numerator in square brackets in **F** becomes:

$$\begin{aligned} [] &= \left\{ \alpha d \left(\frac{d}{2c} T + t_r D \right) - \frac{c \alpha t_r}{T} \left[\frac{ct_r}{2} \left(\frac{\alpha d}{c} \right)^2 + dTD \right] \right\} \left[c^2 - \frac{c^2 \alpha^2 t_r^2}{T^2} \right] - c \frac{c \alpha}{T_3} d^2 \\ &= \alpha d \left[\frac{d}{2c} T + t_r D - \frac{d(\alpha t_r)^2}{2cT} - t_r D \right] \frac{c^2}{T_2} [1 + (\cancel{\alpha t_r})^2 - (\cancel{\alpha t_r})^2] - \frac{c^2 \alpha d^2}{T^3} \\ &= \frac{c^2 \alpha d^2}{T^3} \left[\frac{1}{2} T^2 - \frac{1}{2} (\alpha t_r)^2 - 1 \right] = \frac{c^2 \alpha d^2}{2T^3} [1 + (\alpha t_r)^2 - (\alpha t_r)^2 - 2] = -\frac{c^2 \alpha d^2}{2T^3} \end{aligned}$$

$$\therefore \mathbf{F} = \frac{q^2}{4\pi\epsilon_0} \frac{c^2 \alpha d^2}{[(c\mathcal{Z} - lv)T]^3} \hat{\mathbf{x}}. \text{ It remains to compute the denominator:}$$

$$\begin{aligned} (c\mathcal{Z} - lv)T &= \left\{ c \left[\frac{ct_r}{2} \left(\frac{\alpha d}{c} \right)^2 + dTD \right] - \alpha d \left(\frac{d}{2c} T + t_r D \right) \frac{c \alpha t_r}{T} \right\} T \\ &= \left[\frac{1}{2} \alpha^2 \cancel{t_r} d^2 + cdTD - \frac{1}{2} \alpha^2 \cancel{t_r} d^2 - \frac{cd(\alpha t_r)^2}{T} D \right] T = cdD \left[\underbrace{T^2 - (\alpha t_r)^2}_{1 + (\cancel{\alpha t_r})^2 - (\cancel{\alpha t_r})^2} \right] = dcD \end{aligned}$$

$$\therefore \mathbf{F} = \frac{q^2}{4\pi\epsilon_0} \frac{c^2 d^2 \alpha}{c^3 d^3 D^3} \hat{\mathbf{x}} = \boxed{\frac{q^2}{4\pi\epsilon_0} \frac{\alpha}{cd \left[1 + \left(\frac{\alpha d}{2c} \right)^2 \right]^{3/2}}} \hat{\mathbf{x}} \quad \left(\alpha = \frac{F}{mc} \right)$$

Energy must come from the “reservoir” of energy stored in the electromagnetic fields.

$$(b) F = mc\alpha = \frac{1}{2} \frac{q^2}{4\pi\epsilon_0} \frac{\alpha}{cd \left[1 + \left(\frac{\alpha d}{2c} \right)^2 \right]^{3/2}} \Rightarrow \left[1 + \left(\frac{\alpha d}{2c} \right)^2 \right]^{3/2} = \frac{q^2}{8\pi\epsilon_0 mc^2 d} = \left(\frac{\mu_0 q^2}{8\pi md} \right).$$

(force on one end only)

$$\therefore \alpha = \frac{2c}{d} \sqrt{\left(\frac{\mu_0 q^2}{8\pi md} \right)^{2/3} - 1}, \quad \text{so} \quad \boxed{F = \frac{2mc^2}{d} \sqrt{\left(\frac{\mu_0 q^2}{8\pi md} \right)^{2/3} - 1}}.$$

Problem 12.64

(a) $A^\mu = (V/c, A_x, A_y, A_z)$ is a 4-vector (like $x^\mu = (ct, x, y, z)$), so (using Eq. 12.19): $V = \gamma(\bar{V} + v\bar{A}_x)$. But $\bar{V} = 0$, and

$$\bar{A}_x = \frac{\mu_0}{4\pi} \frac{(\mathbf{m} \times \bar{\mathbf{r}})_x}{\bar{r}^3}$$

Now $(\mathbf{m} \times \bar{\mathbf{r}})_x = m_y \bar{z} - m_z \bar{y} = m_y z - m_z y$. So

$$V = \gamma v \frac{\mu_0}{4\pi} \frac{(m_y z - m_z y)}{\bar{r}^3}$$

Now $\bar{x} = \gamma(x - vt) = \gamma R_x$, $\bar{y} = y = R_y$, $\bar{z} = z = R_z$, where $\mathbf{R}$ is the vector (in $\mathcal{S}$) from the (instantaneous) location of the dipole to the point of observation. Thus

$$\bar{r}^2 = \gamma^2 R_x^2 + R_y^2 + R_z^2 = \gamma^2 (R_x^2 + R_y^2 + R_z^2) + (1 - \gamma^2)(R_y^2 + R_z^2) = \gamma^2 \left(R_2 - \frac{v^2}{c^2} R^2 \sin^2 \theta \right)$$

(where θ is the angle between $\mathbf{R}$ and the x -axis, so that $R_y^2 + R_z^2 = R^2 \sin^2 \theta$).

$$\therefore V = \frac{\mu_0}{4\pi} \frac{v\gamma(m_y R_z - m_z R_y)}{\gamma^3 R^3 \left(1 - \frac{v^2}{c^2} \sin^2 \theta\right)^{3/2}}; \quad \mathbf{v} \cdot (\mathbf{m} \times \mathbf{R}) = v(\mathbf{m} \times \mathbf{R})_x = v(m_y R_z - m_z R_y), \quad \text{so}$$

$$V = \frac{\mu_0}{4\pi} \frac{\mathbf{v} \cdot (\mathbf{m} \times \mathbf{R}) \left(1 - \frac{v^2}{c^2}\right)}{R^3 \left(1 - \frac{v^2}{c^2} \sin^2 \theta\right)^{3/2}},$$

or, using $\mu_0 = \frac{1}{\epsilon_0 c^2}$ and $\mathbf{v} \cdot (\mathbf{m} \times \mathbf{R}) = \mathbf{R} \cdot (\mathbf{v} \times \mathbf{m})$:
$$V = \frac{1}{4\pi\epsilon_0} \frac{\hat{\mathbf{R}} \cdot (\mathbf{v} \times \mathbf{m}) \left(1 - \frac{v^2}{c^2}\right)}{c^2 R^2 \left(1 - \frac{v^2}{c^2} \sin^2 \theta\right)^{3/2}}$$

(b) In the nonrelativistic limit ($v^2 \ll c^2$):

$$V = \frac{1}{4\pi\epsilon_0} \frac{\hat{\mathbf{R}} \cdot (\mathbf{v} \times \mathbf{m})}{c^2 R^2} = \frac{1}{4\pi\epsilon_0} \frac{\hat{\mathbf{R}} \cdot \mathbf{p}}{R^2}, \quad \text{with} \quad \mathbf{p} = \frac{\mathbf{v} \times \mathbf{m}}{c^2},$$

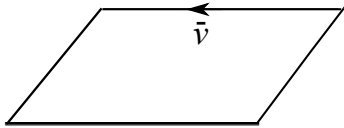
which is the potential of an *electric* dipole.

Problem 12.65

(a) $\mathbf{B} = -\frac{\mu_0}{2} K \hat{\mathbf{y}}$ (Eq. 5.58); $\mathbf{N} = \mathbf{m} \times \mathbf{B}$ (Eq. 6.1), so $\mathbf{N} = -\frac{\mu_0}{2} mK (\hat{\mathbf{z}} \times \hat{\mathbf{y}})$.

$$\boxed{\mathbf{N} = \frac{\mu_0}{2} mK \hat{\mathbf{x}}} = \frac{\mu_0}{2} (\lambda v l^2) (\sigma v) \hat{\mathbf{x}} = \frac{\mu_0}{2} \lambda \sigma v^2 l^2 \hat{\mathbf{x}}.$$

(b)

Charge density in the front side: λ_0 ($\lambda = \gamma\lambda_0$);Charge density on the back side: $\bar{\lambda} = \bar{\gamma}\lambda_0$, where $\bar{v} = \frac{2v}{1+v^2/c^2}$,

$$\text{so } \bar{\gamma} = \frac{1}{\sqrt{1 - \frac{4v^2/c^2}{(1+v^2/c^2)^2}}} = \frac{(1+v^2/c^2)}{\sqrt{1 + 2\frac{v^2}{c^2} + \frac{v^4}{c^4} - 4\frac{v^2}{c^2}}} = \frac{1+v^2/c^2}{\sqrt{1 - 2\frac{v^2}{c^2} + \frac{v^4}{c^4}}} = \frac{(1+v^2/c^2)}{(1-v^2/c^2)} = \gamma^2 \left(1 + \frac{v^2}{c^2}\right)$$

Length of front and back sides in this frame: l/γ . So net charge on back side is:

$$q_+ = \bar{\lambda} \frac{l}{\gamma} = \gamma^2 \left(1 + \frac{v^2}{c^2}\right) \frac{\lambda l}{\gamma} = \left(1 + \frac{v^2}{c^2}\right) \lambda l$$

Net charge on front side is:

$$q_- = \lambda_0 \frac{l}{\gamma} = \frac{\lambda l}{\gamma} = \frac{1}{\gamma^2} \lambda l$$

So dipole moment (note: charges on *sides* are equal):

$$\mathbf{p} = (q_+) \frac{l}{2} \hat{\mathbf{y}} - (q_-) \frac{l}{2} \hat{\mathbf{y}} = \left[\left(1 + \frac{v^2}{c^2}\right) \lambda \frac{l}{2} - \frac{1}{\gamma^2} \lambda \frac{l}{2} \right] \hat{\mathbf{y}} = \frac{\lambda l^2}{2} \left(1 + \frac{v^2}{c^2} - 1 + \frac{v^2}{c^2}\right) \hat{\mathbf{y}} = \boxed{\frac{\lambda l^2 v^2}{c^2} \hat{\mathbf{y}}}.$$

$$\mathbf{E} = \frac{\sigma_0}{2\epsilon_0} \hat{\mathbf{z}}, \text{ where } \sigma = \gamma\sigma_0, \text{ so } \mathbf{N} = \mathbf{p} \times \mathbf{E} = \frac{\lambda l^2 v^2}{c^2} \frac{\sigma}{2\epsilon_0 \gamma} (\hat{\mathbf{y}} \times \hat{\mathbf{z}}) = \boxed{\frac{1}{\gamma} \frac{\mu_0}{2} \lambda \sigma l^2 v^2 \hat{\mathbf{x}}}.$$

So apart from the relativistic factor of γ the torque is the same in both systems — but in $\mathcal{S}$ it is the torque exerted by a *magnetic* field on a *magnetic* dipole, whereas in $\bar{\mathcal{S}}$ it is the torque exerted by an *electric* field on an *electric* dipole.

Problem 12.66

Choose axes so that $\mathbf{E}$ points in the z direction and $\mathbf{B}$ in the yz plane: $\mathbf{E} = (0, 0, E)$; $\mathbf{B} = (0, B \cos \phi, B \sin \phi)$. Go to a frame moving at speed v in the x direction:

$$\bar{\mathbf{E}} = (0, -\gamma v B \sin \phi, \gamma(E + v B \cos \phi)); \quad \bar{\mathbf{B}}(0, \gamma(B \cos \phi + \frac{v}{c^2} E), \gamma B \sin \phi).$$

(I used Eq. 12.109.) Parallel provided $\frac{-\gamma v B \sin \phi}{\gamma(B \cos \phi + \frac{v}{c^2} E)} = \frac{\gamma(E + v B \cos \phi)}{\gamma B \sin \phi}$, or

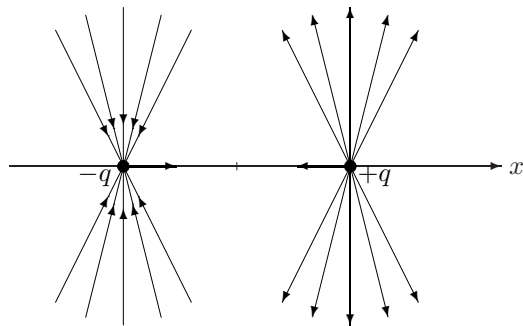
$$-v B^2 \sin^2 \phi = (B \cos \phi + \frac{v}{c^2} E)(E + v B \cos \phi) = E B \cos \phi + v B^2 \cos^2 \phi + \frac{v}{c^2} E^2 + \frac{v^2}{c^2} E B \cos \phi$$

$$0 = v B^2 + \frac{v}{c^2} E^2 + E B \cos \phi \left(1 + \frac{v^2}{c^2}\right); \quad \frac{v}{1 + v^2/c^2} = -\frac{E B \cos \phi}{B^2 + E^2/c^2}$$

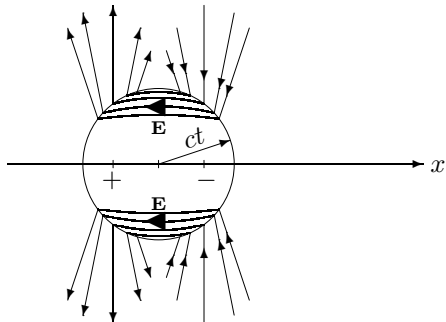
Now $\mathbf{E} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0 & E \\ 0 & B \cos \phi & B \sin \phi \end{vmatrix} = -E B \cos \phi \hat{\mathbf{x}}$. So $\frac{\mathbf{v}}{1 + v^2/c^2} = \frac{\mathbf{E} \times \mathbf{B}}{B^2 + E^2/c^2}$. qed

No, there can be no frame in which $\mathbf{E} \perp \mathbf{B}$, for $(\mathbf{E} \cdot \mathbf{B})$ is invariant, and since it is not zero in $\mathcal{S}$ it can't be zero in $\bar{\mathcal{S}}$.

Problem 12.67



Just before:
Field lines emanate
from *present* position
of particle.



Just after: Field lines *outside* sphere of radius ct emanate from position particle *would* have reached, had it kept going on its original “flight plan”. *Inside* the sphere $E = 0$. On the surface the lines connect up (since they cannot simply *terminate* in empty space), as suggested in the figure.

This produces a dense cluster of tangentially-directed field lines, which expand with the spherical shell. This is a pictorial way of understanding the generation of *electromagnetic radiation*.

Problem 12.68

Equation 12.67 assumes the particle is (instantaneously) at rest in $\mathcal{S}$. Here the particle is at rest in $\bar{\mathcal{S}}$. So $\mathbf{F}_\perp = \frac{1}{\gamma} \bar{\mathbf{F}}_\perp$, $F_\parallel = \bar{F}_\parallel$. Using $\bar{\mathbf{F}} = q\bar{\mathbf{E}}$, then,

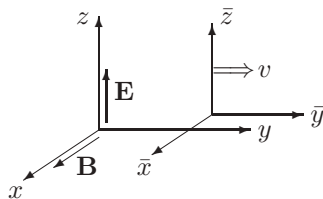
$$F_x = \bar{F}_x = q\bar{E}_x, \quad F_y = \frac{1}{\gamma} \bar{F}_y = \frac{1}{\gamma} q\bar{E}_y, \quad F_z = \frac{1}{\gamma} \bar{F}_z = \frac{1}{\gamma} q\bar{E}_z.$$

Invoking Eq. 12.109:

$$F_x = qE_x, \quad F_y = \frac{1}{\gamma} q\gamma(E_y - vB_z) = q(E_y - vB_z) \quad F_z = \frac{1}{\gamma} q\gamma(E_z + vB_y) = q(E_z + vB_y).$$

But $\mathbf{v} \times \mathbf{B} = -vB_z \hat{\mathbf{x}} + vB_y \hat{\mathbf{z}}$, so $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. *qed*

Problem 12.69



Rewrite Eq. 12.109 with $x \rightarrow y$, $y \rightarrow z$, $z \rightarrow x$:

$$\begin{aligned} \bar{E}_y &= E_y & \bar{E}_z &= \gamma(E_z - vB_x) & \bar{E}_x &= \gamma(E_x + vB_z) \\ \bar{B}_y &= B_y & \bar{B}_z &= \gamma\left(B_z + \frac{v}{c^2}E_x\right) & \bar{B}_x &= \gamma\left(B_x - \frac{v}{c^2}E_z\right) \end{aligned}$$

This gives the fields in system $\bar{\mathcal{S}}$ moving in the y direction at speed v .

Now $\mathbf{E} = (0, 0, E_0)$; $\mathbf{B} = (B_0, 0, 0)$, so $\bar{E}_y = 0$, $\bar{E}_z = \gamma(E_0 - vB_0)$, $\bar{E}_x = 0$.

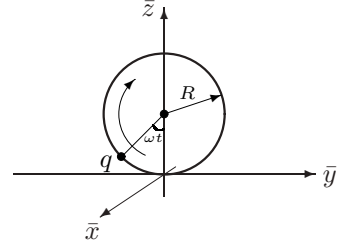
If we want $\bar{\mathbf{E}} = \mathbf{0}$, we must pick v so that $E_0 - vB_0 = 0$; i.e. $\boxed{v = E_0/B_0}$

(The condition $E_0/B_0 < c$ guarantees that there is no problem *getting* to such a system.)

With this, $\bar{B}_y = 0$, $\bar{B}_z = 0$, $\bar{B}_x = \gamma(B_0 - \frac{v}{c^2}E_0) = \gamma B_0(1 - \frac{v^2}{c^2}) = \gamma B_0 \frac{1}{\gamma^2} = \frac{1}{\gamma}B_0$; $\boxed{\bar{\mathbf{B}} = \frac{1}{\gamma}B_0\hat{\mathbf{x}}}$.

The trajectory in $\bar{\mathcal{S}}$: Since the particle started out at rest at the origin in $\mathcal{S}$, it started out with velocity $-v\hat{\mathbf{y}}$ in $\bar{\mathcal{S}}$. According to Eq. 12.71 it will move in a circle of radius R , given by

$$p = qBR, \text{ or } \gamma mv = q\left(\frac{1}{\gamma}B_0\right)R \Rightarrow \boxed{R = \frac{m\gamma^2 v}{qB_0}}.$$



The actual trajectory is given by $\boxed{\bar{x} = 0; \bar{y} = -R \sin \omega \bar{t}; \bar{z} = R(1 - \cos \omega \bar{t})}$; where $\boxed{\omega = \frac{v}{R}}$.

The trajectory in $\mathcal{S}$: The Lorentz transformations Eqs. 12.18 and 12.19, for the case of relative motion in the y -direction, read:

$$\begin{aligned} \bar{x} &= x & x &= \bar{x} \\ \bar{y} &= \gamma(y - vt) & y &= \gamma(\bar{y} + v\bar{t}) \\ \bar{z} &= z & z &= \bar{z} \\ \bar{t} &= \gamma\left(t - \frac{v}{c^2}y\right) & t &= \gamma\left(\bar{t} + \frac{v}{c^2}\bar{y}\right) \end{aligned}$$

So the trajectory in $\mathcal{S}$ is given by:

$$\begin{aligned} x &= 0; y = \gamma(-R \sin \omega \bar{t} + v\bar{t}) = \gamma\left\{-R \sin\left[\omega\gamma\left(t - \frac{v}{c^2}y\right)\right] + v\gamma\left(t - \frac{v}{c^2}y\right)\right\}, \text{ or} \\ \underbrace{y\left(1 + \gamma^2 \frac{v^2}{c^2}\right)}_{\gamma^2 y(1 - \frac{v^2}{c^2} + \frac{v^2}{c^2}) = \gamma^2 y} &= \gamma^2 vt - \gamma R \sin\left[\omega\gamma\left(t - \frac{v}{c^2}y\right)\right] \Big\} (y - vt)\gamma = -R \sin\left[\omega\gamma\left(t - \frac{v}{c^2}y\right)\right]; \\ z &= R(1 - \cos^2 \omega \bar{t}) = R\left[1 - \cos \omega\gamma\left(t - \frac{v}{c^2}y\right)\right]. \end{aligned}$$

So: $\boxed{x = 0; y = vt - \frac{R}{\gamma} \sin\left[\omega\gamma\left(t - \frac{v}{c^2}y\right)\right]; z = R - R \cos\left[\omega\gamma\left(t - \frac{v}{c^2}y\right)\right]}.$

We can get rid of the trigonometric terms by the usual trick:

$$\left. \begin{aligned} \gamma(y - vt) &= -R \sin\left[\omega\gamma\left(t - \frac{v}{c^2}y\right)\right] \\ z - R &= -R \cos\left[\omega\gamma\left(t - \frac{v}{c^2}y\right)\right] \end{aligned} \right\} \Rightarrow \boxed{\gamma^2(y - vt)^2 + (z - R)^2 = R^2}.$$

Absent the γ^2 , this would be the cycloid we found back in Ch. 5 (Eq. 5.9). The γ^2 makes it, as it were, an *elliptical* cycloid — same picture as Fig. 5.7, but with the horizontal axis stretched out.

Problem 12.70

(a) $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$ suggests $\mathbf{E} \rightarrow \frac{1}{\epsilon_0} \mathbf{D}$
 $\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}$ suggests $\mathbf{B} \rightarrow \mu_0 \mathbf{H}$ } but it's a little cleaner if we divide by μ_0 while we're at it, so that

$\mathbf{E} \rightarrow \frac{1}{\mu_0 \epsilon_0} \mathbf{D} = c^2 \mathbf{D}$, $\mathbf{B} \rightarrow \mathbf{H}$. Then:

$$D^{\mu\nu} = \begin{Bmatrix} 0 & cD_x & cD_y & cD_z \\ -cD_x & 0 & H_z & -H_y \\ -cD_y & -H_z & 0 & H_x \\ -cD_z & H_y & -H_x & 0 \end{Bmatrix}$$

Then (following the derivation in Sect. 12.3.4):

$$\frac{\partial}{\partial x^\nu} D^{0\nu} = c \nabla \cdot \mathbf{D} = c \rho_f = J_f^0; \quad \frac{\partial}{\partial x^\nu} D^{1\nu} = \frac{1}{c} \frac{\partial}{\partial t} (-c D_x) + (\nabla \times \mathbf{H})_x = (J_f)_x; \quad \text{so} \quad \boxed{\frac{\partial D_{\mu\nu}}{\partial x^\nu} = J_f^\mu},$$

where $\boxed{J_f^\mu = (c \rho_f, \mathbf{J}_f)}$. Meanwhile, the homogeneous Maxwell equations $(\nabla \cdot \mathbf{B} = 0, \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t})$ are unchanged,

and hence $\boxed{\frac{\partial G^{\mu\nu}}{\partial x^\nu} = 0}$.

(b)
$$\boxed{H^{\mu\nu} = \begin{Bmatrix} 0 & H_x & H_y & H_z \\ -H_x & 0 & -c D_z & c D_y \\ -H_y & c D_z & 0 & -c D_x \\ -H_z & -c D_y & c D_x & 0 \end{Bmatrix}}$$

(c) If the material is at rest, $\eta_\nu = (-c, 0, 0, 0)$, and the sum over ν collapses to a single term:

$$D^{\mu 0} \eta_0 = c^2 \epsilon F^{\mu 0} \eta_0 \Rightarrow D^{\mu 0} = c^2 \epsilon F^{\mu 0} \Rightarrow -c \mathbf{D} = -c^2 \epsilon \frac{\mathbf{E}}{c} \Rightarrow \mathbf{D} = \epsilon \mathbf{E} \text{ (Eq. 4.32), } \checkmark$$

$$H^{\mu 0} \eta_0 = \frac{1}{\mu} G^{\mu 0} \eta_0 \Rightarrow H^{\mu 0} = \frac{1}{\mu} G^{\mu 0} \Rightarrow -\mathbf{H} = -\frac{1}{\mu} \mathbf{B} \Rightarrow \mathbf{H} = \frac{1}{\mu} \mathbf{B} \text{ (Eq. 6.31). } \checkmark$$

(d) In general, $\eta_\nu = \gamma(-c, \mathbf{u})$, so, for $\mu = 0$:

$$D^{0\nu} \eta_\nu = D^{01} \eta_1 + D^{02} \eta_2 + D^{03} \eta_3 = c D_x (\gamma u_x) + c D_y (\gamma u_y) + c D_z (\gamma u_z) = \gamma c (\mathbf{D} \cdot \mathbf{u}),$$

$$F^{0\nu} \eta_\nu = F^{01} \eta_1 + F^{02} \eta_2 + F^{03} \eta_3 = \frac{E_x}{c} (\gamma u_x) + \frac{E_y}{c} (\gamma u_y) + \frac{E_z}{c} (\gamma u_z) = \frac{\gamma}{c} (\mathbf{E} \cdot \mathbf{u}), \text{ so}$$

$$D^{0\nu} \eta_\nu = c^2 \epsilon F^{0\nu} \eta_\nu \Rightarrow \gamma c (\mathbf{D} \cdot \mathbf{u}) = c^2 \epsilon \left(\frac{\gamma}{c} \right) (\mathbf{E} \cdot \mathbf{u}) \Rightarrow \mathbf{D} \cdot \mathbf{u} = \epsilon (\mathbf{E} \cdot \mathbf{u}). \quad [1]$$

$$H^{0\nu} \eta_\nu = H^{01} \eta_1 + H^{02} \eta_2 + H^{03} \eta_3 = H_x (\gamma u_x) + H_y (\gamma u_y) + H_z (\gamma u_z) = \gamma (\mathbf{H} \cdot \mathbf{u}),$$

$$G^{0\nu} \eta_\nu = G^{01} \eta_1 + G^{02} \eta_2 + G^{03} \eta_3 = B_x (\gamma u_x) + B_y (\gamma u_y) + B_z (\gamma u_z) = \gamma (\mathbf{B} \cdot \mathbf{u}), \text{ so}$$

$$H^{0\nu} \eta_\nu = \frac{1}{\mu} G^{0\nu} \eta_\nu \Rightarrow \gamma (\mathbf{H} \cdot \mathbf{u}) = \frac{1}{\mu} \gamma (\mathbf{B} \cdot \mathbf{u}) \Rightarrow \mathbf{H} \cdot \mathbf{u} = \frac{1}{\mu} (\mathbf{B} \cdot \mathbf{u}). \quad [2]$$

Similarly, for $\mu = 1$:

$$\begin{aligned} D^{1\nu} \eta_\nu &= D^{10} \eta_0 + D^{12} \eta_2 + D^{13} \eta_3 = (-c D_x) (-\gamma c) + H_z (\gamma u_y) + (-H_y) (\gamma u_z) = \gamma (c^2 D_x + u_y H_z - u_z H_y) \\ &= \gamma [c^2 \mathbf{D} + (\mathbf{u} \times \mathbf{H})]_x, \end{aligned}$$

$$\begin{aligned} F^{1\nu} \eta_\nu &= F^{10} \eta_0 + F^{12} \eta_2 + F^{13} \eta_3 = \frac{-E_x}{c} (-\gamma c) + B_z (\gamma u_y) + (-B_y) (\gamma u_z) = \gamma (E_x + u_y B_z - u_z B_y) \\ &= \gamma [\mathbf{E} + (\mathbf{u} \times \mathbf{B})]_x, \quad \text{so} \quad D^{1\nu} \eta_\nu = c^2 \epsilon F^{1\nu} \eta_\nu \Rightarrow \end{aligned}$$

$$\gamma [c^2 \mathbf{D} + (\mathbf{u} \times \mathbf{H})]_x = c^2 \epsilon \gamma [\mathbf{E} + (\mathbf{u} \times \mathbf{B})]_x \Rightarrow \mathbf{D} + \frac{1}{c^2} (\mathbf{u} \times \mathbf{H}) = \epsilon [\mathbf{E} + (\mathbf{u} \times \mathbf{B})]. \quad [3]$$

$$\begin{aligned} H^{1\nu} \eta_\nu &= H^{10} \eta_0 + H^{12} \eta_2 + H^{13} \eta_3 = (-H_x) (-\gamma c) + (-c D_z) (\gamma u_y) + (c D_y) (\gamma u_z) \\ &= \gamma c (H_x - u_y D_z + u_z D_y) = \gamma c [\mathbf{H} - (\mathbf{u} \times \mathbf{D})]_x, \end{aligned}$$

$$\begin{aligned}
G^{1\nu}\eta_\nu &= G^{10}\eta_0 + G^{12}\eta_2 + G^{13}\eta_3 = (-B_x)(-\gamma c) + \left(-\frac{E_z}{c}\right)(\gamma u_y) + \left(\frac{E_y}{c}\right)(\gamma u_z) \\
&= \frac{\gamma}{c}(c^2 B_x - u_y E_z + u_z E_y) = \frac{\gamma}{c} [c^2 \mathbf{B} - (\mathbf{u} \times \mathbf{E})]_x, \quad \text{so} \quad H^{1\nu}\eta_\nu = \frac{1}{\mu} G^{1\nu}\eta_\nu \Rightarrow \\
\gamma c [\mathbf{H} - (\mathbf{u} \times \mathbf{D})]_x &= \frac{1}{\mu} \frac{\gamma}{c} [c^2 \mathbf{B} - (\mathbf{u} \times \mathbf{E})]_x \Rightarrow \mathbf{H} - (\mathbf{u} \times \mathbf{D}) = \frac{1}{\mu} \left[\mathbf{B} - \frac{1}{c^2} (\mathbf{u} \times \mathbf{E}) \right]. \tag{4}
\end{aligned}$$

Use Eq. [4] as an expression for $\mathbf{H}$, plug this into Eq. [3], and solve for $\mathbf{D}$:

$$\begin{aligned}
\mathbf{D} + \frac{1}{c^2} \mathbf{u} \times \left\{ (\mathbf{u} \times \mathbf{D}) + \frac{1}{\mu} \left[\mathbf{B} - \frac{1}{c^2} (\mathbf{u} \times \mathbf{E}) \right] \right\} &= \epsilon [\mathbf{E} + (\mathbf{u} \times \mathbf{B})]; \\
\mathbf{D} + \frac{1}{c^2} [(\mathbf{u} \cdot \mathbf{D})\mathbf{u} - u^2 \mathbf{D}] &= \epsilon [\mathbf{E} + (\mathbf{u} \times \mathbf{B})] - \frac{1}{\mu c^2} (\mathbf{u} \times \mathbf{B}) + \frac{1}{\mu c^4} [\mathbf{u} \times (\mathbf{u} \times \mathbf{E})].
\end{aligned}$$

Using Eq. [1] to rewrite $(\mathbf{u} \cdot \mathbf{D})$:

$$\begin{aligned}
\mathbf{D} \left(1 - \frac{u^2}{c^2} \right) &= -\frac{\epsilon}{c^2} (\mathbf{E} \cdot \mathbf{u})\mathbf{u} + \epsilon [\mathbf{E} + (\mathbf{u} \times \mathbf{B})] - \frac{1}{\mu c^2} (\mathbf{u} \times \mathbf{B}) + \frac{1}{\mu c^4} [(\mathbf{E} \cdot \mathbf{u})\mathbf{u} - u^2 \mathbf{E}] \\
&= \epsilon \left\{ \left[1 - \frac{u^2}{\epsilon \mu c^4} \right] \mathbf{E} - \frac{1}{c^2} \left[1 - \frac{1}{\epsilon \mu c^2} \right] (\mathbf{E} \cdot \mathbf{u})\mathbf{u} + (\mathbf{u} \times \mathbf{B}) \left[1 - \frac{1}{\epsilon \mu c^2} \right] \right\}.
\end{aligned}$$

Let $\boxed{\gamma \equiv \frac{1}{\sqrt{1 - u^2/c^2}}, \quad v \equiv \frac{1}{\sqrt{\epsilon \mu}}.}$ Then

$$\boxed{\mathbf{D} = \gamma^2 \epsilon \left\{ \left(1 - \frac{u^2 v^2}{c^4} \right) \mathbf{E} + \left(1 - \frac{v^2}{c^2} \right) \left[(\mathbf{u} \times \mathbf{B}) - \frac{1}{c^2} (\mathbf{E} \cdot \mathbf{u})\mathbf{u} \right] \right\}.}$$

Now use Eq. [3] as an expression for $\mathbf{D}$, plug this into Eq. [4], and solve for $\mathbf{H}$:

$$\begin{aligned}
\mathbf{H} - \mathbf{u} \times \left\{ -\frac{1}{c^2} (\mathbf{u} \times \mathbf{H}) + \epsilon [\mathbf{E} + (\mathbf{u} \times \mathbf{B})] \right\} &= \frac{1}{\mu} \left[\mathbf{B} - \frac{1}{c^2} (\mathbf{u} \times \mathbf{E}) \right]; \\
\mathbf{H} + \frac{1}{c^2} [(\mathbf{u} \cdot \mathbf{H})\mathbf{u} - u^2 \mathbf{H}] &= \frac{1}{\mu} \left[\mathbf{B} - \frac{1}{c^2} (\mathbf{u} \times \mathbf{E}) \right] + \epsilon (\mathbf{u} \times \mathbf{E}) + \epsilon [\mathbf{u} \times (\mathbf{u} \times \mathbf{B})].
\end{aligned}$$

Using Eq. [2] to rewrite $(\mathbf{u} \cdot \mathbf{H})$:

$$\begin{aligned}
\mathbf{H} \left(1 - \frac{u^2}{c^2} \right) &= -\frac{1}{\mu c^2} (\mathbf{B} \cdot \mathbf{u})\mathbf{u} + \frac{1}{\mu} \left[\mathbf{B} - \frac{1}{c^2} (\mathbf{u} \times \mathbf{E}) \right] + \epsilon (\mathbf{u} \times \mathbf{E}) + \epsilon [(\mathbf{B} \cdot \mathbf{u})\mathbf{u} - u^2 \mathbf{B}] \\
&= \frac{1}{\mu} \left\{ [1 - \mu \epsilon u^2] \mathbf{B} + \left[\epsilon \mu - \frac{1}{c^2} \right] [(\mathbf{u} \times \mathbf{E}) + (\mathbf{B} \cdot \mathbf{u})\mathbf{u}] \right\}.
\end{aligned}$$

$$\boxed{\mathbf{H} = \frac{\gamma^2}{\mu} \left\{ \left(1 - \frac{u^2}{v^2} \right) \mathbf{B} + \left(\frac{1}{v^2} - \frac{1}{c^2} \right) [(\mathbf{u} \times \mathbf{E}) + (\mathbf{B} \cdot \mathbf{u})\mathbf{u}] \right\}.}$$

Problem 12.71

We know that (proper) power transforms as the zeroth component of a 4-vector $K^0 = \frac{1}{c} \frac{dW}{d\tau}$. The Larmor formula says that for $v = 0$, $\frac{dW}{d\tau} = \frac{1}{4\pi\epsilon_0} \frac{2}{3} \frac{q^2}{c^3} a^2$ (Eq. 11.70). Can we think of a 4-vector whose zeroth component reduces to this when the velocity is zero?

Well, a^2 smells like $(\alpha^\nu \alpha_\nu)$, but how do we get a 4-vector in here? How about η^μ , whose zeroth component is just c , when $v = 0$? Try, then:

$$K^\mu = \frac{1}{4\pi\epsilon_0} \frac{2}{3} \frac{q^2}{c^5} (\alpha^\nu \alpha_\nu) \eta_\mu$$

This has the right transformation properties, but we must check that it does reduce to the Larmor formula when $v = 0$:

$\frac{dW}{dt} = \frac{1}{\gamma} \frac{dW}{d\tau} = \frac{1}{\gamma} c K^0 = \frac{1}{\gamma} c \frac{\mu_0 q^2}{6\pi c^3} (\alpha^\nu \alpha_\nu) \eta^0$, but $\eta^0 = c\gamma$, so $\frac{dW}{dt} = \frac{\mu_0 q^2}{6\pi c} (\alpha^\nu \alpha_\nu)$. [Incidentally, this tells us that the power itself (as opposed to *proper* power) is a *scalar*. If this had been obvious from the start, we could simply have looked for a Lorentz *scalar* that generalizes the Larmor formula.]

In Prob. 12.39(b) we calculated $(\alpha^\nu \alpha_\nu)$ in terms of *ordinary* velocity and acceleration:

$$\begin{aligned} \alpha^\nu \alpha_\nu &= \gamma^4 \left[a^2 + \frac{(\mathbf{v} \cdot \mathbf{a})^2}{(c^2 - v^2)} \right] = \gamma^6 \left[a^2 \gamma^{-2} + \frac{1}{c^2} (\mathbf{v} \cdot \mathbf{a})^2 \right] \\ &= \gamma^6 \left[a^2 \left(1 - \frac{v^2}{c^2} \right) + \frac{1}{c^2} (\mathbf{v} \cdot \mathbf{a})^2 \right] = \gamma^6 \left\{ a^2 - \frac{1}{c^2} [v^2 a^2 - (\mathbf{v} \cdot \mathbf{a})^2] \right\}. \end{aligned}$$

Now $\mathbf{v} \cdot \mathbf{a} = va \cos \theta$, where θ is the angle between $\mathbf{v}$ and $\mathbf{a}$, so:

$$v^2 a^2 - (\mathbf{v} \cdot \mathbf{a})^2 = v^2 a^2 (1 - \cos^2 \theta) = v^2 a^2 \sin^2 \theta = |\mathbf{v} \times \mathbf{a}|^2.$$

$$\alpha^\nu \alpha_\nu = \gamma^6 \left(a^2 - \left| \frac{\mathbf{v} \times \mathbf{a}}{c} \right|^2 \right).$$

$$\frac{dW}{dt} = \frac{\mu_0 q^2}{6\pi c} \gamma^6 \left(a^2 - \left| \frac{\mathbf{v} \times \mathbf{a}}{c} \right|^2 \right), \text{ which is Liénard's formula (Eq. 11.73).}$$

Problem 12.72

(a) It's inconsistent with the constraint $\eta_\mu K^\mu = 0$ (Prob. 12.39(d)).

(b) We want to find a 4-vector b^μ with the property that $(\frac{d\alpha^\mu}{d\tau} + b^\mu) \eta_\mu = 0$. How about $b^\mu = \kappa (\frac{d\alpha^\nu}{d\tau} \eta_\nu) \eta^\mu$? Then $(\frac{d\alpha^\nu}{d\tau} + b^\mu) \eta_\mu = \frac{d\alpha^\mu}{d\tau} \eta_\mu + \kappa \frac{d\alpha^\nu}{d\tau} \eta_\nu (\eta^\mu \eta_\mu)$. But $\eta^\mu \eta_\mu = -c^2$, so this becomes $(\frac{d\alpha^\mu}{d\tau} \eta_\mu) - c^2 \kappa (\frac{d\alpha^\nu}{d\tau} \eta_\nu)$, which *is* zero,

if we pick $\kappa = 1/c^2$. This suggests $K_{\text{rad}}^\mu = \frac{\mu_0 q^2}{6\pi c} \left(\frac{d\alpha^\mu}{d\tau} + \frac{1}{c^2} \frac{d\alpha^\nu}{d\tau} \eta_\nu \eta^\mu \right)$. Note that $\eta^\mu = (c, \mathbf{v})\gamma$, so the spatial

components of b^μ vanish in the nonrelativistic limit $v \ll c$, and hence this still reduces to the Abraham-Lorentz formula. [Incidentally, $\alpha^\nu \eta_\nu = 0 \Rightarrow \frac{d}{d\tau} (\alpha^\nu \eta_\nu) = 0 \Rightarrow \frac{d\alpha^\nu}{d\tau} \eta_\nu + \alpha^\nu \frac{d\eta_\nu}{d\tau} = 0$, so $\frac{d\alpha^\nu}{d\tau} \eta_\nu = -\alpha^\nu \alpha_\nu$, and hence b^μ can just as well be written $-\frac{1}{c^2} (\alpha^\nu \alpha_\nu) \eta^\mu$.]

Problem 12.73

Define the electric current 4-vector as before: $J_e^\mu = (c\rho_e, \mathbf{J}_e)$, and the magnetic current analogously: $J_m^\mu = (c\rho_m, \mathbf{J}_m)$. The fundamental laws are then

$$\partial_\nu F^{\mu\nu} = \mu_0 J_e^\mu, \quad \partial_\nu G^{\mu\nu} = \frac{\mu_0}{c} J_m^\mu, \quad K^\mu = \left(q_e F^{\mu\nu} + \frac{q_m}{c} G^{\mu\nu} \right) \eta_\nu.$$

The first of these reproduces $\nabla \cdot \mathbf{E} = (1/\epsilon_0)\rho_e$ and $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_e + \mu_0 \epsilon_0 \partial \mathbf{E} / \partial t$, just as before; the second yields $\nabla \cdot \mathbf{B} = (\mu_0/c)(c\rho_m) = \mu_0 \rho_m$ and $-(1/c)[\partial \mathbf{B} / \partial t + \nabla \times \mathbf{E}] = (\mu_0/c) \mathbf{J}_m$, or $\nabla \times \mathbf{E} = -\mu_0 \mathbf{J}_m - \partial \mathbf{B} / \partial t$ (generalizing Sec. 12.3.4). These are Maxwell's equations with magnetic charge (Eq. 7.44). The third says

$$K^1 = \frac{q_e}{\sqrt{1-u^2/c^2}} [\mathbf{E} + (\mathbf{u} \times \mathbf{B})]_x + \frac{q_m}{c} \left[\frac{-c}{\sqrt{1-u^2/c^2}} (-B_x) + \frac{u_y}{\sqrt{1-u^2/c^2}} \left(-\frac{E_z}{c} \right) + \frac{u_z}{\sqrt{1-u^2/c^2}} \left(\frac{E_y}{c} \right) \right],$$

$$\mathbf{K} = \frac{1}{\sqrt{1-u^2/c^2}} \left\{ q_e [\mathbf{E} + (\mathbf{u} \times \mathbf{B})] + q_m \left[\mathbf{B} - \frac{1}{c^2} (\mathbf{u} \times \mathbf{E}) \right] \right\}, \quad \text{or}$$

$$\mathbf{F} = q_e [\mathbf{E} + (\mathbf{u} \times \mathbf{B})] + q_m \left[\mathbf{B} - \frac{1}{c^2} (\mathbf{u} \times \mathbf{E}) \right],$$

which is the generalized Lorentz force law (Eq. 7.69).

4th ed	Same?	3rd ed
1.1	✓	1.1
1.2	✓	1.2
1.3	✓	1.3
1.4	✓	1.4
1.5	✓	1.5
1.6	✓	1.6
1.7	✓	1.7
1.8	✓	1.8
1.9	✓	1.9
1.10	✓	1.10
1.11	✓	1.11
1.12	✓	1.12
1.13	✓	1.13
1.14	✓	1.14
1.15	✓	1.15
1.16	✓	1.16
1.17	✓	1.17
1.18	✓	1.18
1.19	new	–
1.20	✓	1.19
1.21	✓	1.20
1.22	✓	1.21
1.23	✓	1.22
1.24	✓	1.23
1.25	✓	1.24
1.26	✓	1.25
1.27	✓	1.26
1.28	✓	1.27
1.29	✓	1.28
1.30	✓	1.29
1.31	✓	1.30
1.32	✓	1.31
1.33	✓	1.32
1.34	✓	1.33
1.35	✓	1.34
1.36	✓	1.35
1.37	✓	1.36
1.38	✓	1.37
1.39	✓	1.38
1.40	✓	1.39
1.41	✓	1.40
1.42	✓	1.41
1.43	✓	1.42
1.44	✓	1.43
1.45	✓	1.44

4th ed	Same?	3rd ed
1.46	✓	1.45
1.47	✓	1.46
1.48	✓	1.47
1.49	✓	1.48
1.50	✓	1.49
1.51	✓	1.50
1.52	✓	1.51
1.53	✓	1.52
1.54	✓	1.53
1.55	✓	1.54
1.56	✓	1.55
1.57	✓	1.56
1.58	✓	1.57
1.59	✓	1.58
1.60	✓	1.59
1.61	✓	1.60
1.62	✓	1.61
1.63	✓	1.62
1.64	new	–
2.1	✓	2.1
2.2	mod	2.2
2.3	✓	2.3
2.4	✓	2.4
2.5	✓	2.5
2.6	✓	2.6
2.7	✓	2.7
2.8	✓	2.8
2.9	✓	2.9
2.10	✓	2.10
2.11	✓	2.11
2.12	✓	2.12
2.13	✓	2.13
2.14	✓	2.14
2.15	mod	2.15
2.16	✓	2.16
2.17	✓	2.17
2.18	✓	2.18
2.19	✓	2.19
2.20	✓	2.20
2.21	✓	2.21
2.22	✓	2.22
2.23	✓	2.23
2.24	✓	2.24
2.25	✓	2.25
2.26	✓	2.26

4th ed	Same?	3rd ed
2.27	✓	2.27
2.28	✓	2.28
2.29	✓	2.29
2.30	✓	2.30
2.31	✓	2.31
2.32	new	–
2.33	new	–
2.34	✓	2.32
2.35	✓	2.33
2.36	✓	2.34
2.37	new	–
2.38	✓	2.35
2.39	✓	2.36
2.40	new	–
2.41	✓	2.37
2.42	✓	2.38
2.43	✓	2.39
2.44	✓	2.40
2.45	✓	2.41
2.46	new	–
2.47	✓	2.43
2.48	✓	2.44
2.49	✓	2.45
2.50	✓	2.46
2.51	new	–
2.52	✓	2.47
2.53	✓	2.48
2.54	mod	2.49
2.55	✓	2.50
2.56	✓	2.51
2.57	✓	2.52
2.58	new	–
2.59	new	–
2.60	new	–
2.61	new	–
3.1	✓	3.1
3.2	✓	3.2
3.3	✓	3.3
3.4	new	–
3.5	✓	3.4
3.6	✓	3.5
3.7	✓	3.6
3.8	✓	3.7
3.9	✓	3.8
3.10	✓	3.9

4th ed	Same?	3rd ed
3.11	✓	3.10
3.12	✓	3.11
3.13	✓	3.12
3.14	✓	3.13
3.15	✓	3.14
3.16	mod	3.15
3.17	✓	3.16
3.18	✓	3.17
3.19	✓	3.18
3.20	✓	3.19
3.21	✓	3.20
3.22	✓	3.21
3.23	✓	3.22
3.24	✓	3.23
3.25	✓	3.24
3.26	✓	3.25
3.27	✓	3.26
3.28	new	–
3.29	✓	3.27
3.30	✓	3.28
3.31	✓	3.29
3.32	✓	3.30
3.33	✓	3.31
3.34	✓	3.32
3.35	new	–
3.36	✓	3.33
3.37	new	–
3.38	new	–
3.39	✓	3.35
3.40	✓	3.36
3.41	new	–
3.42	new	–
3.43	✓	3.37
3.44	✓	3.38
3.45	✓	3.39
3.46	✓	3.40
3.47	✓	3.41
3.48	✓	3.42
3.49	new	–
3.50	✓	3.43
3.51	✓	3.44
3.52	mod	3.45
3.53	✓	3.46
3.54	✓	3.47
3.55	✓	3.48

4th ed	Same?	3rd ed
3.56	✓	3.49
3.57	new	–
3.58	new	–
4.1	✓	4.1
4.2	✓	4.2
4.3	✓	4.3
4.4	✓	4.4
4.5	✓	4.5
4.6	✓	4.6
4.7	✓	4.7
4.8	✓	4.8
4.9	✓	4.9
4.10	✓	4.10
4.11	✓	4.11
4.12	✓	4.12
4.13	✓	4.13
4.14	✓	4.14
4.15	✓	4.15
4.16	mod	4.16
4.17	✓	4.17
4.18	✓	4.18
4.19	✓	4.19
4.20	✓	4.20
4.21	✓	4.21
4.22	✓	4.22
4.23	✓	4.23
4.24	✓	4.24
4.25	✓	4.25
4.26	✓	4.26
4.27	✓	4.27
4.28	✓	4.28
4.29	✓	4.29
4.30	✓	4.30
4.31	new	–
4.32	new	–
4.33	✓	4.31
4.34	new	–
4.35	✓	4.32
4.36	✓	4.33
4.37	✓	4.34
4.38	✓	4.35
4.39	✓	4.36
4.40	✓	4.37
4.41	✓	4.38
4.42	✓	4.39

4th ed	Same?	3rd ed
4.43	✓	4.40
5.1	✓	5.1
5.2	✓	5.2
5.3	✓	5.3
5.4	✓	5.4
5.5	✓	5.5
5.6	✓	5.6
5.7	✓	5.7
5.8	✓	5.8
5.9	✓	5.9
5.10	✓	5.10
5.11	✓	5.11
5.12	new	–
5.13	✓	5.12
5.14	✓	5.13
5.15	✓	5.14
5.16	✓	5.15
5.17	✓	5.16
5.18	✓	5.17
5.19	✓	5.18
5.20	✓	5.19
5.21	✓	5.20
5.22	✓	5.21
5.23	✓	5.22
5.24	✓	5.23
5.25	✓	5.24
5.26	✓	5.25
5.27	✓	5.26
5.28	✓	5.27
5.29	✓	5.28
5.30	✓	5.29
5.31	✓	5.30
5.32	✓	5.31
5.33	✓	5.32
5.34	✓	5.33
5.35	✓	5.34
5.36	✓	5.37
5.37	mod	5.35, 5.36
5.38	✓	5.60
5.39	new	–
5.40	✓	5.38
5.41	✓	5.39
5.42	✓	5.40
5.43	✓	5.41
5.44	✓	5.42

4th ed	Same?	3rd ed
5.45	✓	5.43
5.46	✓	5.44
5.47	✓	5.46
5.48	mod	5.47
5.49	✓	5.48
5.50	✓	5.49
5.51	new	–
5.52	✓	5.50
5.53	✓	5.51
5.54	✓	5.52
5.55	✓	5.53
5.56	✓	5.54
5.57	✓	5.55
5.58	✓	5.56
5.59	✓	5.57
5.60	✓	5.58
5.61	✓	5.59
5.62	✓	5.61
6.1	✓	6.1
6.2	✓	6.2
6.3	✓	6.3
6.4	✓	6.4
6.5	✓	6.5
6.6	✓	6.6
6.7	✓	6.7
6.8	✓	6.8
6.9	✓	6.9
6.10	✓	6.10
6.11	✓	6.11
6.12	✓	6.12
6.13	✓	6.13
6.14	✓	6.14
6.15	✓	6.15
6.16	✓	6.16
6.17	✓	6.17
6.18	✓	6.18
6.19	✓	6.19
6.20	✓	6.20
6.21	✓	6.21
6.22	✓	6.22
6.23	✓	6.25
6.24	new	–
6.25	✓	6.23

4th ed	Same?	3rd ed
6.26	✓	6.24
6.27	✓	6.26
6.28	✓	6.27
6.29	✓	6.28
7.1	✓	7.1
7.2	✓	7.2
7.3	✓	7.3
7.4	✓	7.4
7.5	✓	7.5
7.6	✓	7.6
7.7	✓	7.7
7.8	✓	7.8
7.9	✓	7.9
7.10	✓	7.10
7.11	✓	7.11
7.12	✓	7.12
7.13	✓	7.13
7.14	✓	7.14
7.15	✓	7.15
7.16	✓	7.16
7.17	✓	7.17
7.18	✓	7.18
7.19	✓	7.19
7.20	new	–
7.21	new	–
7.22	✓	7.20
7.23	✓	7.21
7.24	✓	7.22
7.25	✓	7.23
7.26	✓	7.24
7.27	✓	7.25
7.28	✓	7.26
7.29	✓	7.27
7.30	✓	7.28
7.31	✓	7.29
7.32	✓	7.30
7.33	new	–
7.34	✓	7.31
7.35	✓	7.32
7.36	✓	7.33
7.37	✓	7.34
7.38	✓	7.35
7.39	✓	7.36

4th ed	Same?	3rd ed
7.40	✓	7.37
7.41	✓	7.39
7.42	✓	7.41
7.43	✓	7.57
7.44	✓	7.42
7.45	✓	7.43
7.46	✓	7.44
7.47	✓	7.45
7.48	mod	7.46
7.49	mod	7.47
7.50	✓	7.48
7.51	new	–
7.52	✓	7.49
7.53	✓	7.50
7.54	new	–
7.55	✓	7.51
7.56	✓	7.52
7.57	✓	7.53
7.58	✓	7.54
7.59	new	–
7.60	✓	7.55
7.61	✓	7.56
7.62	✓	7.58
7.63	✓	7.59
7.64	✓	7.60
8.1	✓	8.1
8.2	✓	8.2
8.3	✓	8.3
8.4	✓	8.4
8.5	new	–
8.6	mod	8.6
8.7	mod	8.5
8.8	✓	8.7
8.9	new	–
8.10	✓	8.8
8.11	new	–
8.12	new	–
8.13	✓	8.9
8.14	new	–
8.15	new	–
8.16	✓	8.10
8.17	✓	8.11
8.18	new	–

4th ed	Same?	3rd ed	4th ed	Same?	3rd ed	4th ed	Same?	3rd ed
8.19	✓	8.12	10.1	✓	10.1	11.13	new	–
8.20	new	–	10.2	✓	10.2	11.14	✓	11.14
8.21	mod	8.13	10.3	mod	10.3, 10.5	11.15	✓	11.15
8.22	✓	8.14	10.4	✓	10.4	11.16	✓	11.16
8.23	✓	8.15	10.5	✓	10.6	11.17	✓	11.17
8.24	new	–	10.6	✓	10.7	11.18	new	–
9.1	✓	9.1	10.7	new	–	11.19	✓	11.19
9.2	✓	9.2	10.8	new	–	11.20	mod	11.20
9.3	✓	9.3	10.9	new	–	11.21	new	–
9.4	✓	9.4	10.10	✓	10.8	11.22	✓	11.21
9.5	✓	9.5	10.11	✓	10.9	11.23	✓	11.22
9.6	✓	9.6	10.12	✓	10.10	11.24	✓	11.11
9.7	✓	9.7	10.13	✓	10.11	11.25	✓	11.23
9.8	✓	9.8	10.14	✓	10.12	11.26	new	–
9.9	✓	9.9	10.15	✓	10.13	11.27	new	–
9.10	✓	9.10	10.16	✓	10.14	11.28	✓	11.24
9.11	new	–	10.17	✓	10.15	11.29	✓	11.26
9.12	✓	9.11	10.18	✓	10.16	11.30	✓	11.27
9.13	✓	9.12	10.19	✓	10.17	11.31	✓	11.28
9.14	✓	9.13	10.20	✓	10.18	11.32	✓	11.29
9.15	✓	9.14	10.21	mod	12.43a	11.33	✓	11.30
9.16	✓	9.15	10.22	✓	10.19	11.34	✓	11.31
9.17	✓	9.16	10.23	✓	10.20	11.35	new	–
9.18	✓	9.17	10.24	✓	10.21	12.1	✓	12.1
9.19	✓	9.18	10.25	✓	10.22	12.2	✓	12.2
9.20	✓	9.19	10.26	new	–	12.3	✓	12.3
9.21	✓	9.20	10.27	✓	10.23	12.4	✓	12.4
9.22	✓	9.21	10.28	✓	10.24	12.5	✓	12.5
9.23	✓	9.22	10.29	new	–	12.6	✓	12.6
9.24	✓	9.23	10.30	new	–	12.7	✓	12.7
9.25	✓	9.24	10.31	✓	10.25	12.8	✓	12.8
9.26	new	–	10.32	✓	10.26	12.9	✓	12.9
9.27	✓	9.26	10.33	new	–	12.10	✓	12.10
9.28	✓	9.27	10.34	new	–	12.11	✓	12.11
9.29	✓	9.28	11.1	✓	11.1	12.12	✓	12.12
9.30	✓	9.29	11.2	✓	11.2	12.13	✓	12.13
9.31	✓	9.30	11.3	✓	11.3	12.14	mod	12.14
9.32	✓	9.31	11.4	✓	11.4	12.15	✓	12.15
9.33	✓	9.32	11.5	✓	11.5	12.16	✓	12.16
9.34	new	–	11.6	✓	11.6	12.17	✓	12.17
9.35	✓	9.33	11.7	✓	11.7	12.18	✓	12.18
9.36	✓	9.34	11.8	new	–	12.19	✓	11.19
9.37	✓	9.35	11.9	✓	11.8	12.20	✓	12.20
9.38	✓	9.36	11.10	✓	11.9	12.21	✓	12.21
9.39	✓	9.37	11.11	✓	11.12	12.22	✓	12.22
9.40	✓	9.38	11.12	✓	11.10	12.23	✓	12.23

4th ed	Same?	3rd ed
12.24	✓	12.24
12.25	✓	12.25
12.26	mod	12.26
12.27	new	—
12.28	✓	12.27
12.29	✓	12.28
12.30	✓	12.29
12.31	✓	12.30
12.32	✓	12.31
12.33	✓	12.32
12.34	✓	12.33
12.35	✓	12.34
12.36	✓	12.35
12.37	✓	12.36
12.38	✓	12.37
12.39	✓	12.38
12.40	✓	12.39
12.41	✓	12.40
12.42	✓	12.41
12.43	✓	12.42
12.44	new	—
12.45	✓	12.44
12.46	✓	12.45
12.47	✓	12.46
12.48	✓	12.47

4th ed	Same?	3rd ed
12.49	✓	12.48
12.50	✓	12.49
12.51	✓	12.50
12.52	✓	12.51
12.53	✓	12.52
12.54	✓	12.53
12.55	✓	12.54
12.56	✓	12.55
12.57	✓	12.56
12.58	new	—
12.59	✓	12.57
12.60	✓	12.58
12.61	✓	12.59
12.62	✓	12.60
12.63	✓	12.61
12.64	✓	12.62
12.65	✓	12.63
12.66	✓	12.64
12.67	✓	12.65
12.68	✓	12.66
12.69	✓	12.67
12.70	✓	12.68
12.71	✓	12.69
12.72	✓	12.70
12.73	✓	12.71